

Hysteresis Indices and Beta of IBW sites: Curry, Silverado, and Lake Marguerite

2026-03-26

Aims to plot hysteresis per analyte per storm per site

Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.
+HI = clockwise (source limitation or nearstream source)
-HI = counterclockwise (transport limitation or distal source)
- *Beta* is the slope of the CQ relationship on the rising limb of the storm
CIs on beta are bootstrapped 95% CI

NOTES

- Lake Marguerite does not have a lot of chems to work with. Do we know if there are more chems than the 624_runoff_chems.csv from the dropbox?
- Similarly, there is a gap of data in Silverado 2014-2018, was there less sampling in that time or is the data elsewhere?
- individual analyte outliers > 3sd above mean concentration per storm were removed for hysteresis and beta calculations, but remain for individual storm plotting at the bottom of the report
- Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

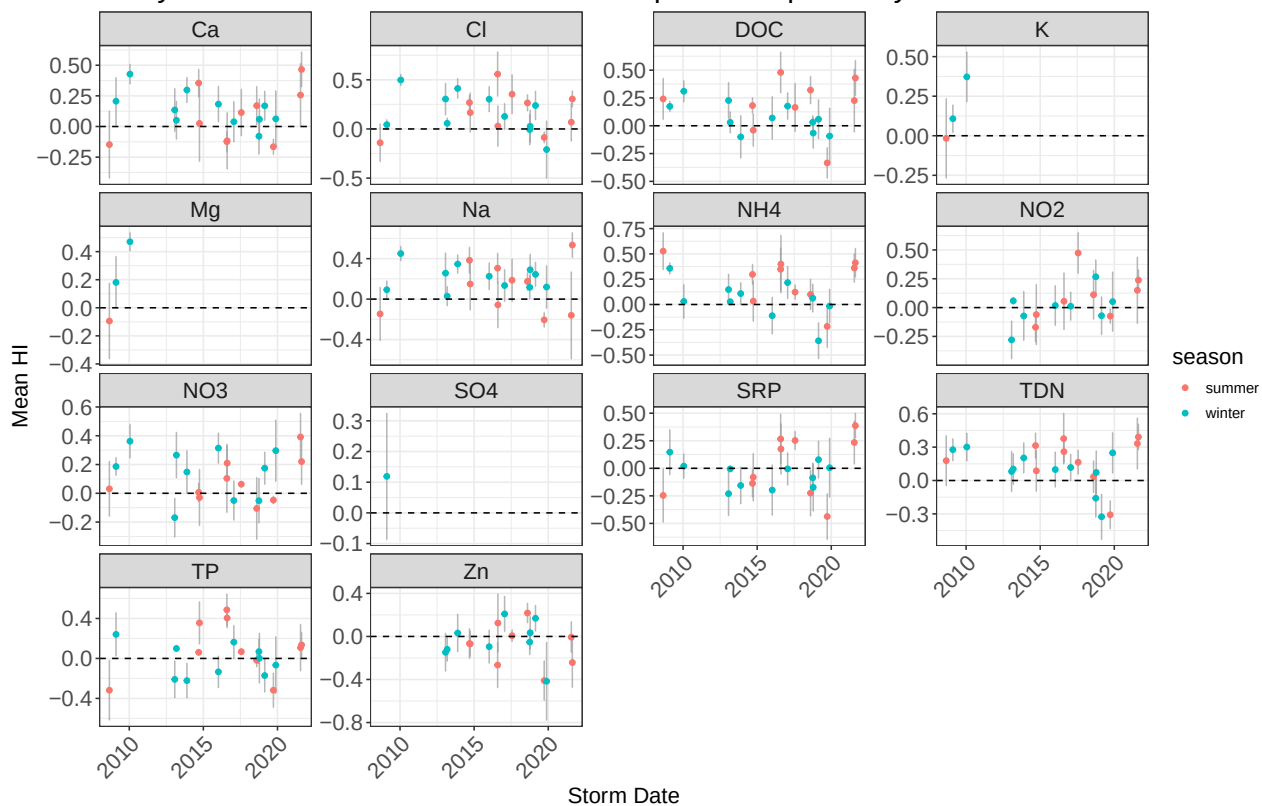
Data munging/cleaning

Workflow before this rmd, in order:

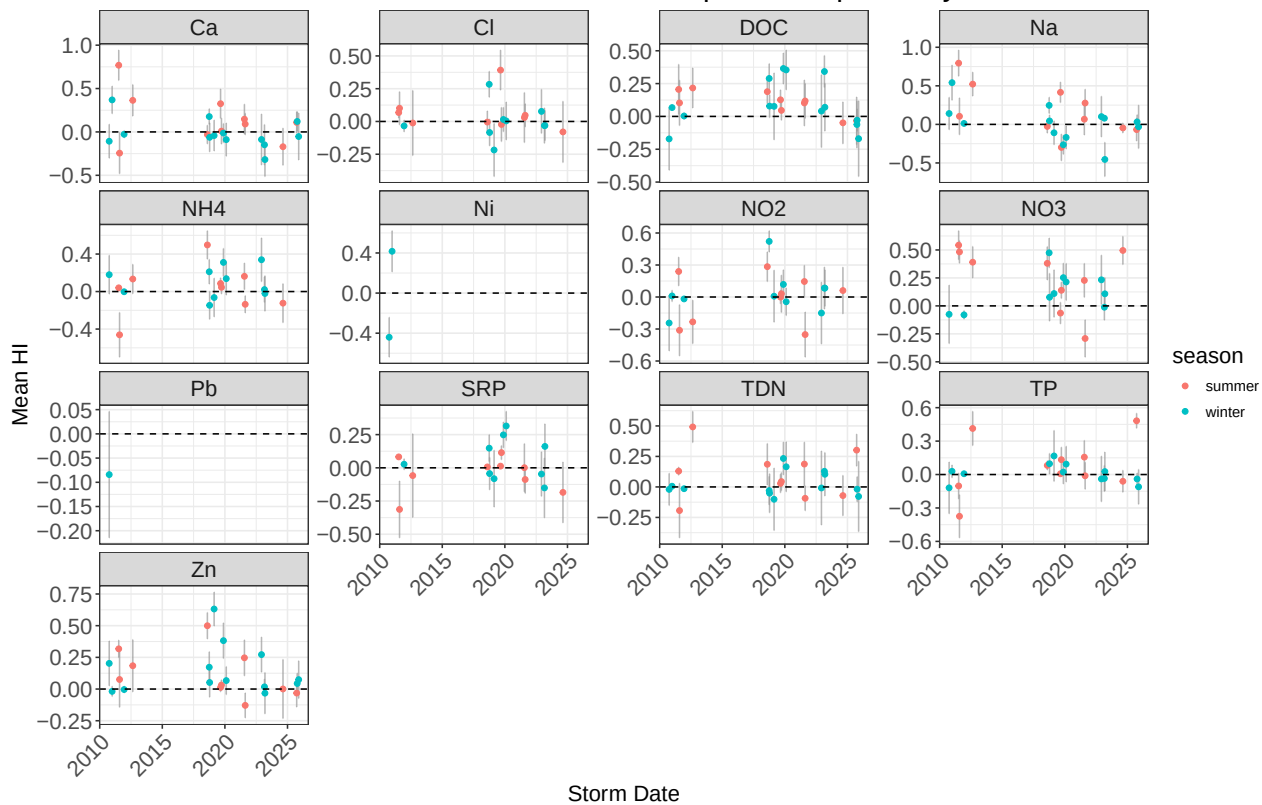
- q_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- precipitation.R: import precip data and merge with Q to calculate pre-storm precip metrics
- merge_Q_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis_calc_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data. also calculates beta

Plotting Hysteresis

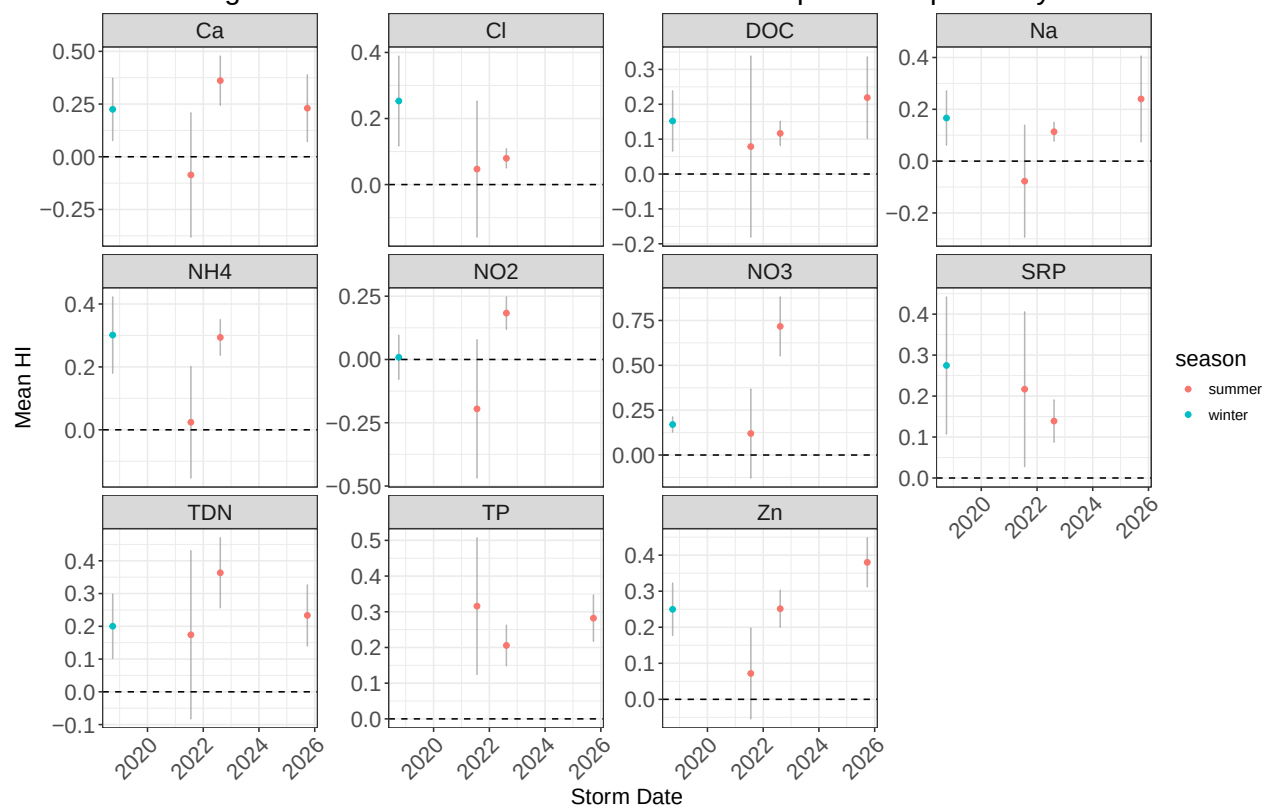
Curry – mean HI with confidence intervals per storm per analyte



Silverado – mean HI with confidence intervals per storm per analyte



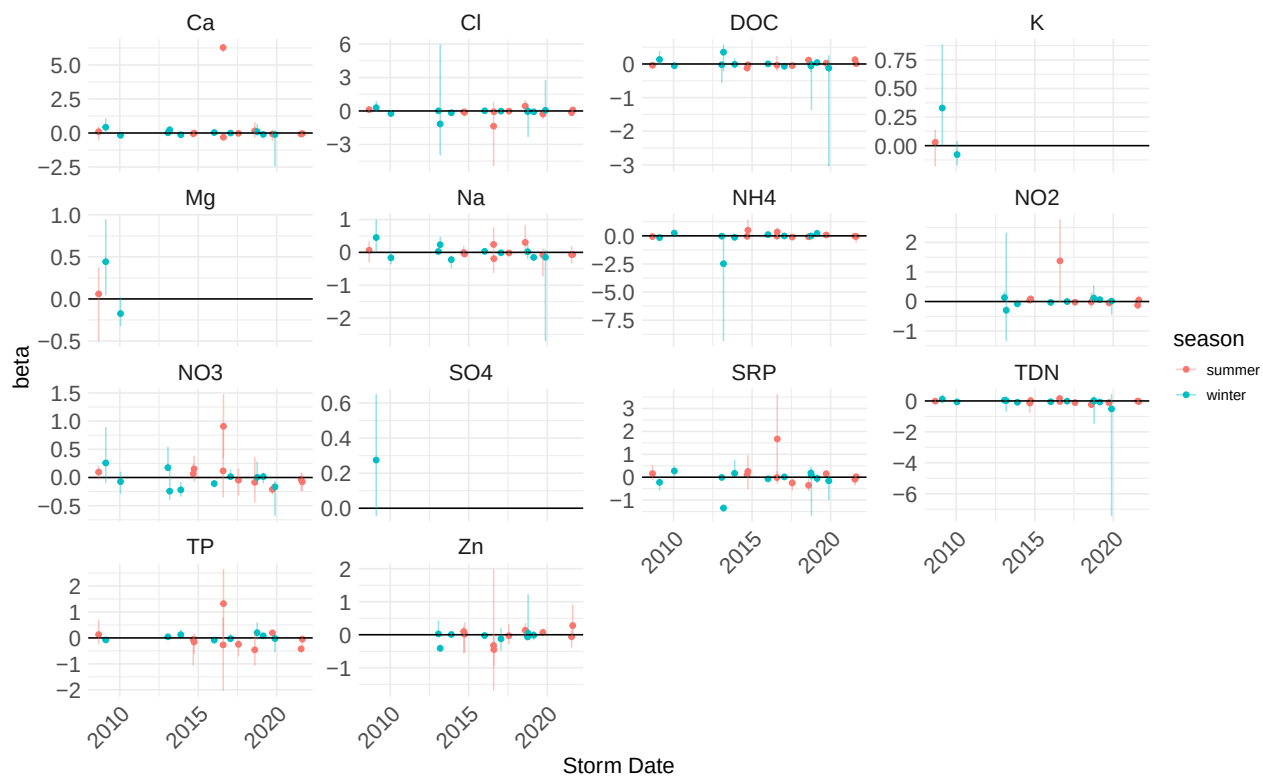
Lake Marguerite – mean HI with confidence intervals per storm per analyte



Plotting Beta

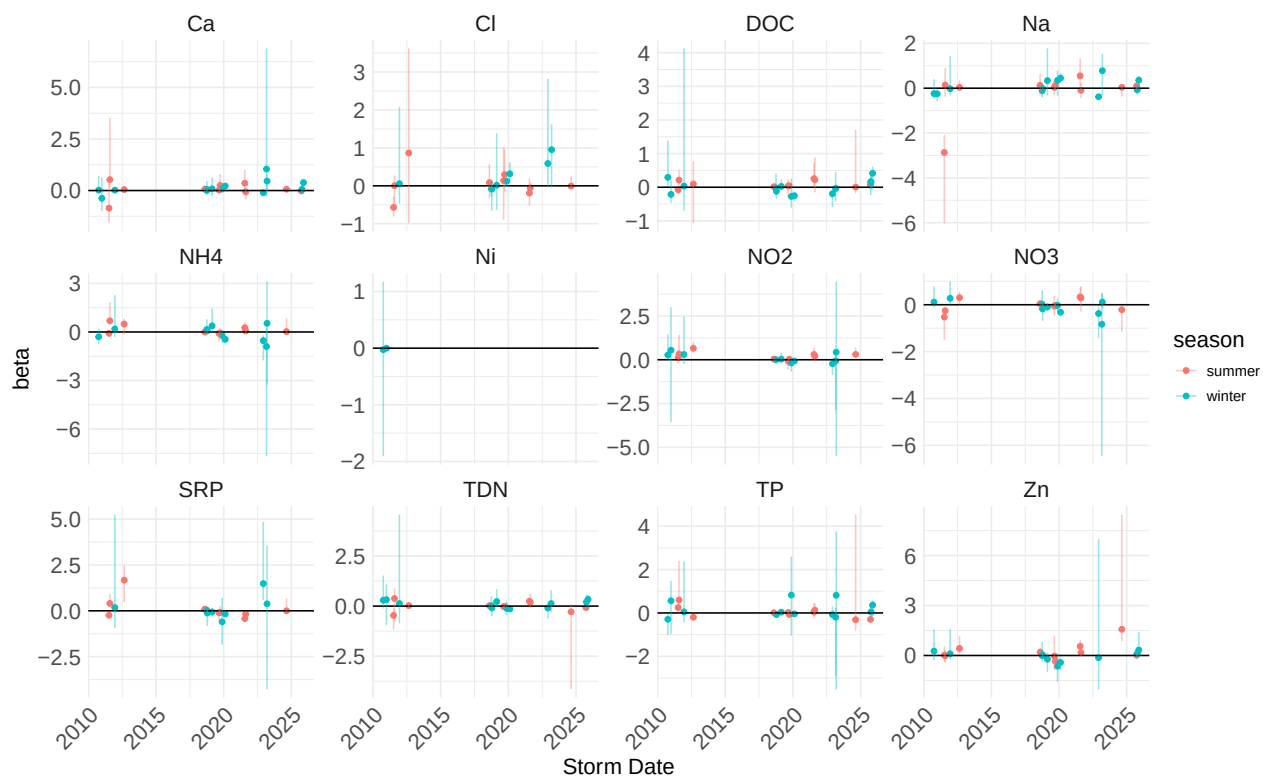
Curry Beta per storm per analyte

Bars are 95% confidence intervals



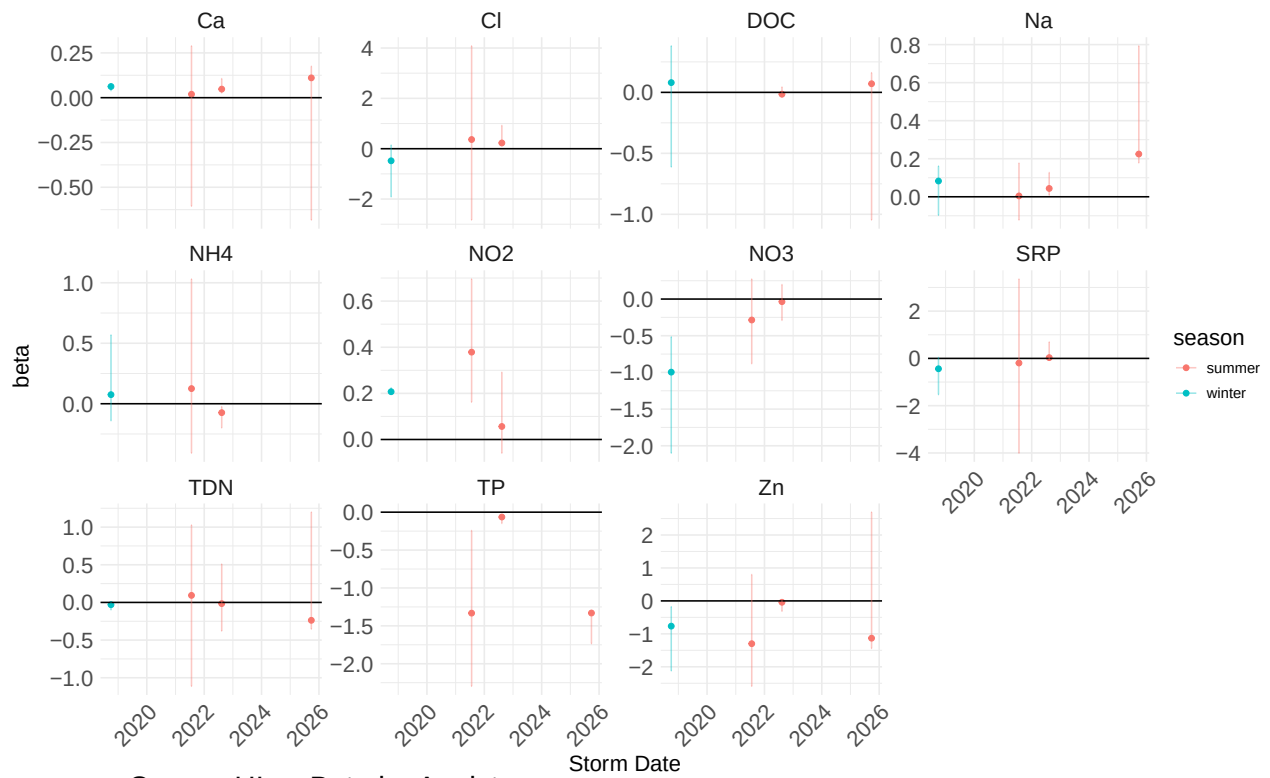
Silverado Beta per storm per analyte

Bars are 95% confidence intervals



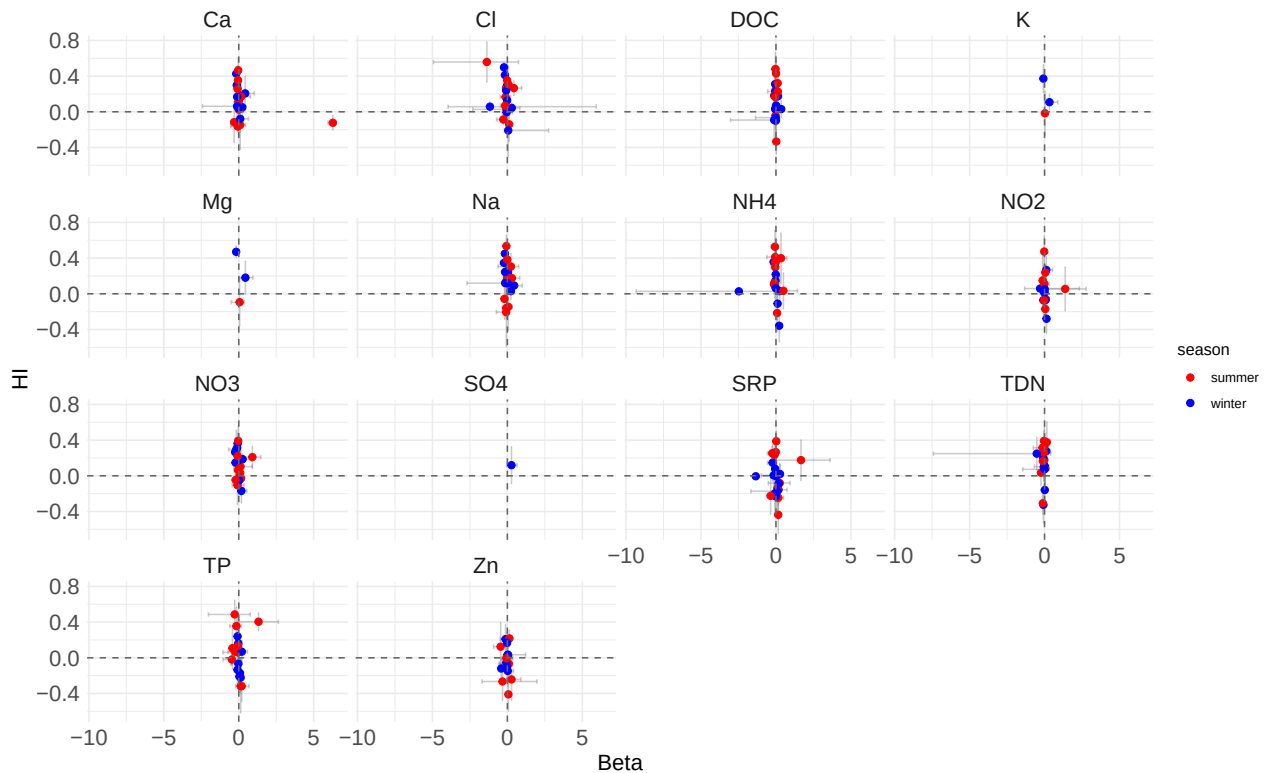
Lake Marguerite Beta per storm per analyte

Bars are 95% confidence intervals



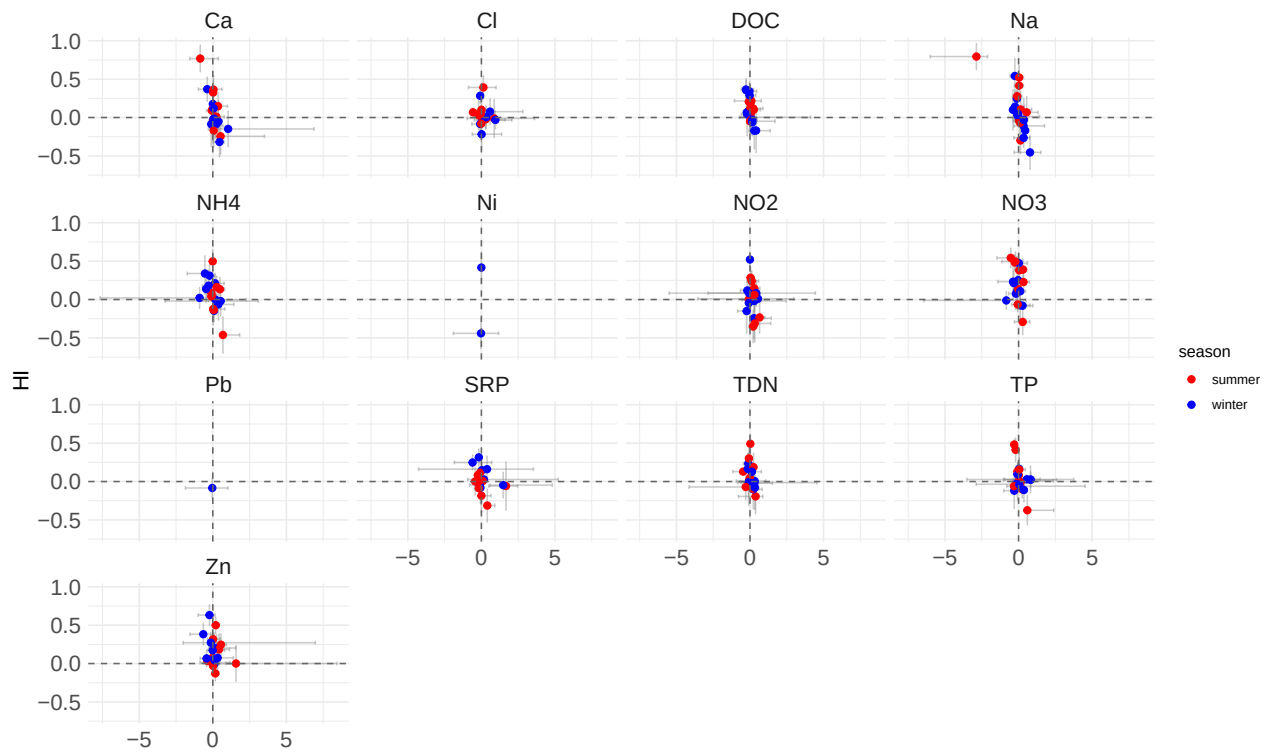
Curry – HI vs Beta by Analyte

Each dot is an individual storm



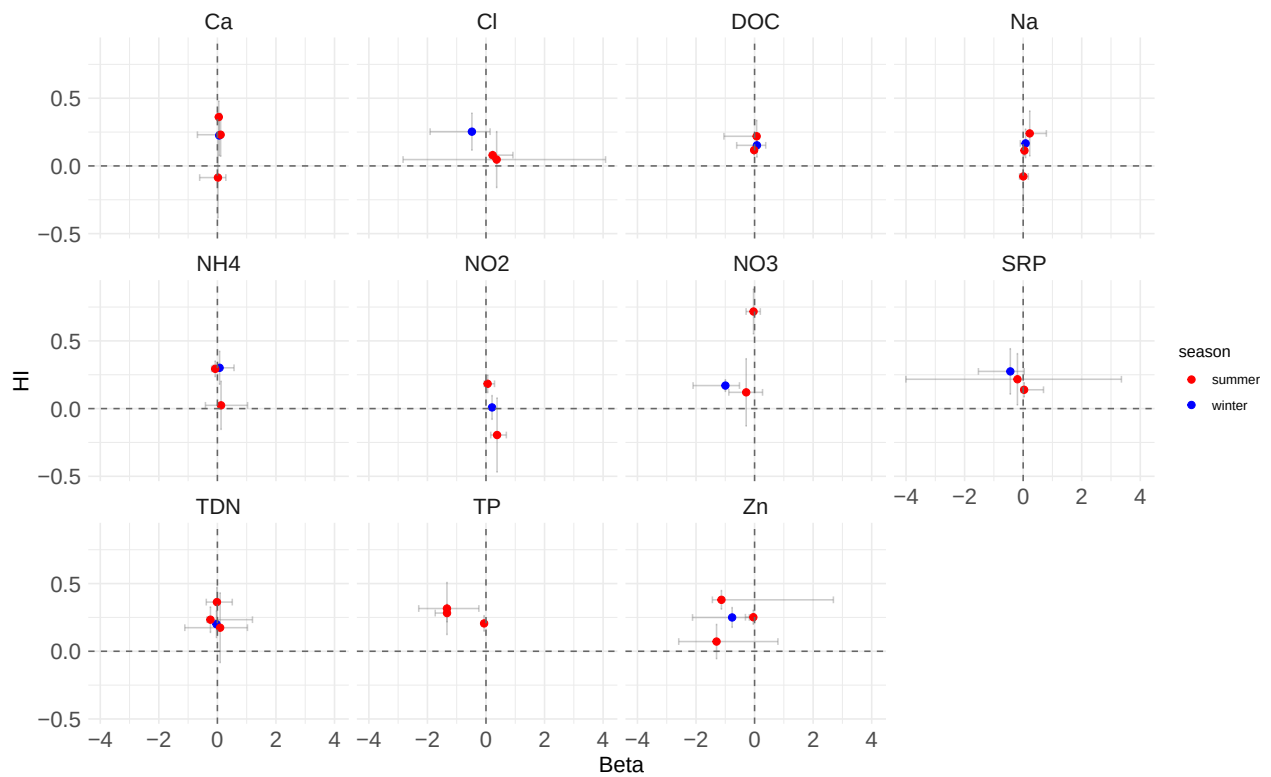
Silverado – HI vs Beta by Analyte

Each dot is an individual storm

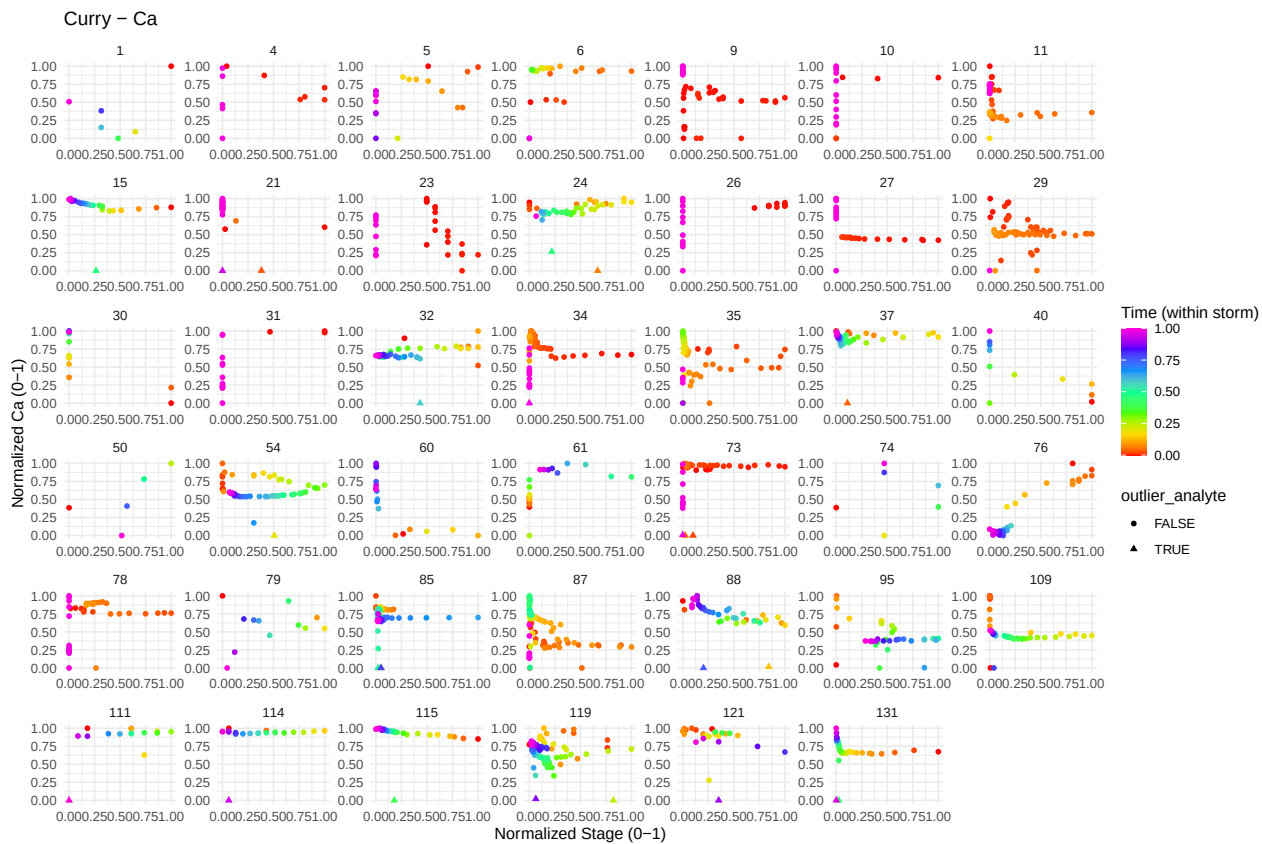


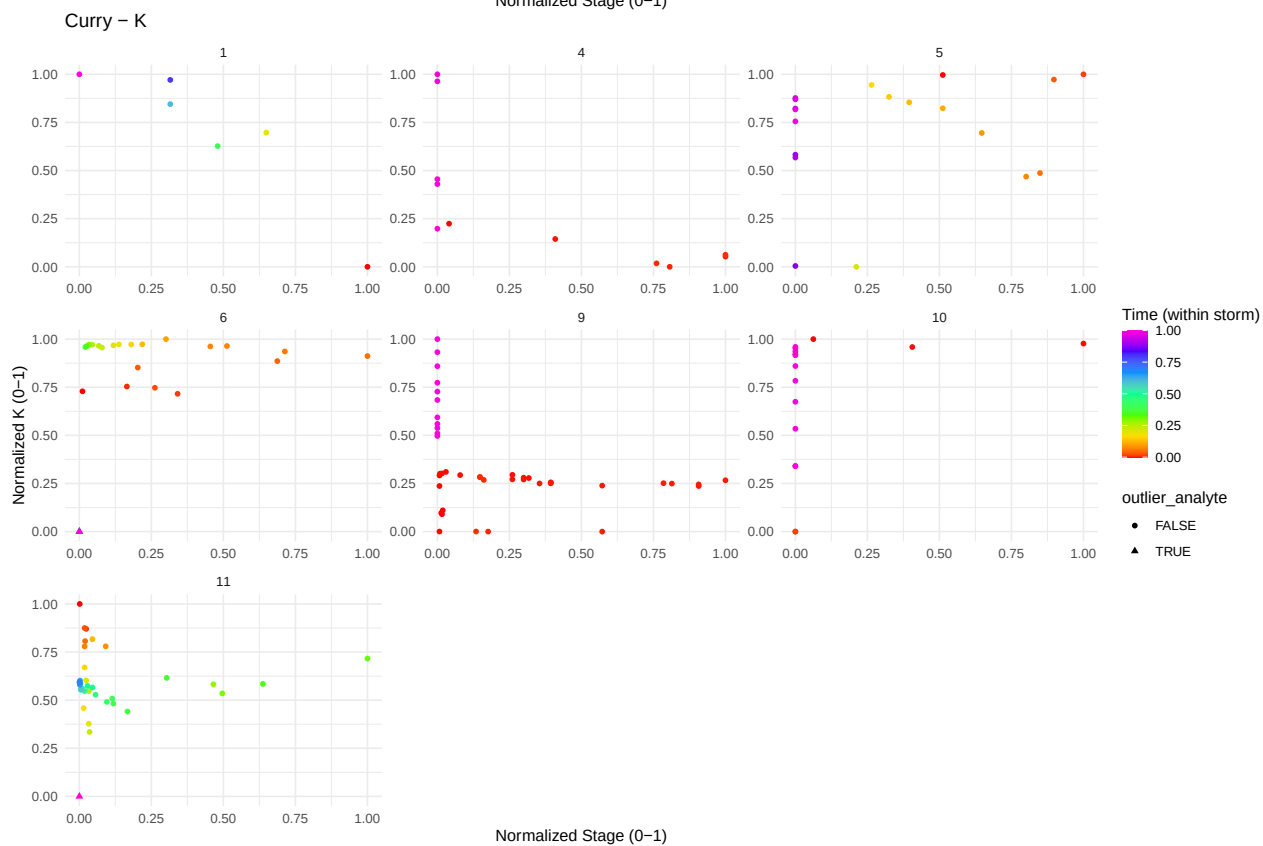
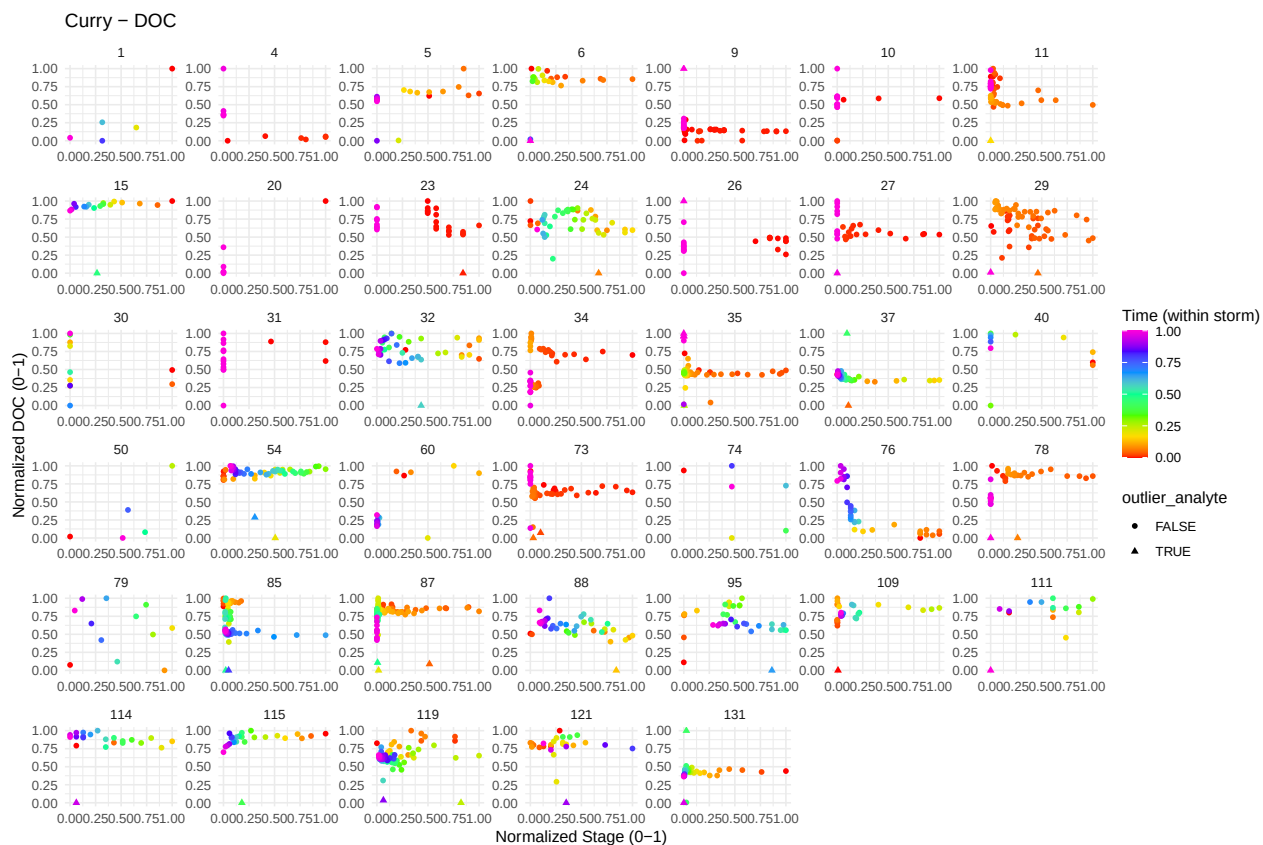
Lake Margarita – HI vs Beta by Analyte

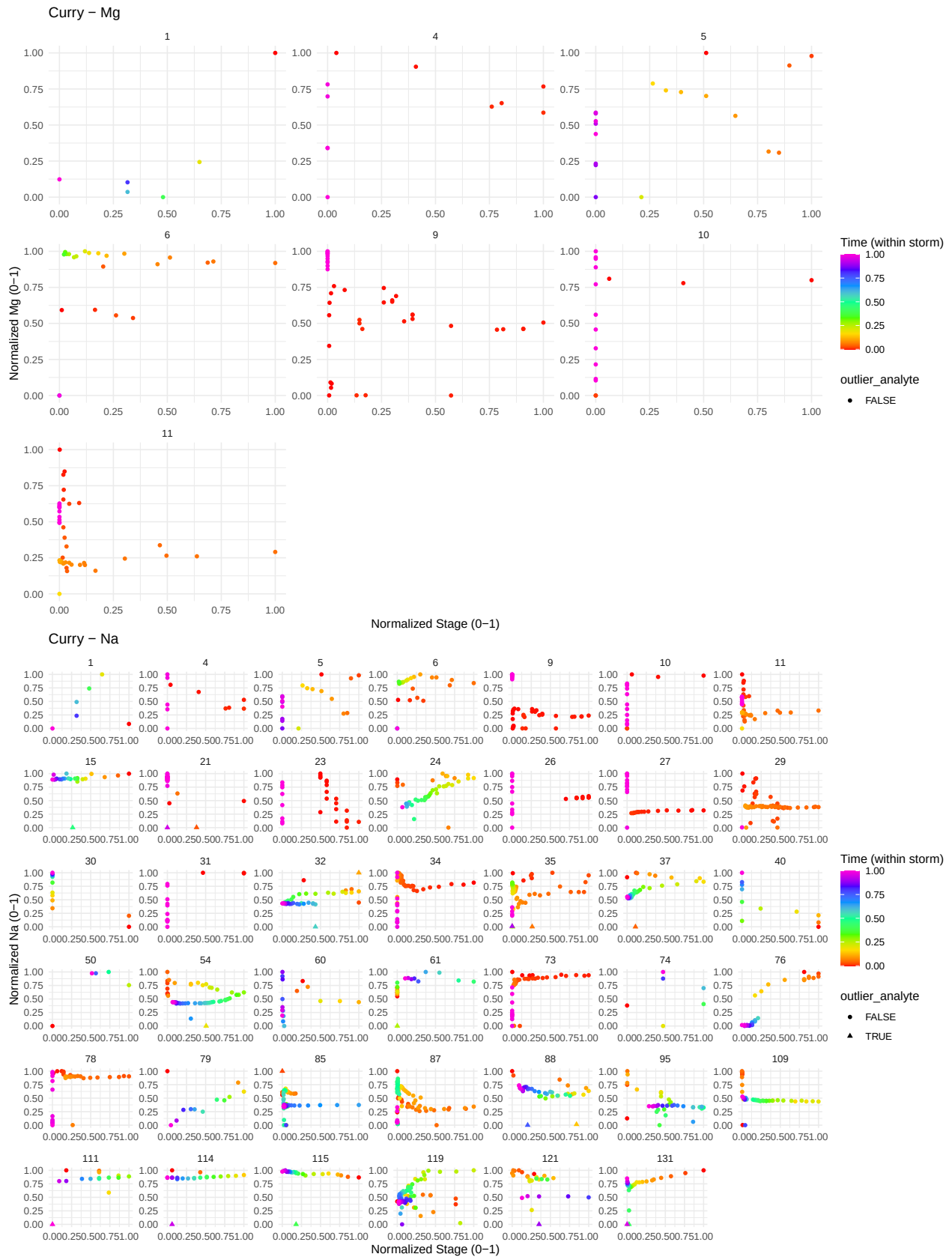
Each dot is an individual storm

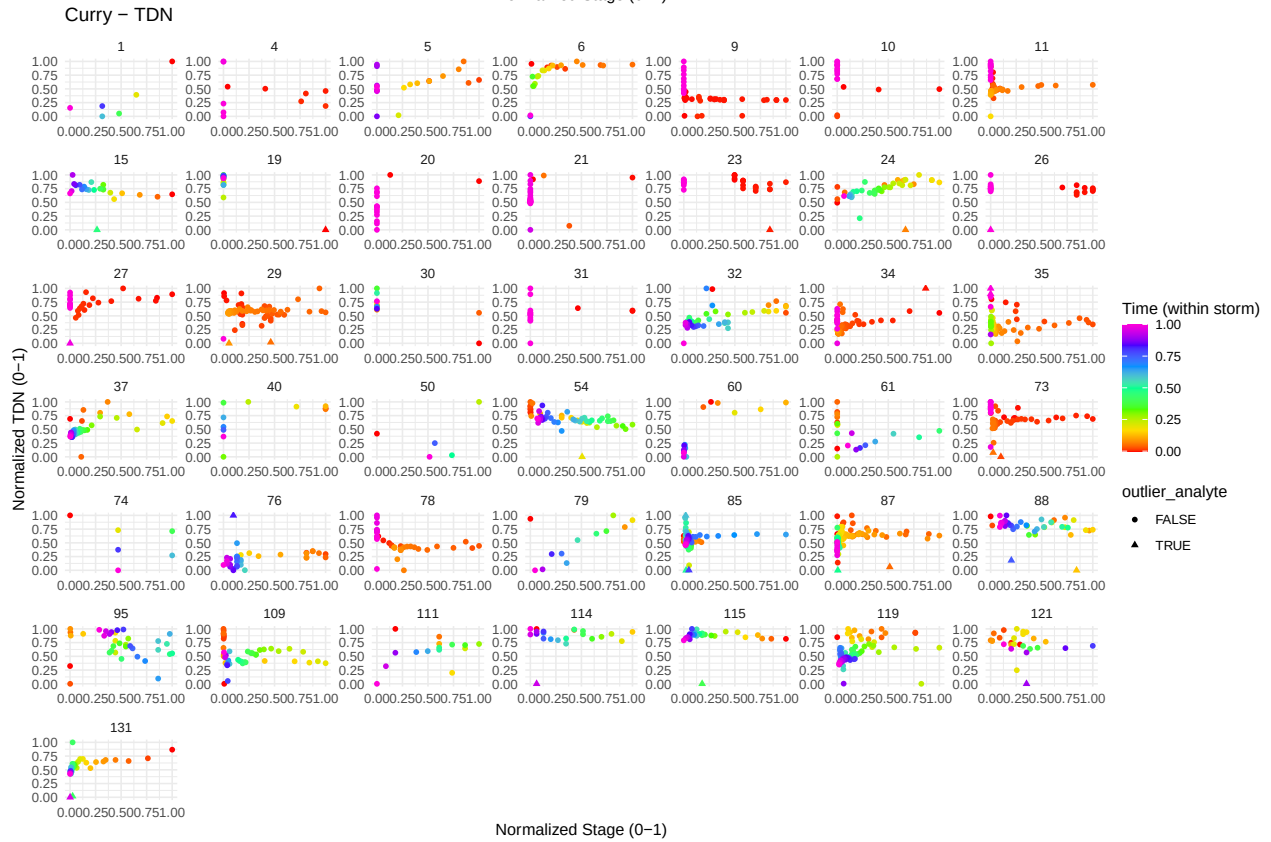
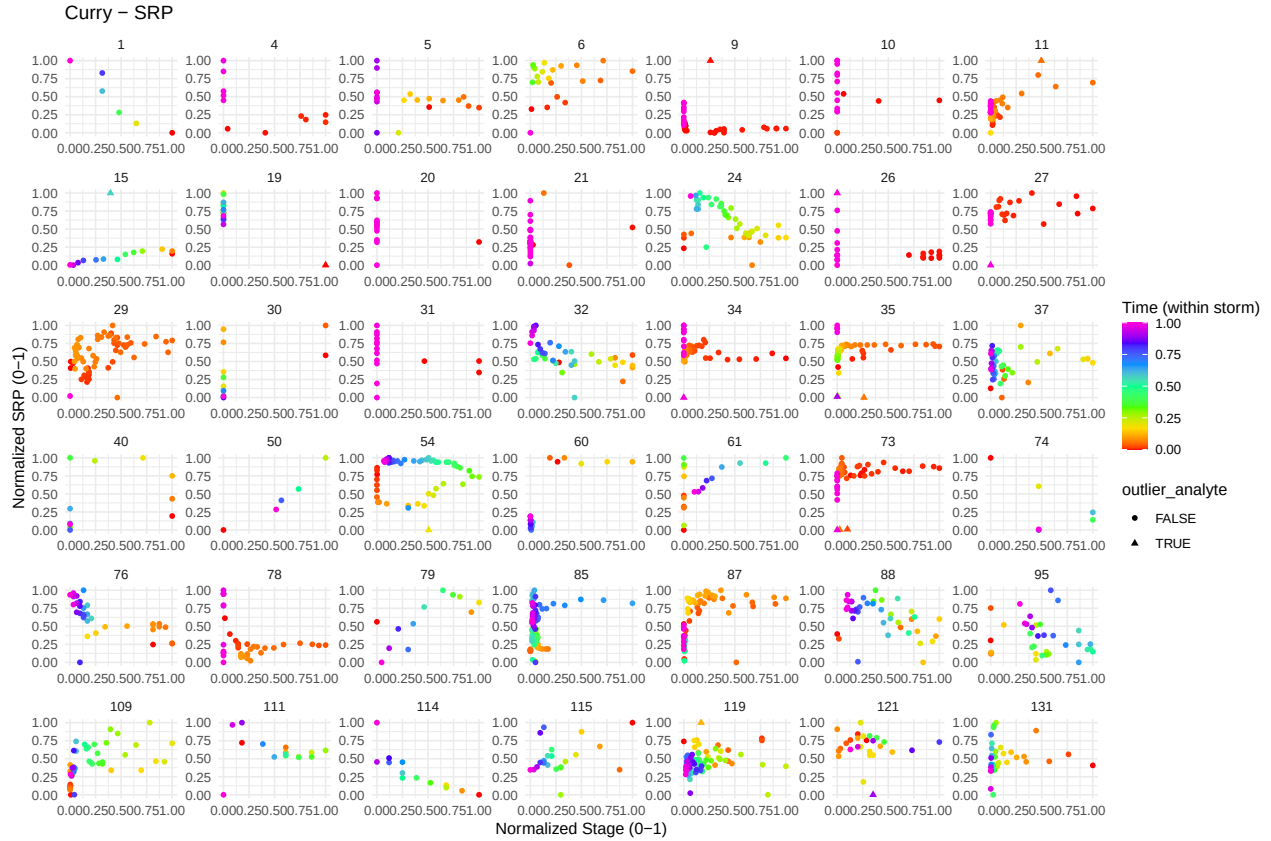


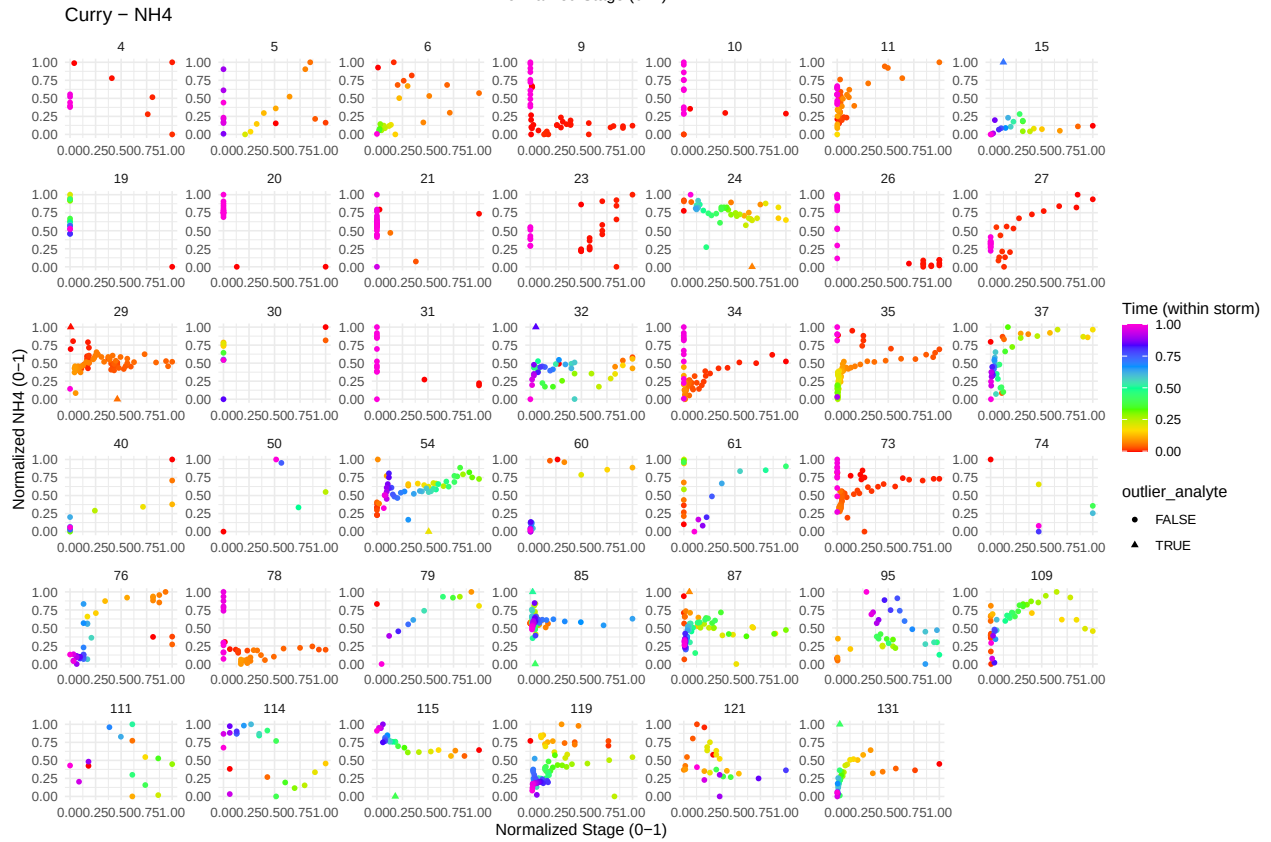
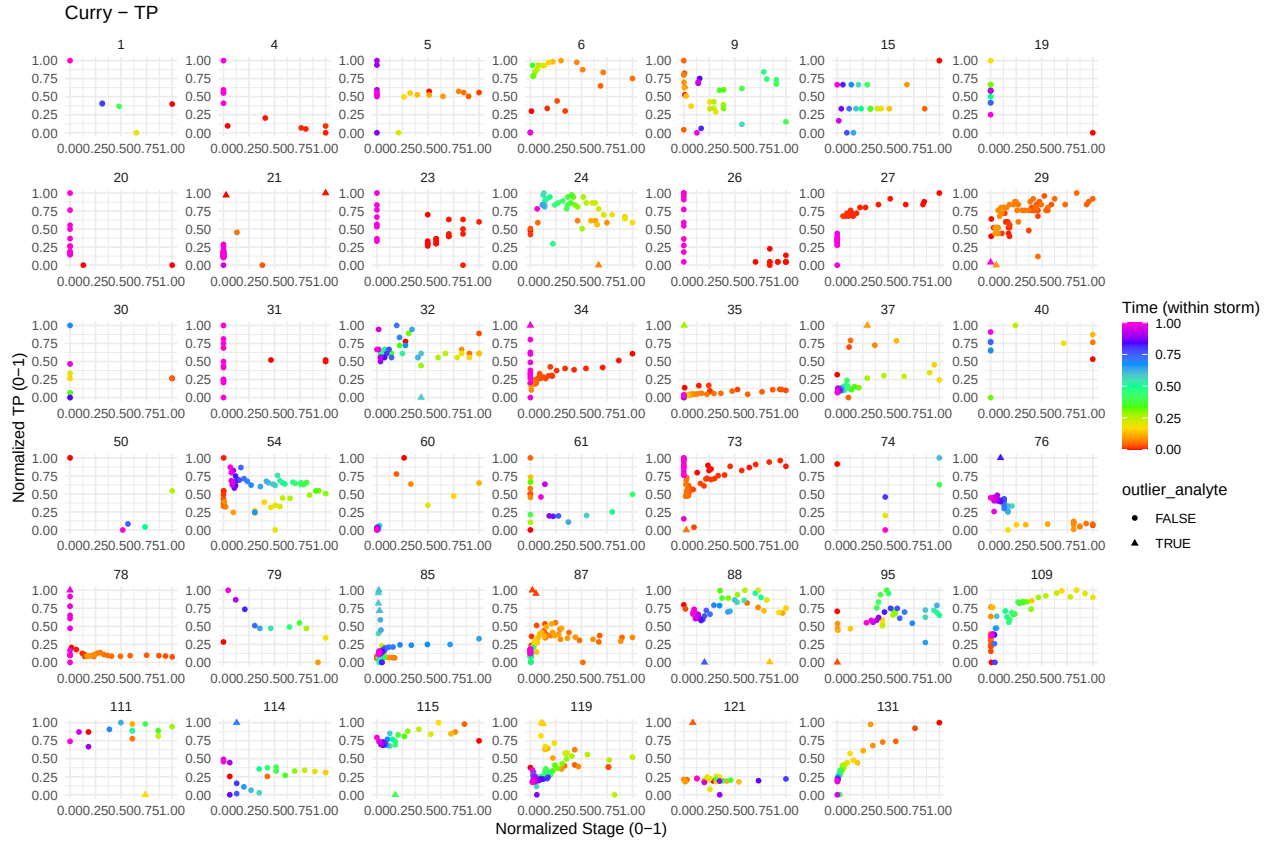
Plotting CQ

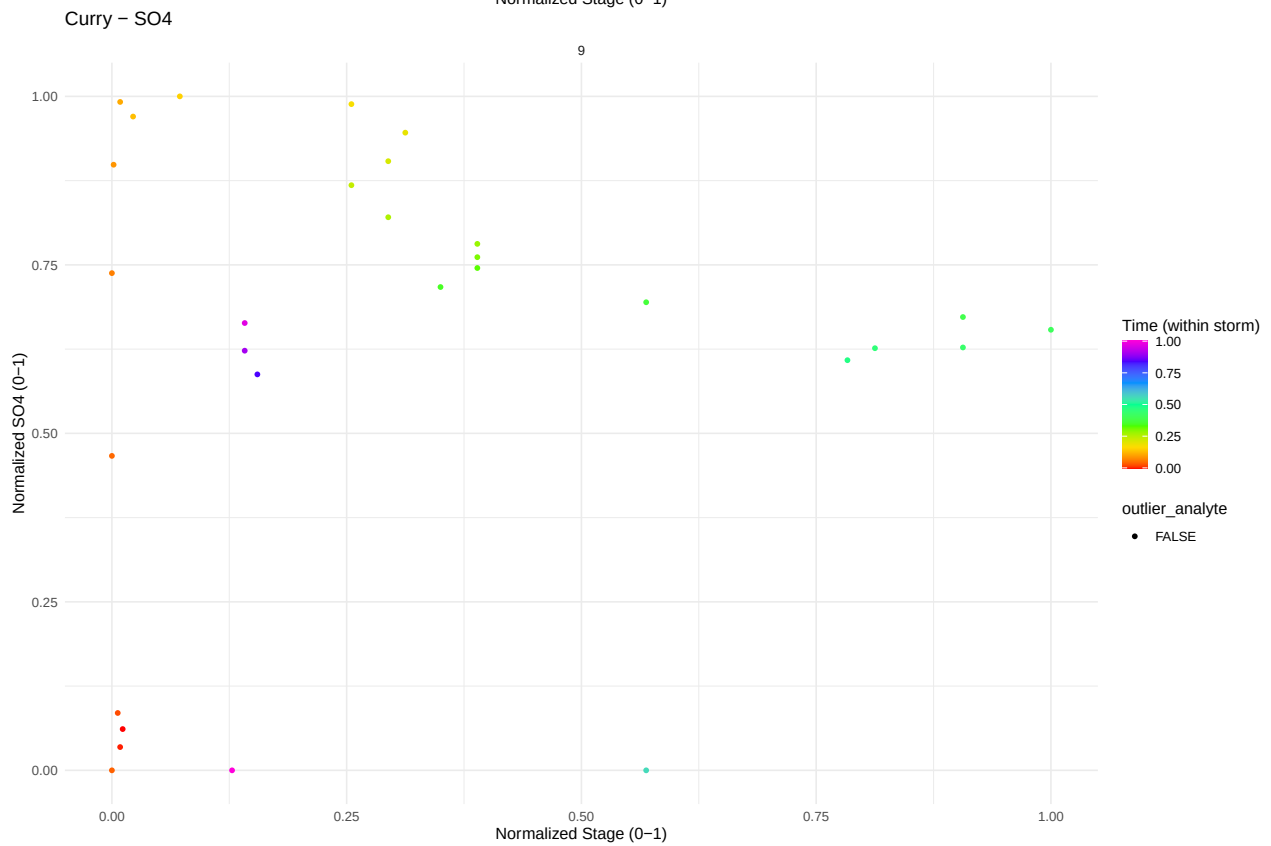
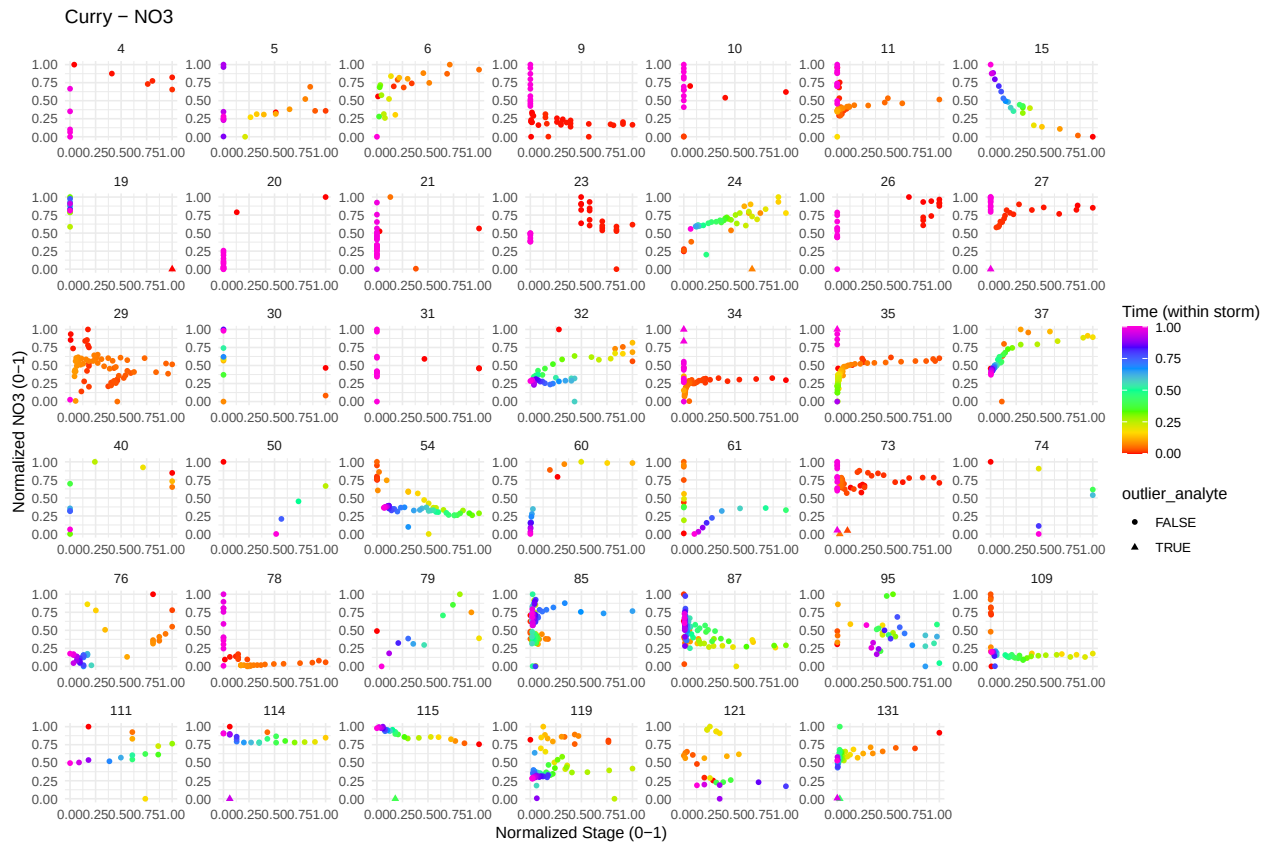


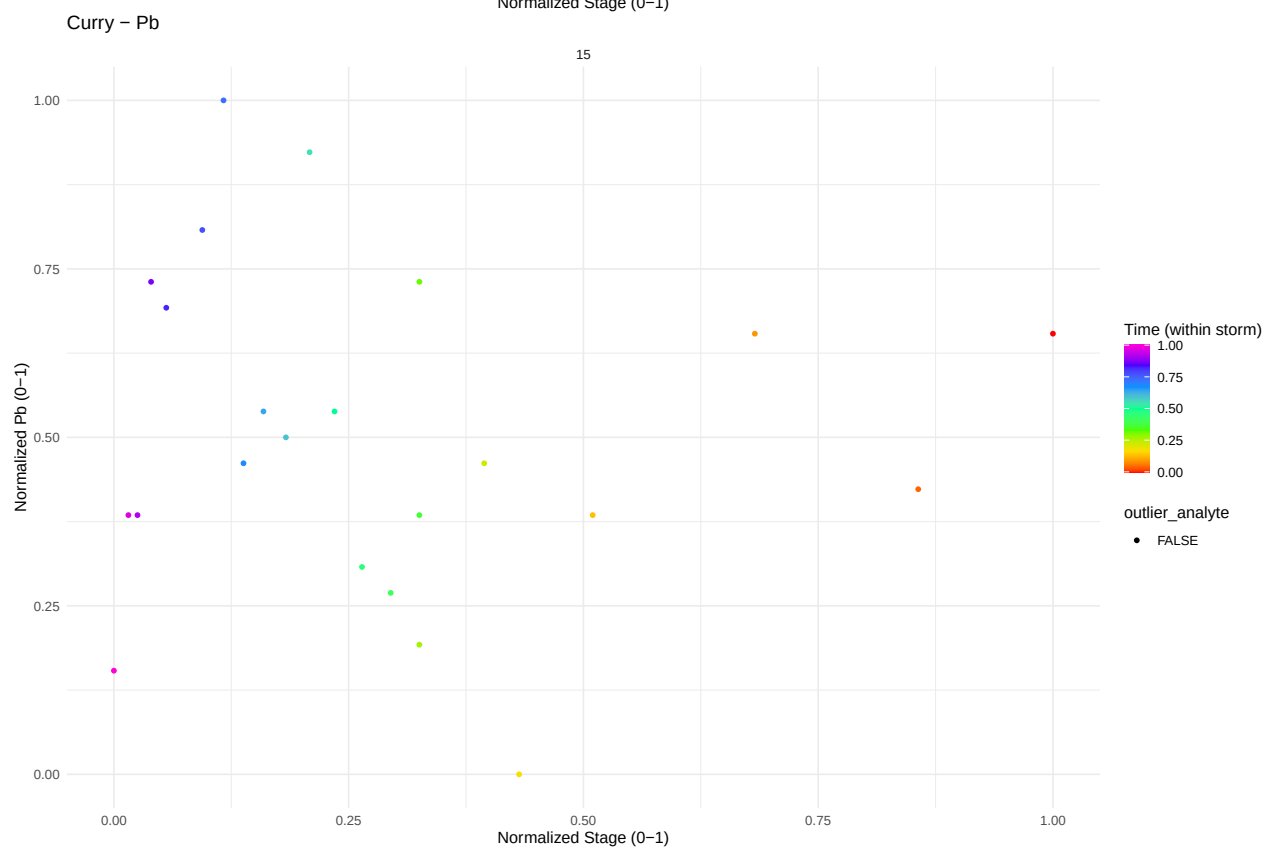
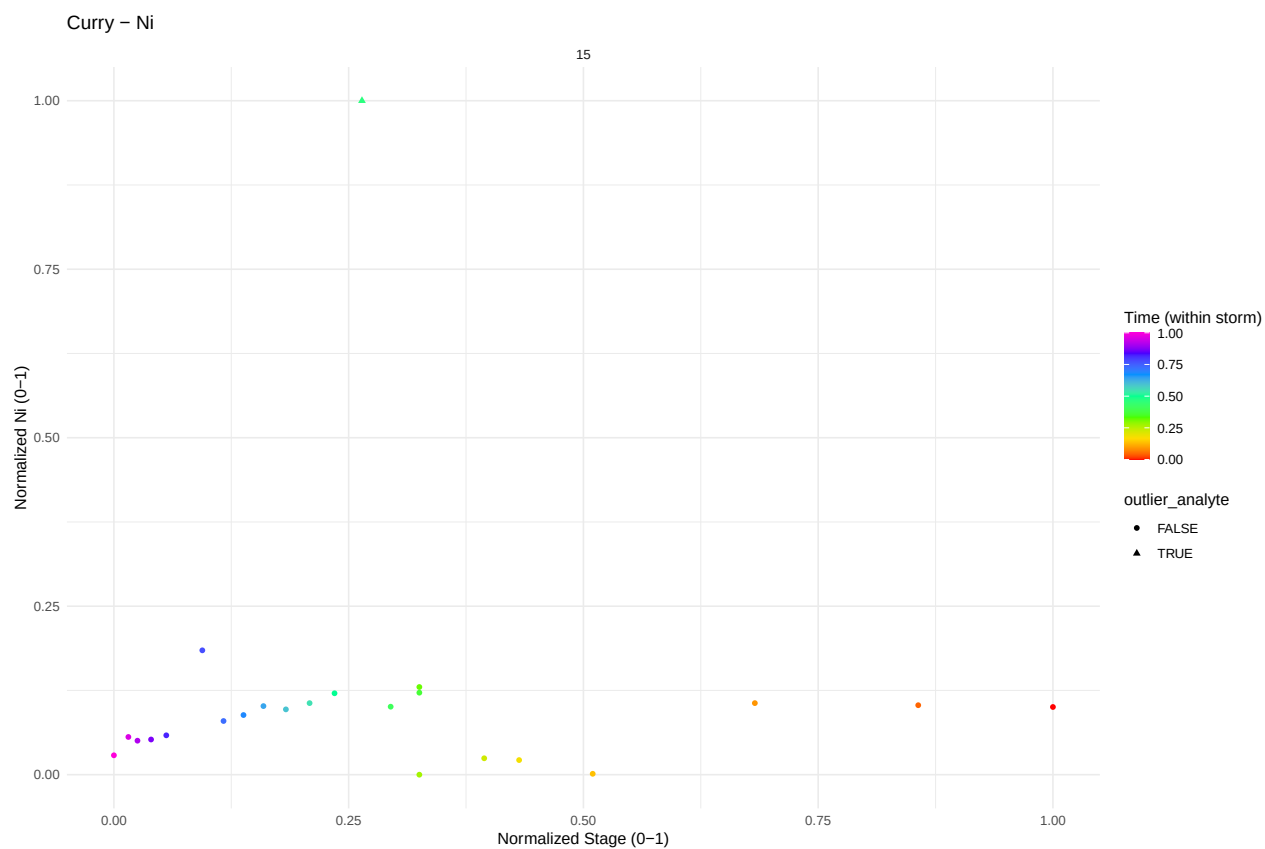


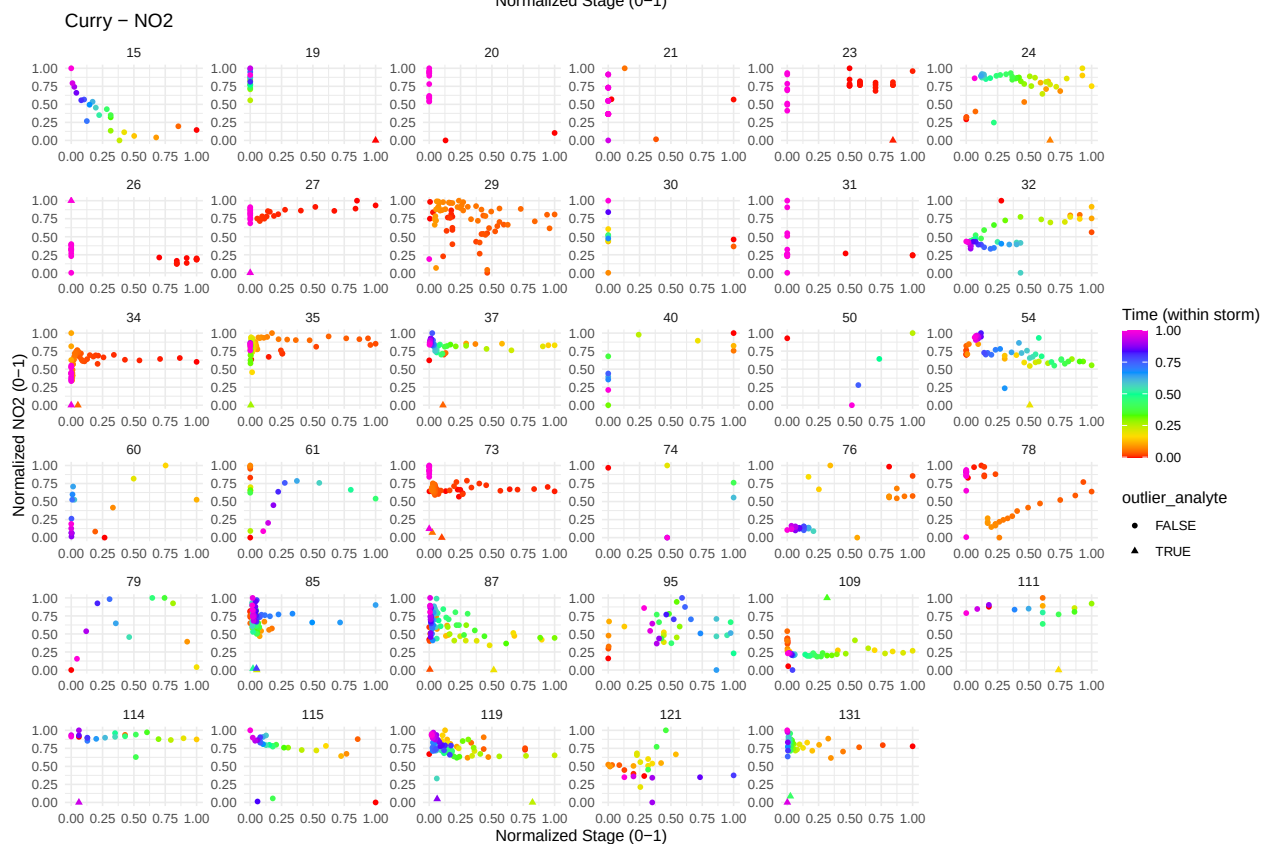
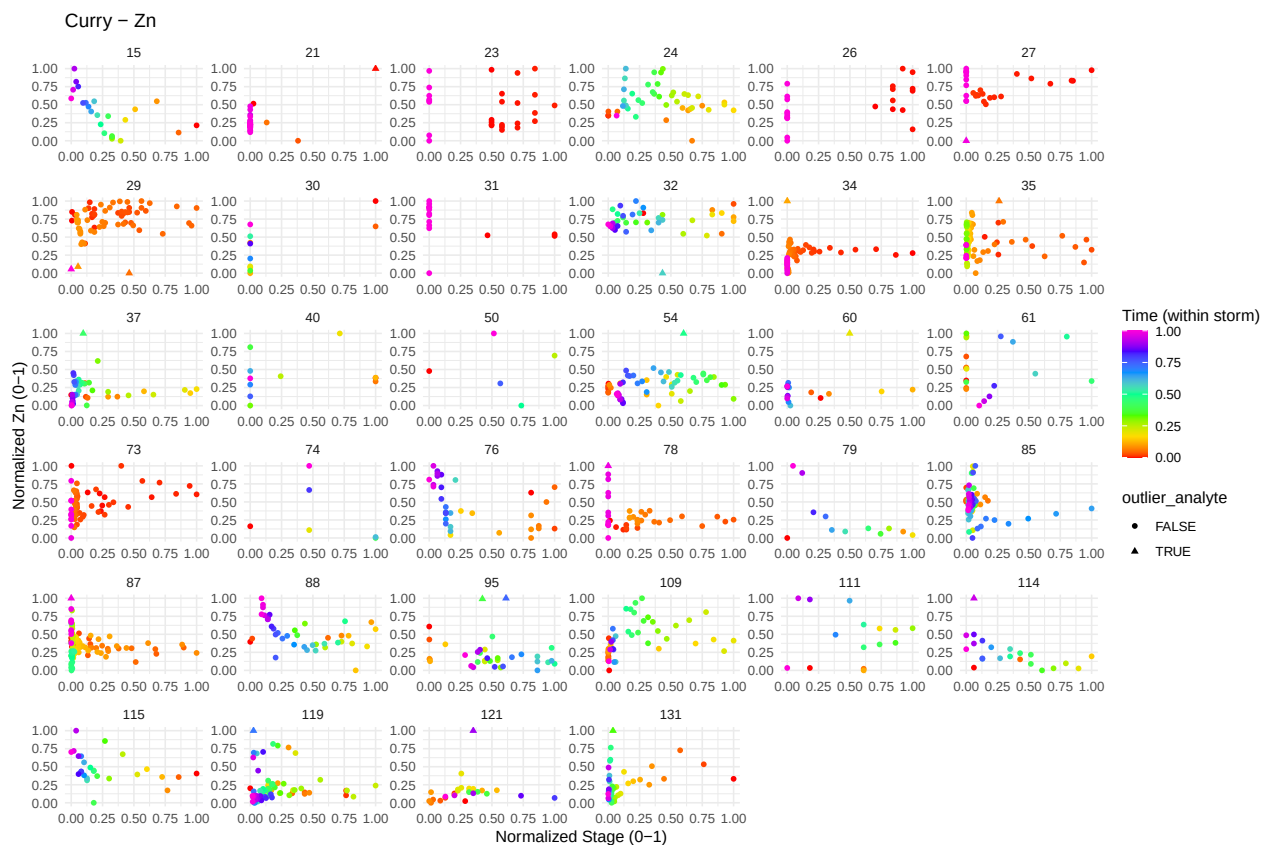


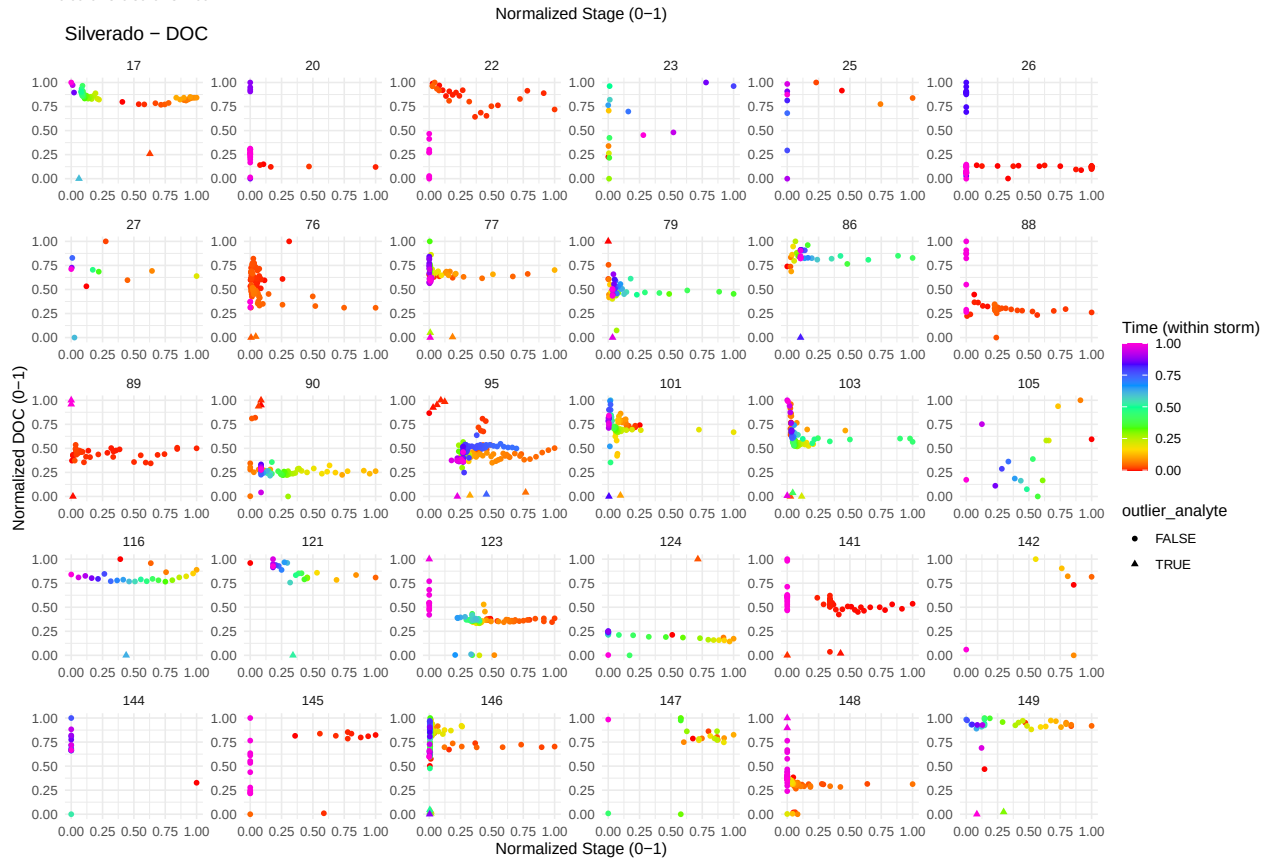
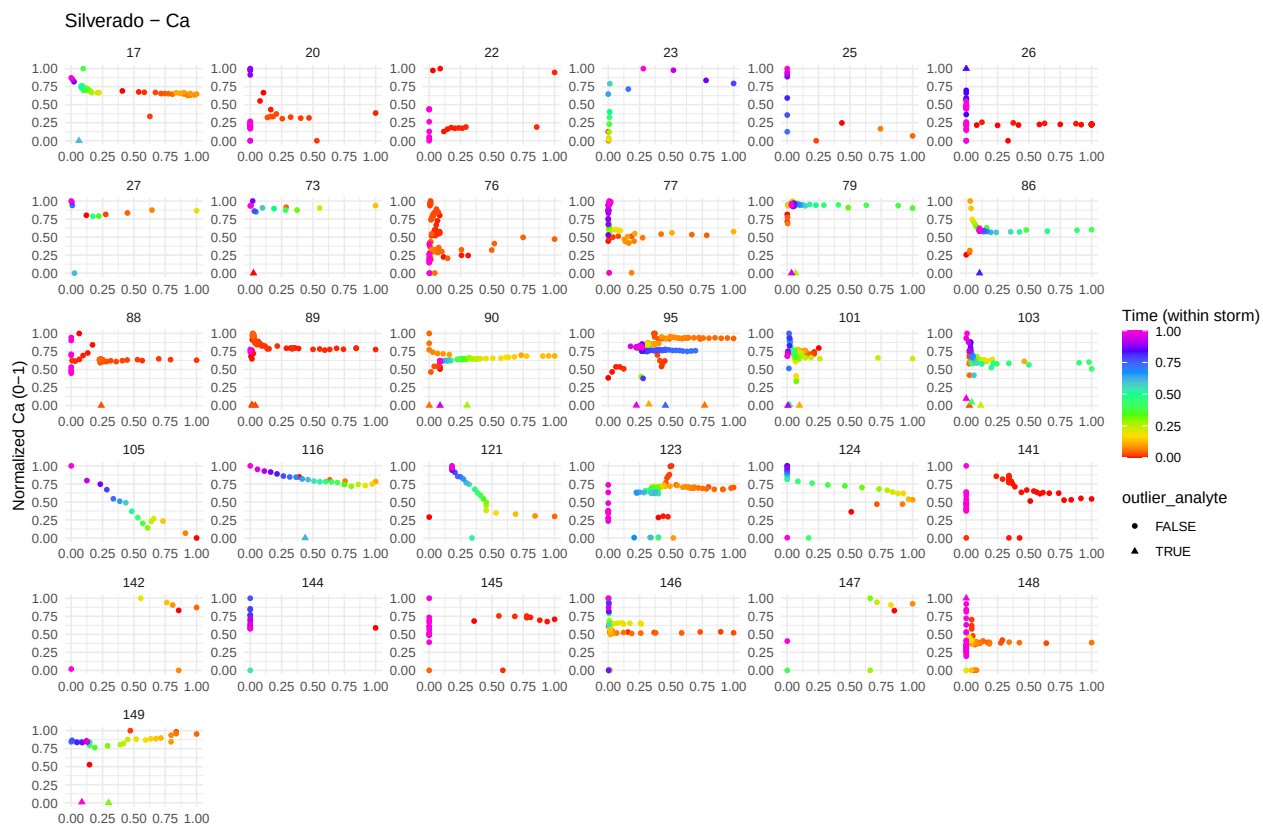


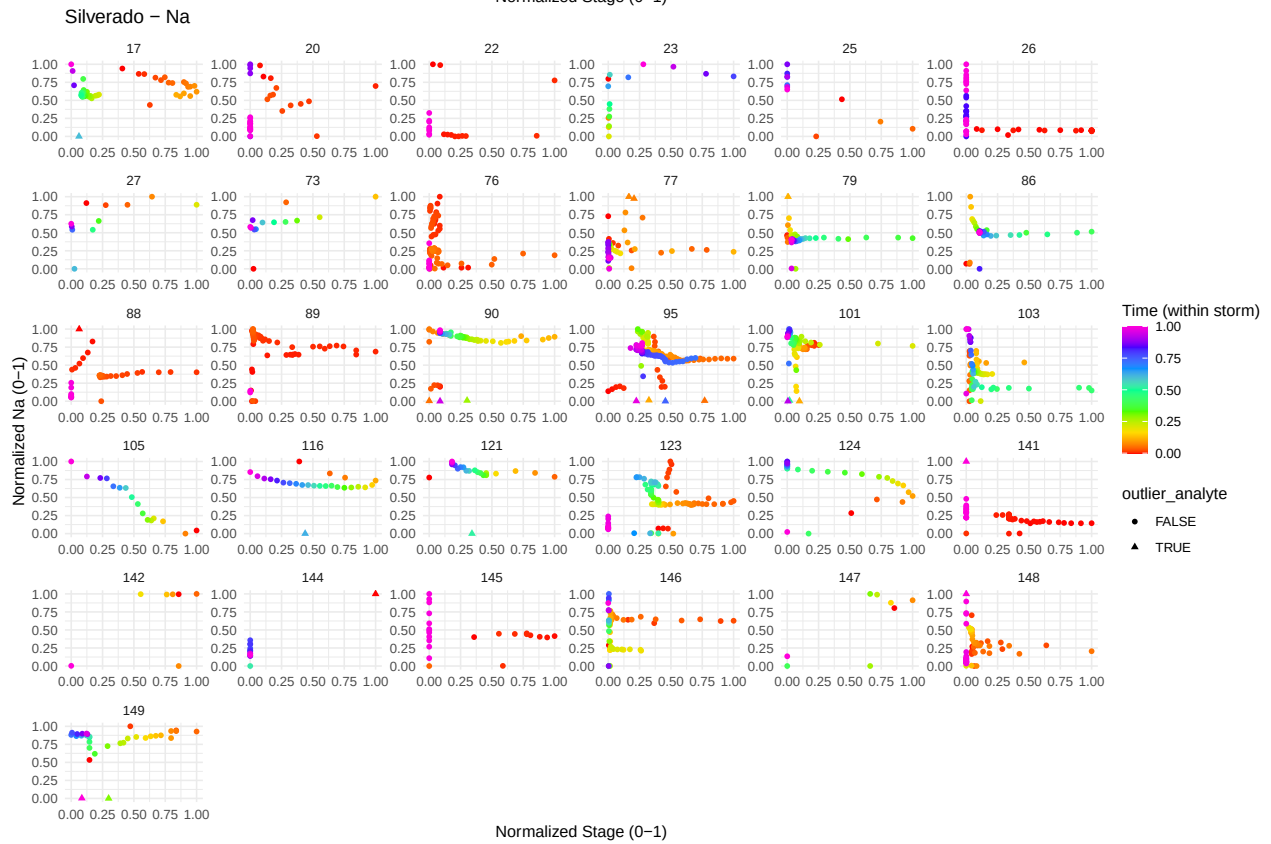
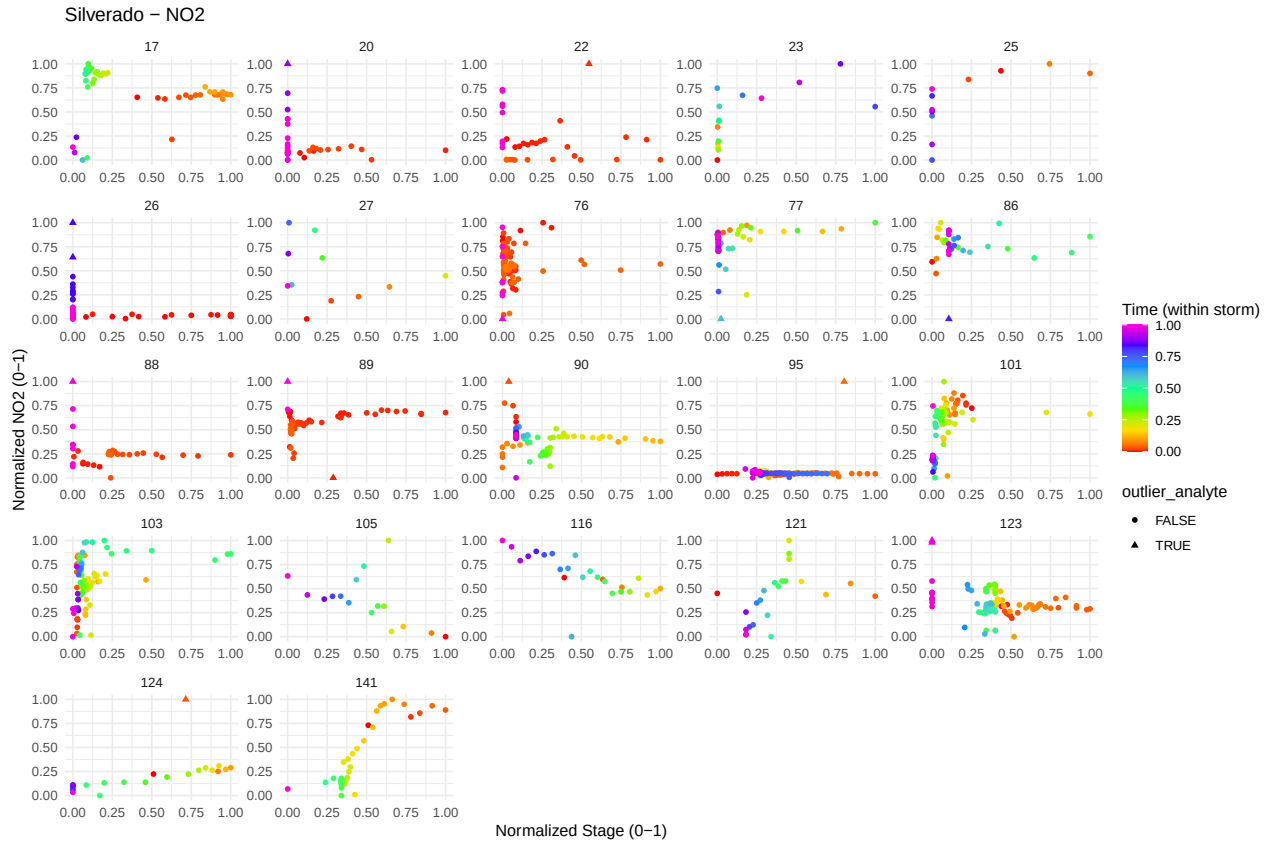


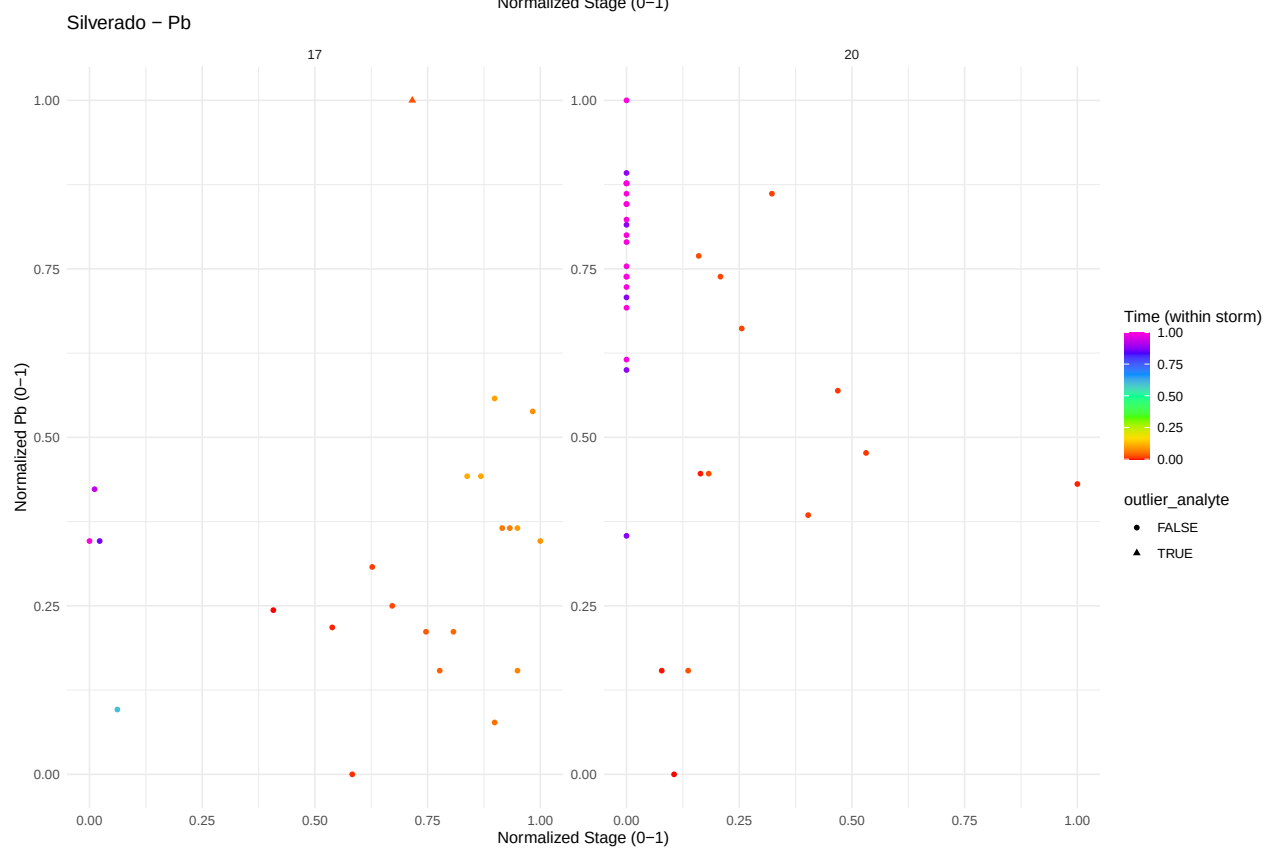
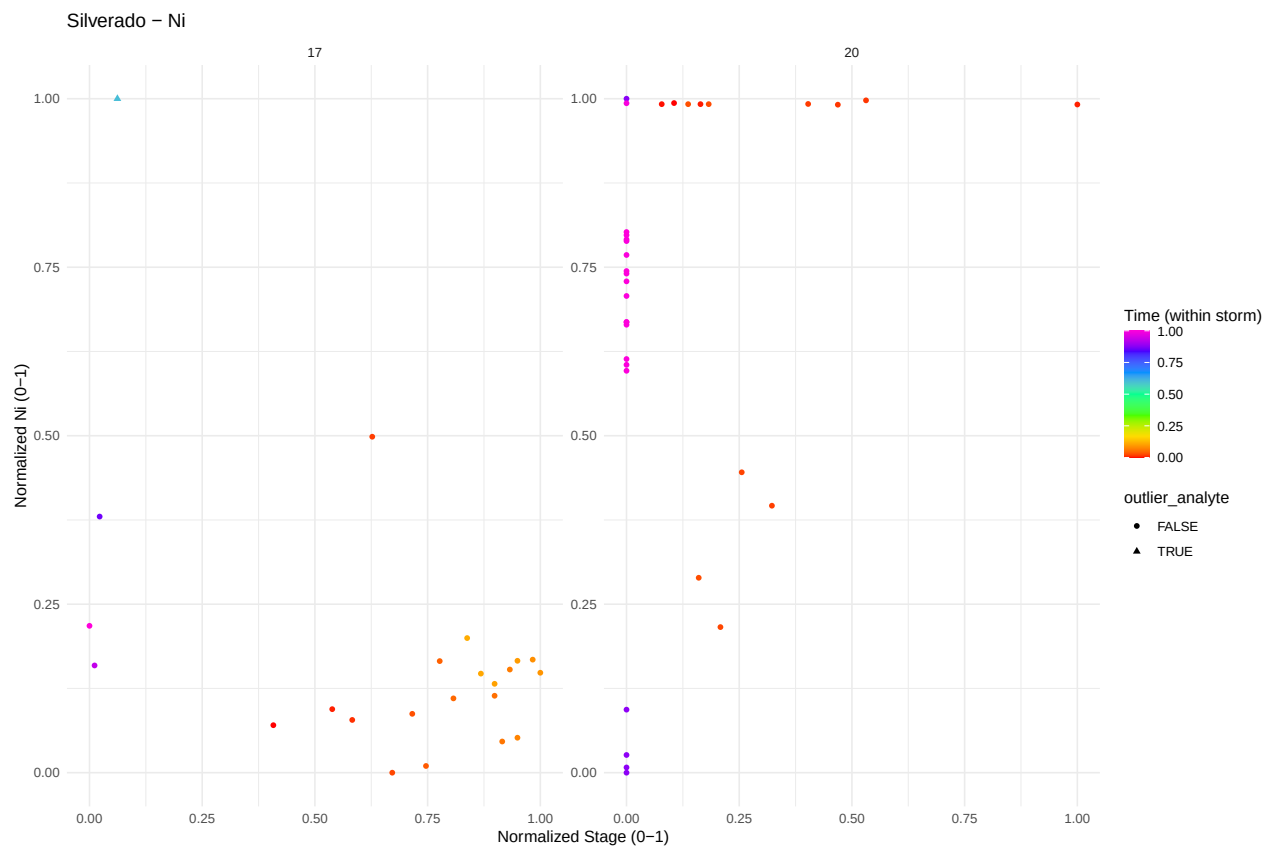












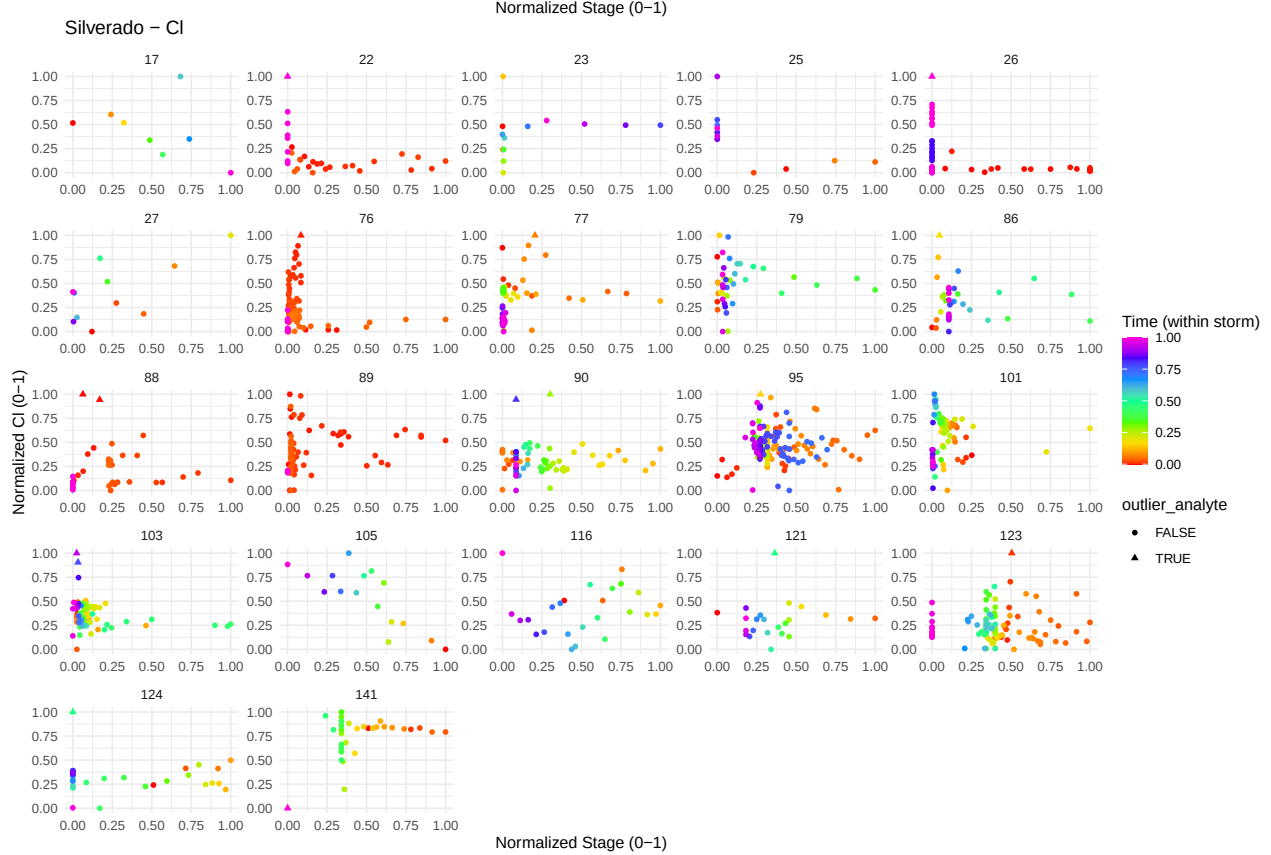
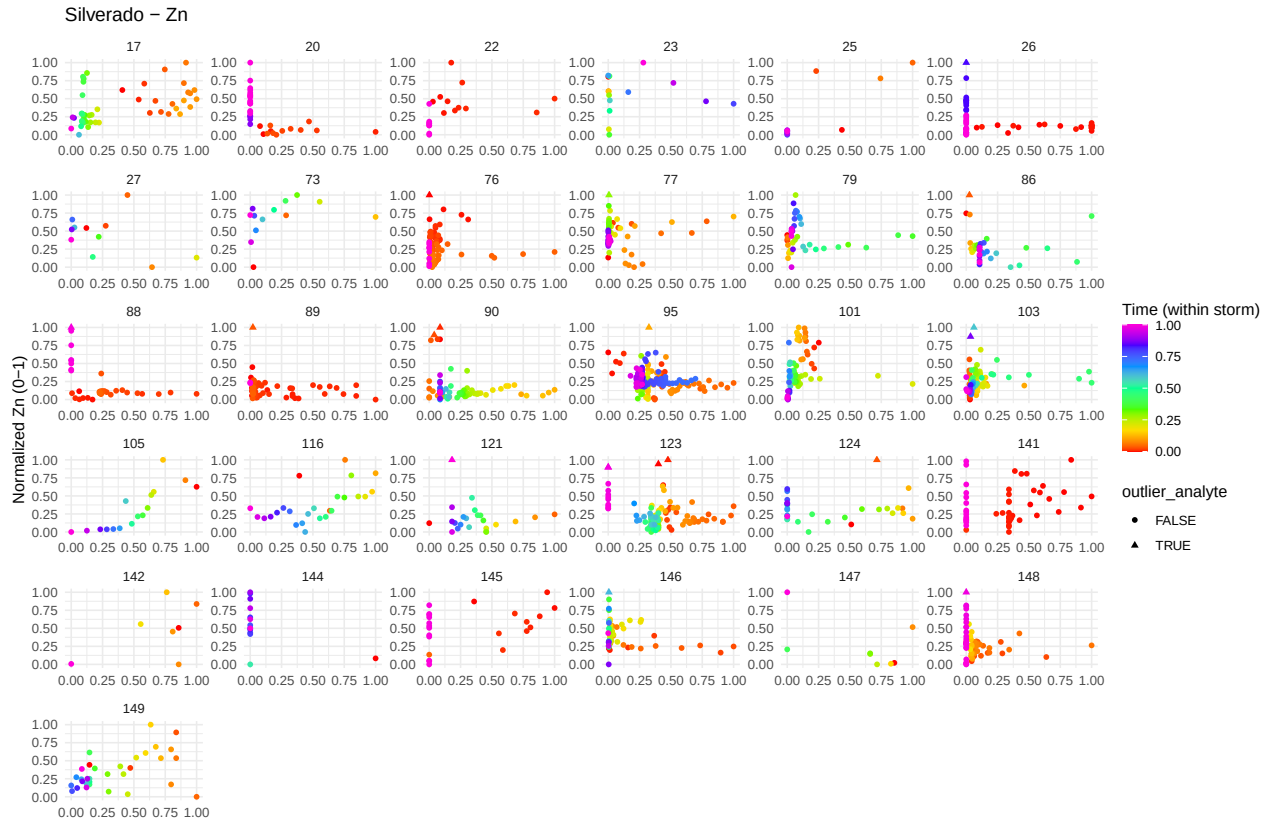
Silverado – TDN

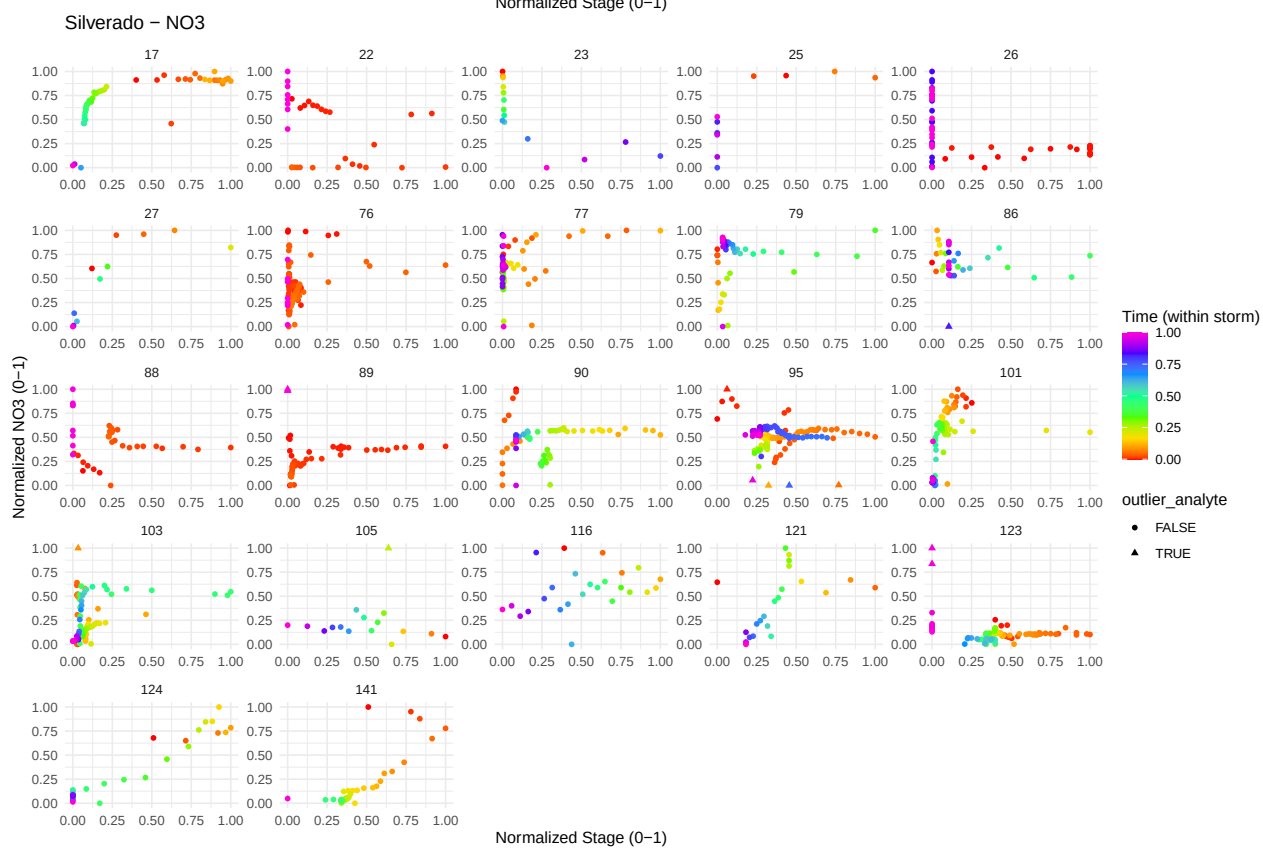
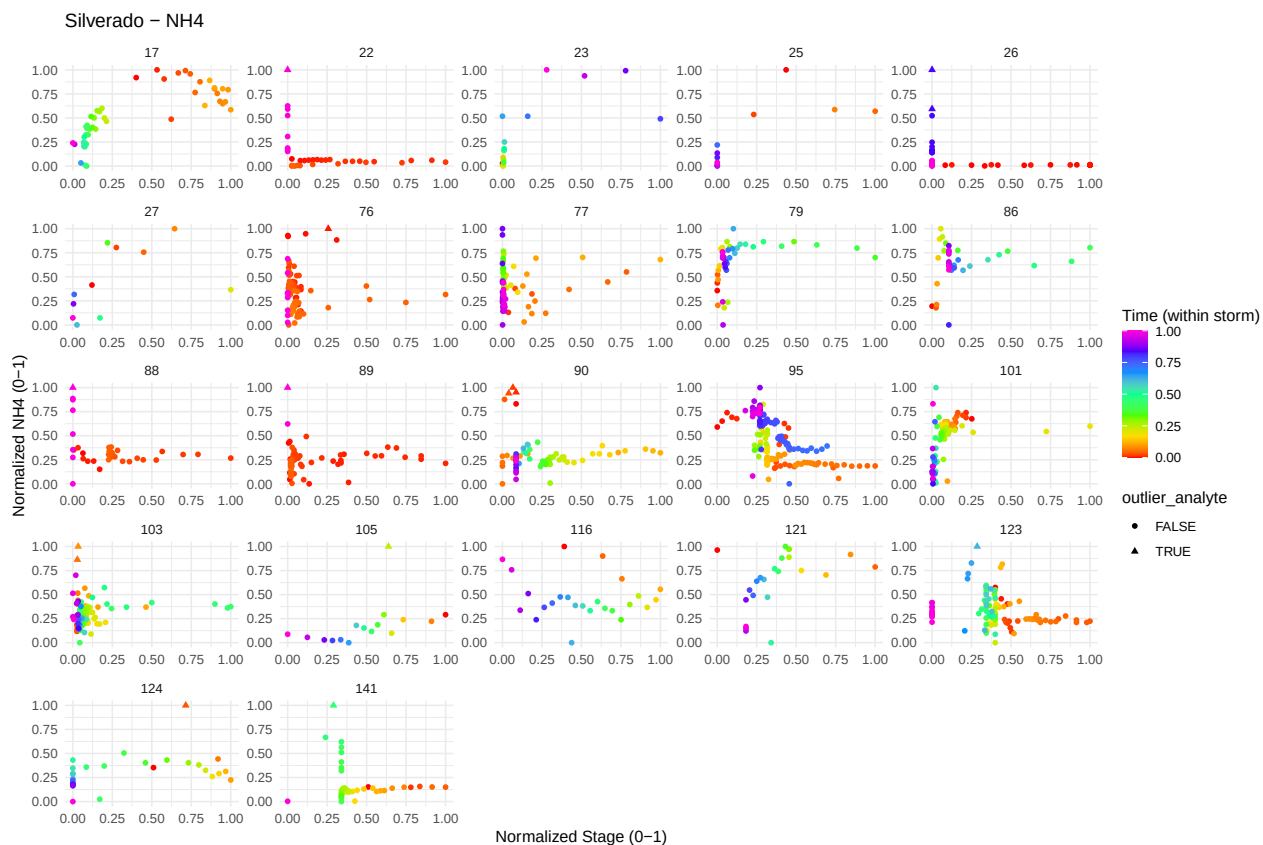


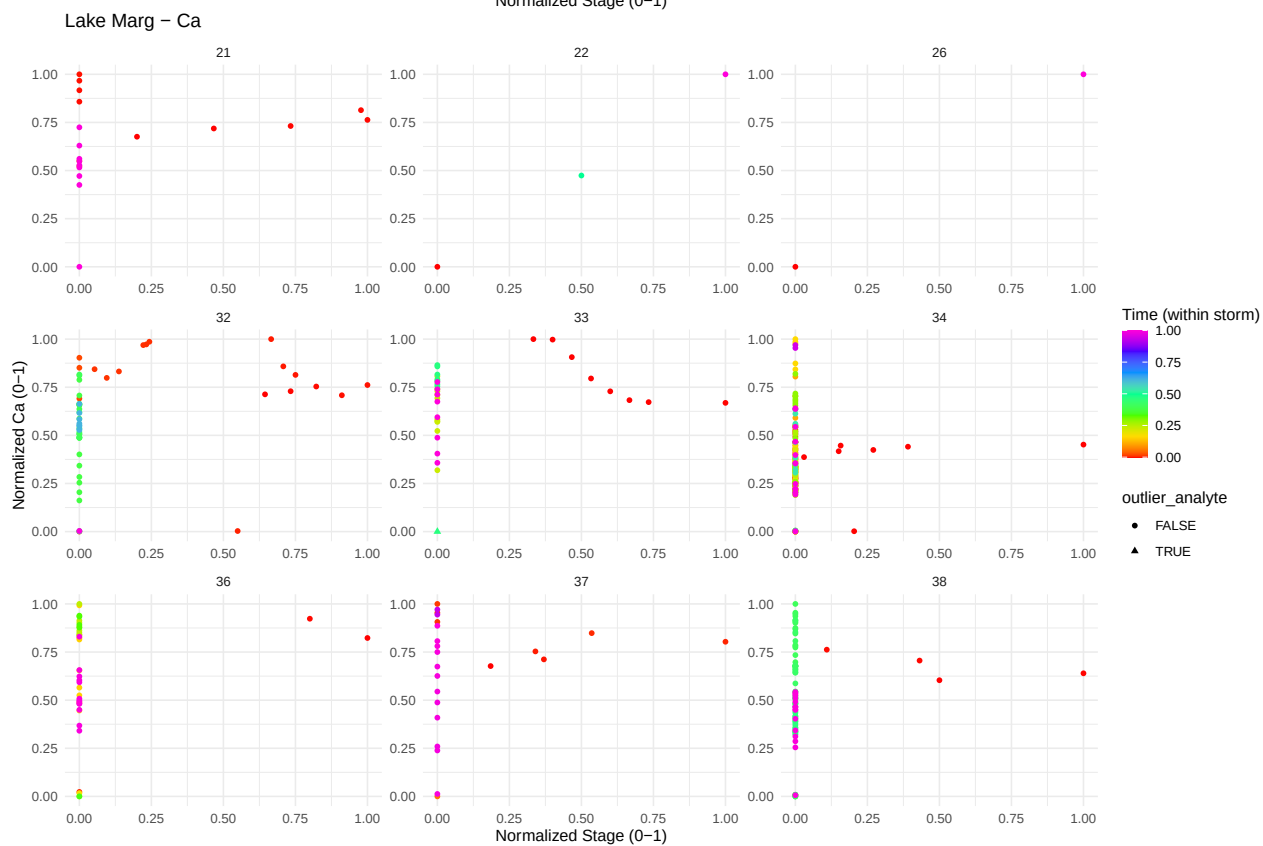
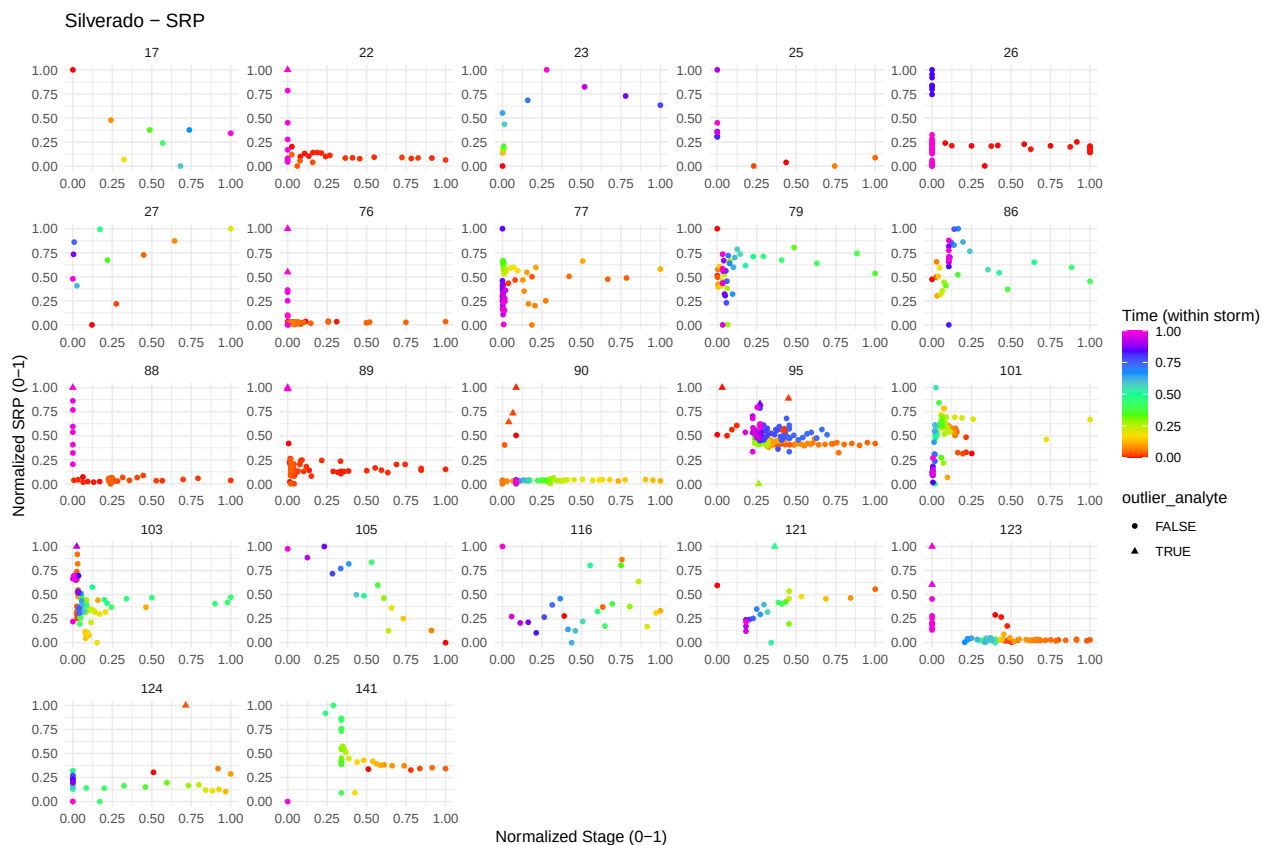
Normalized Stage (0-1)

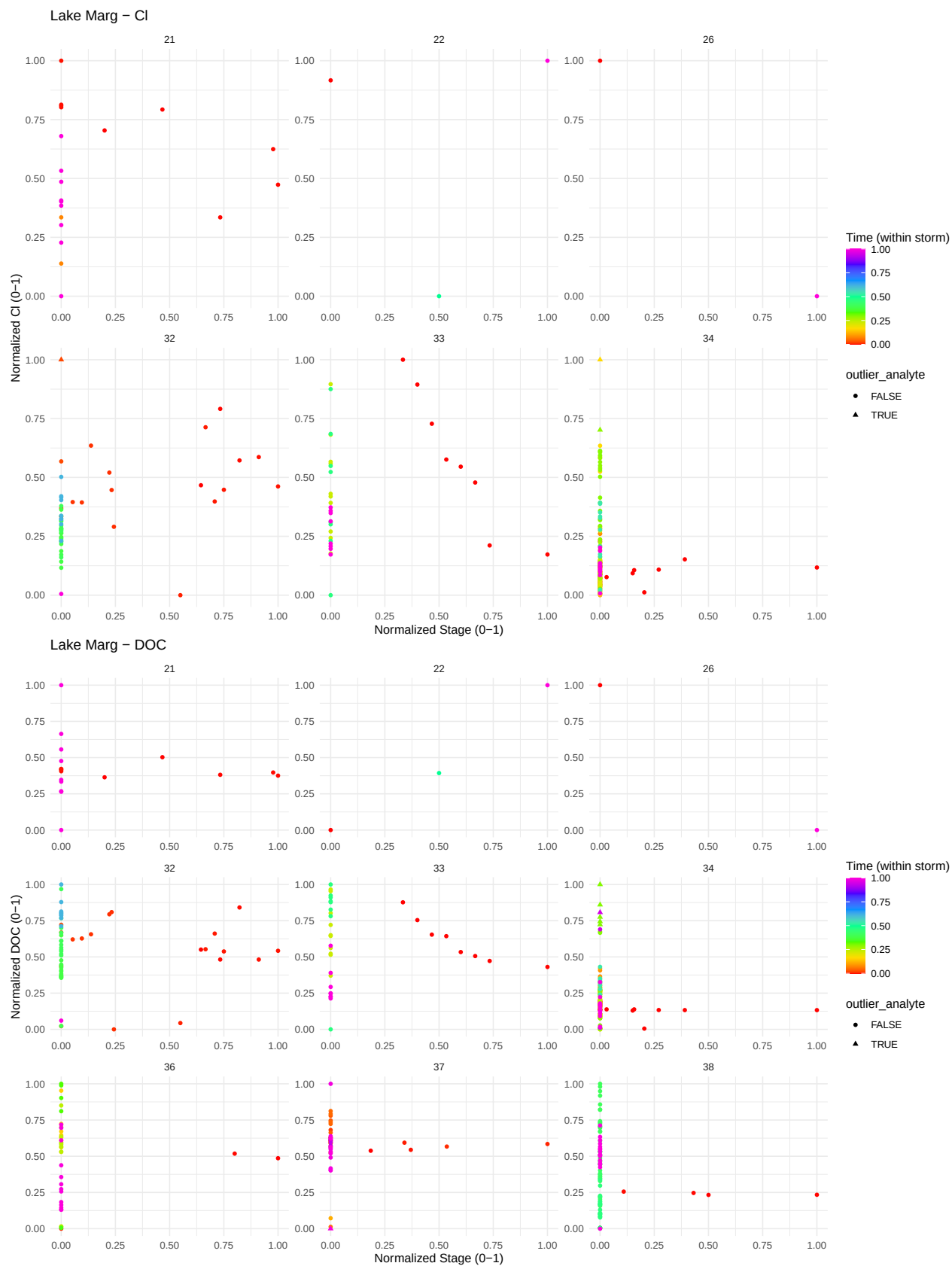
Silverado – TP

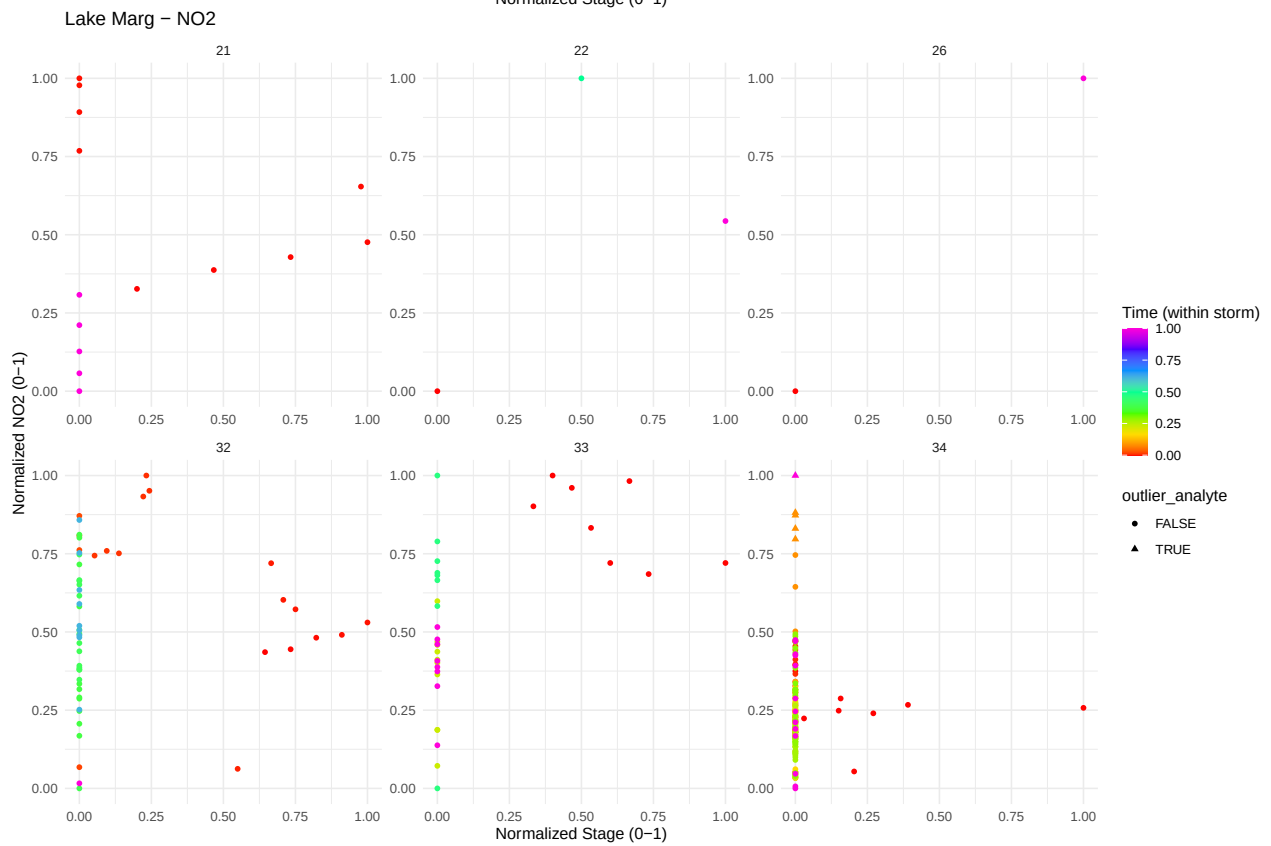
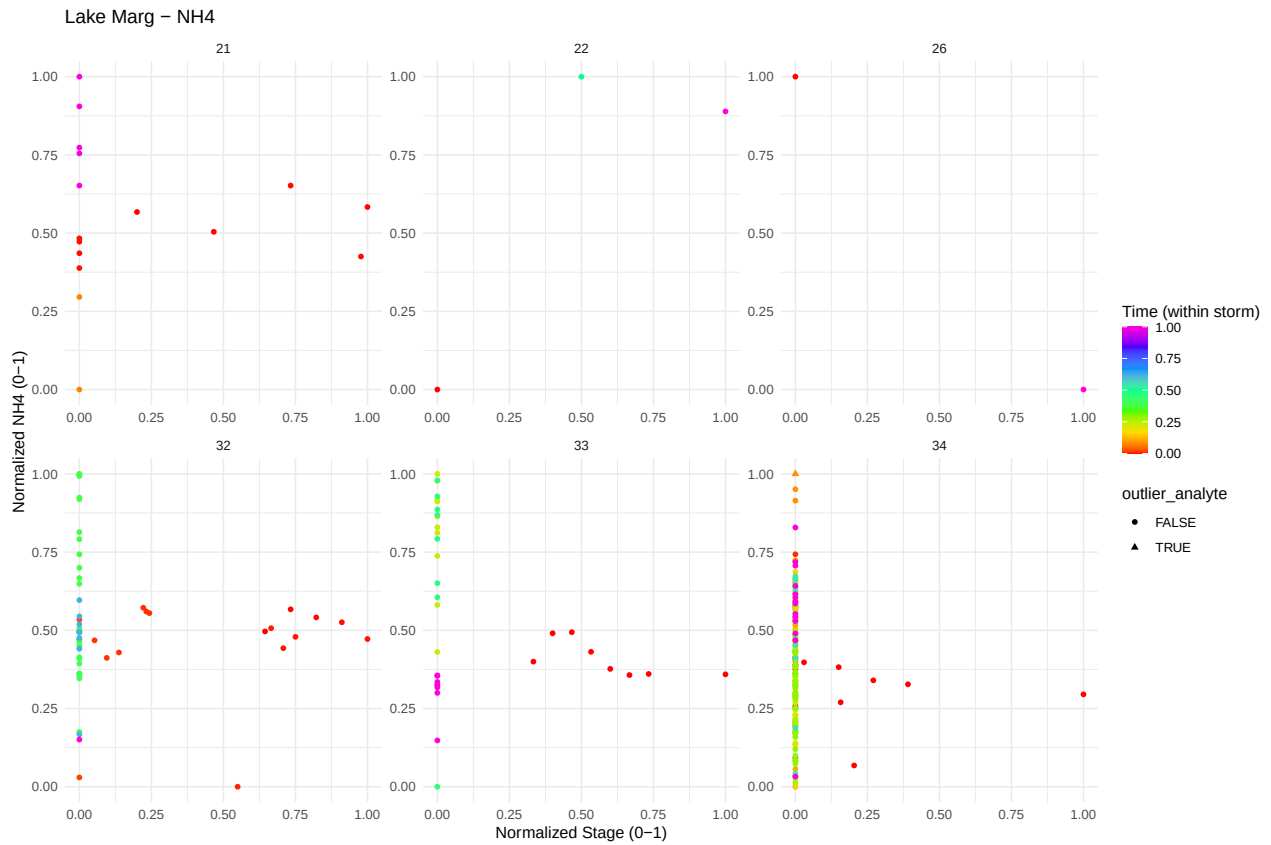


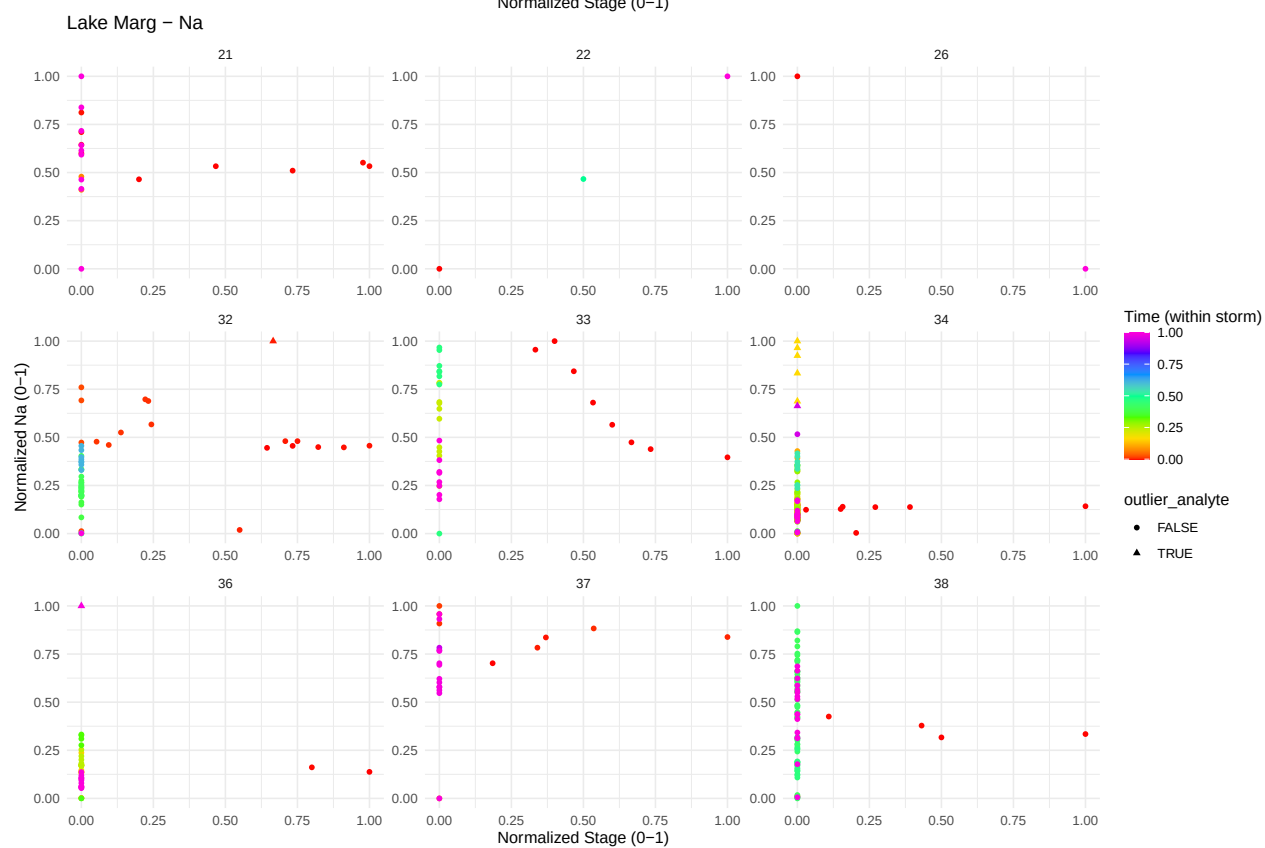
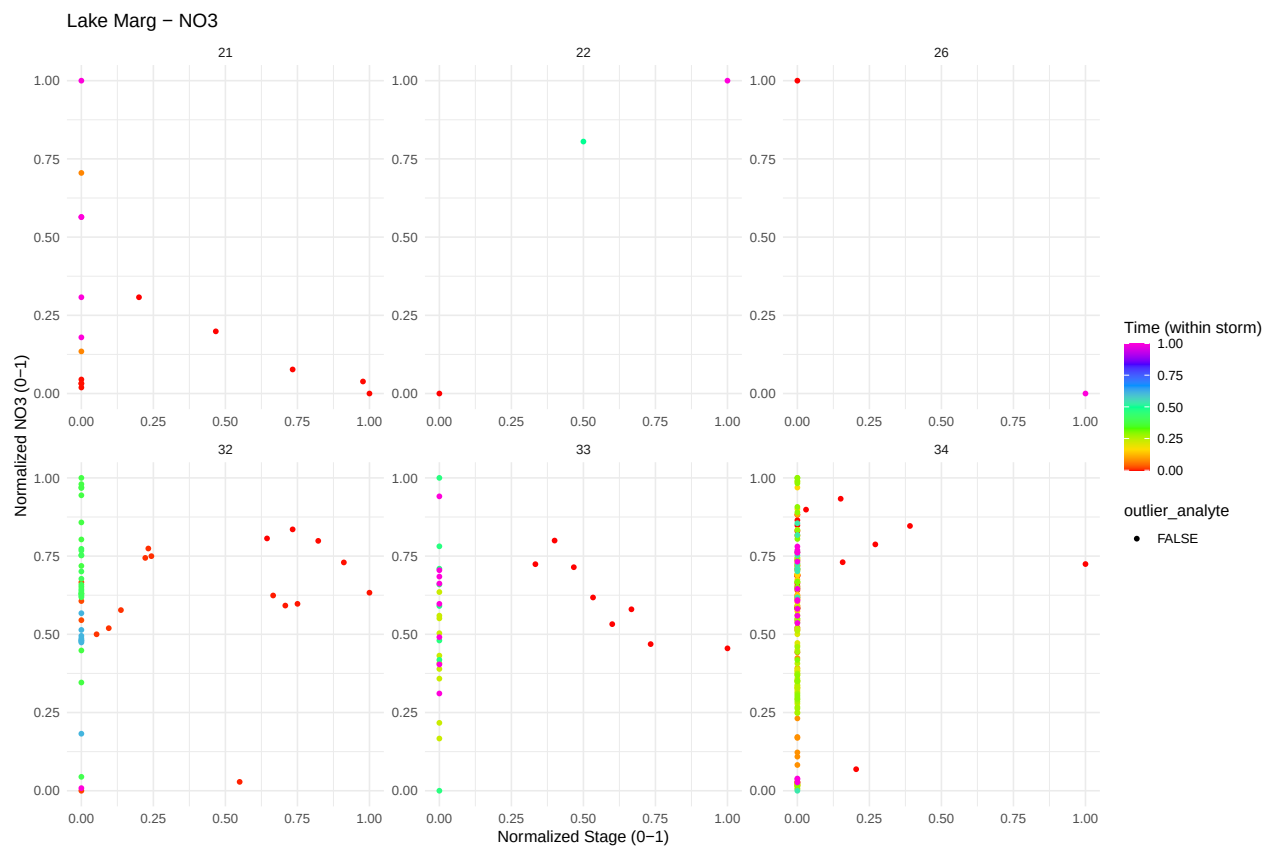


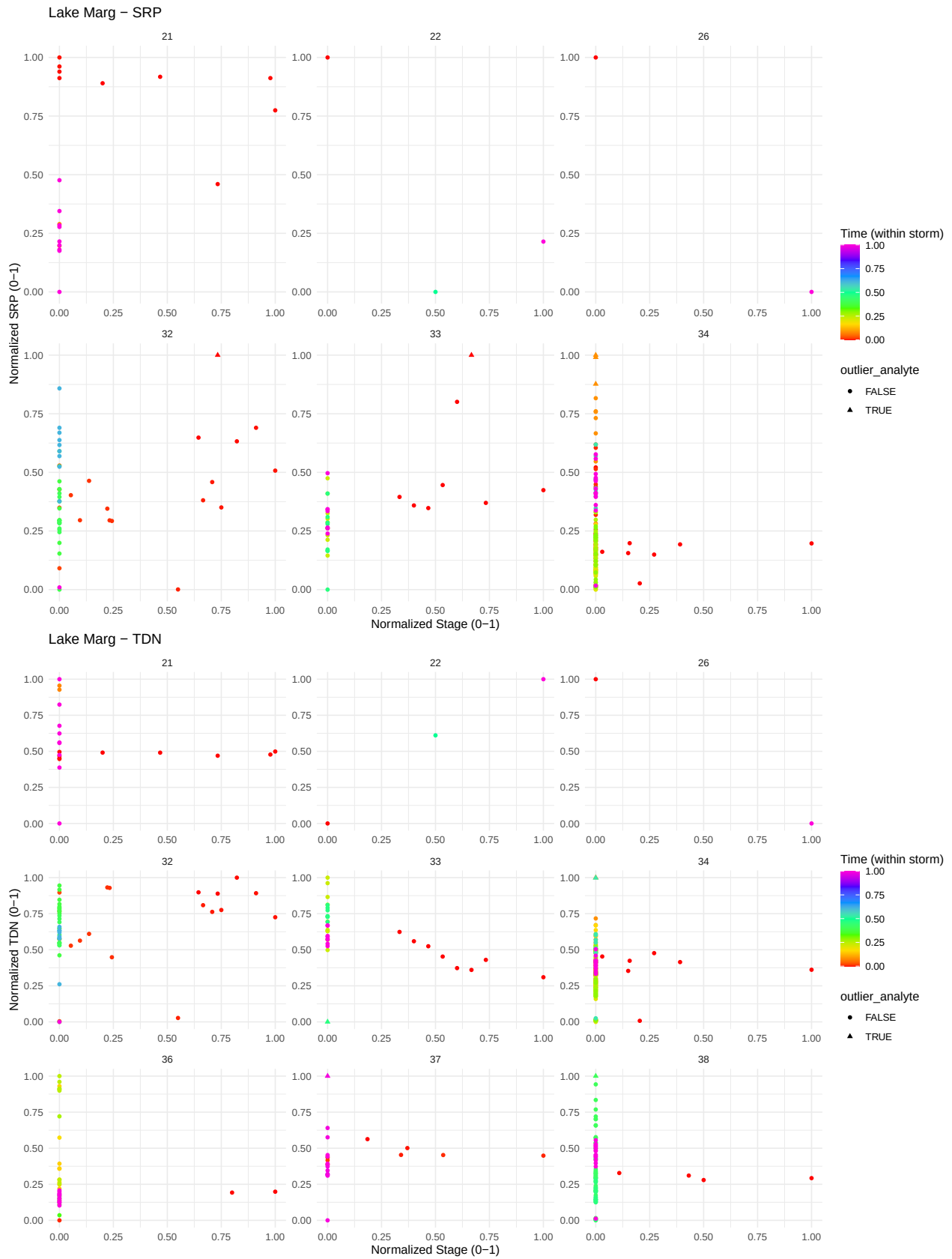


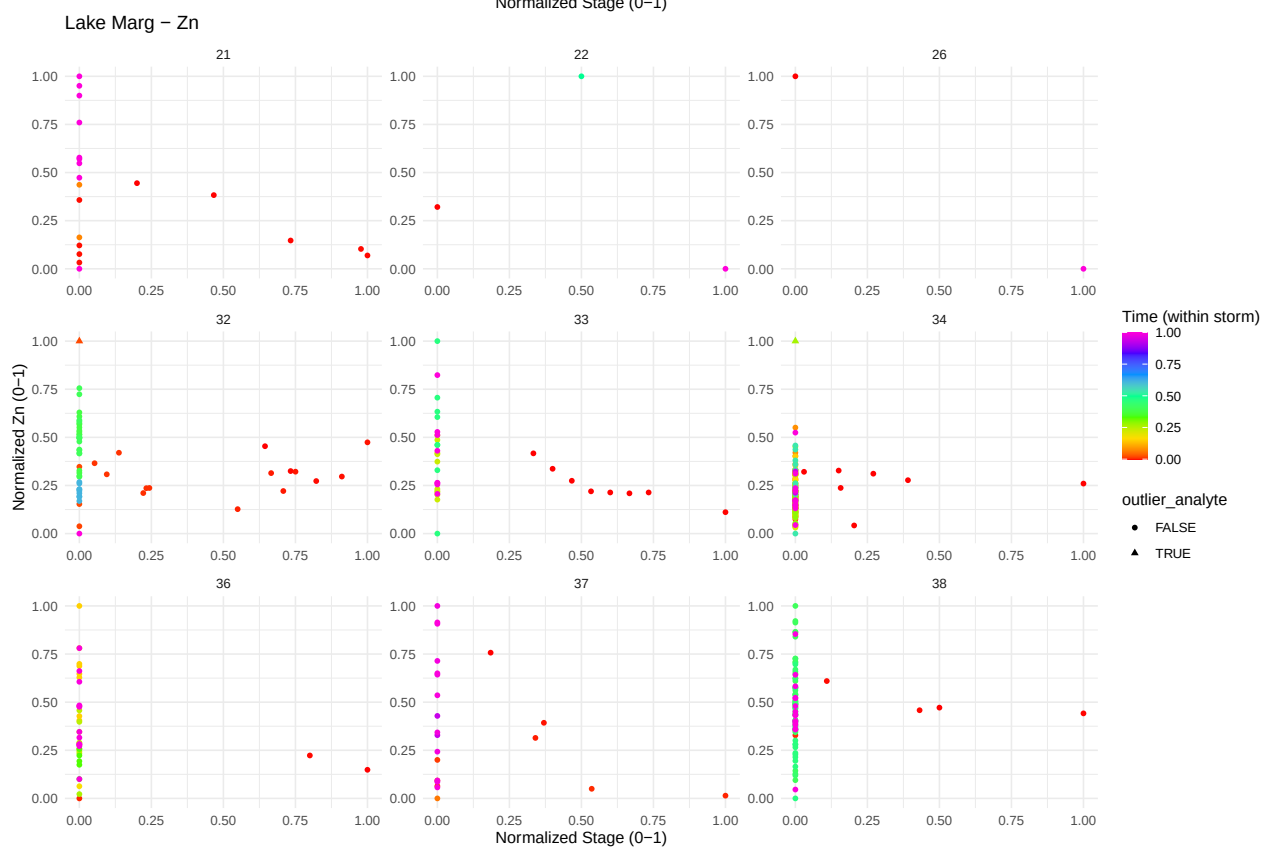
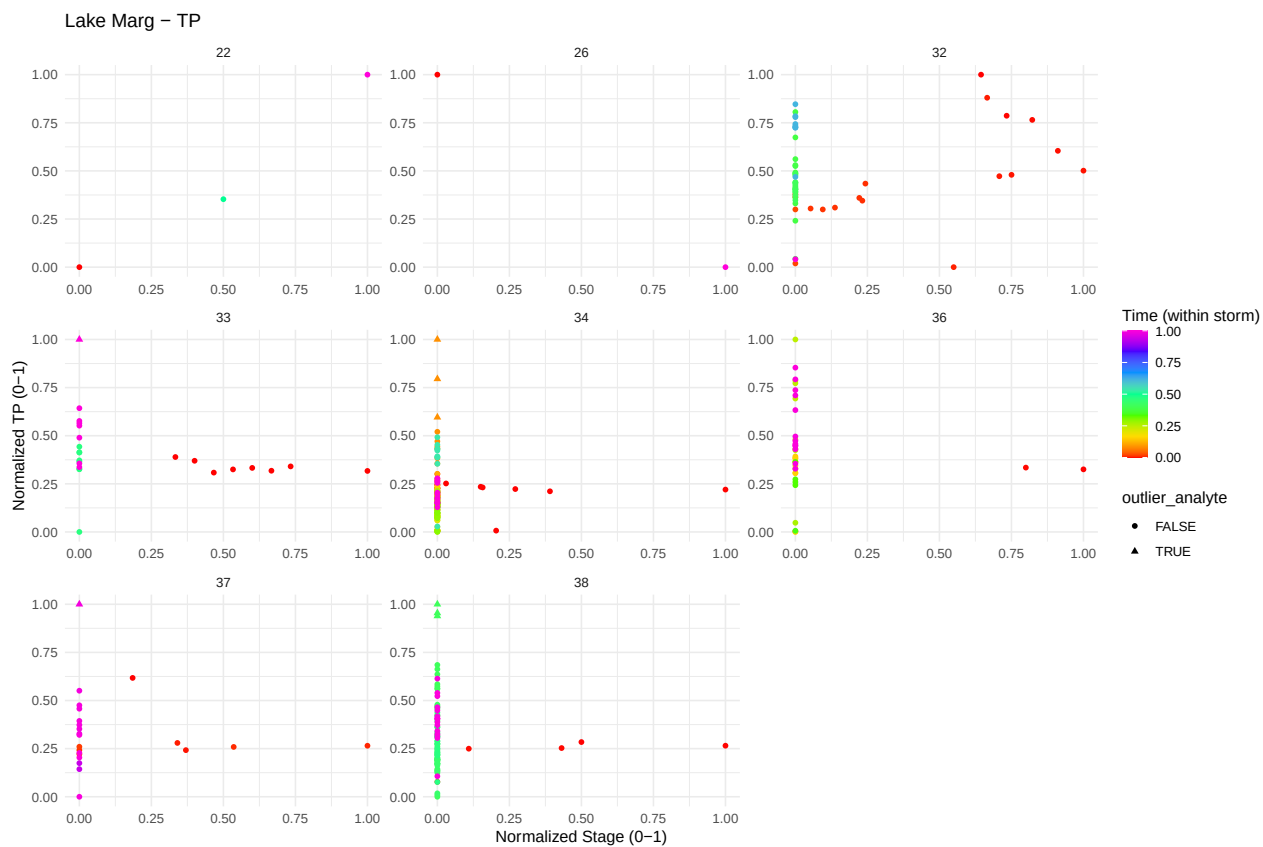










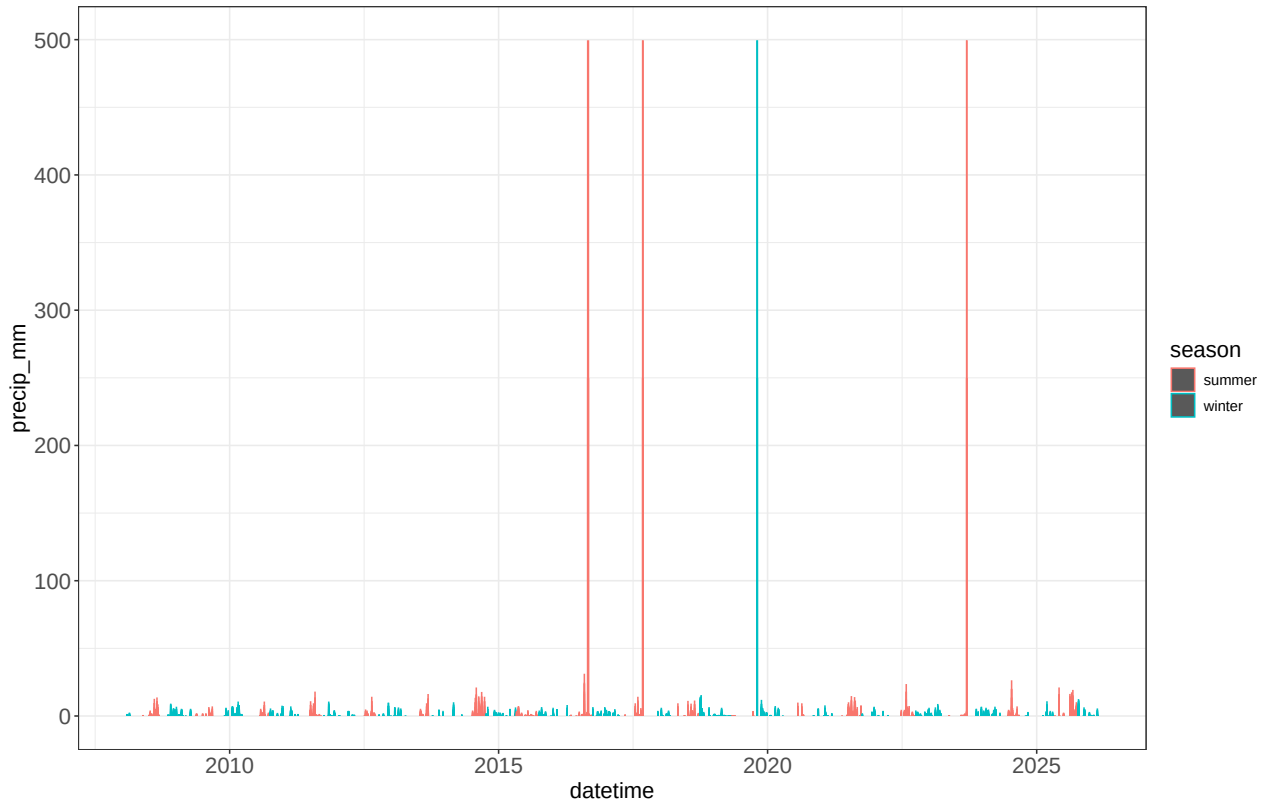


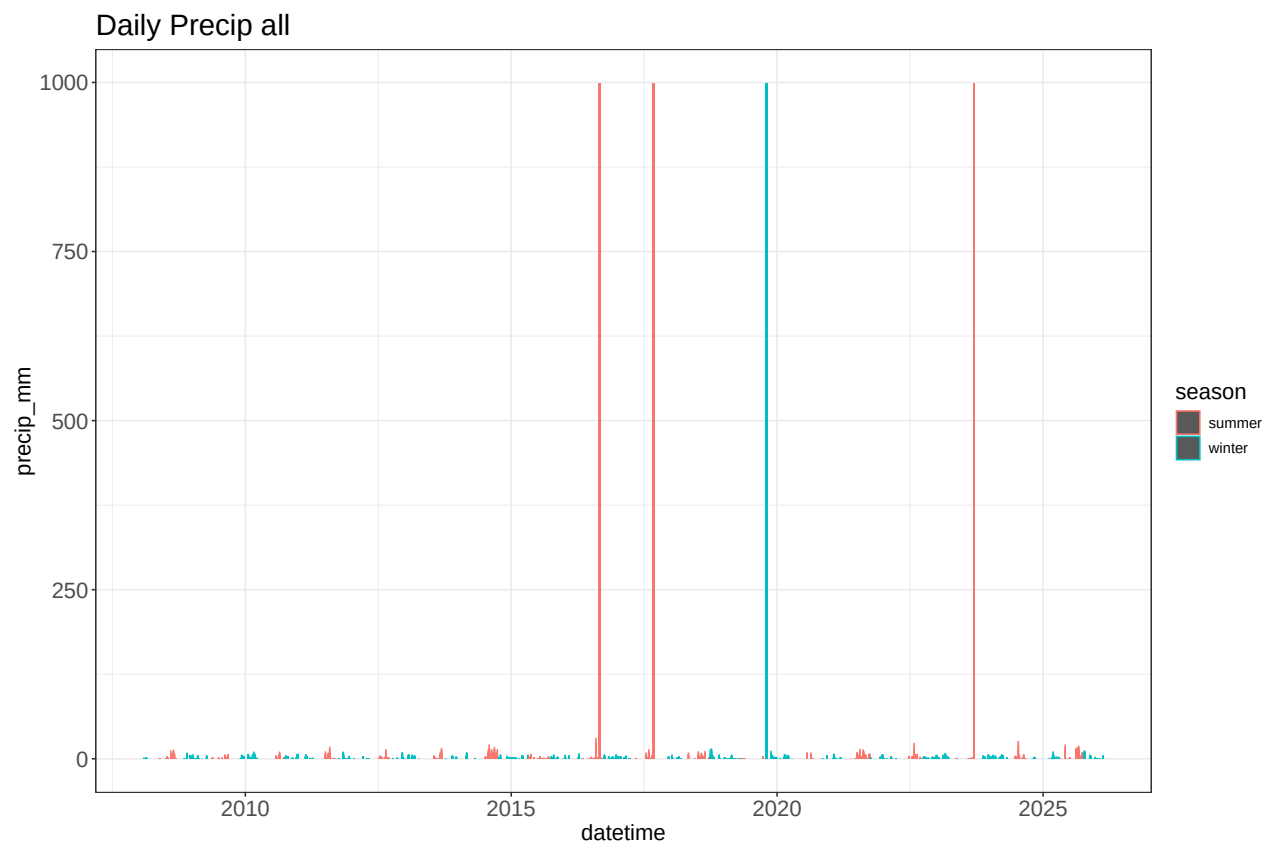
Precipitation

Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

Time series of precip

Daily Precip (excluding large outliers)



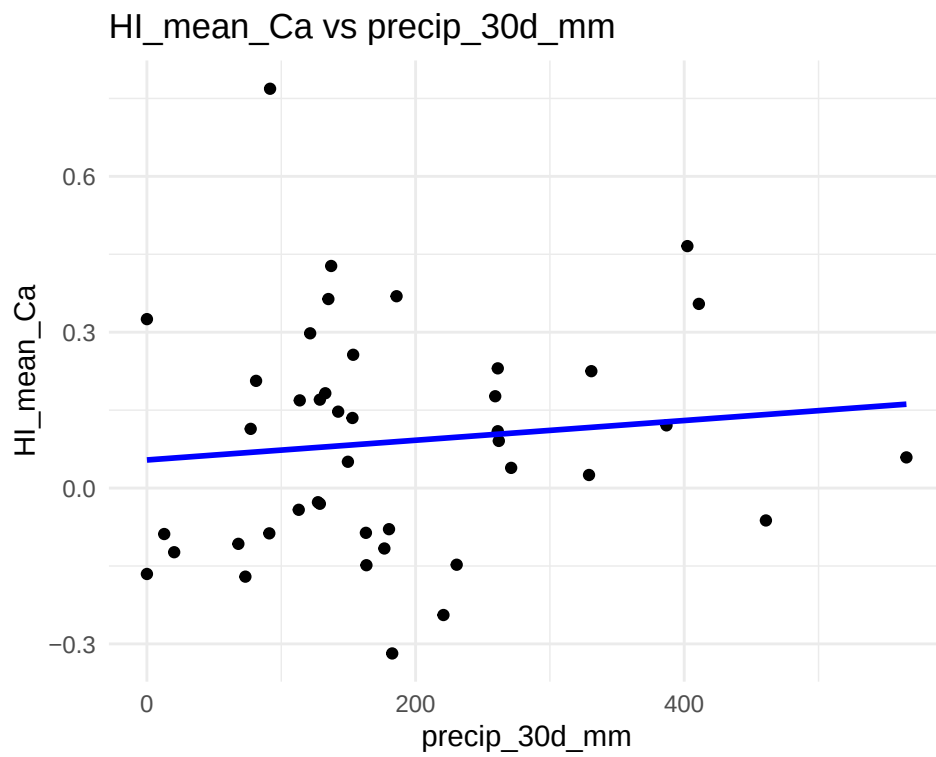
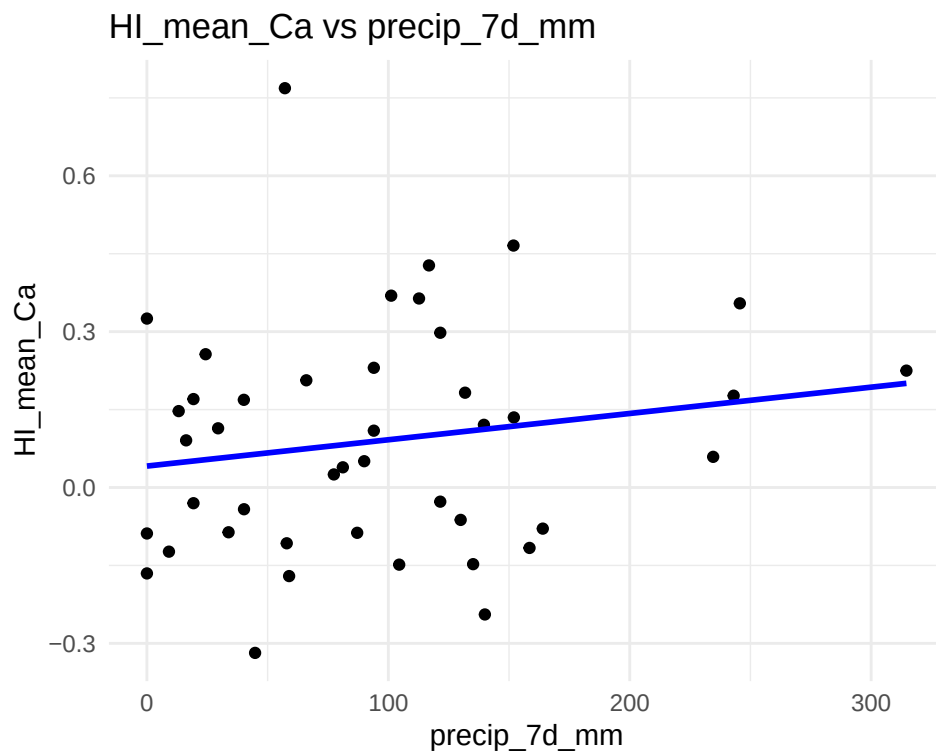


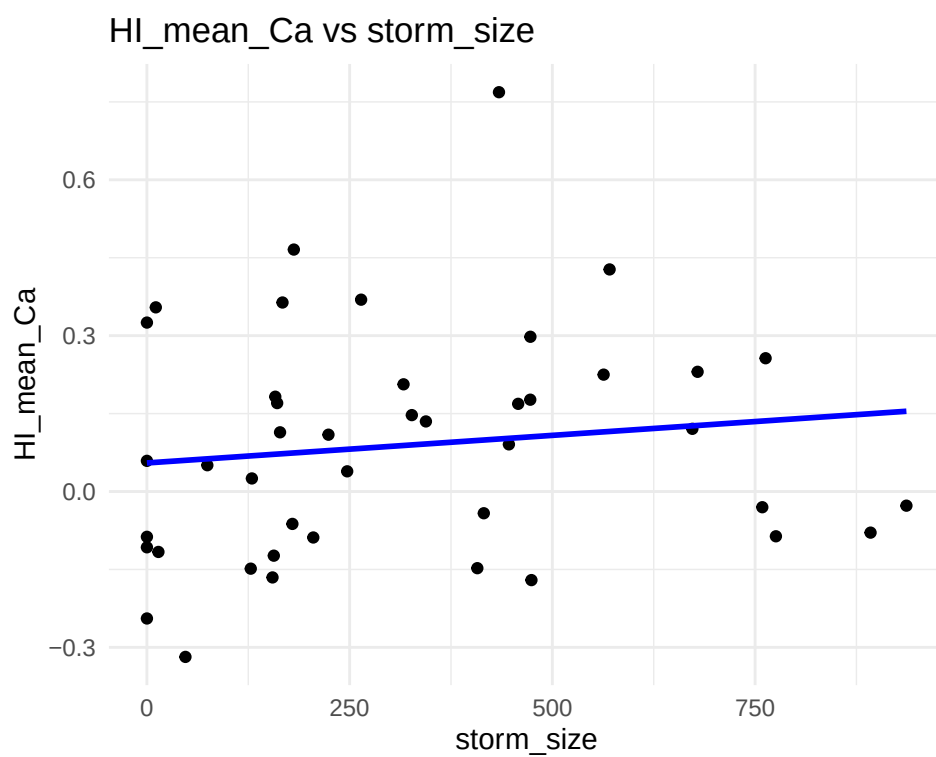
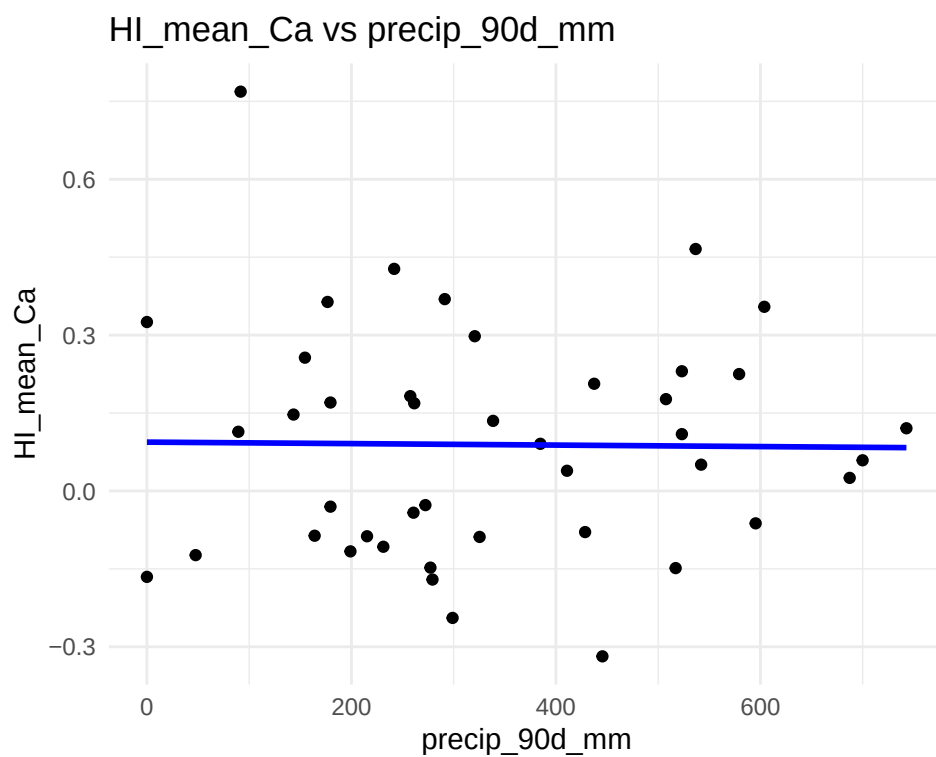
Scatterplot matrices

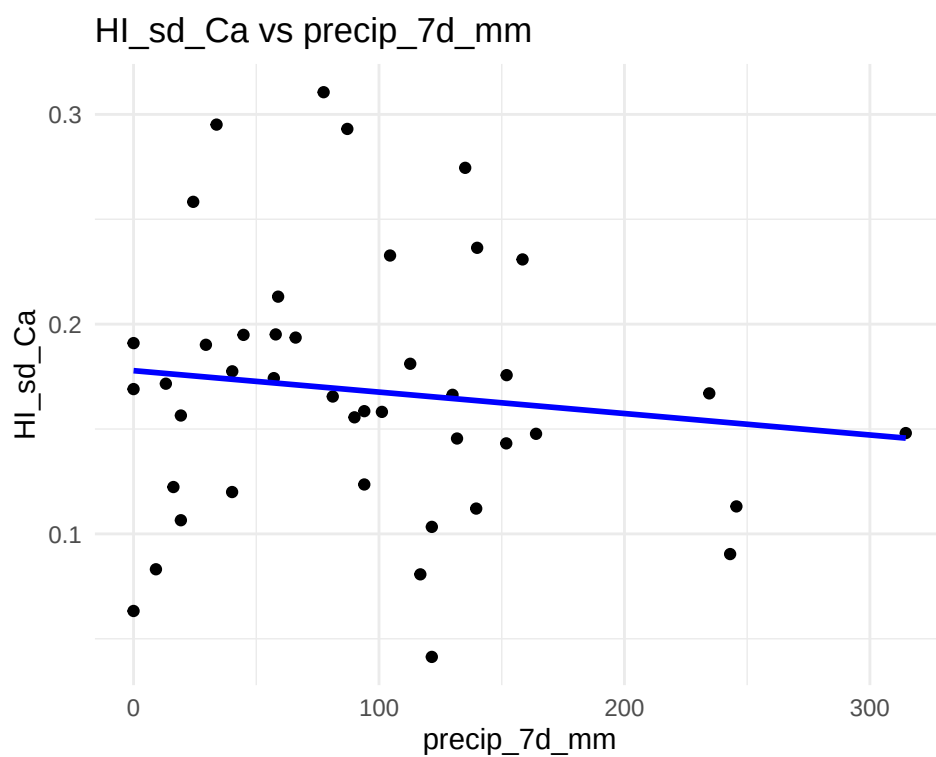
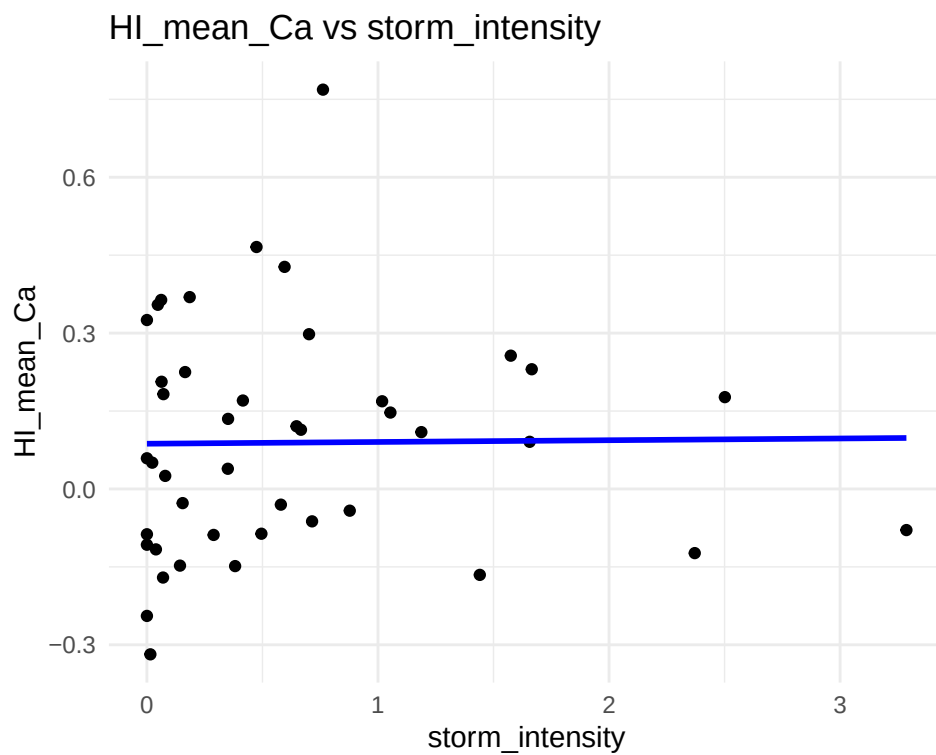
	normalized slope	max width	peak Q	precip 7d	precip 30d	precip 90d	storm size	storm intensity
HI-Ca	-0.54	0.73	0.22	0.17	0.14	0.03	0.25	0.16
HI-Ca (SD)	0.24	-0.36	-0.33	-0.11	-0.05	-0.09	-0.30	-0.45
HI-Cl	-0.67	0.78	0.27	0.08	-0.03	-0.08	-0.10	0.06
HI-Cl (SD)	0.29	0.05	0.03	-0.09	-0.05	-0.24	-0.06	-0.05
HI-DOC	-0.49	0.70	0.17	-0.02	0.01	-0.18	0.13	0.16
HI-DOC (SD)	0.19	0.13	-0.12	-0.30	-0.29	-0.40	-0.15	0.07
HI-K	-1.00	0.50	0.50	-0.50	-0.50	-0.50	0.50	0.50
HI-K (SD)	0.50	0.50	0.50	1.00	1.00	-0.50	0.50	0.50
HI-Mg	-0.50	1.00	0.50	-0.50	-0.50	-0.50	0.50	0.50
HI-Mg (SD)	0.50	-1.00	-0.50	0.50	0.50	0.50	-0.50	-0.50
HI-NH4	-0.62	0.83	0.39	0.16	0.09	-0.06	0.01	-0.08
HI-NH4 (SD)	0.12	0.09	0.06	-0.08	-0.11	-0.21	-0.30	-0.19
HI-NO3	-0.62	0.83	-0.14	0.10	-0.06	-0.12	0.14	-0.03
HI-NO3 (SD)	0.15	0.22	0.13	-0.22	0.09	-0.01	0.00	0.14
HI-Na	-0.72	0.69	0.17	0.25	0.10	0.02	-0.10	0.01
HI-Na (SD)	0.15	0.09	-0.02	-0.09	0.14	-0.19	-0.26	-0.19
HI-SRP	-0.59	0.83	-0.17	-0.18	-0.08	-0.11	0.23	0.17
HI-SRP (SD)	0.44	-0.40	0.15	0.13	0.05	-0.04	-0.25	-0.20
HI-TDN	-0.61	0.70	0.20	0.06	0.10	-0.05	0.05	0.01
HI-TDN (SD)	0.04	0.31	0.12	-0.20	-0.12	-0.15	-0.05	0.01
HI-TP	-0.63	0.82	0.16	-0.13	0.18	0.07	0.07	0.20
HI-TP (SD)	0.13	0.13	0.11	0.00	-0.03	-0.04	-0.10	-0.11
HI-SO4	?	?	?	?	?	?	?	?
HI-SO4 (SD)	?	?	?	?	?	?	?	?
HI-NO2	-0.62	0.77	0.11	-0.02	0.04	-0.06	0.26	0.36
HI-NO2 (SD)	0.16	-0.14	0.04	-0.11	-0.09	-0.20	-0.11	-0.07
HI-Zn	-0.63	0.80	-0.12	-0.05	-0.09	-0.16	0.16	-0.03
HI-Zn (SD)	0.15	-0.14	0.22	0.15	-0.03	-0.10	-0.29	-0.18
HI-Ni	-1.00	1.00	-1.00	1.00	1.00	1.00	1.00	1.00
HI-Ni (SD)	-1.00	1.00	-1.00	1.00	1.00	1.00	1.00	1.00
HI-Pb	?	?	?	?	?	?	?	?
HI-Pb (SD)	?	?	?	?	?	?	?	?

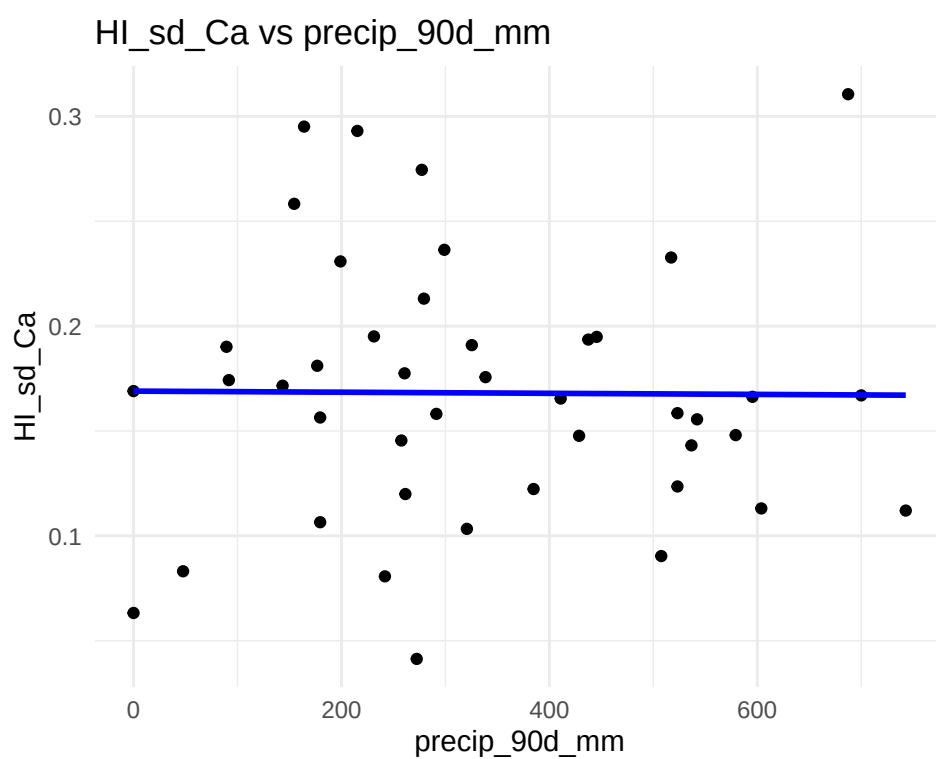
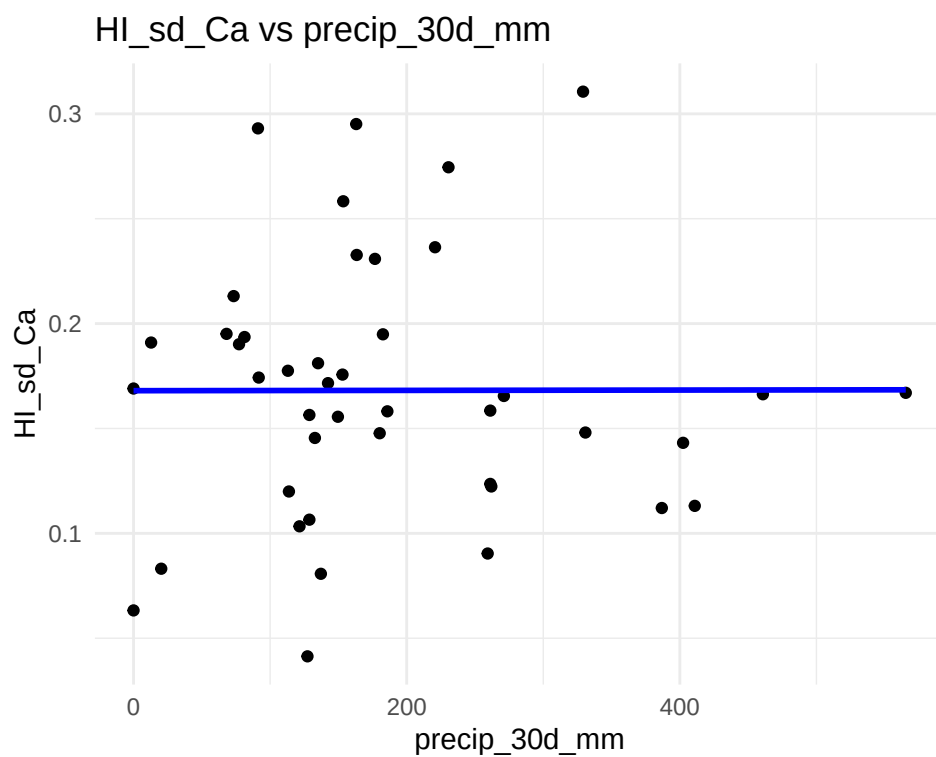
	normalized slope	max width	peak Q	precip 7d	precip 30d	precip 90d	storm size	storm intensity
beta-Ca	0.47	-0.37	-0.21	-0.16	-0.10	0.11	-0.04	-0.05
beta-Cl	0.42	-0.12	-0.36	-0.07	-0.08	0.01	-0.16	-0.43
beta-DOC	0.48	-0.43	-0.15	-0.22	0.00	-0.03	0.06	-0.01
beta-K	0.50	-1.00	-1.00	-0.50	-0.50	1.00	-1.00	-1.00
beta-Mg	1.00	-0.50	-1.00	-0.50	-0.50	1.00	-1.00	-1.00
beta-NH4	0.56	-0.13	0.11	0.13	0.33	-0.01	0.17	0.11
beta-NO3	0.55	-0.42	0.43	-0.01	0.12	-0.05	0.00	0.01
beta-Na	0.42	-0.44	-0.17	-0.24	-0.10	0.06	-0.04	-0.02
beta-SRP	0.50	-0.21	0.24	0.35	0.15	0.04	-0.12	-0.27
beta-TDN	0.30	-0.04	-0.08	0.01	0.14	0.06	-0.01	-0.11
beta-TP	0.51	-0.36	-0.10	0.17	0.05	0.04	-0.01	-0.07
beta-SO4	?	?	?	?	?	?	?	?
beta-NO2	0.44	-0.21	-0.08	0.23	0.31	0.02	0.14	-0.16
beta-Zn	0.15	-0.24	0.15	-0.06	0.03	0.01	-0.04	-0.08
beta-Ni	-1.00	1.00	-1.00	1.00	1.00	1.00	1.00	1.00
beta-Pb	?	?	?	?	?	?	?	?

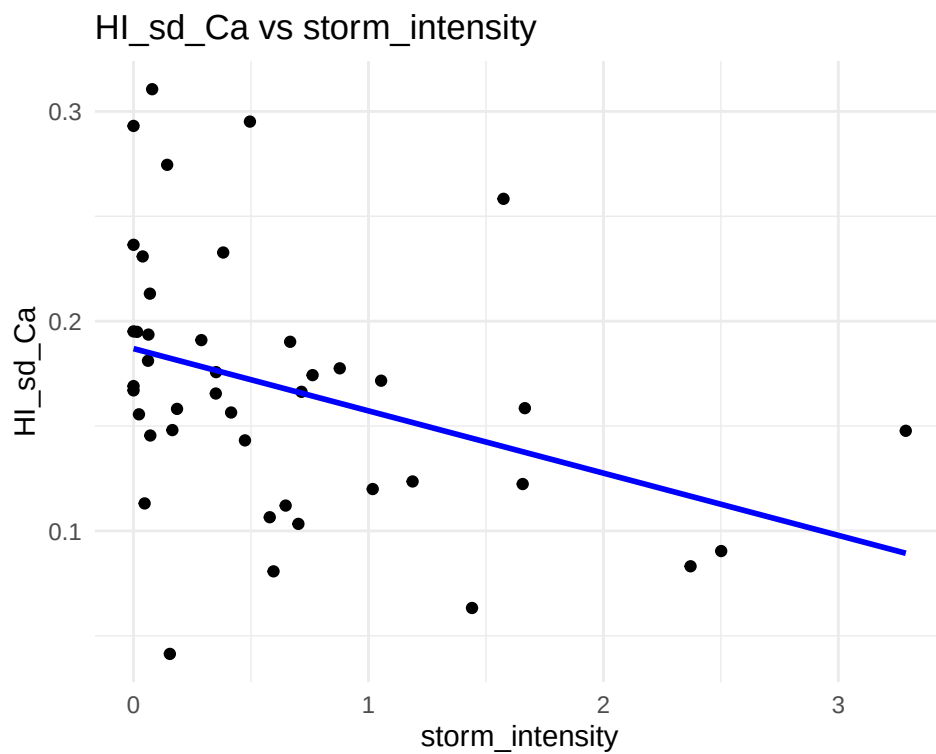
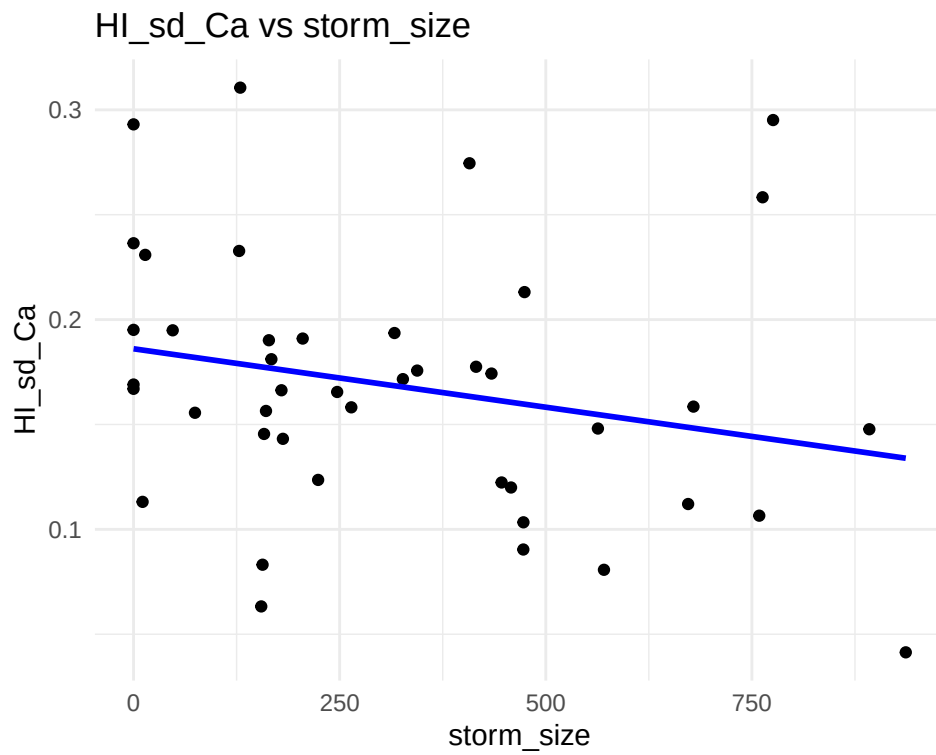
Individual scatterplots

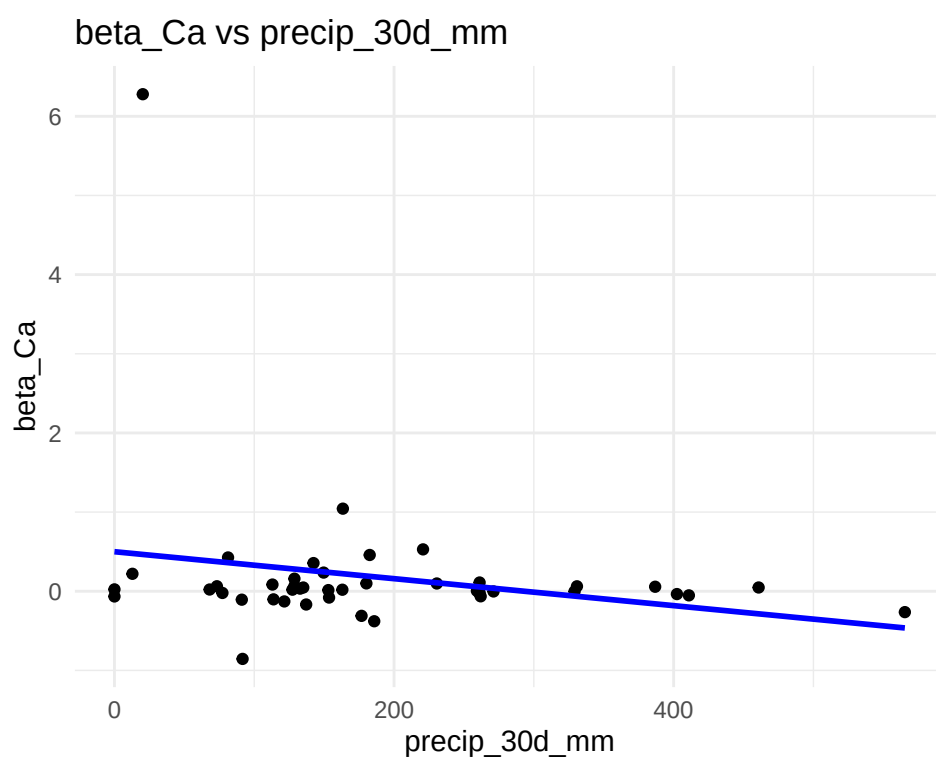
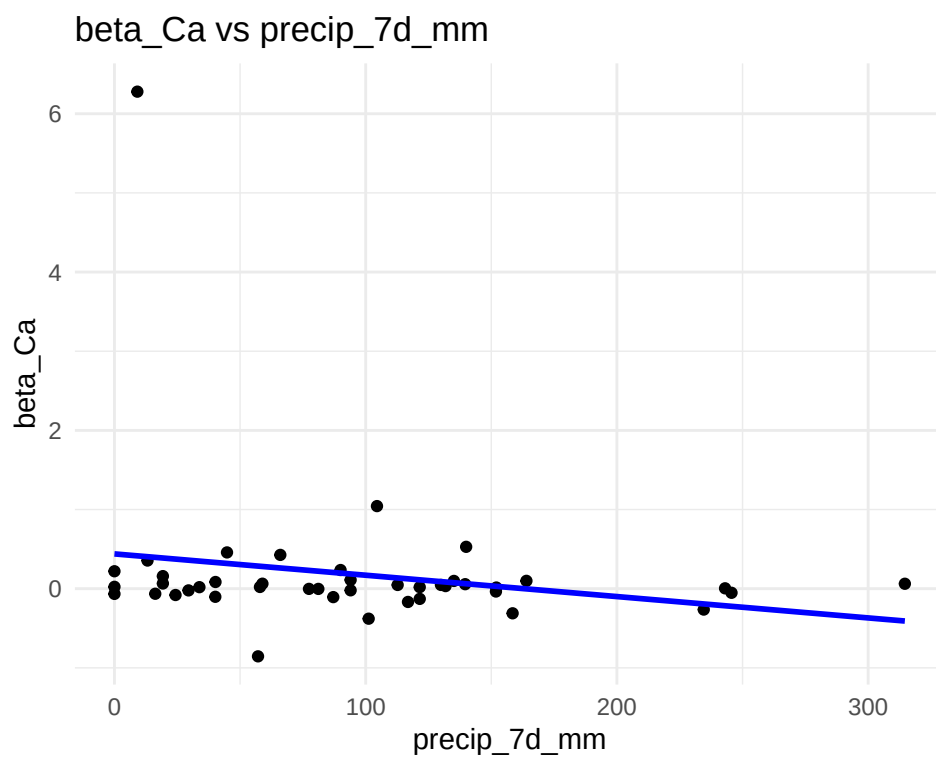


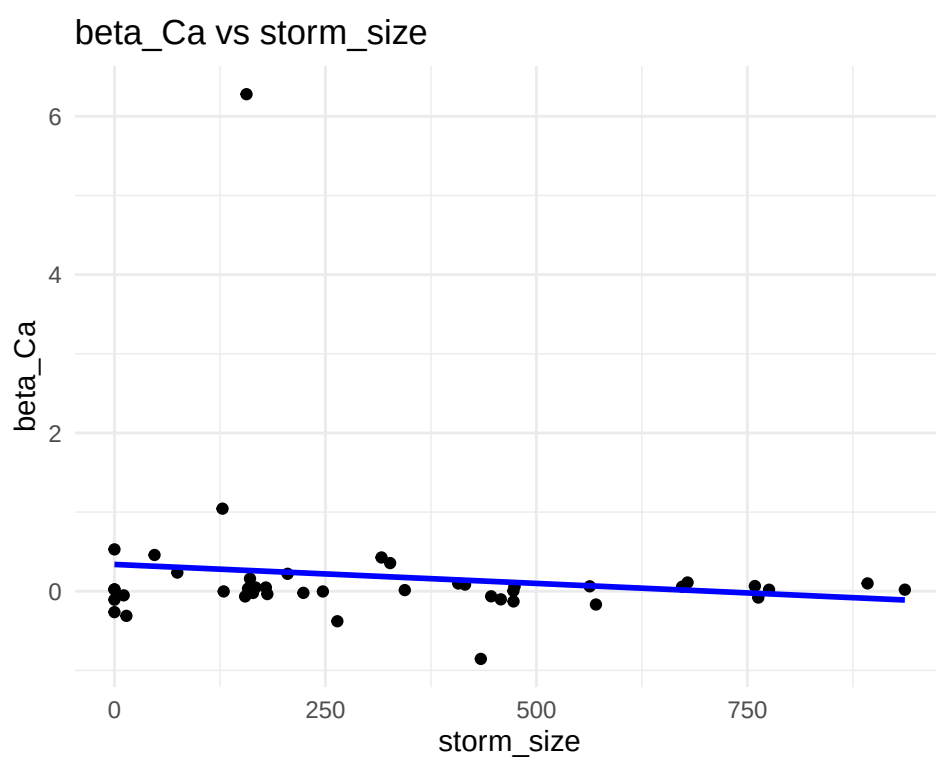
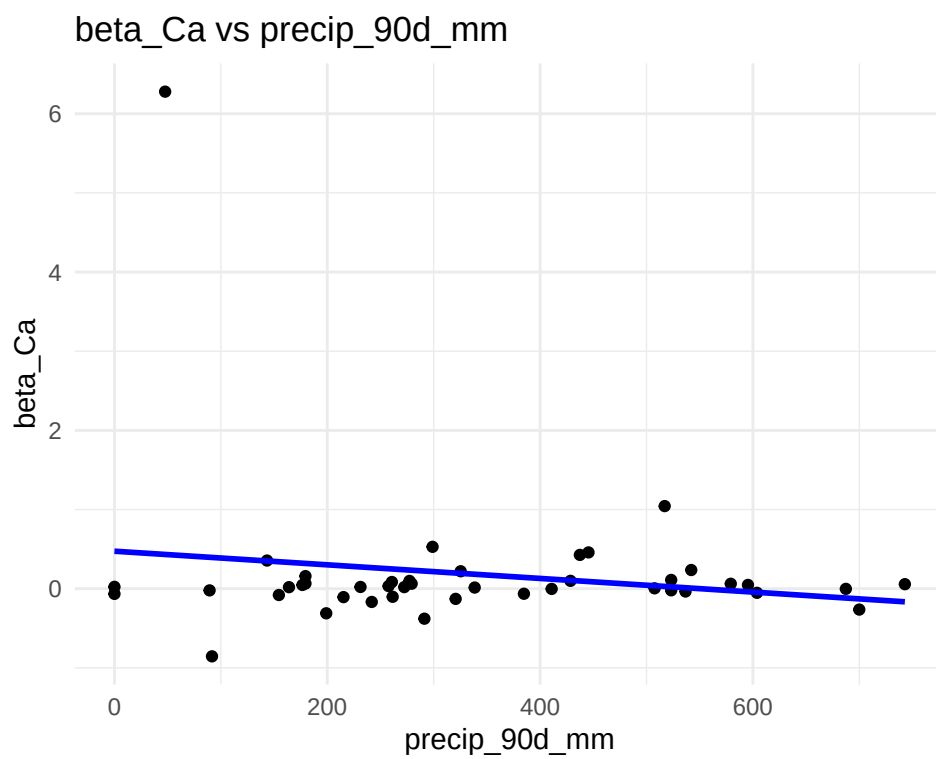


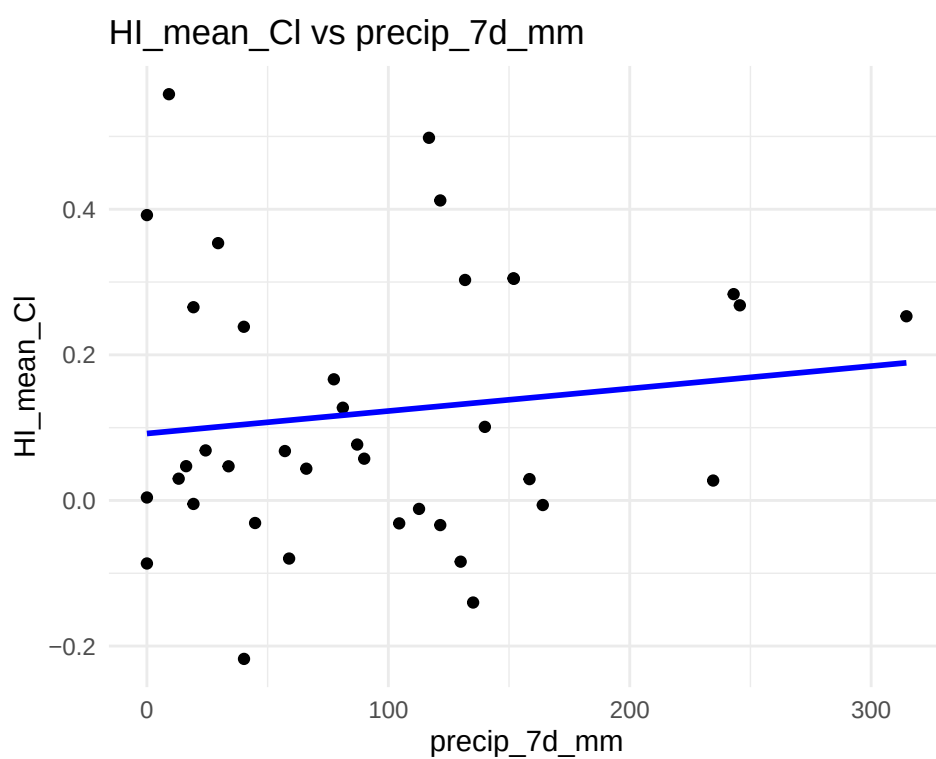
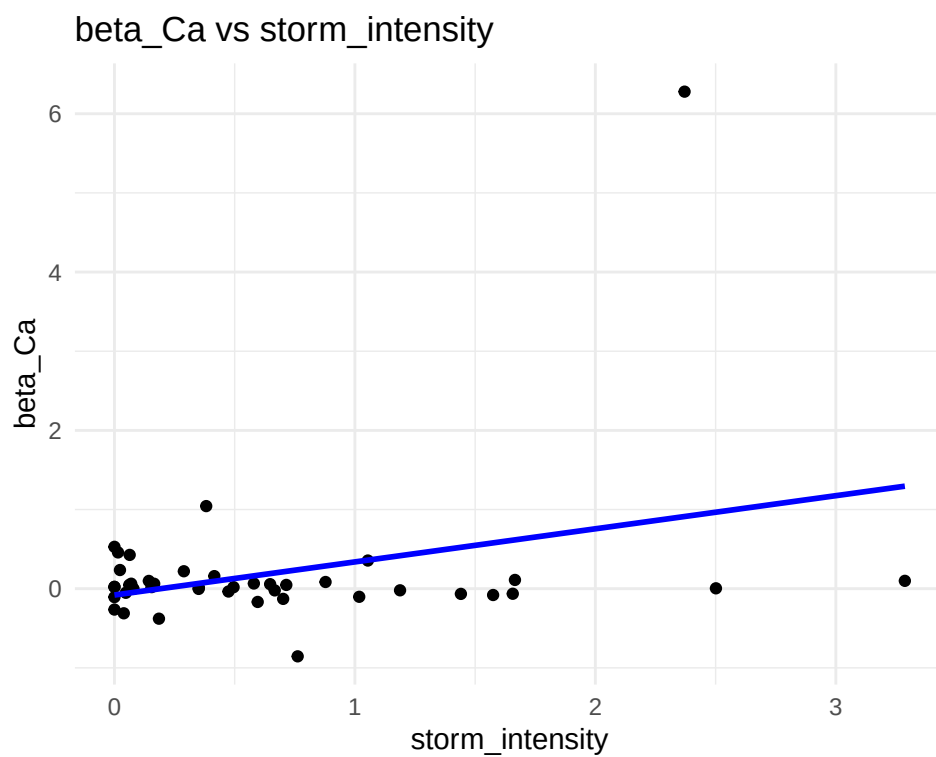


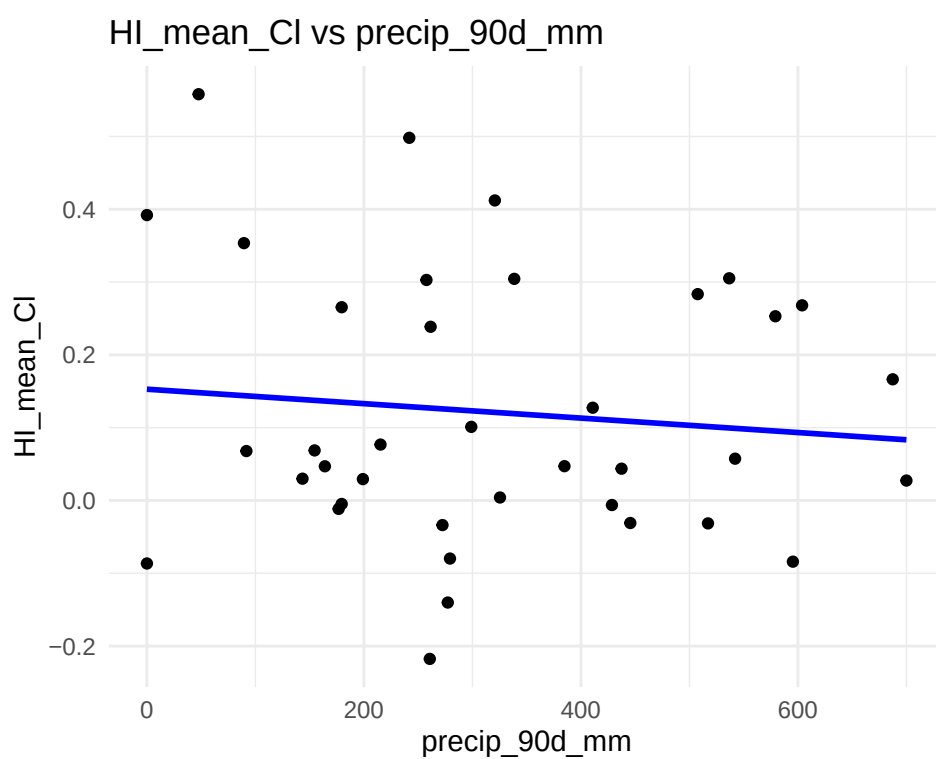
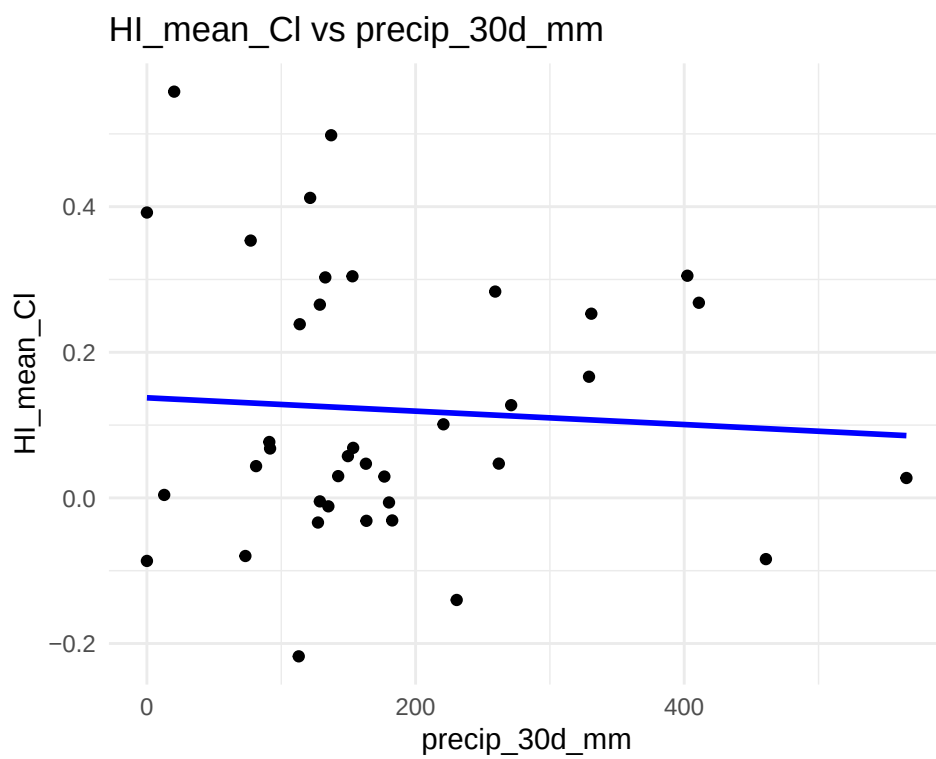




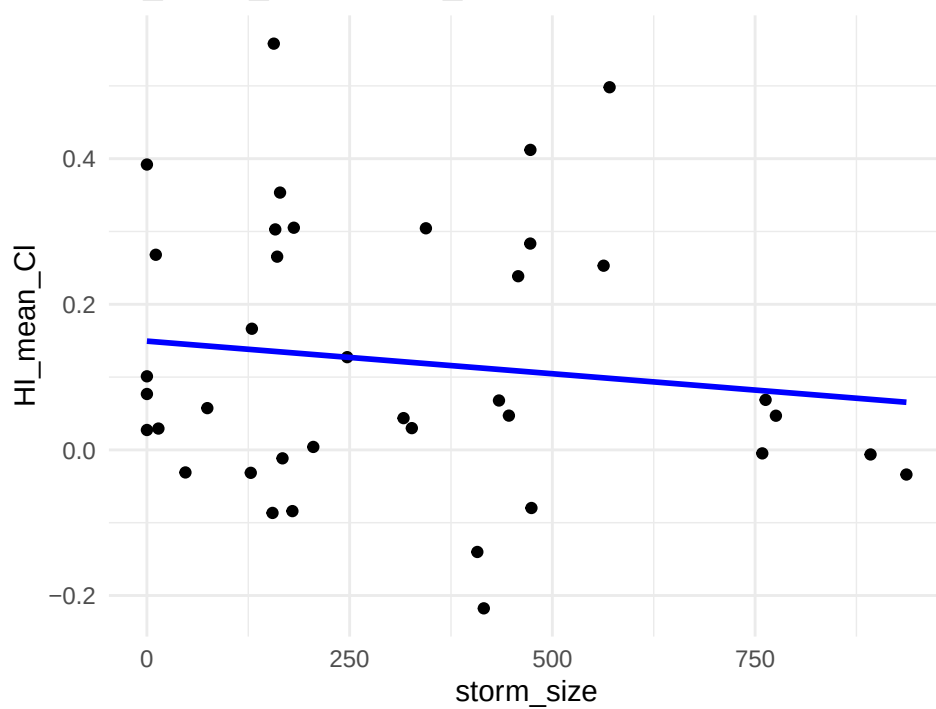




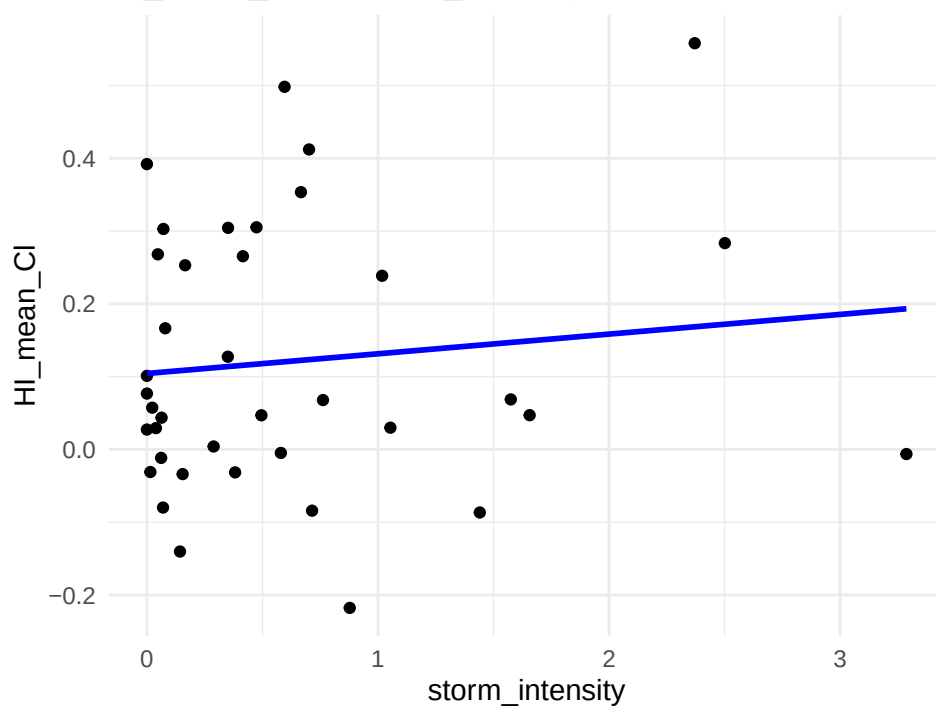


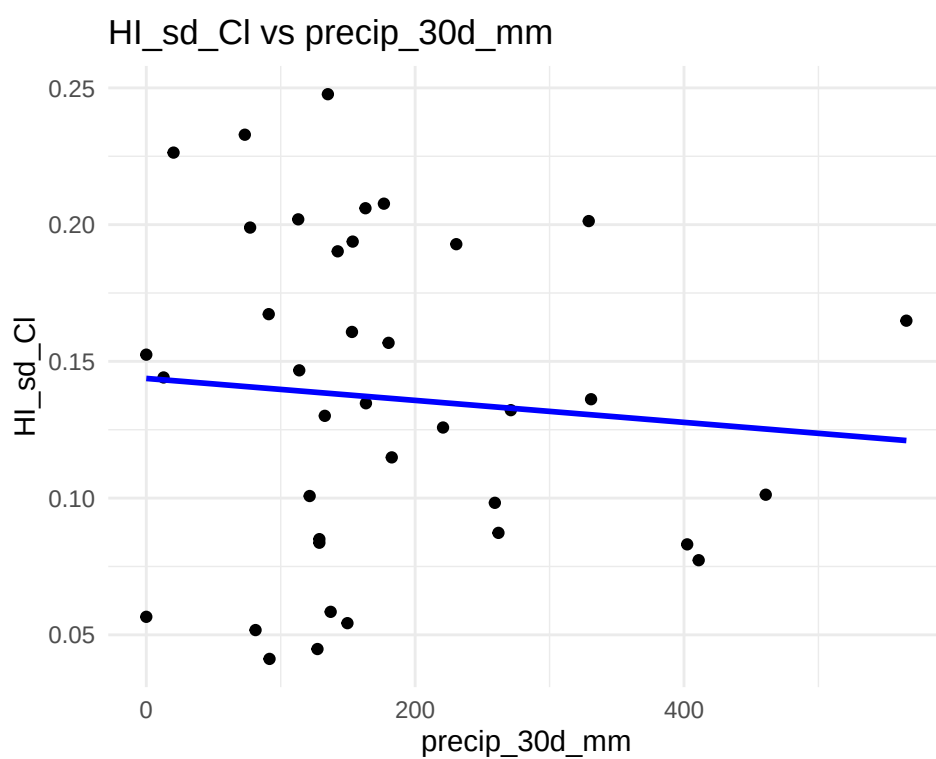
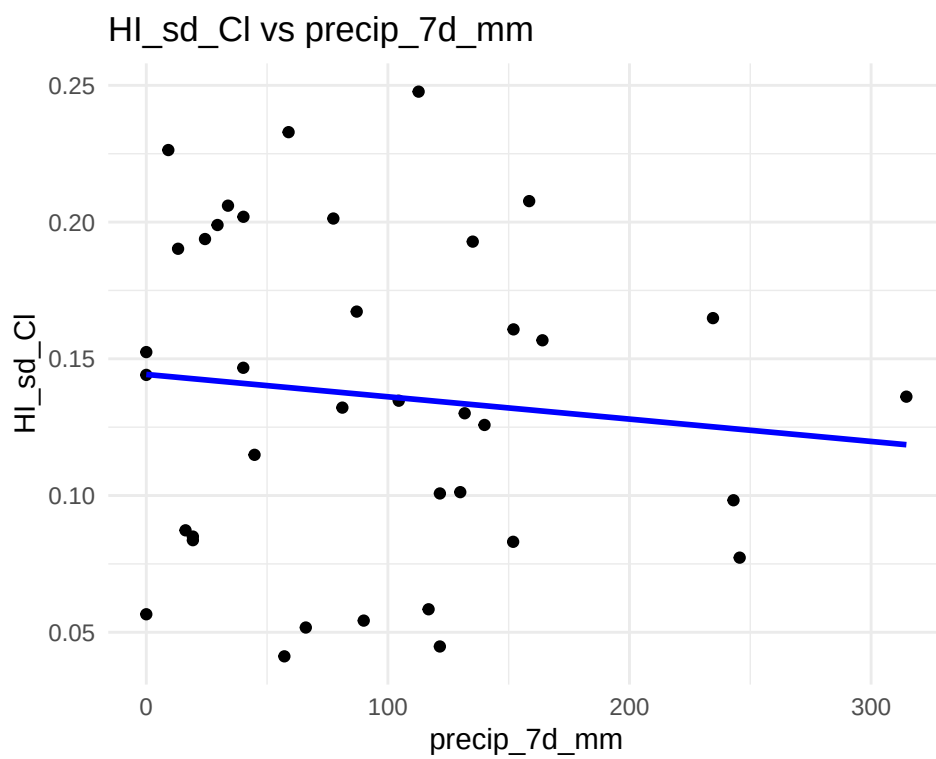


HI_mean_Cl vs storm_size

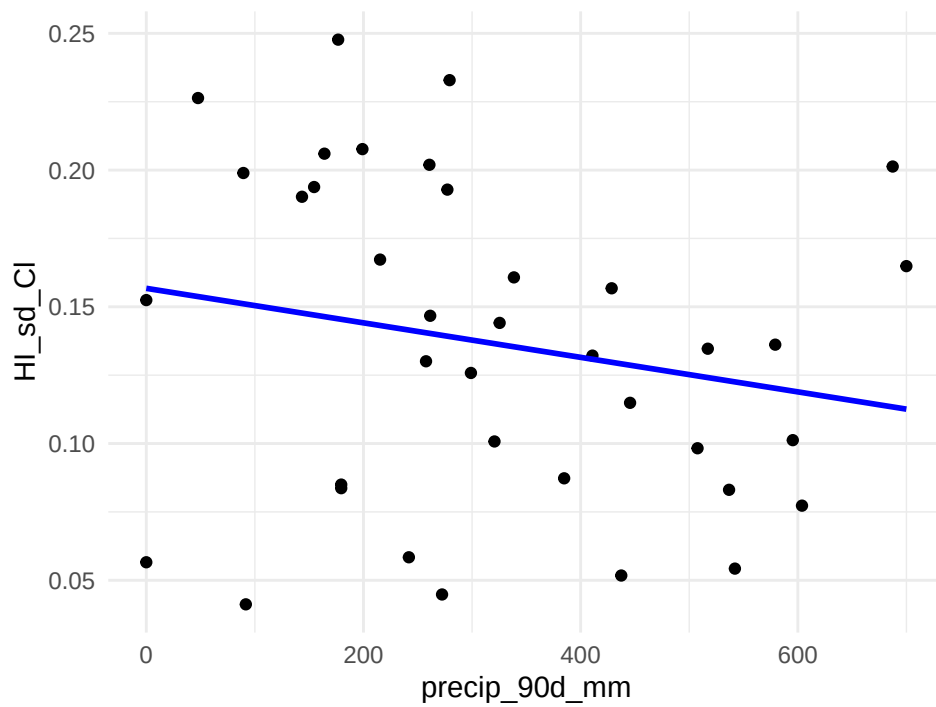


HI_mean_Cl vs storm_intensity

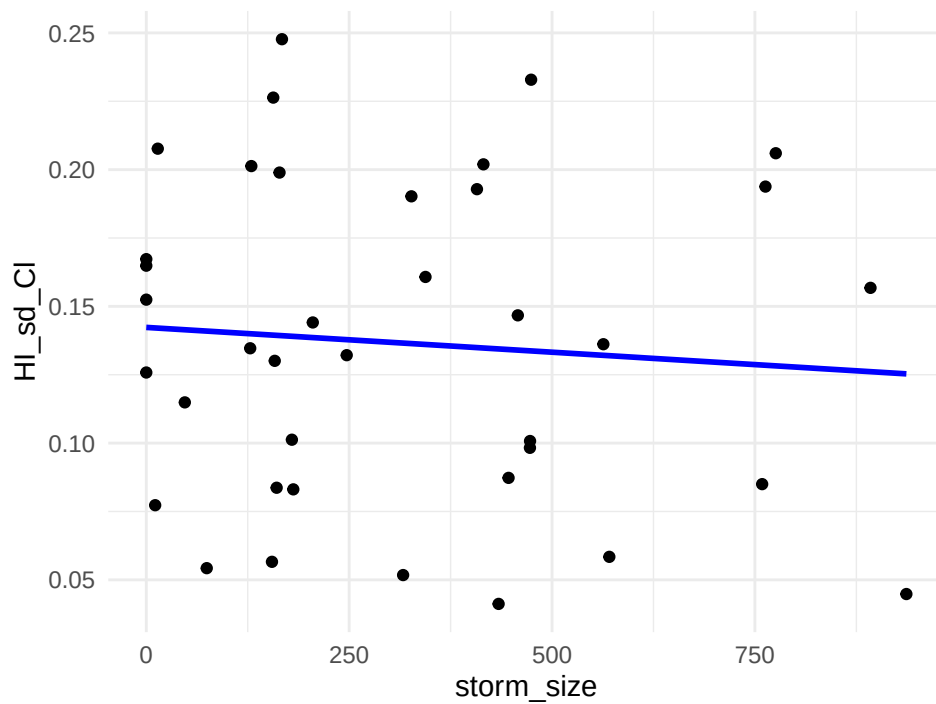


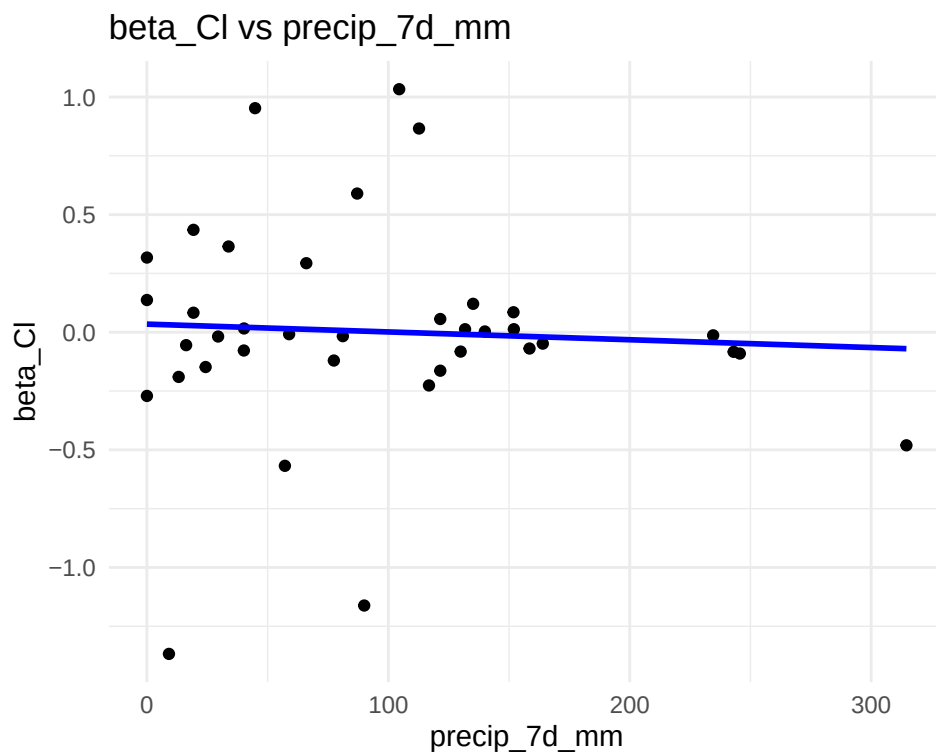
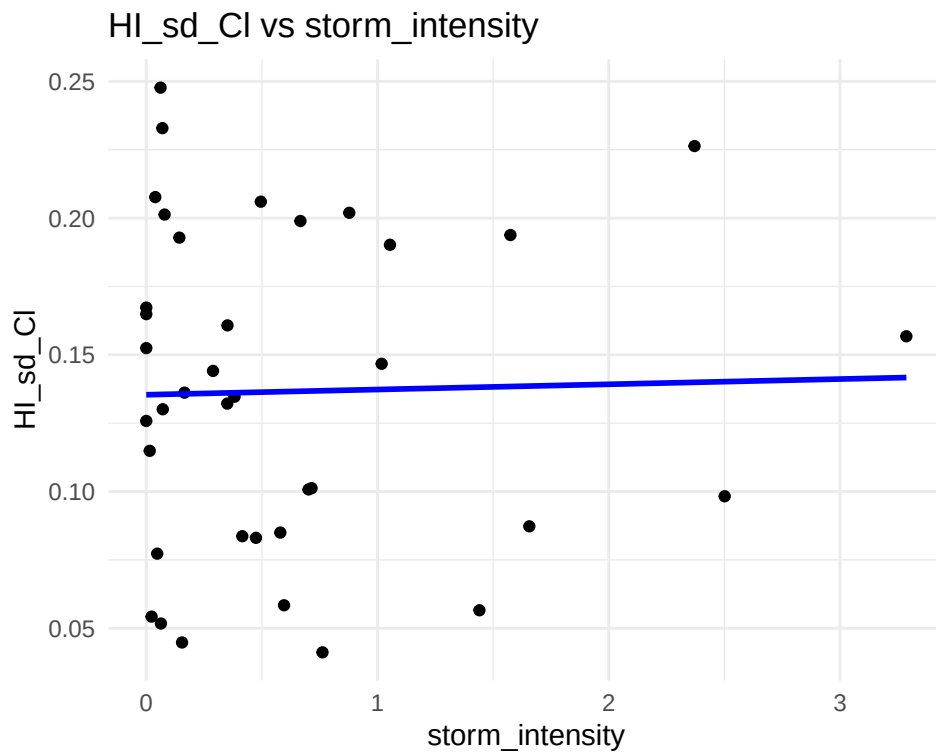


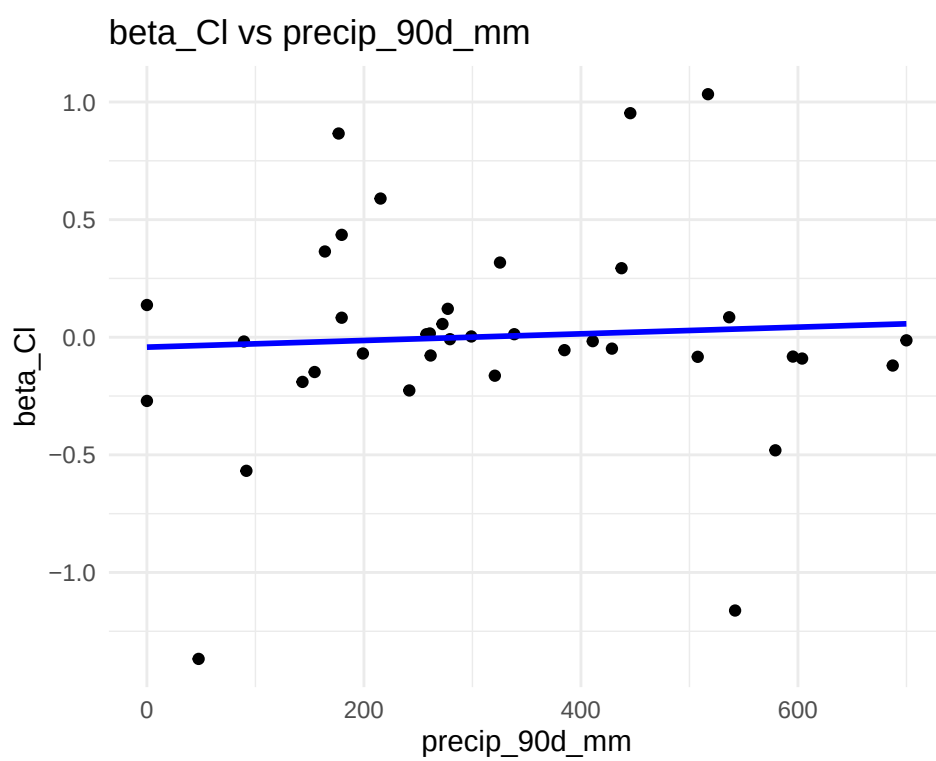
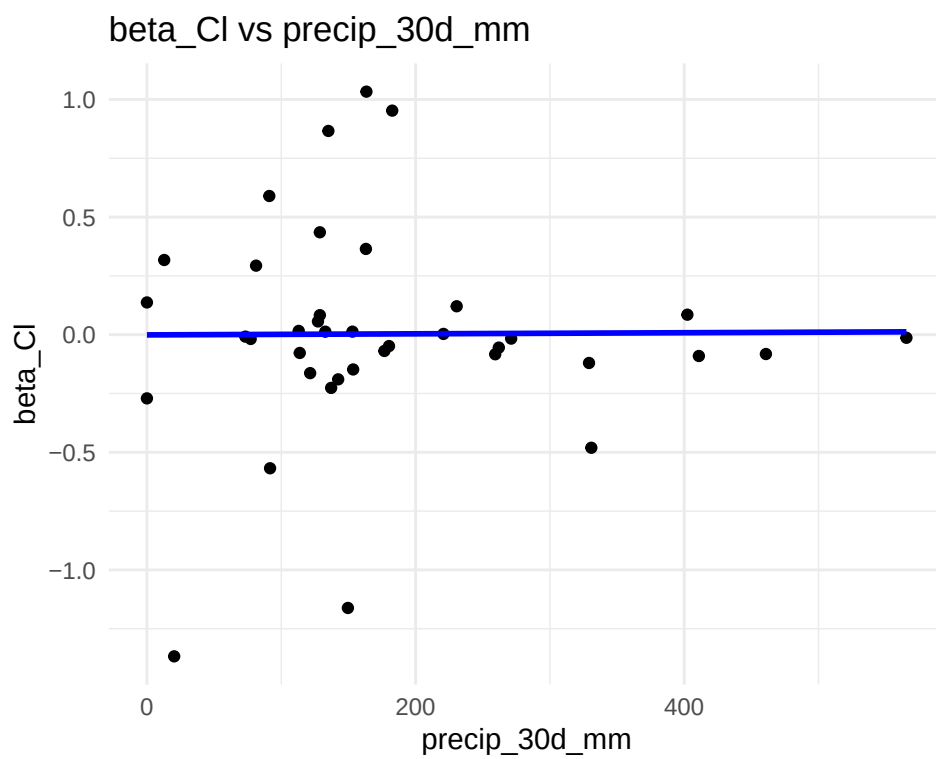
HI_sd_CI vs precip_90d_mm

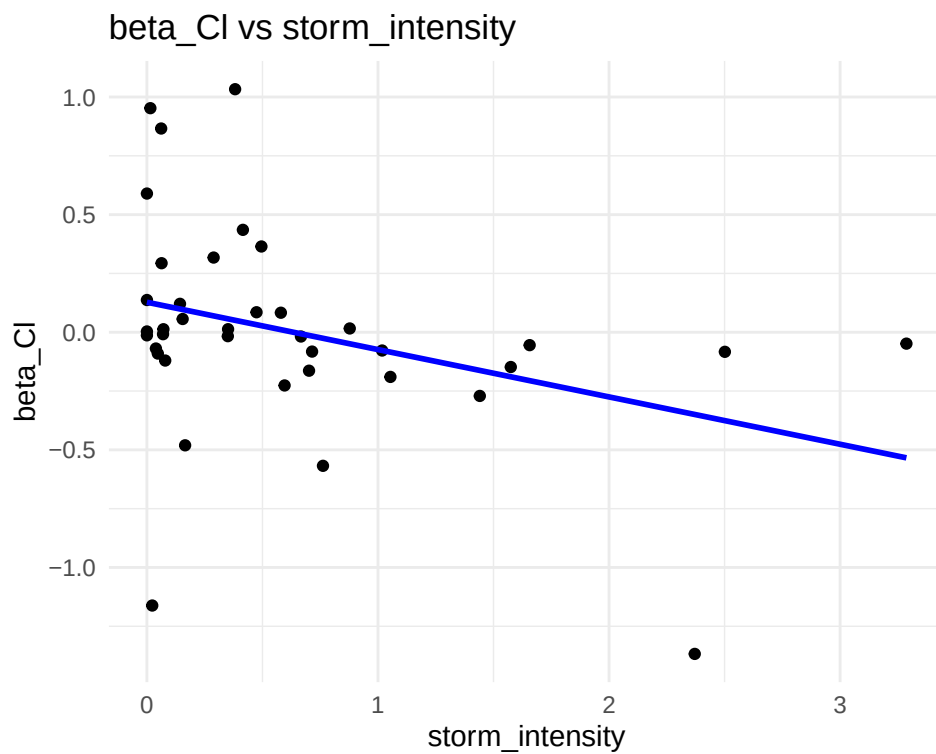
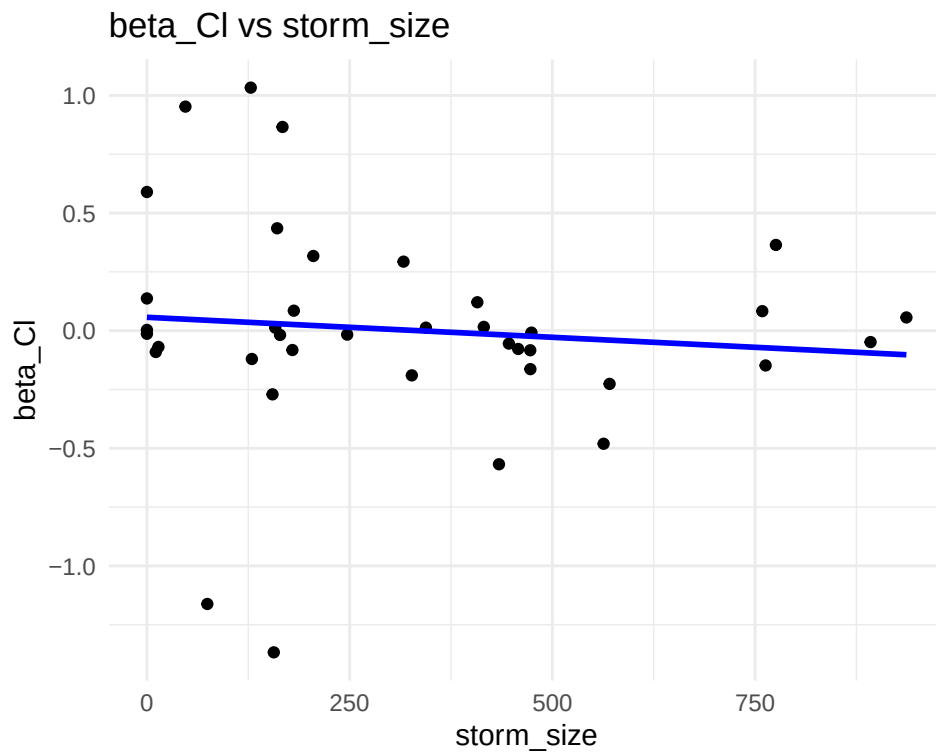


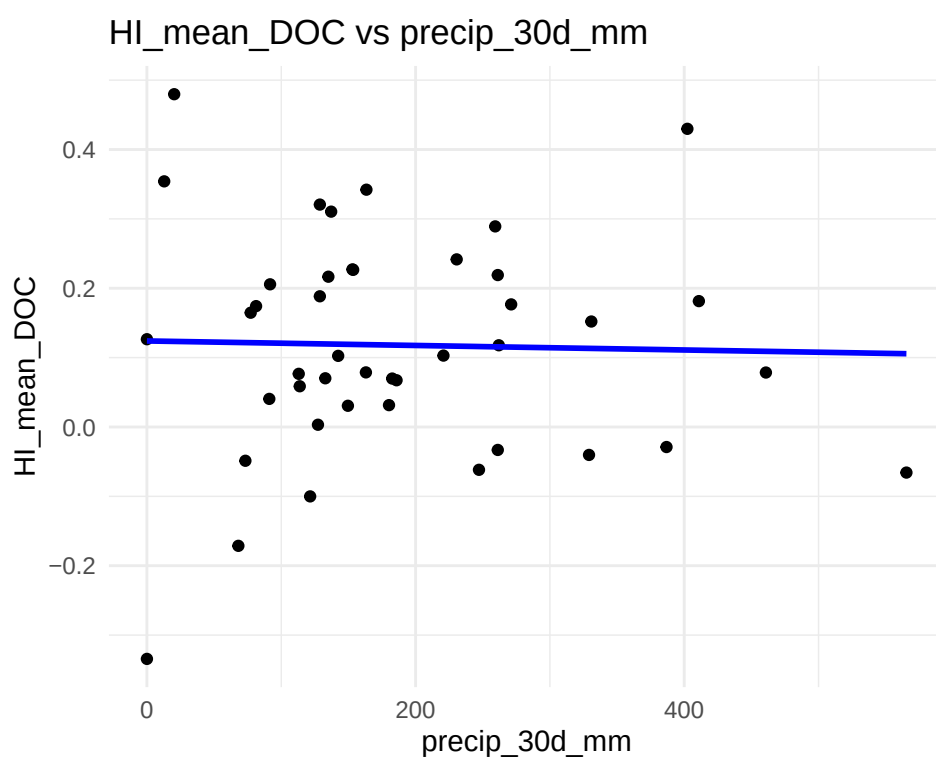
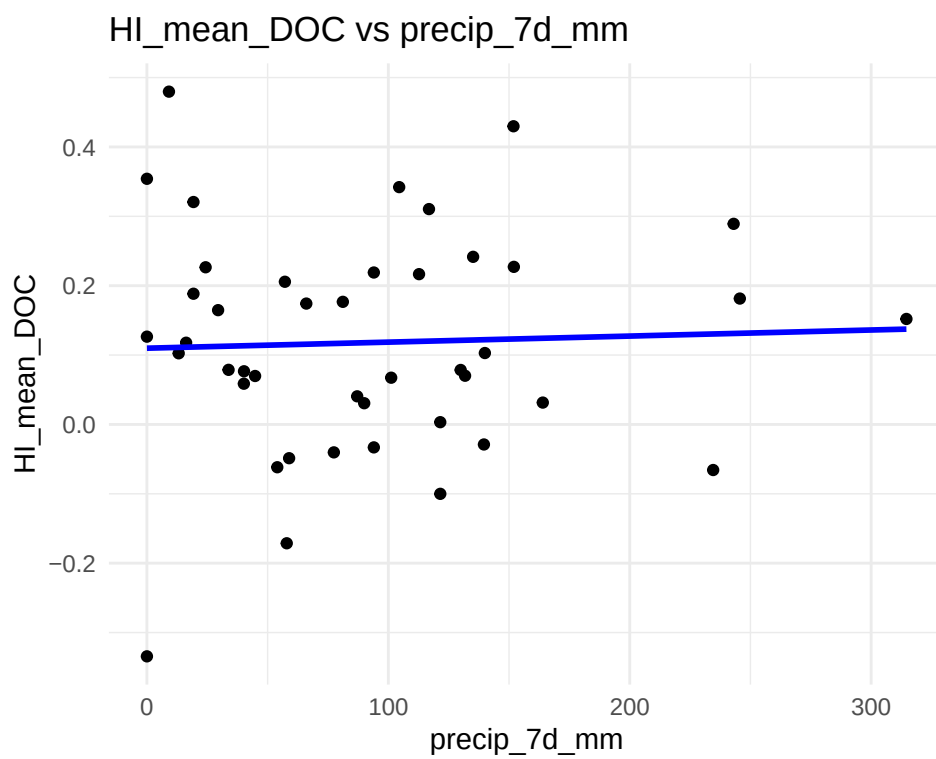
HI_sd_CI vs storm_size

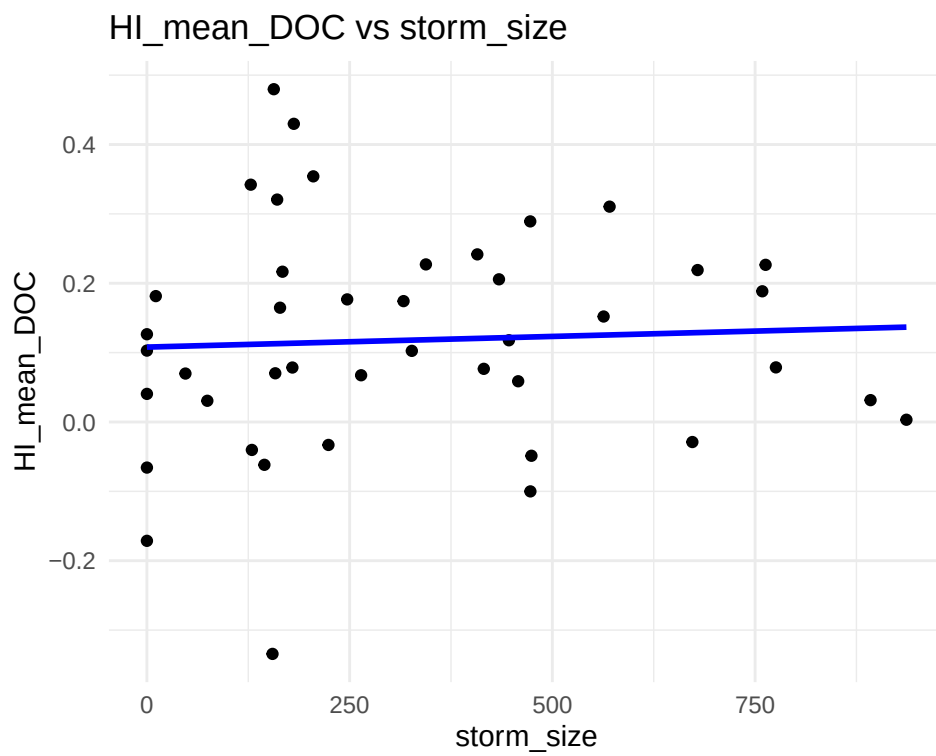
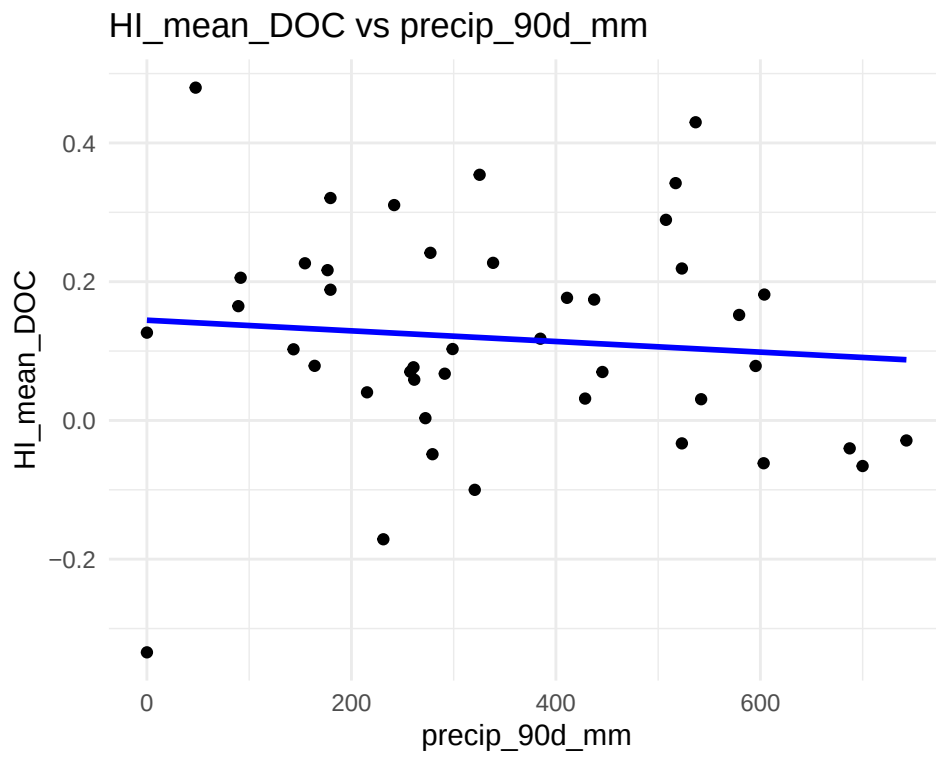


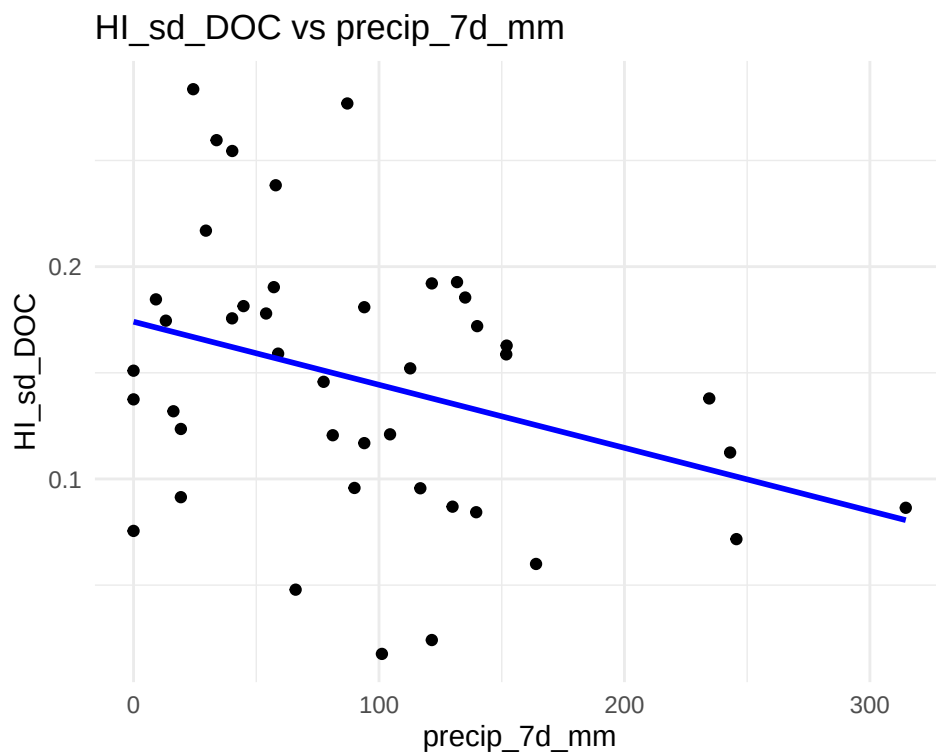
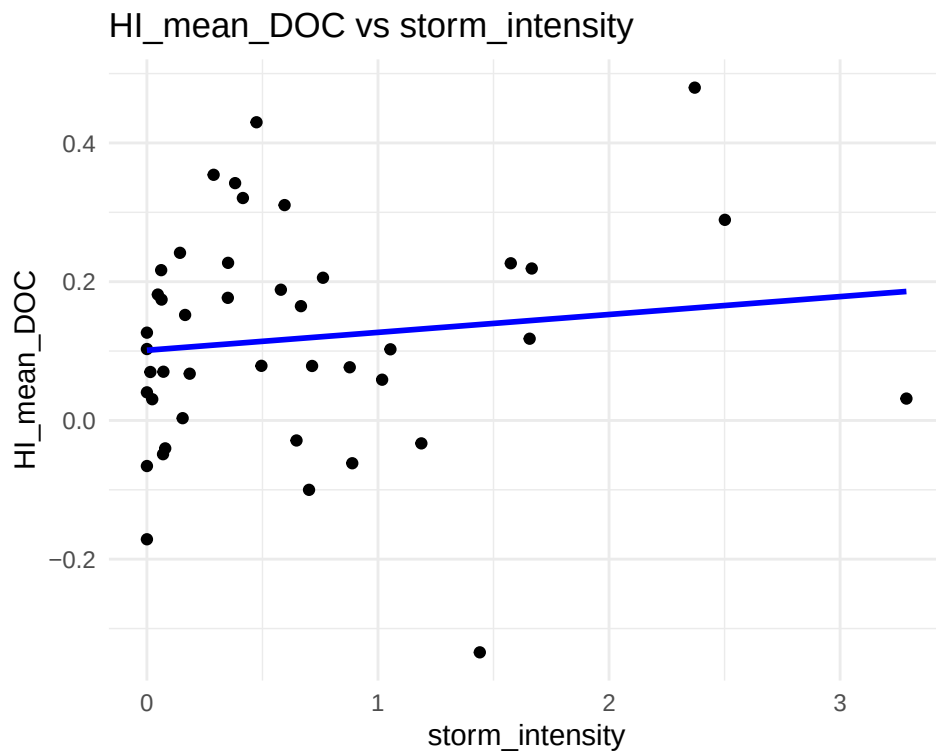


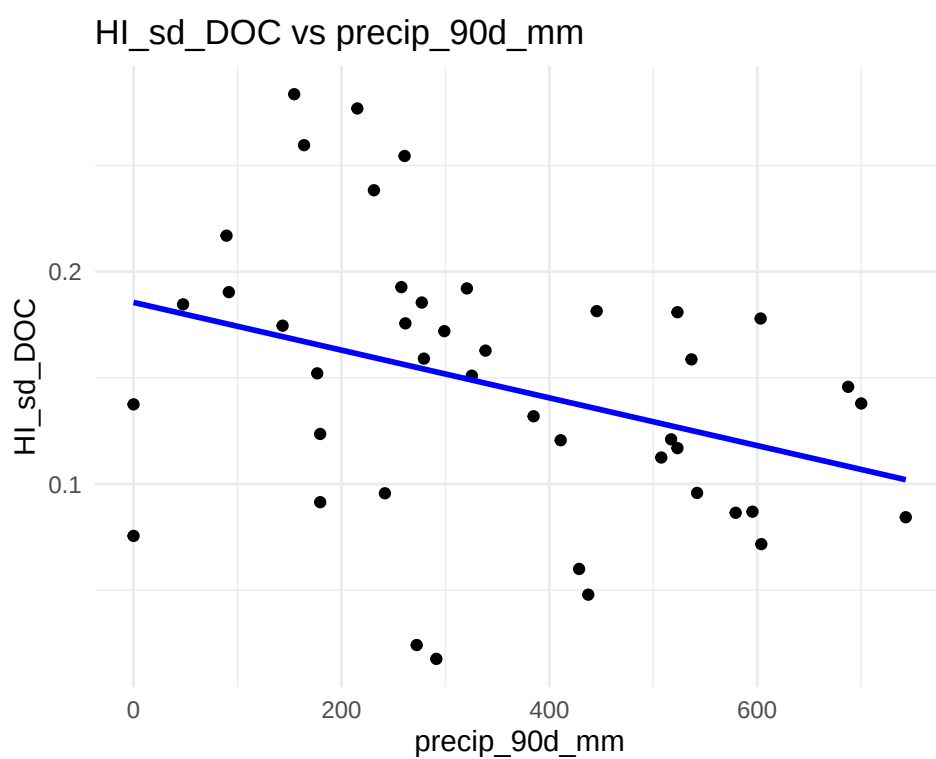
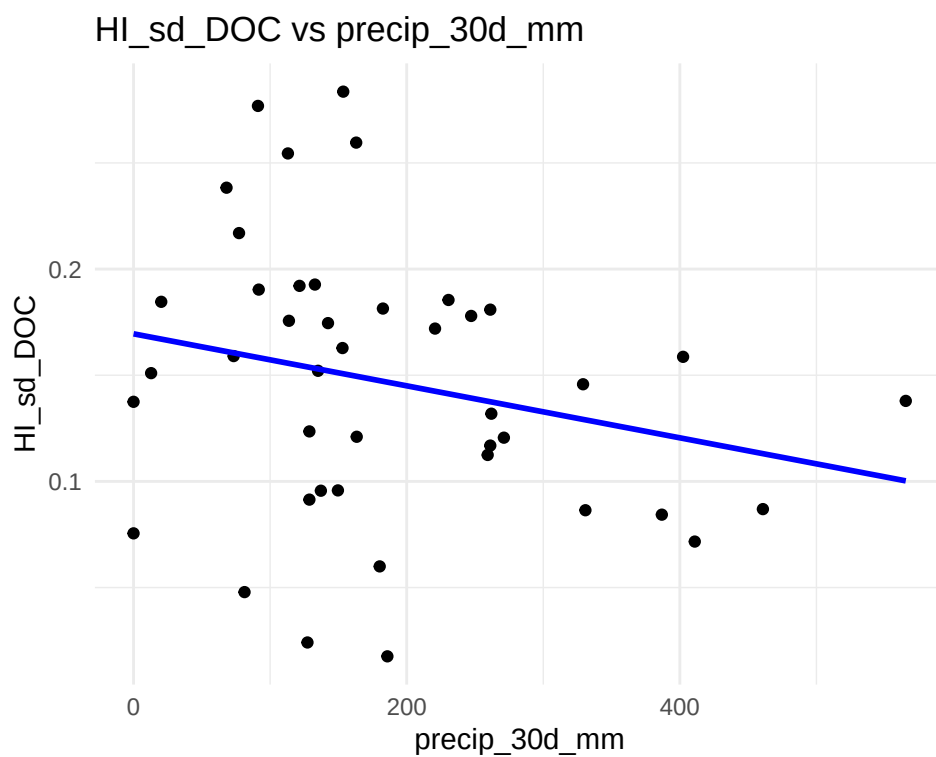


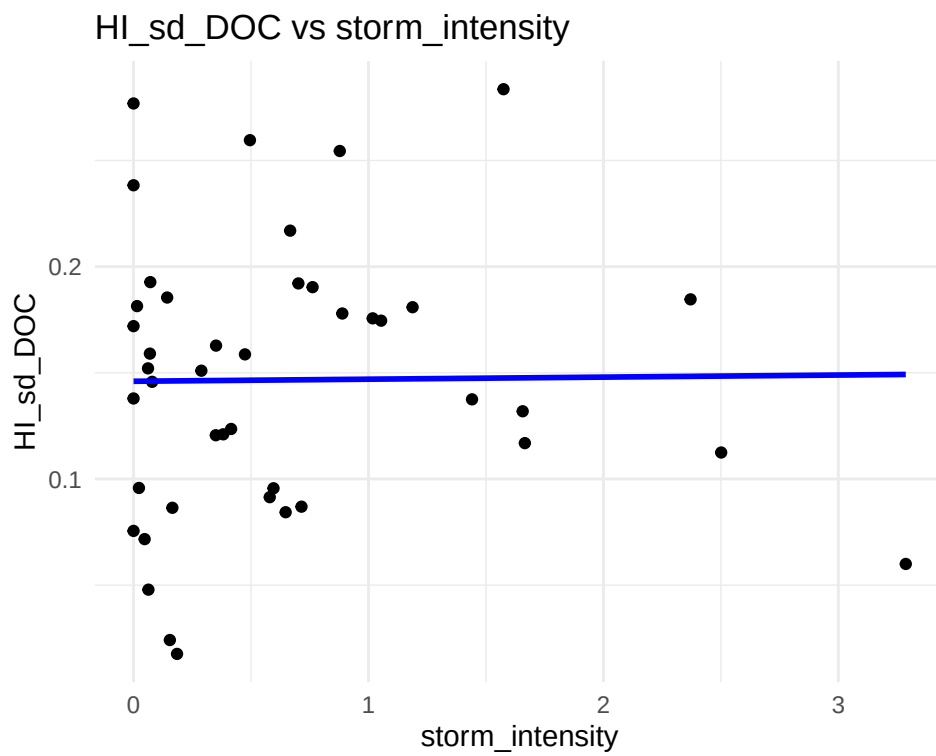
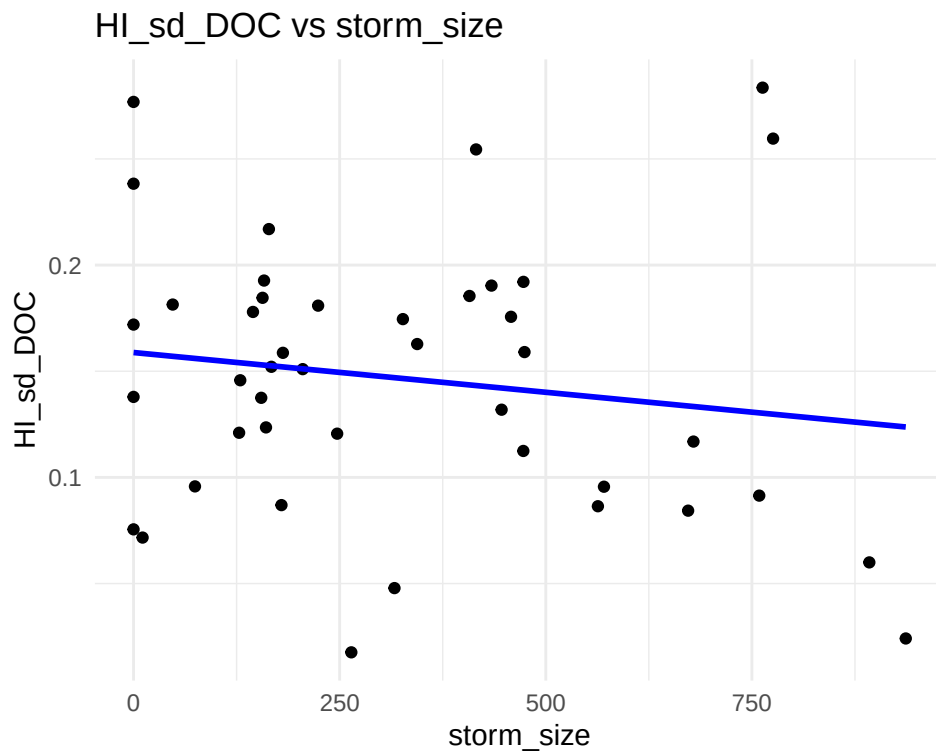




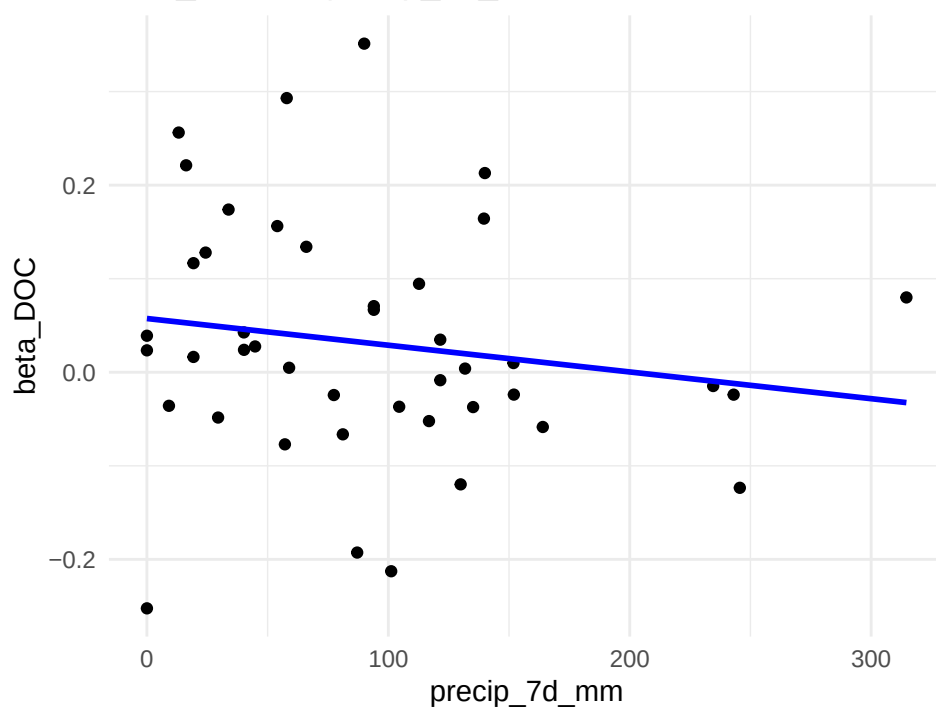




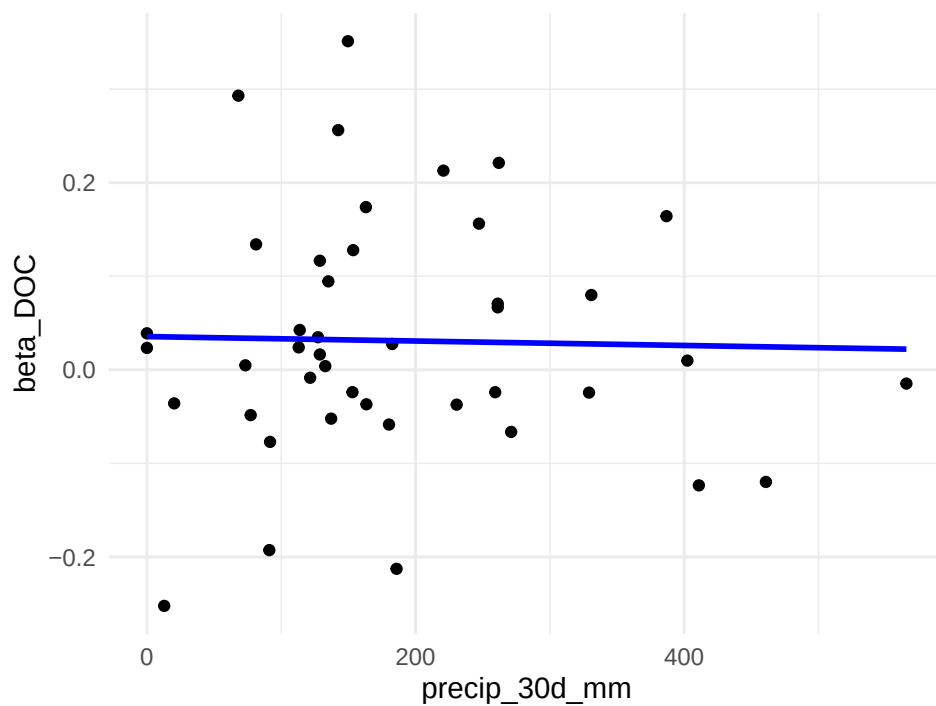




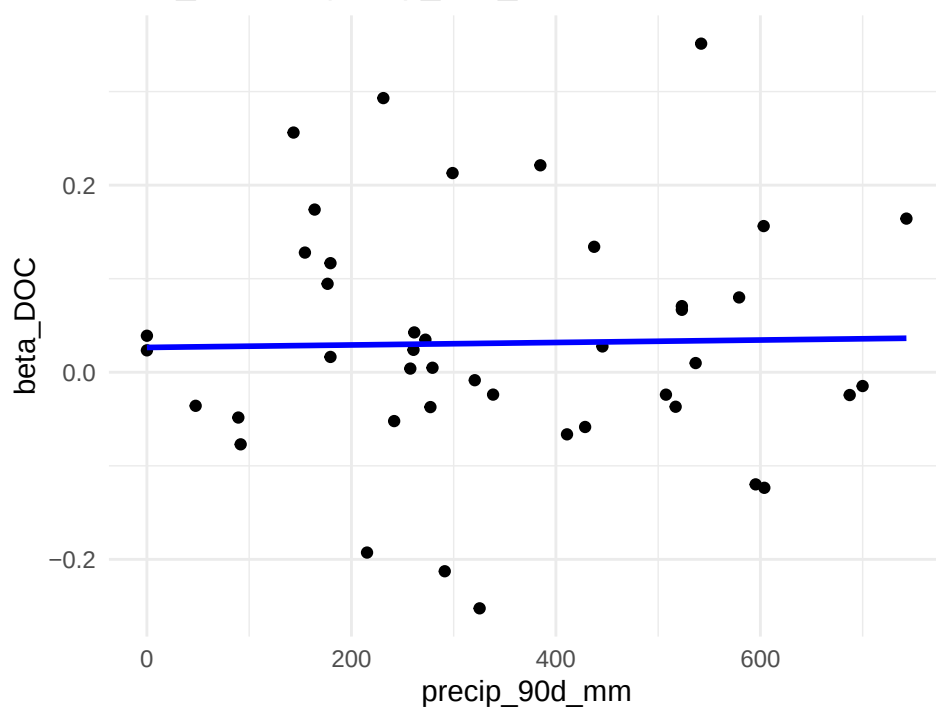
beta_DOC vs precip_7d_mm



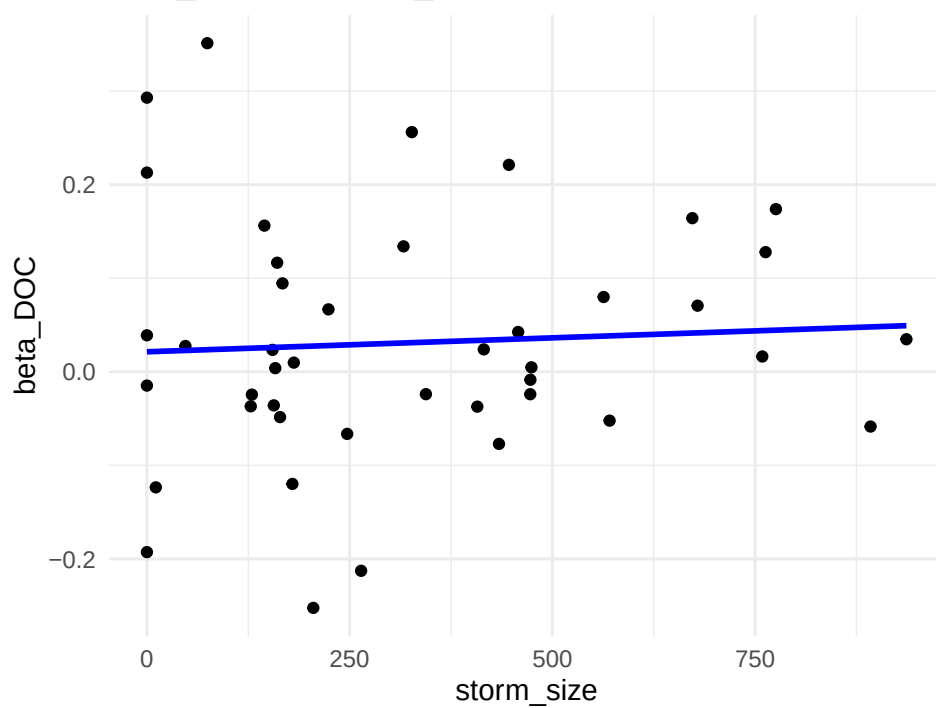
beta_DOC vs precip_30d_mm

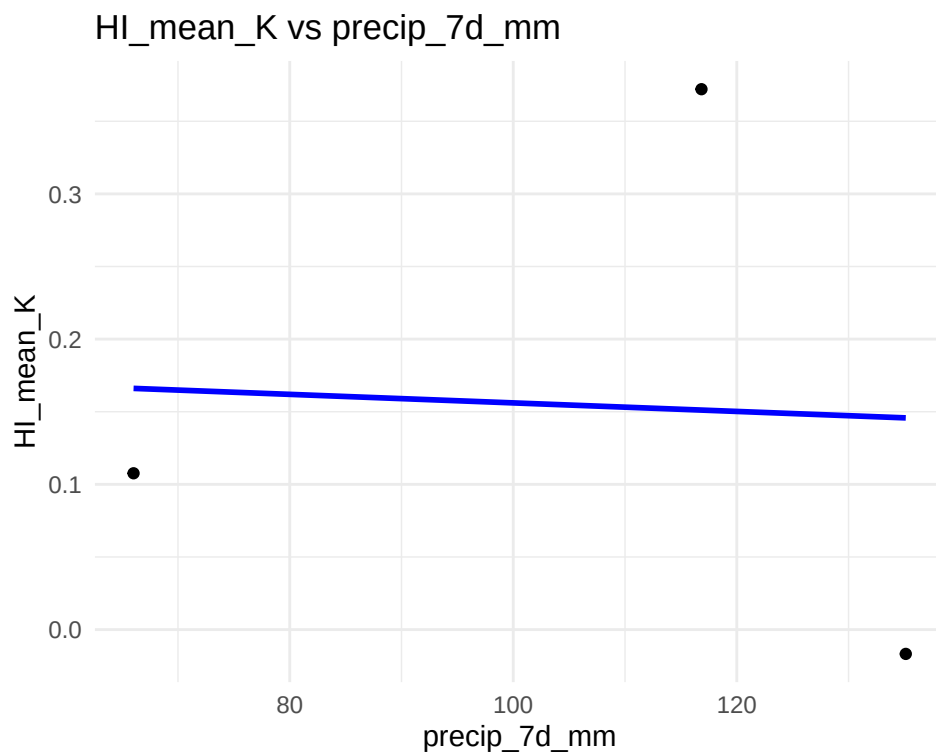
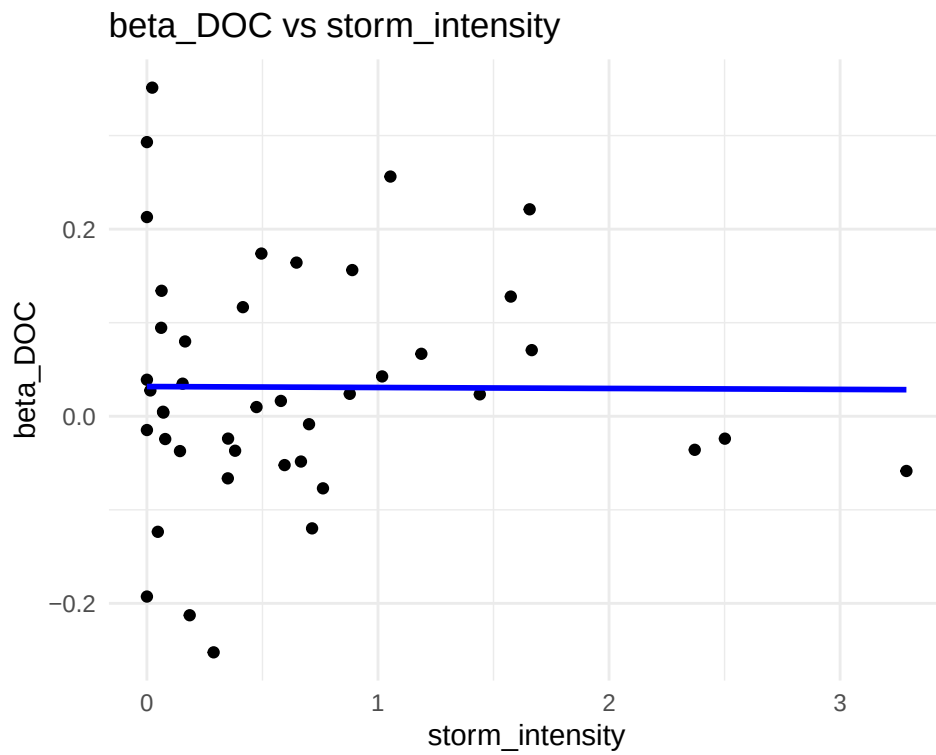


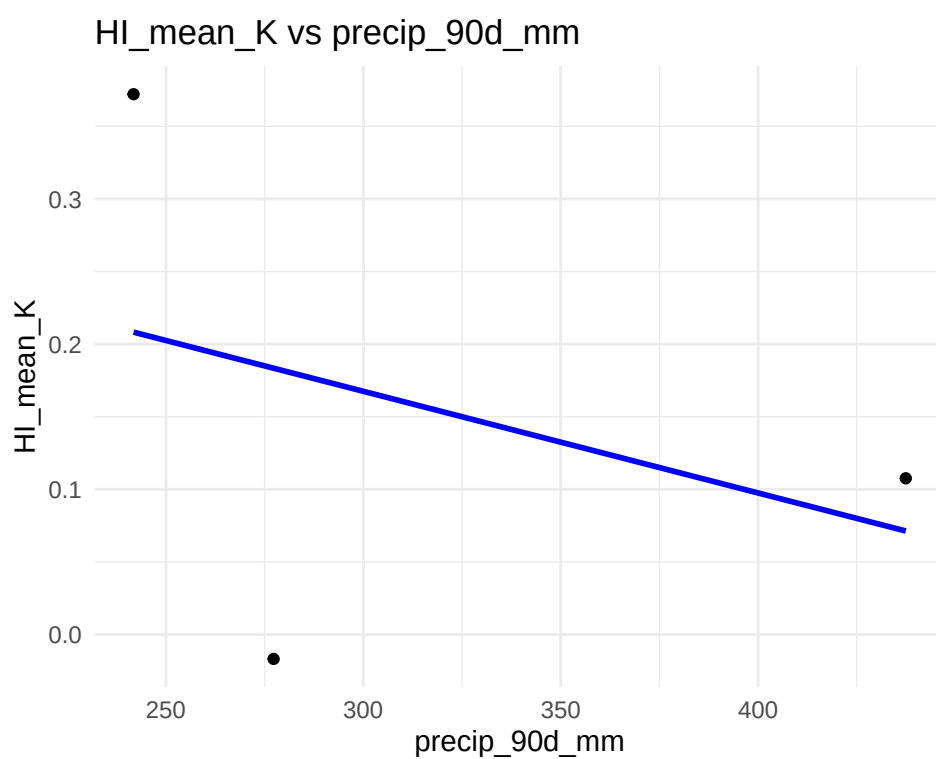
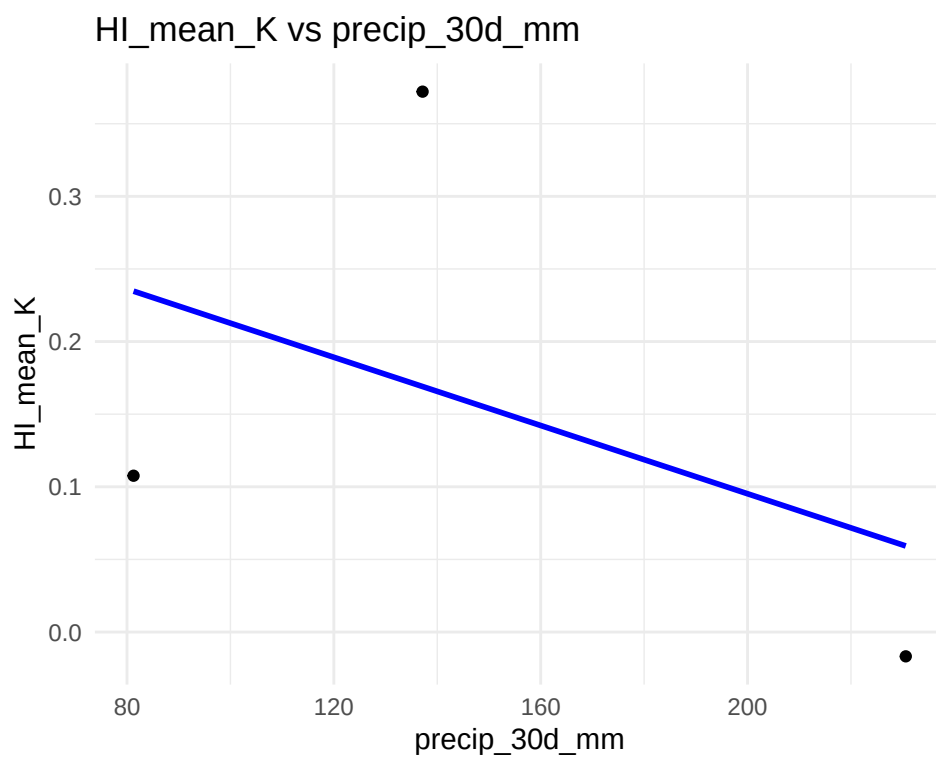
beta_DOC vs precip_90d_mm

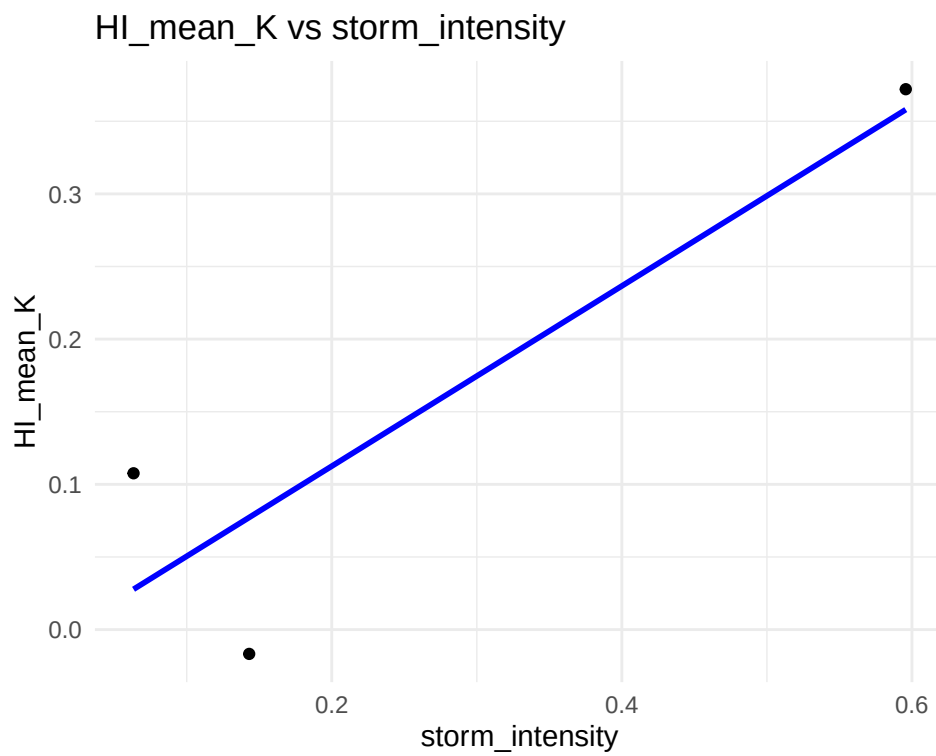
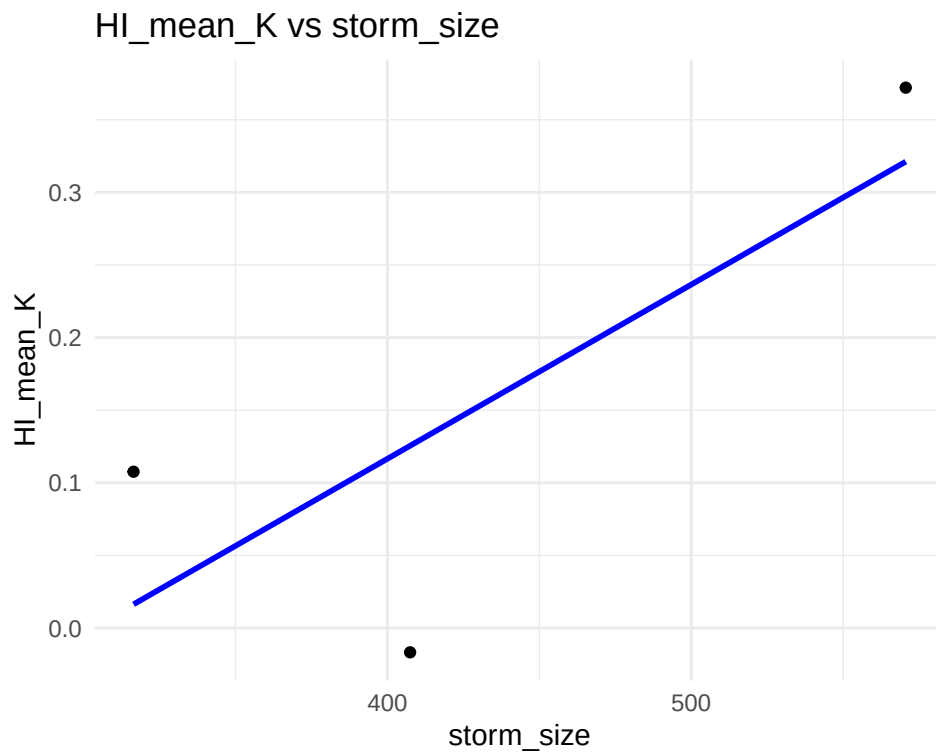


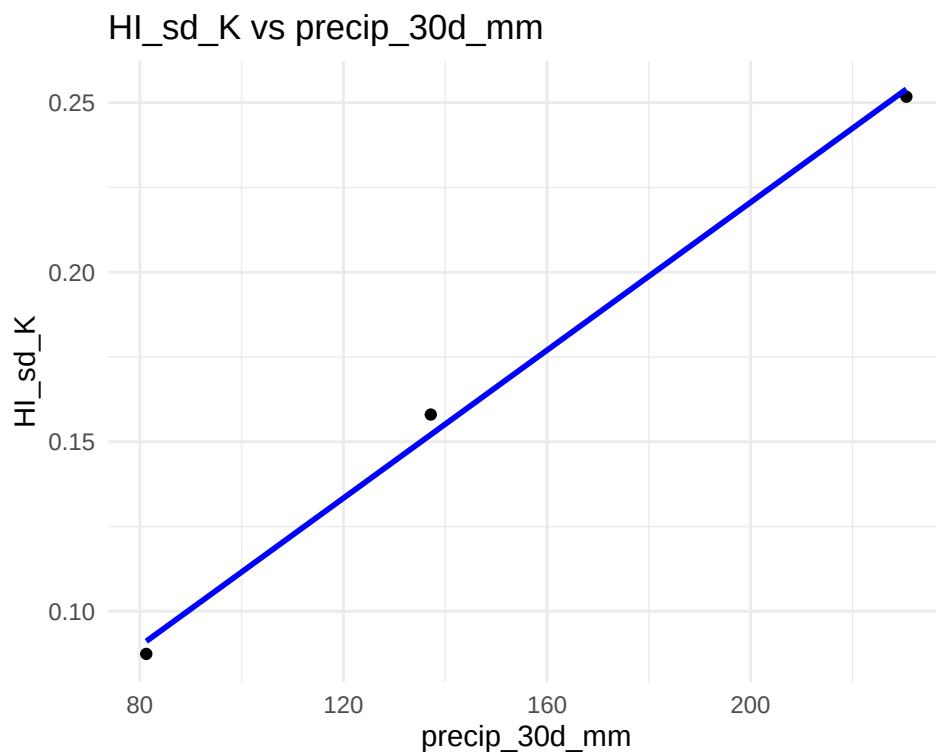
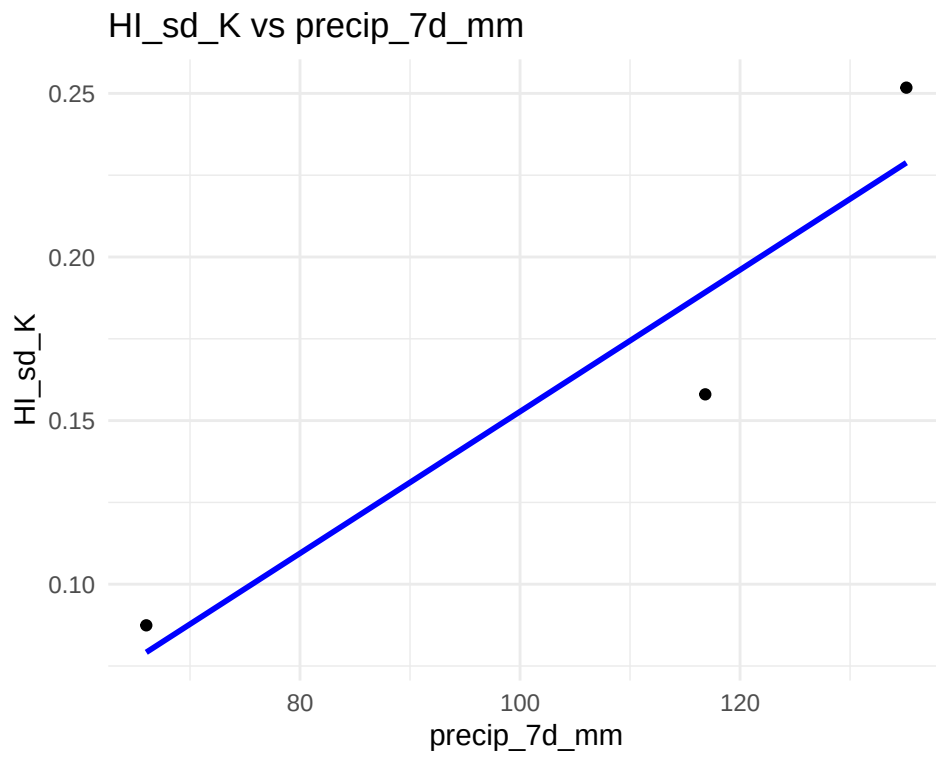
beta_DOC vs storm_size

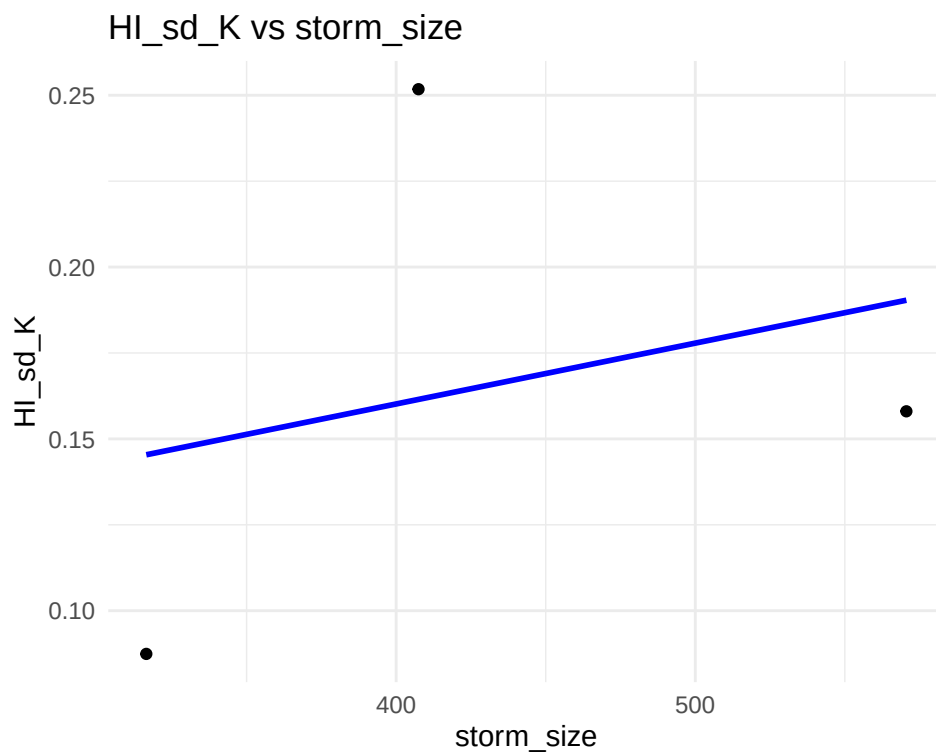
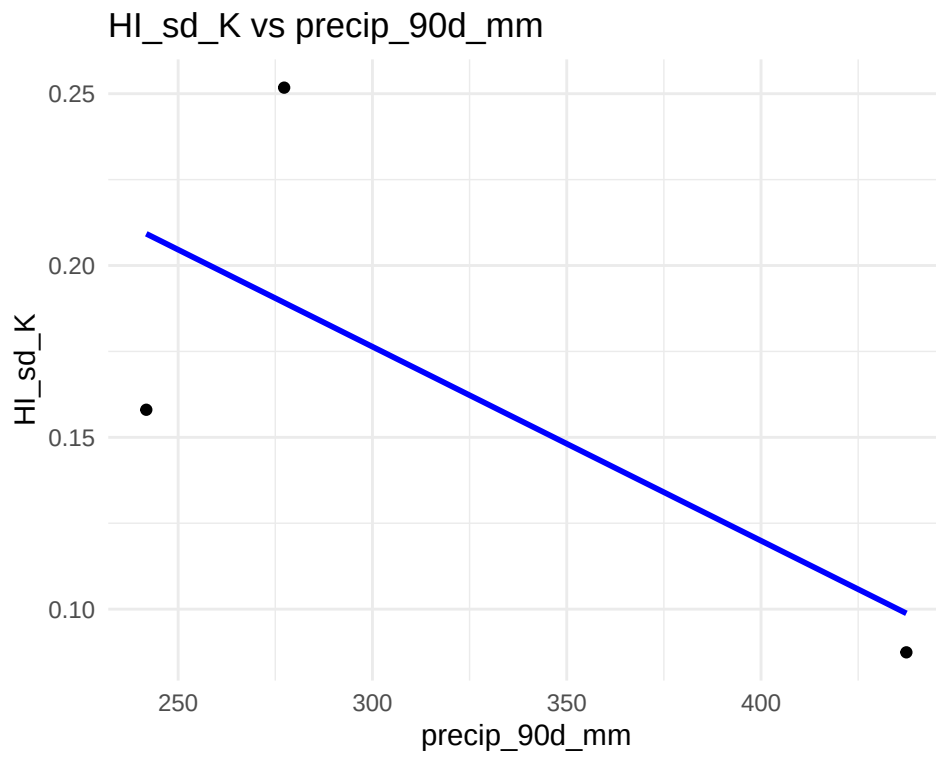


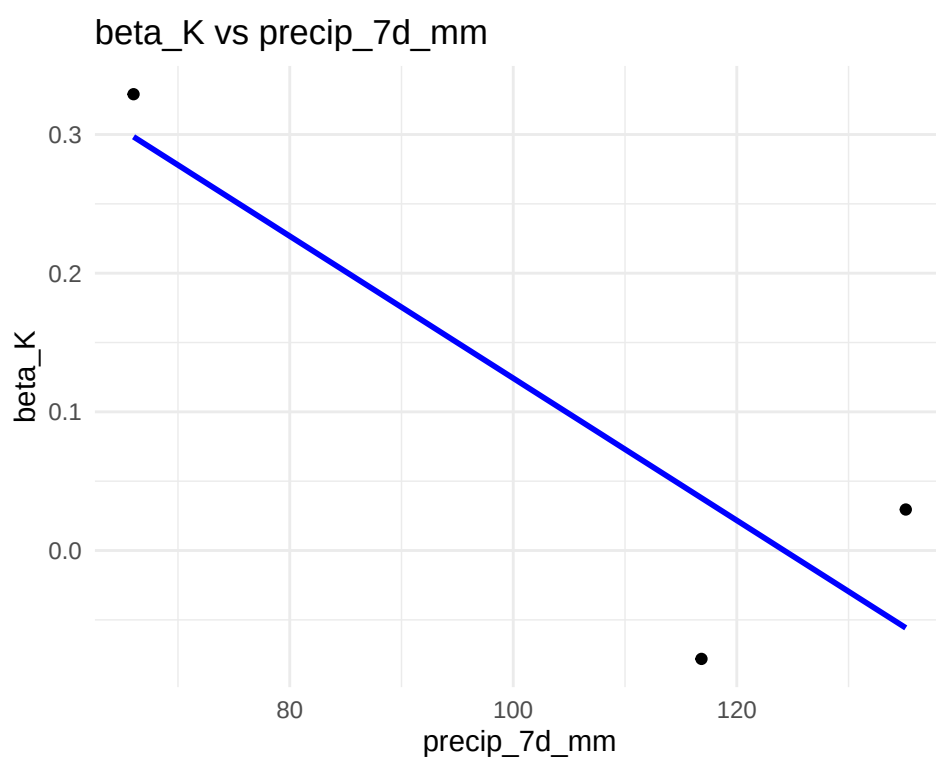
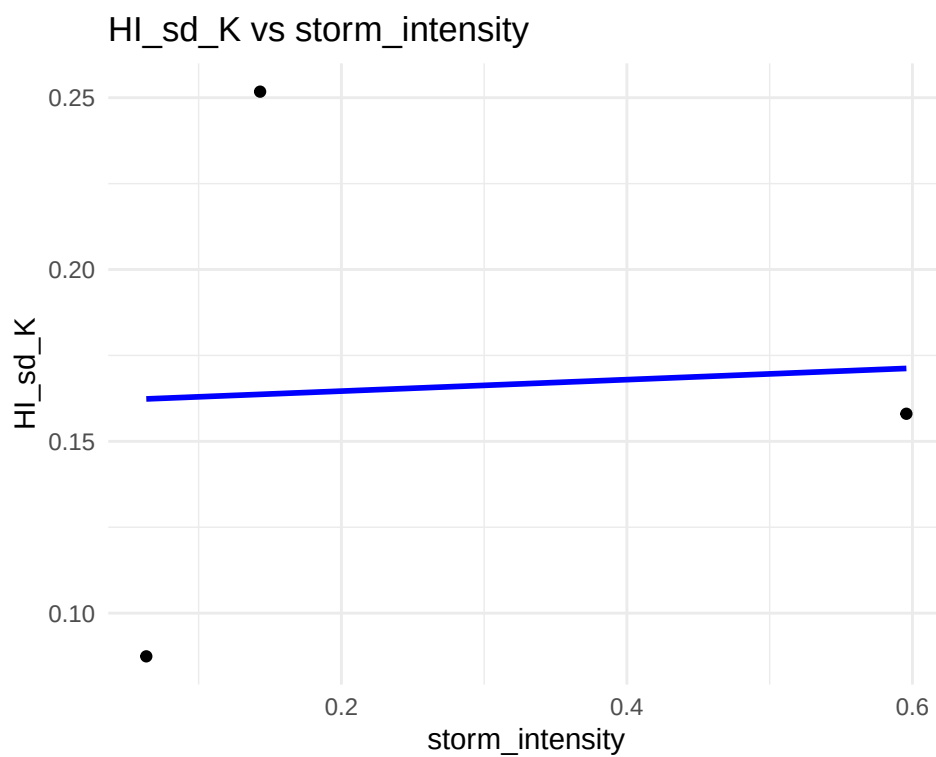


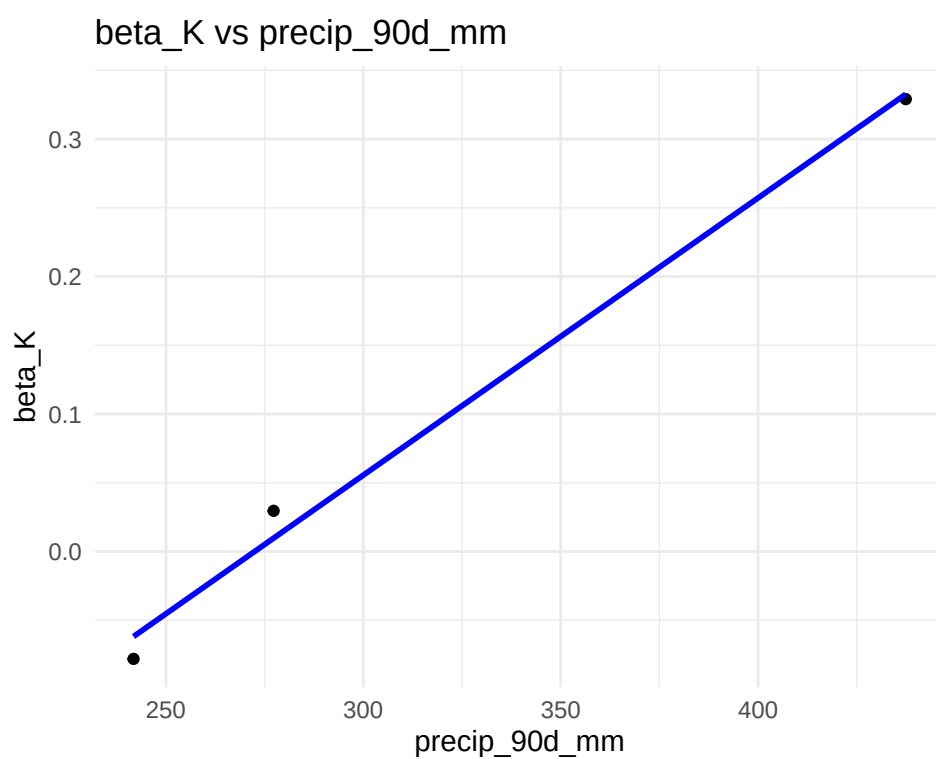
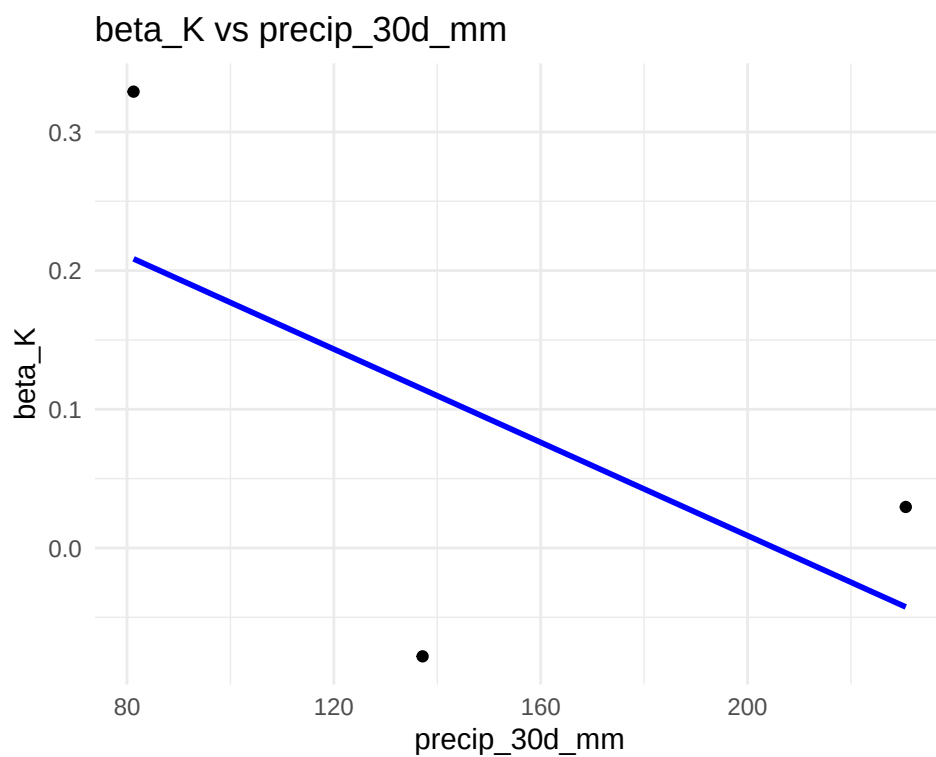


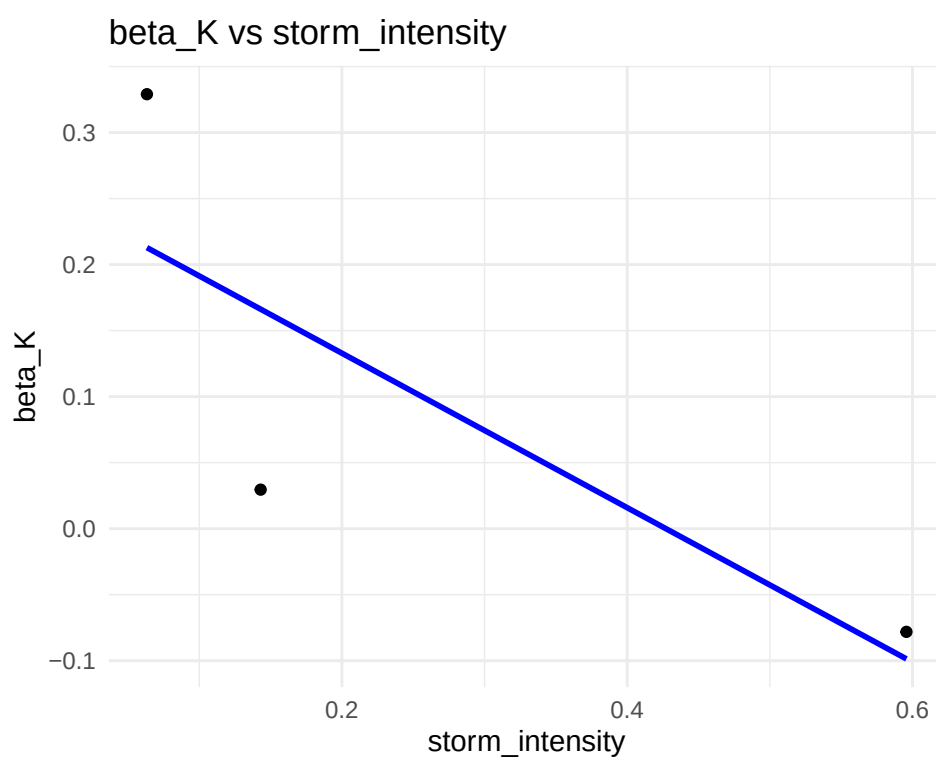
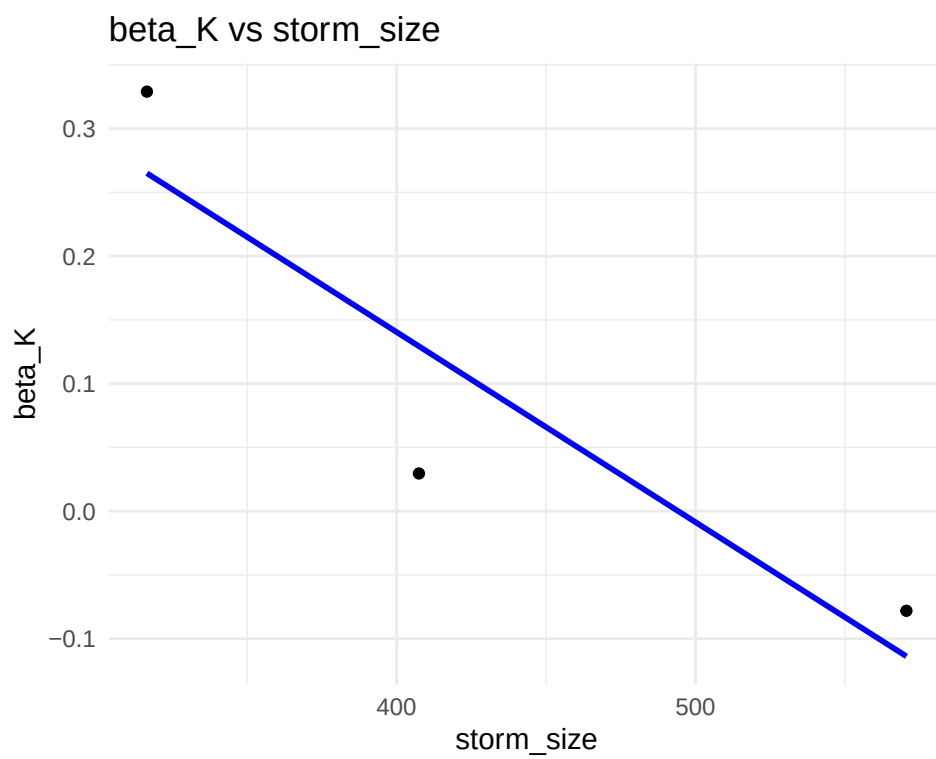


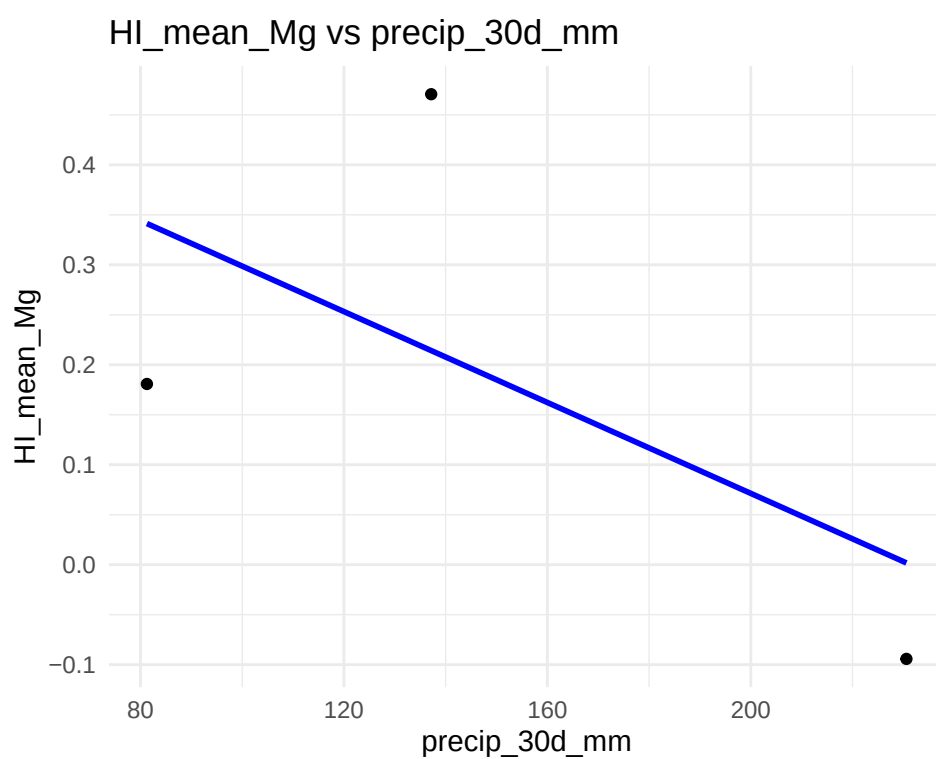
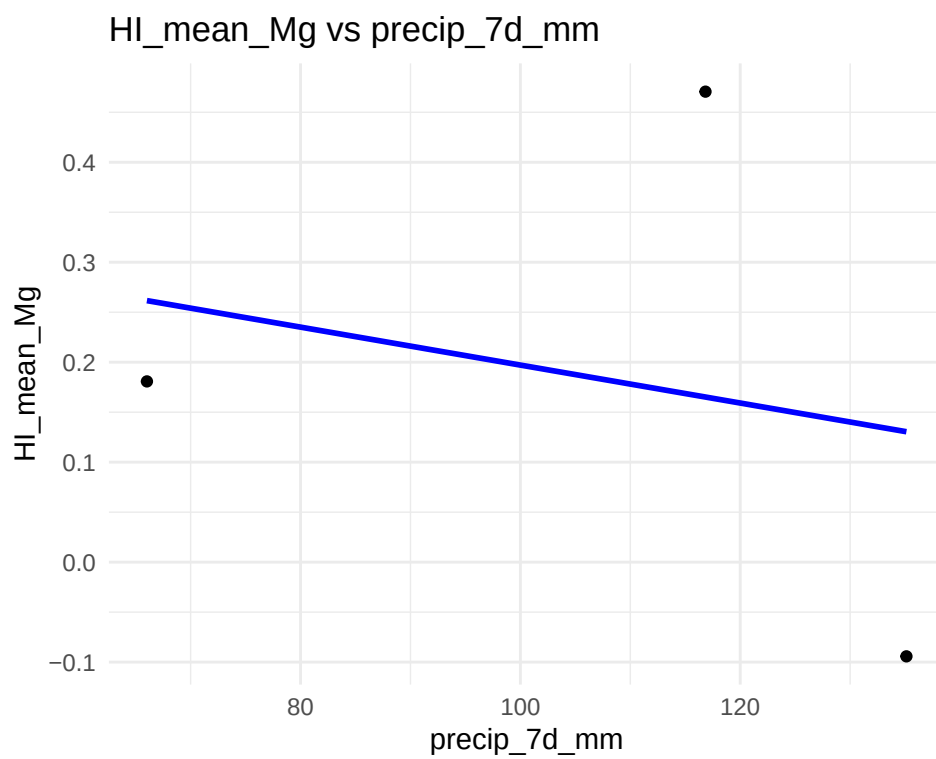


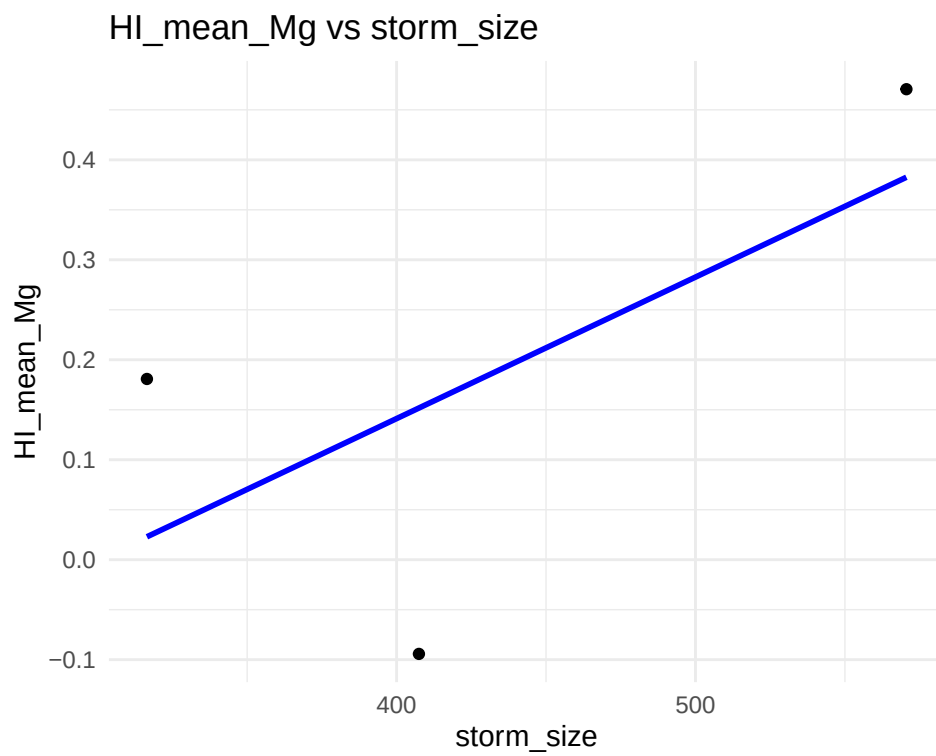
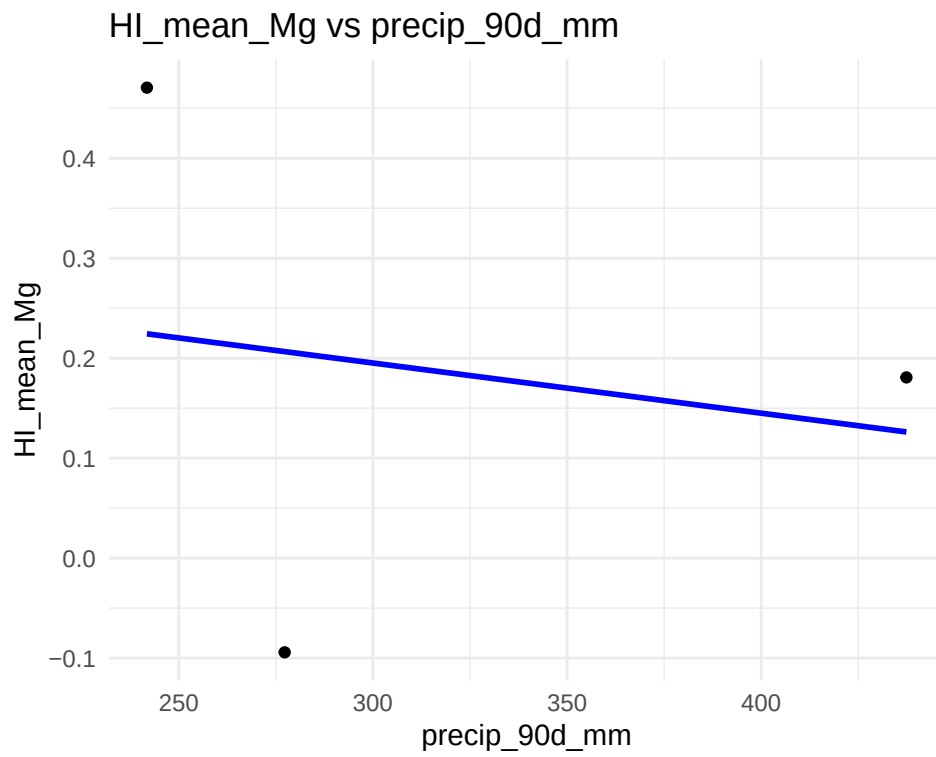


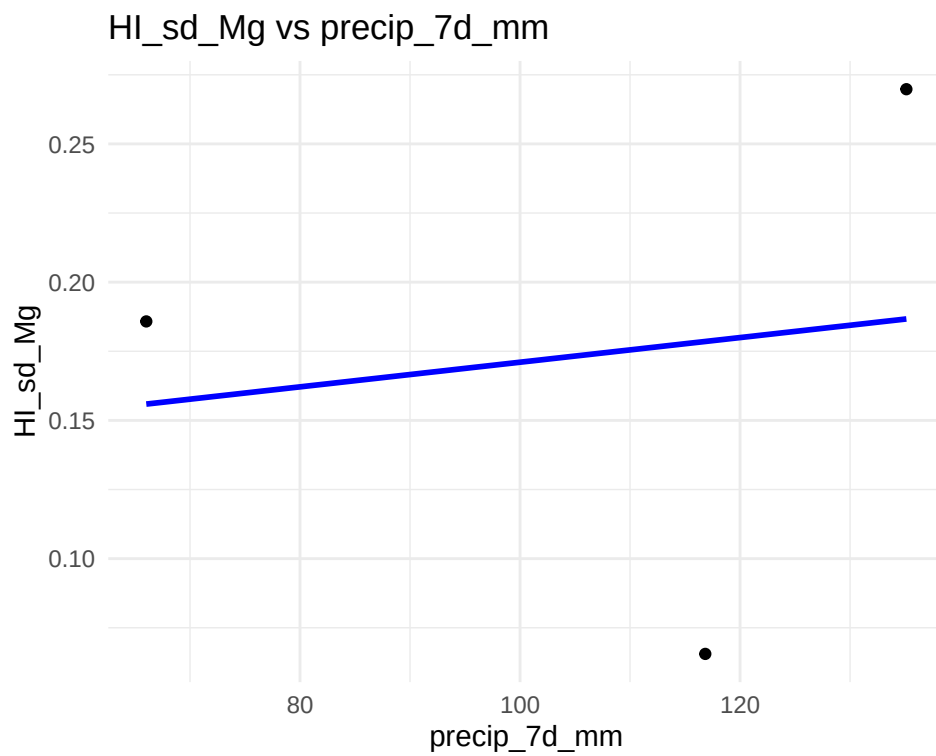
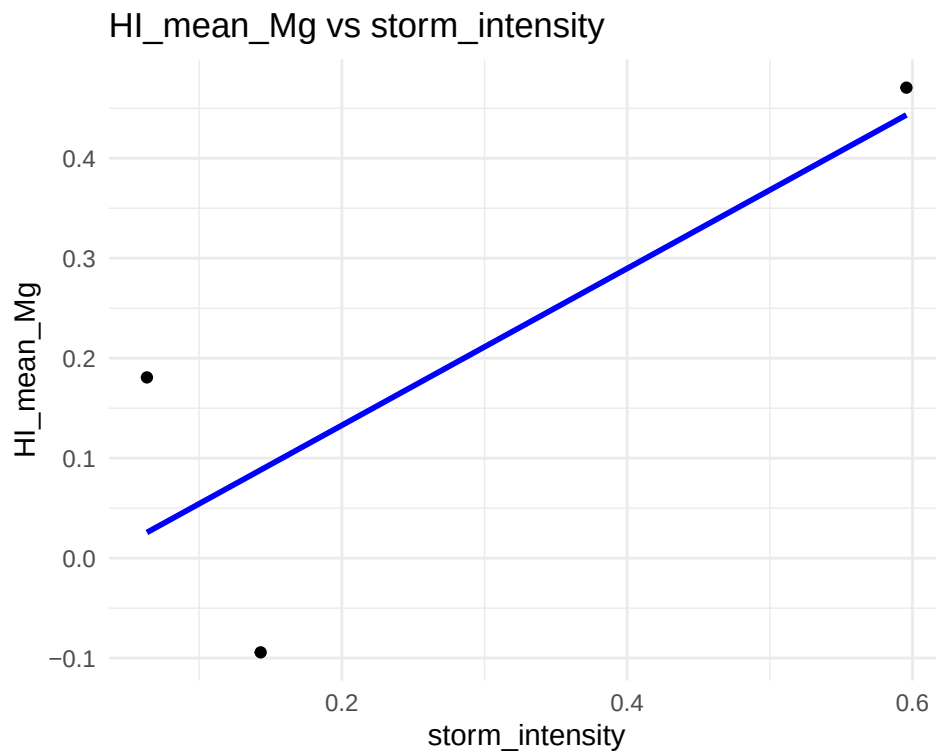


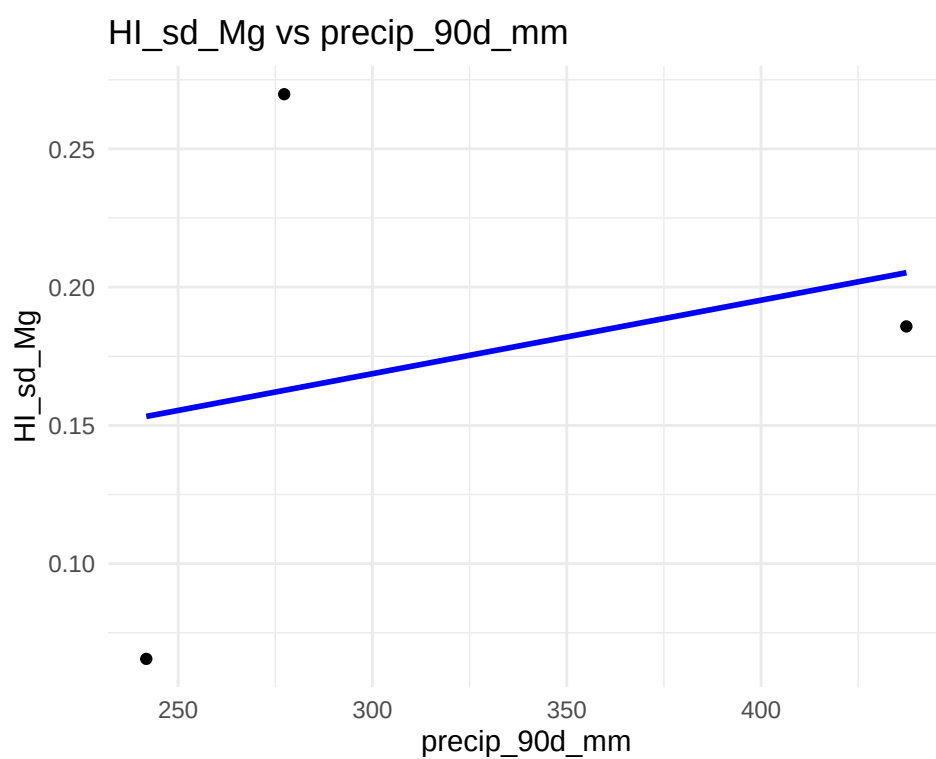
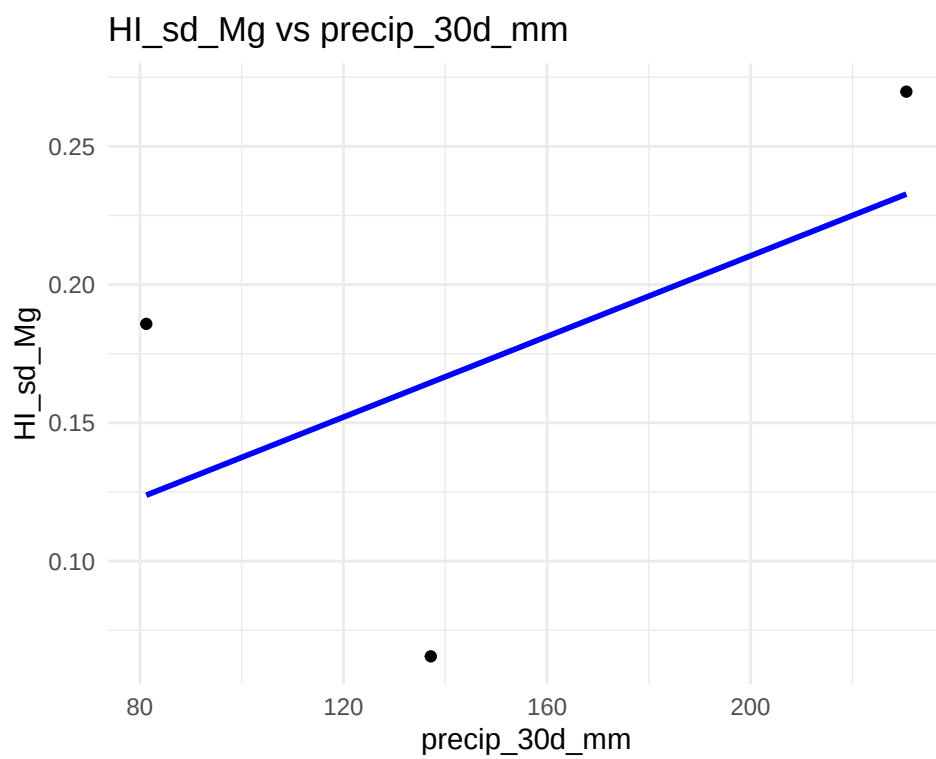


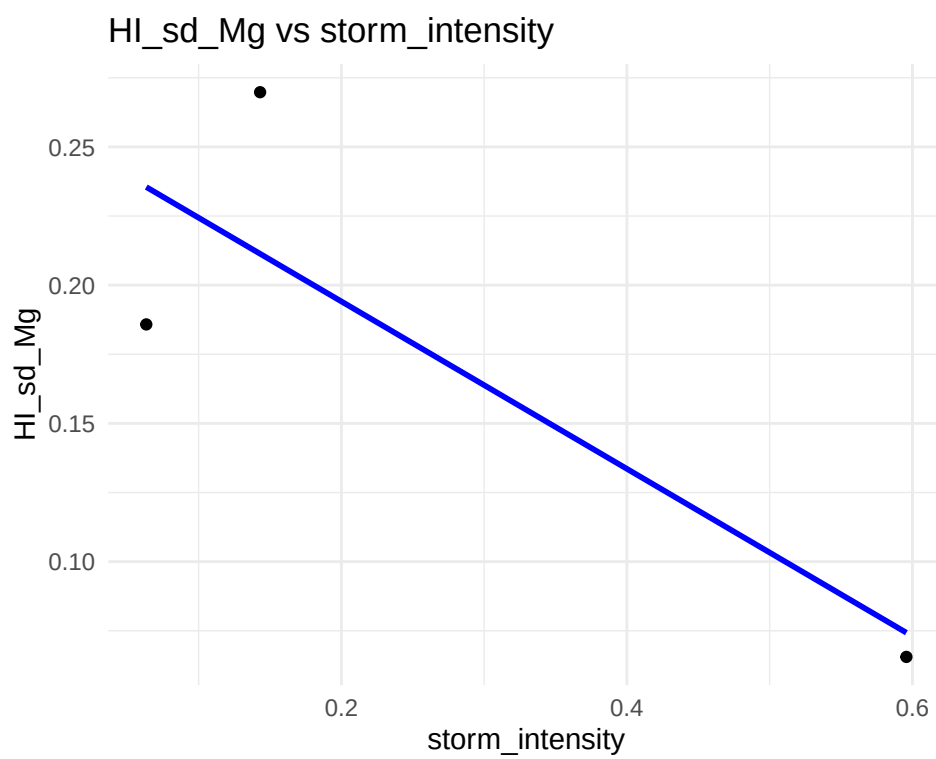
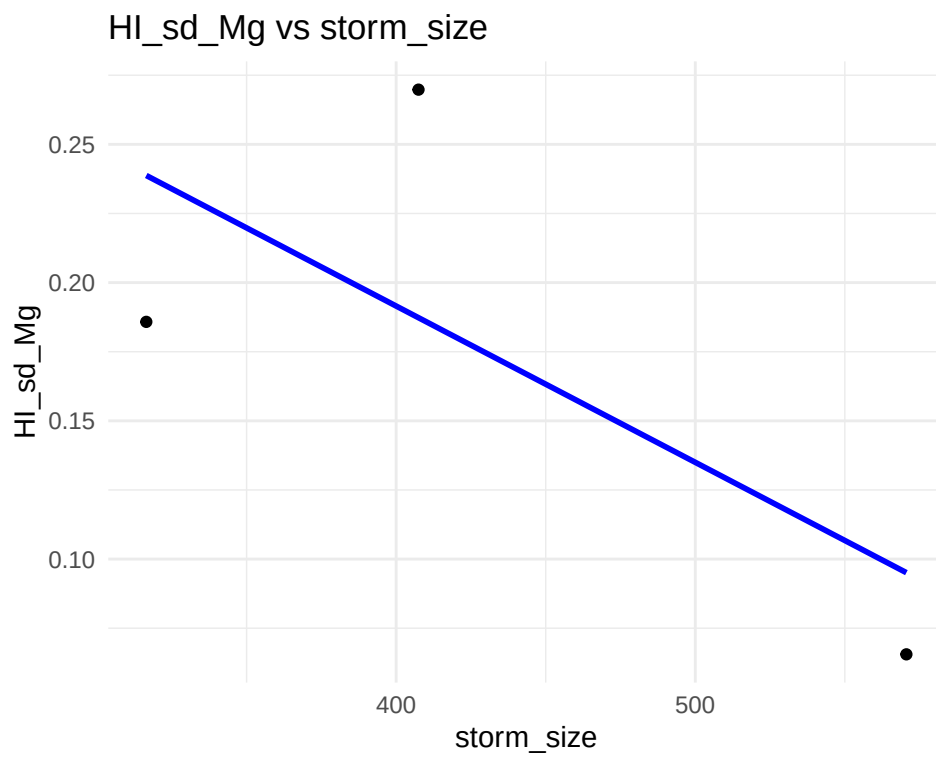


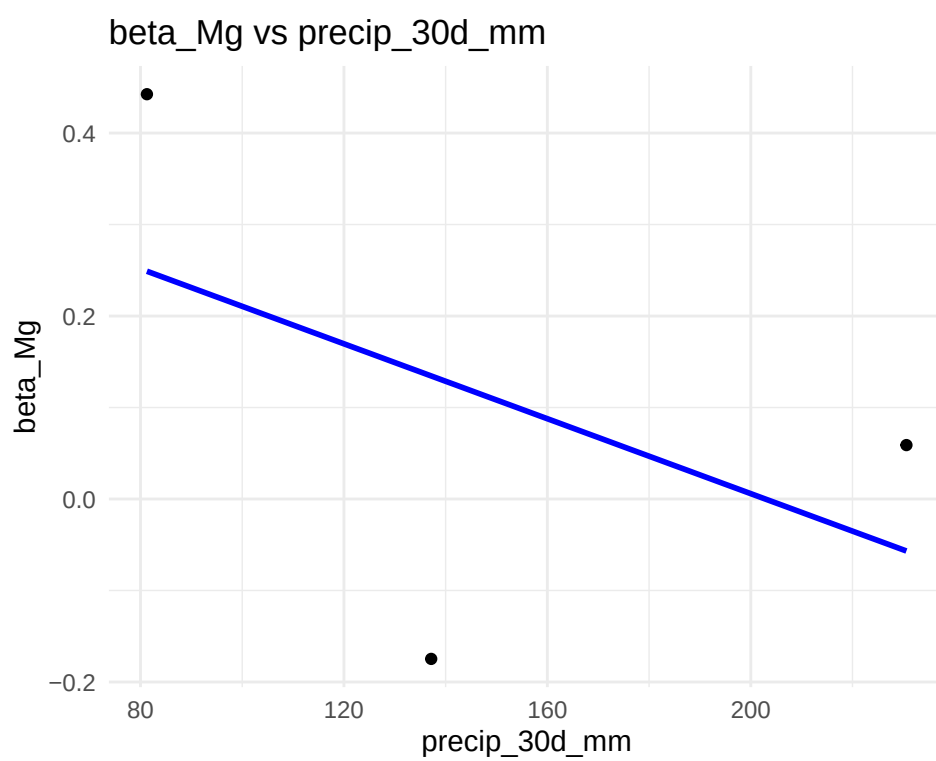
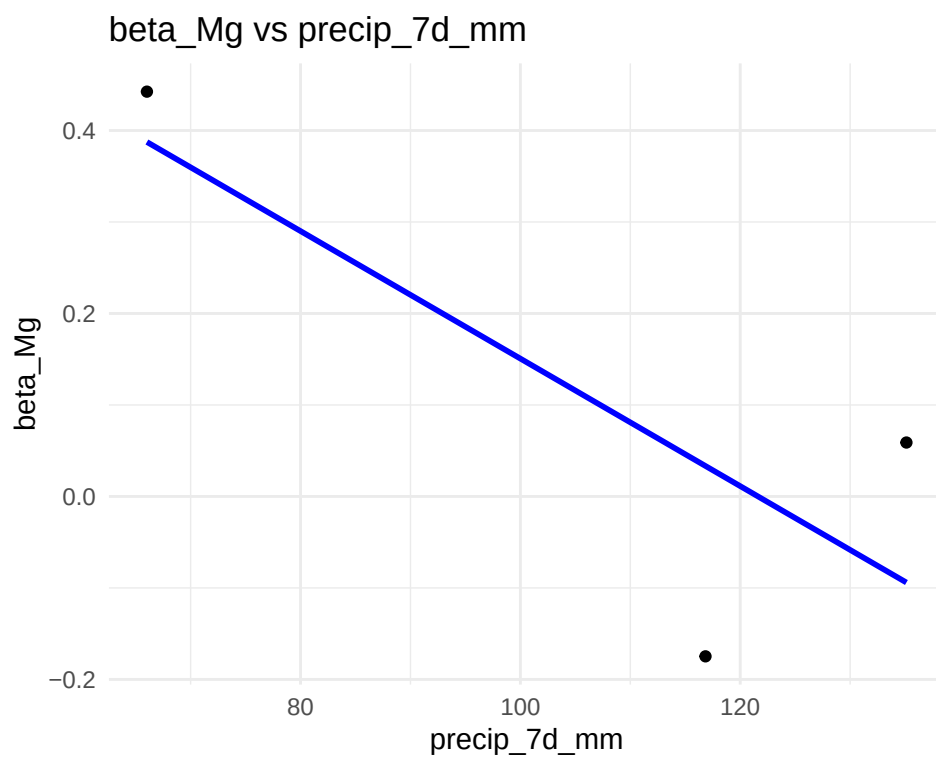


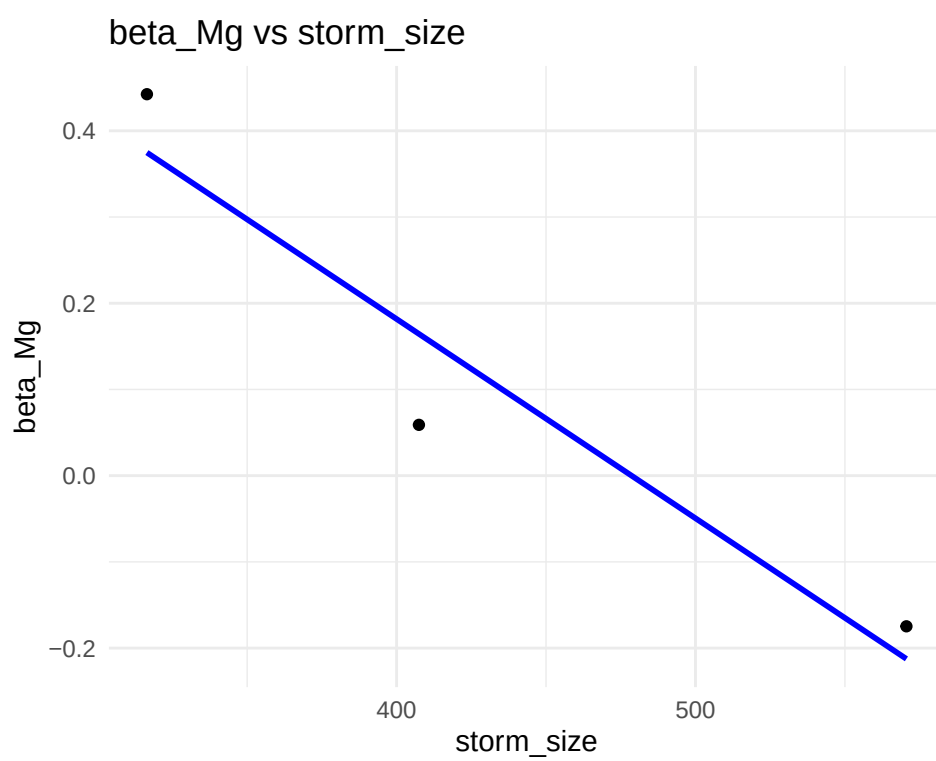
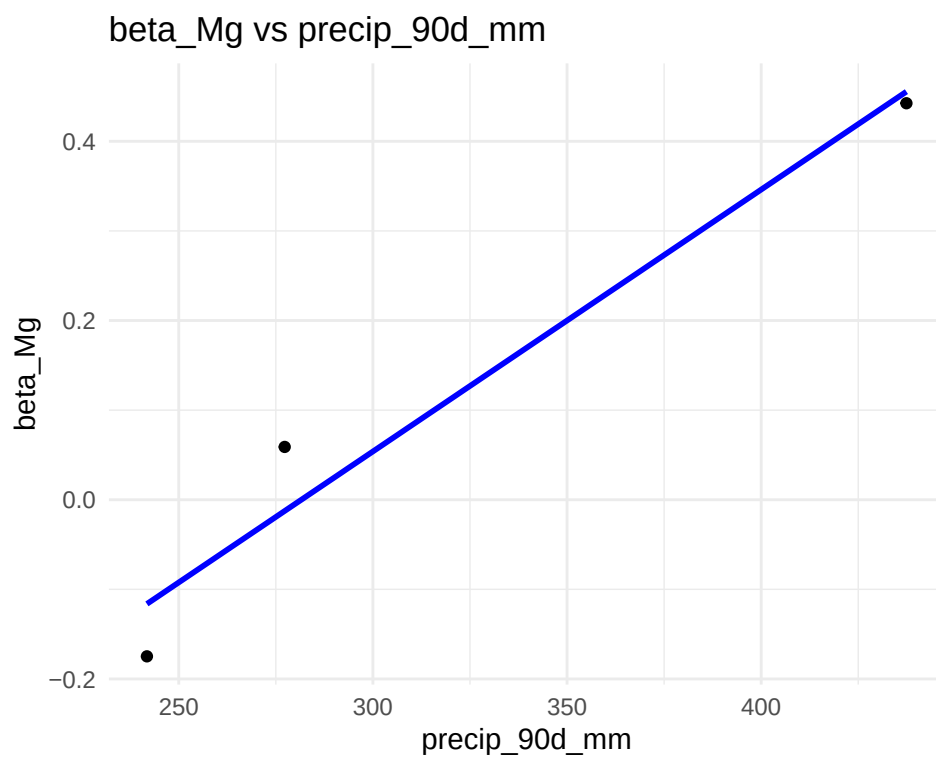


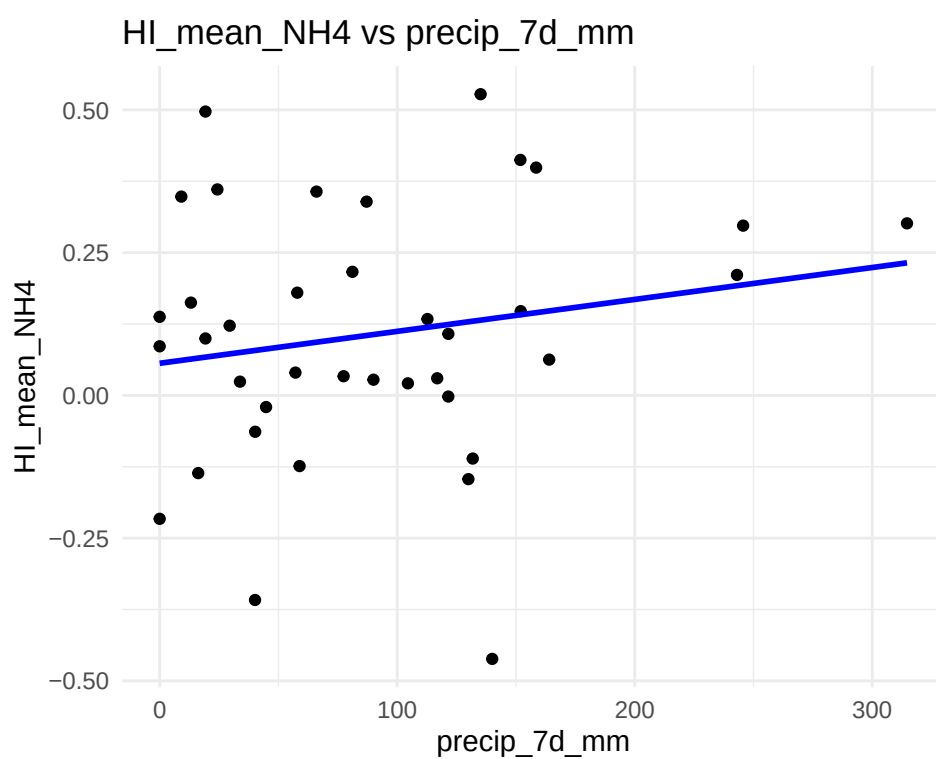
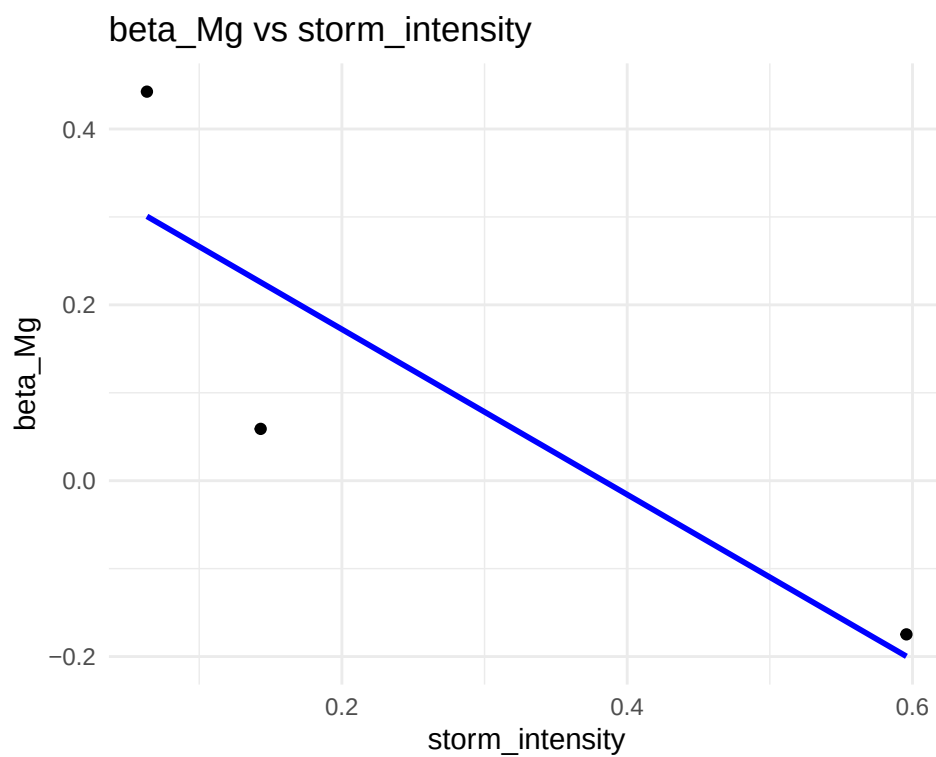


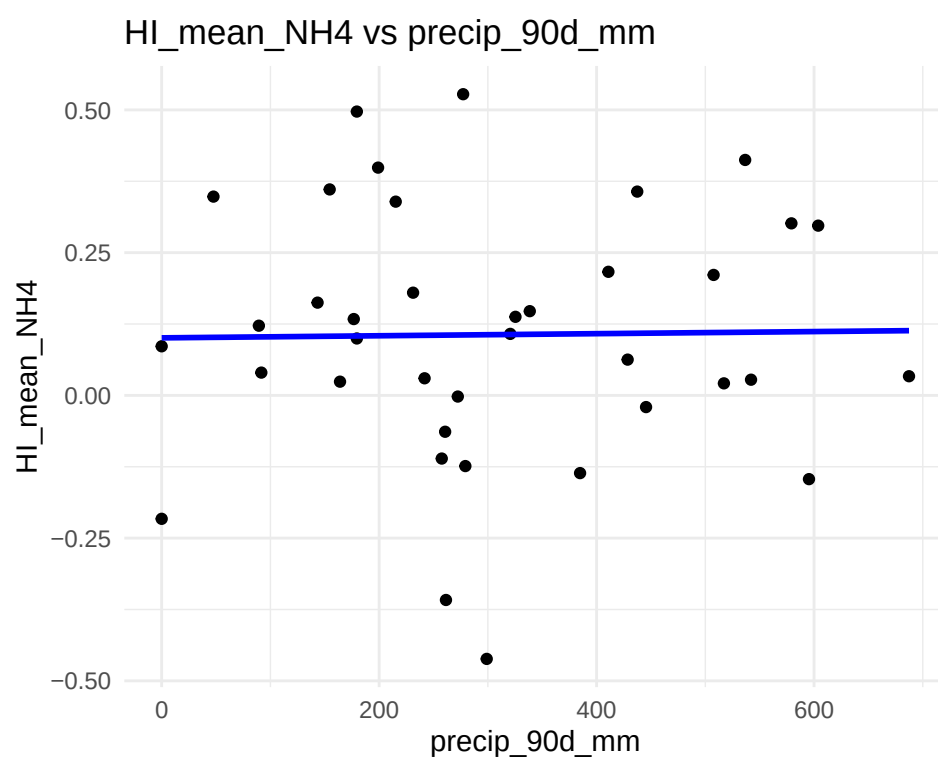
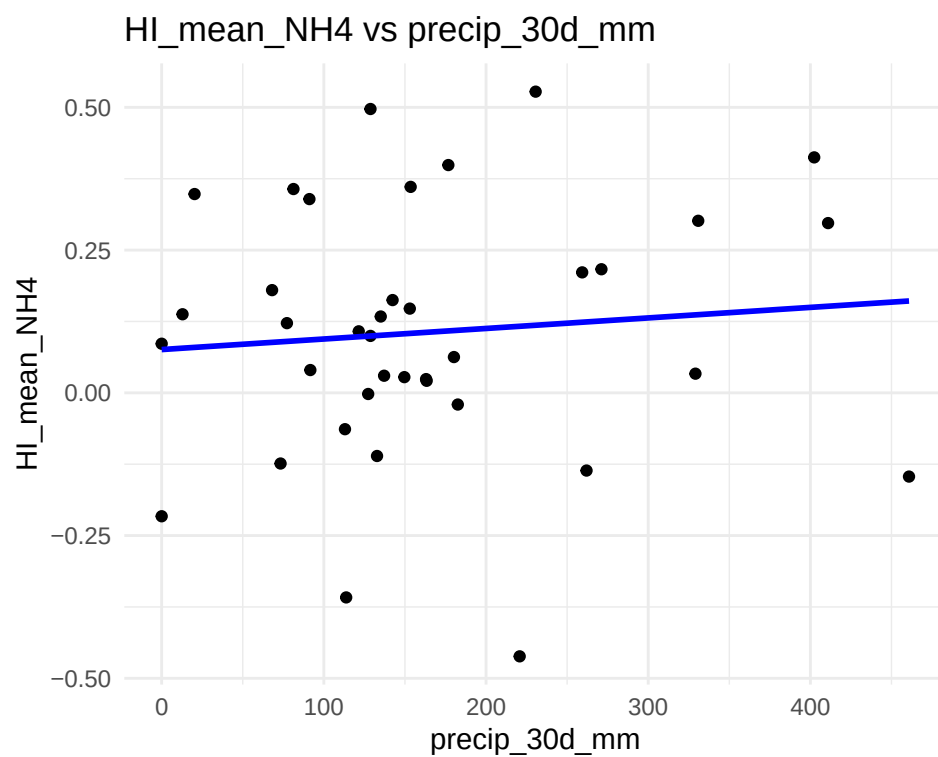




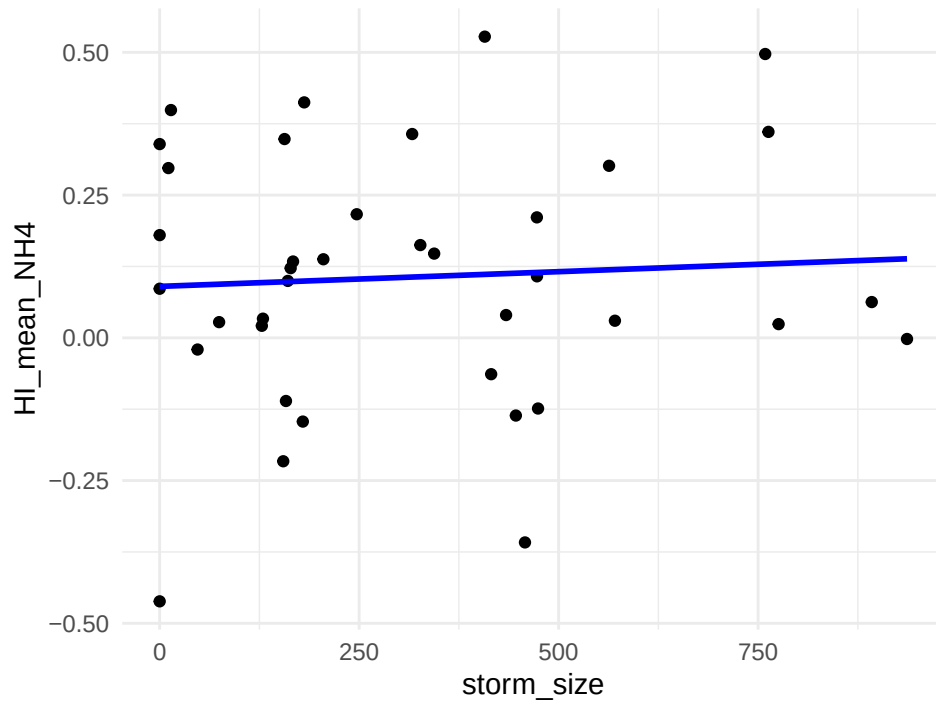




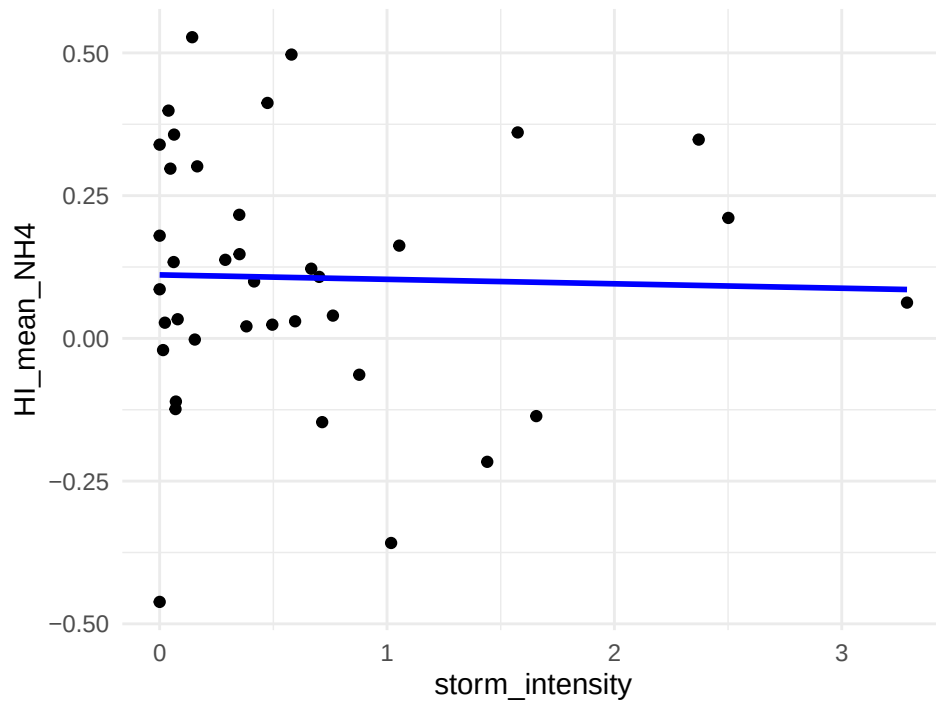




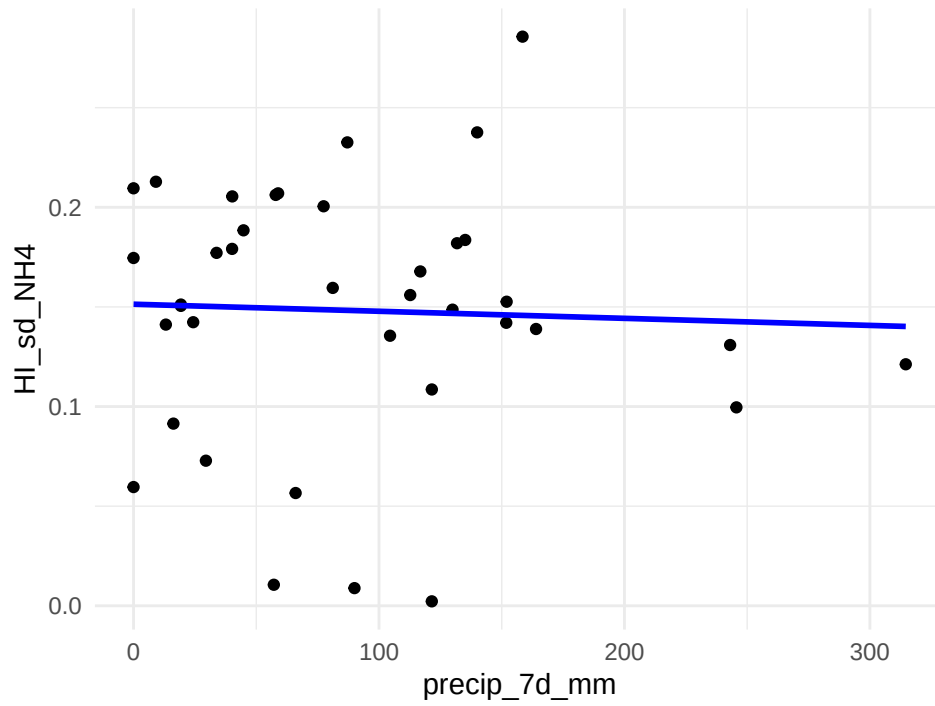
HI_mean_NH4 vs storm_size



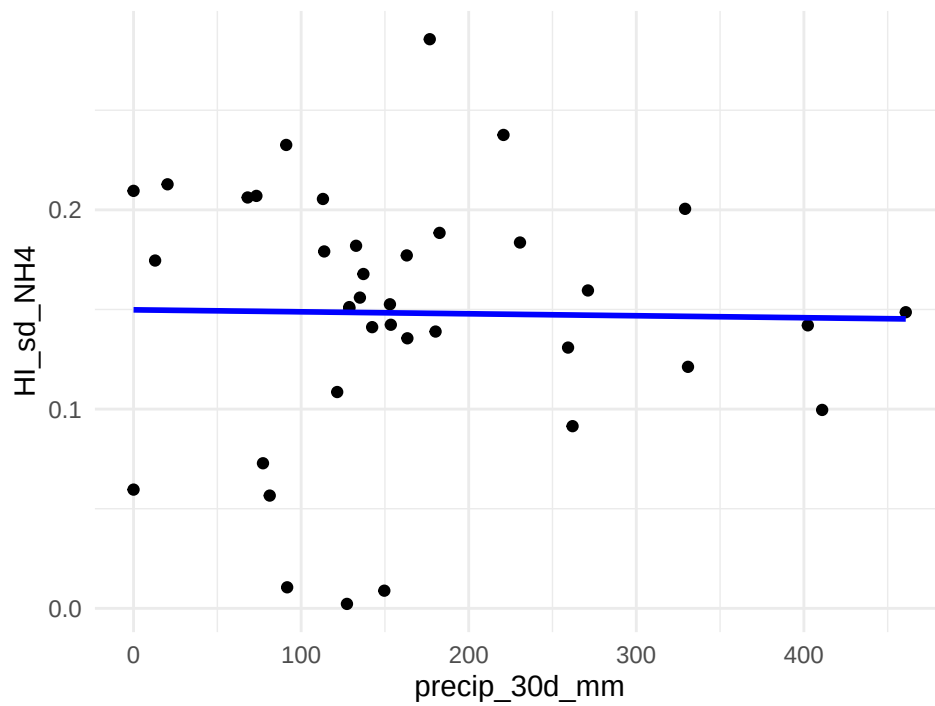
HI_mean_NH4 vs storm_intensity



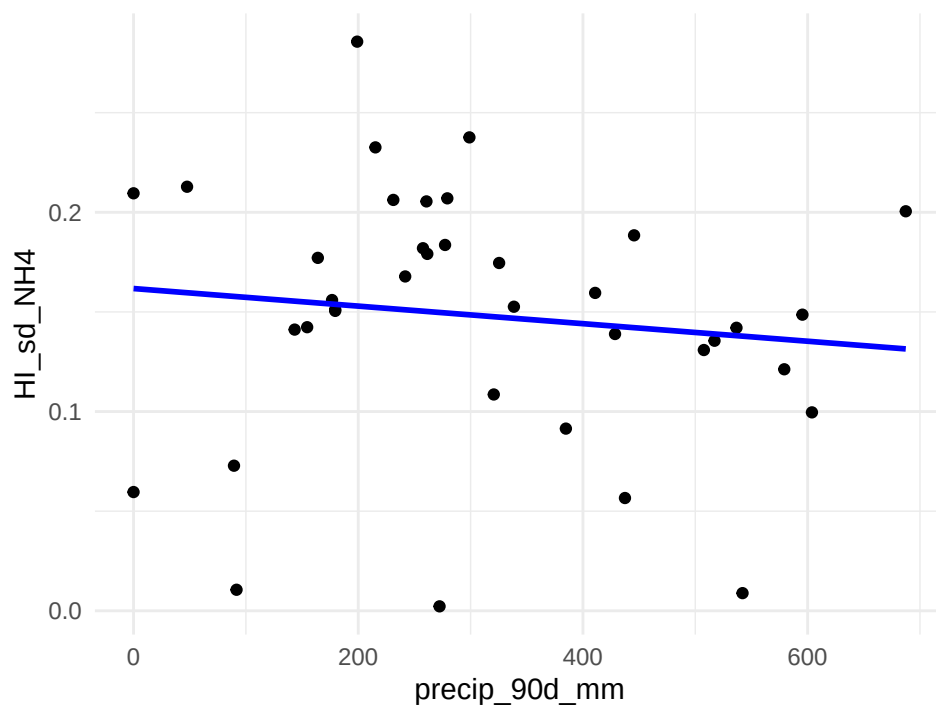
HI_sd_NH4 vs precip_7d_mm



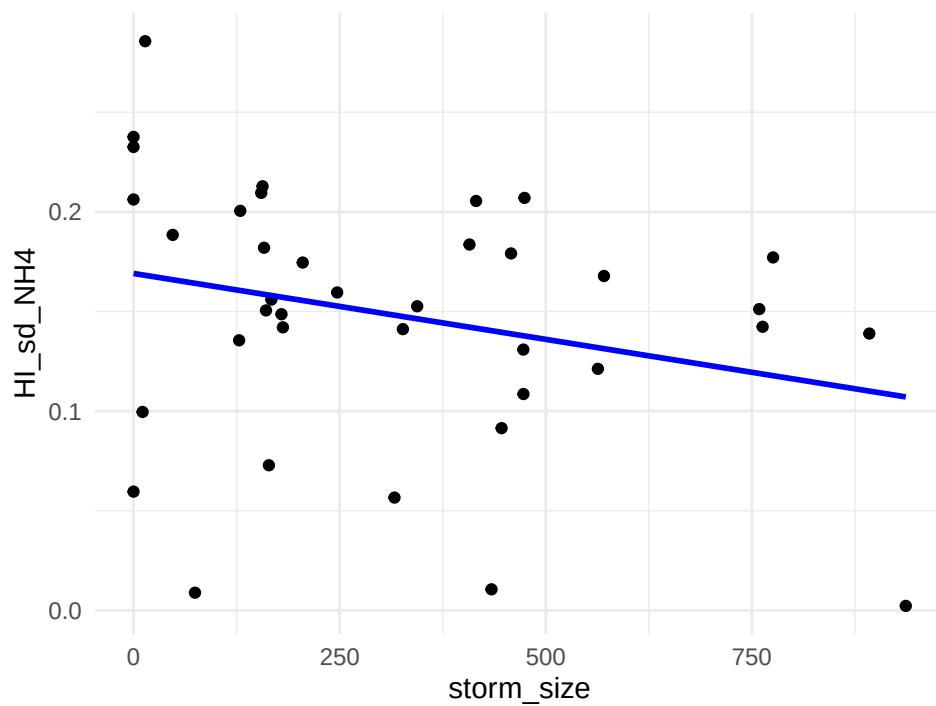
HI_sd_NH4 vs precip_30d_mm

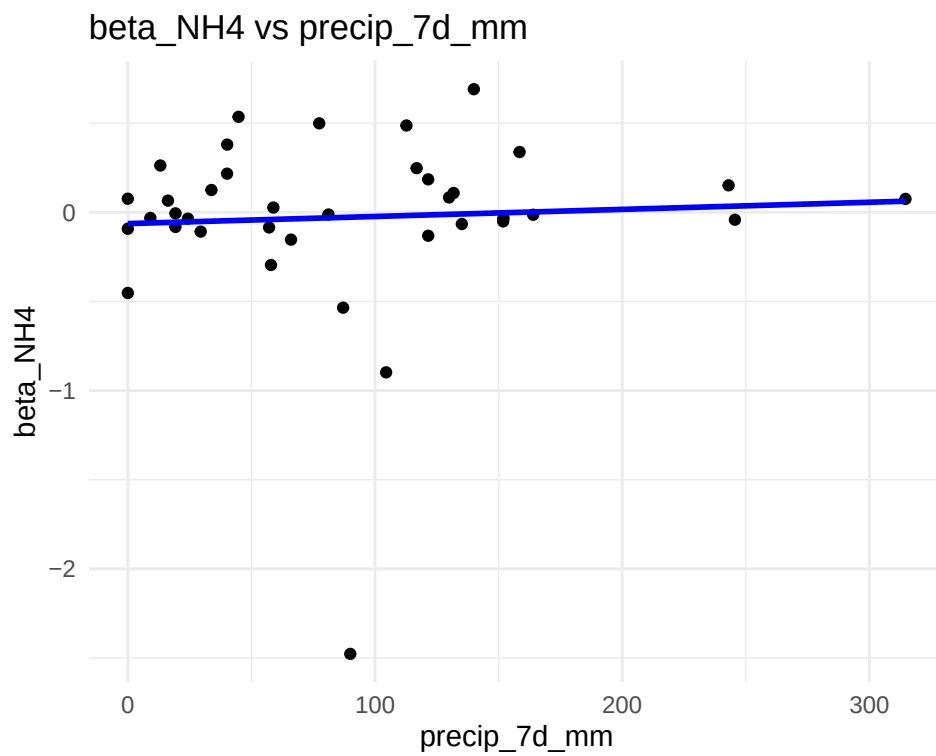
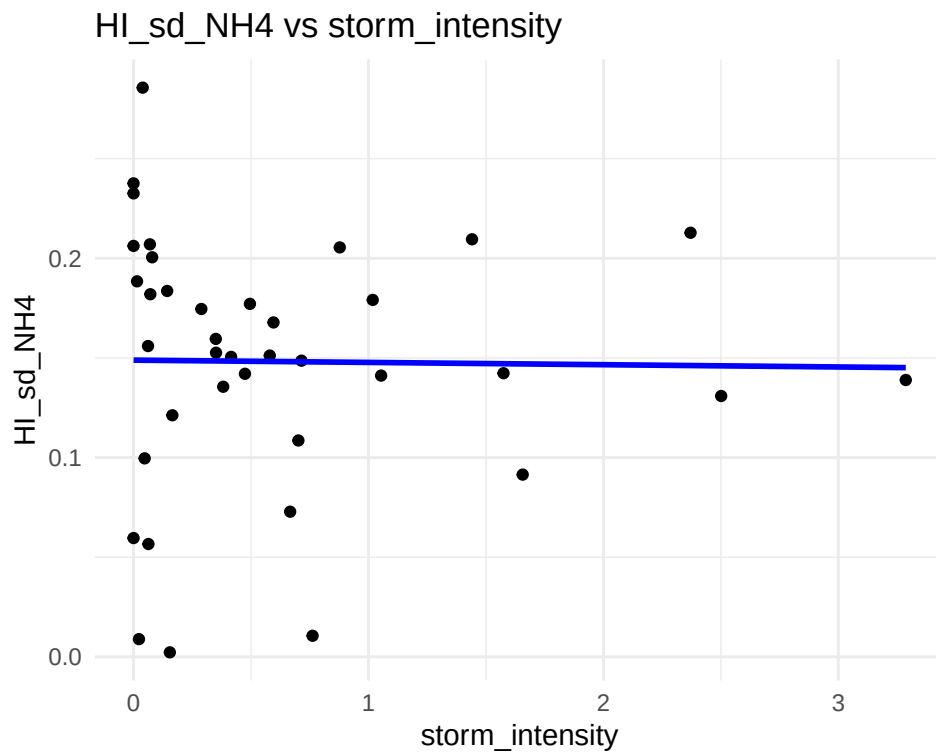


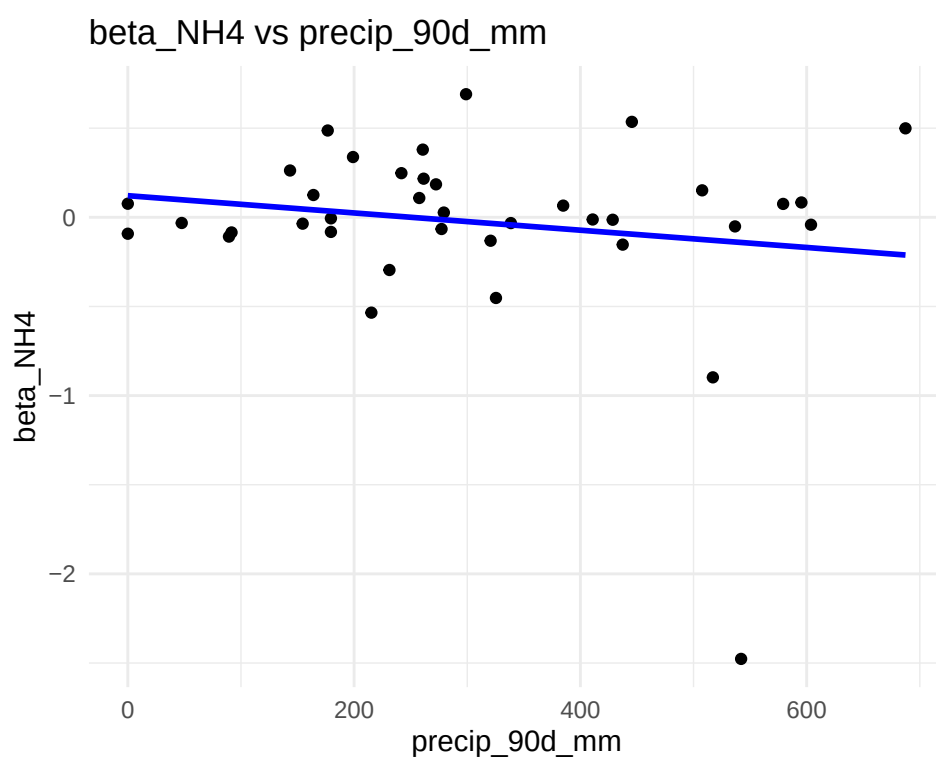
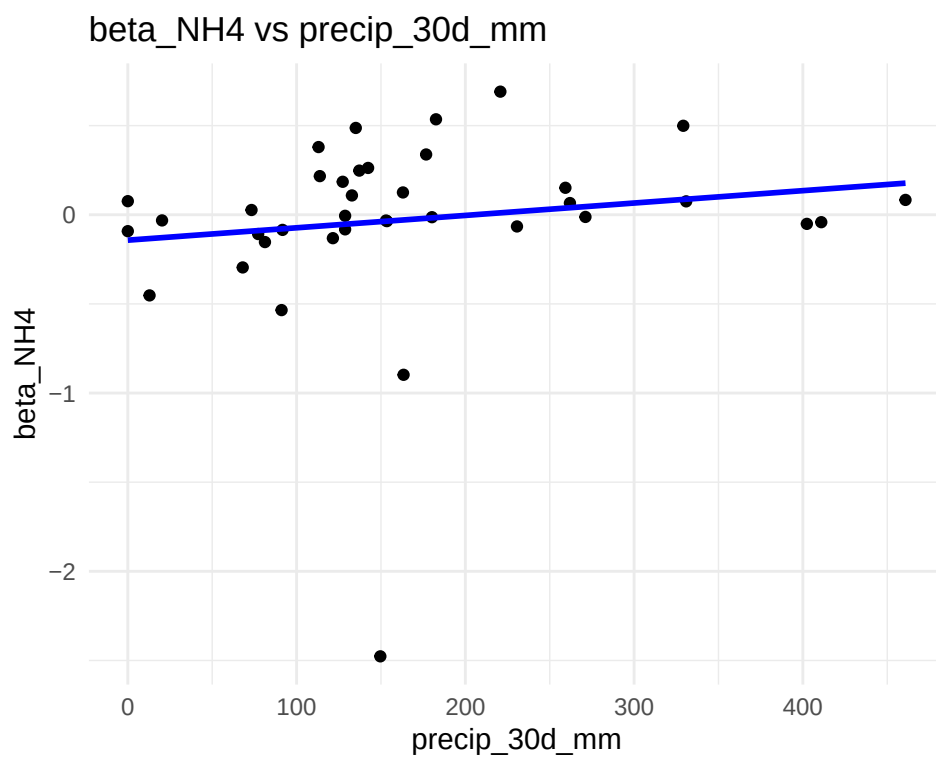
HI_sd_NH4 vs precip_90d_mm

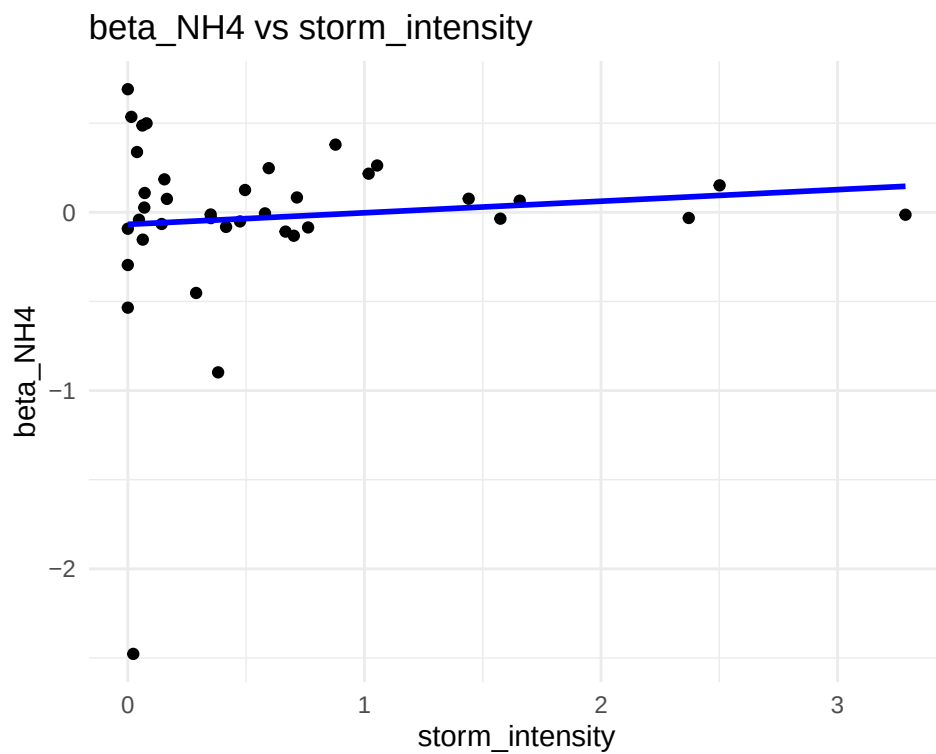
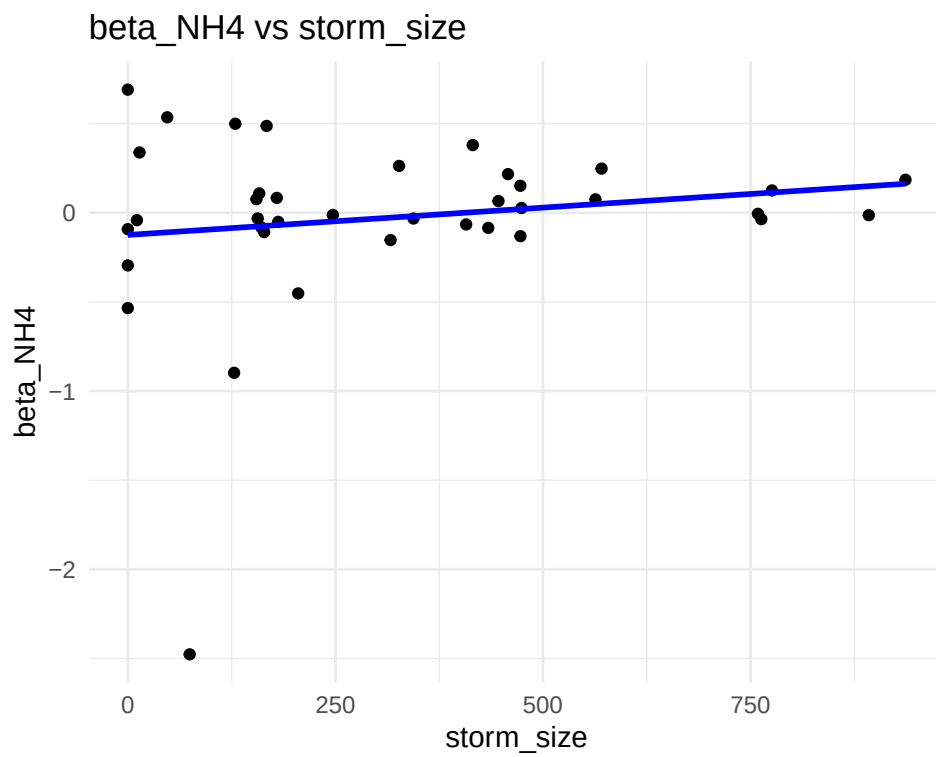


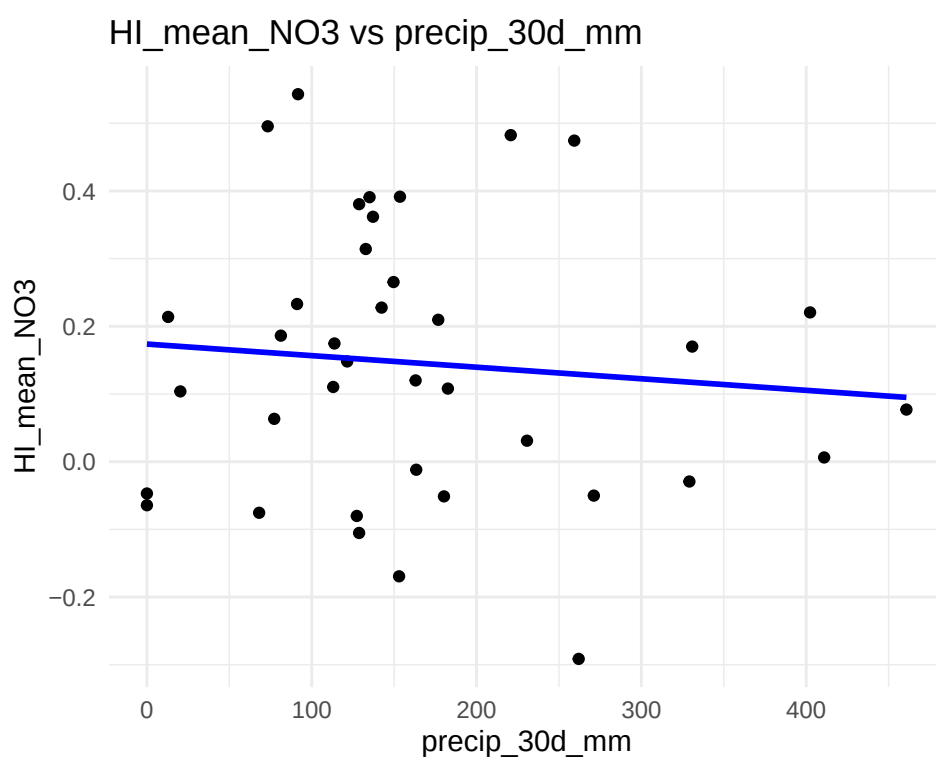
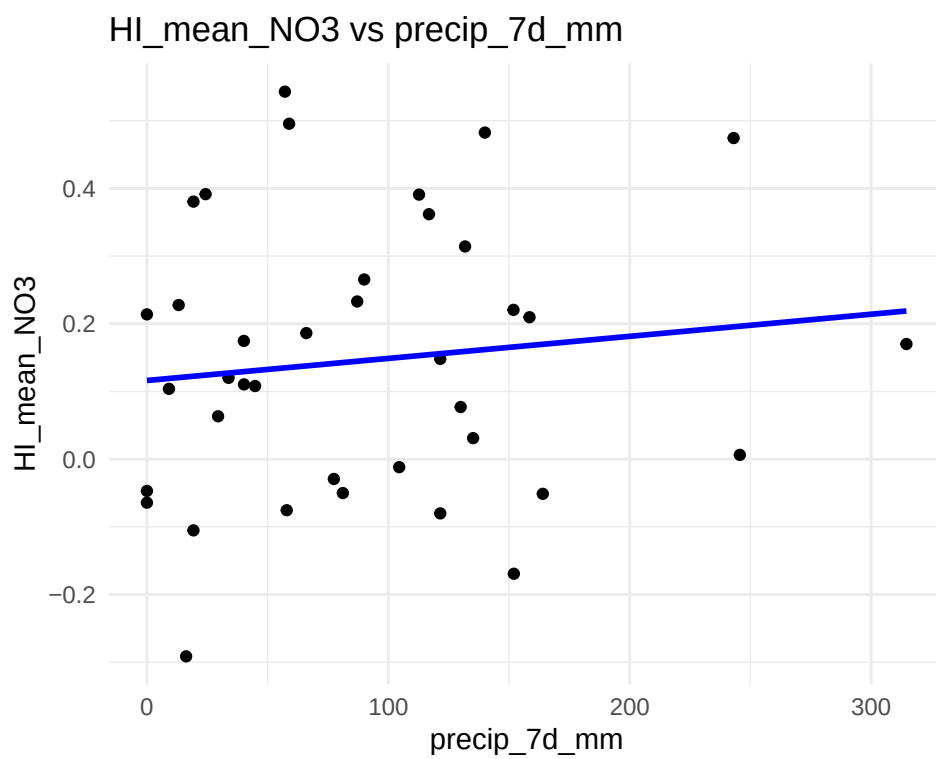
HI_sd_NH4 vs storm_size



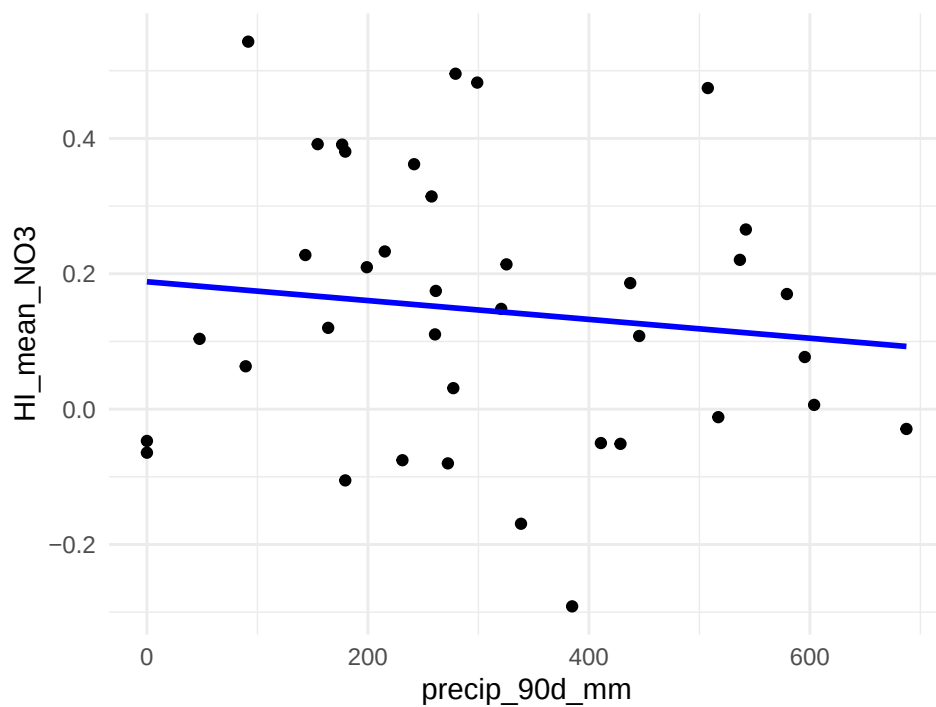




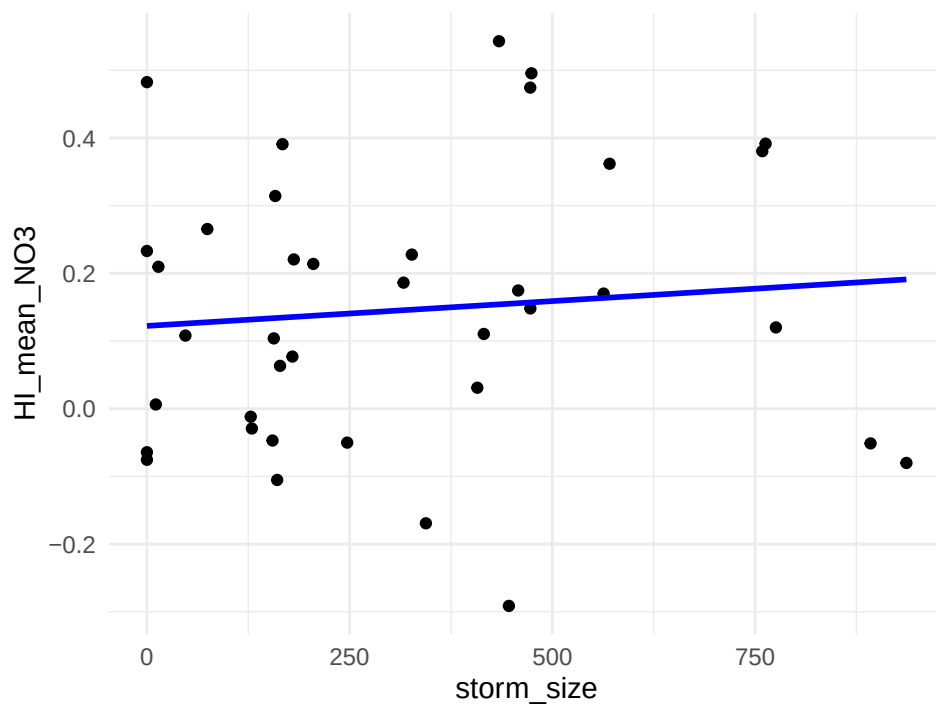


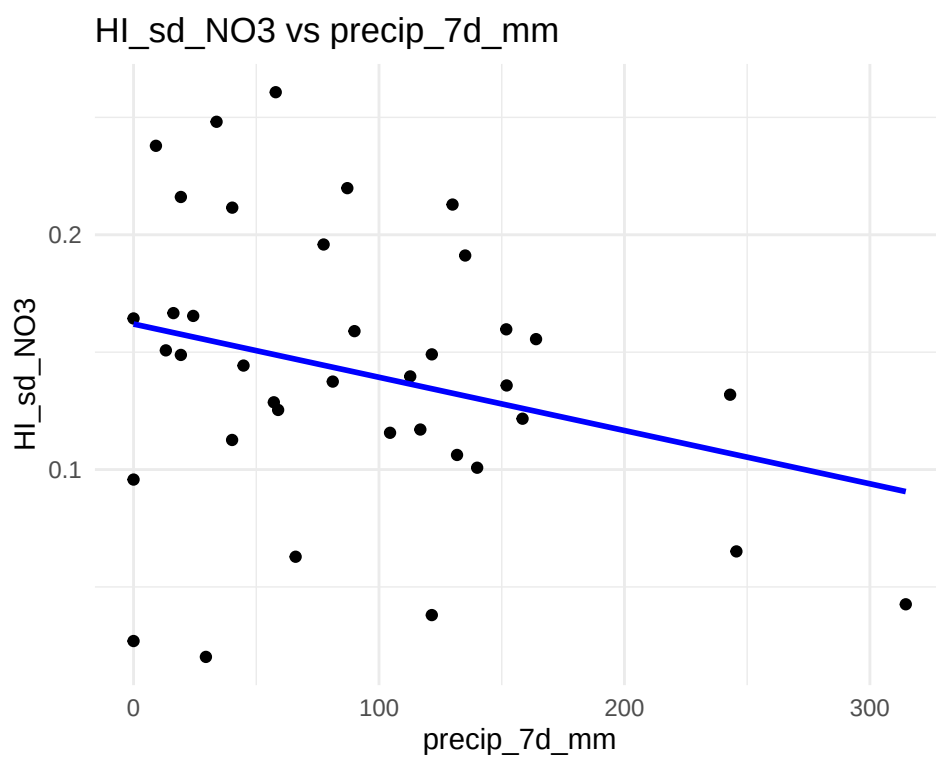
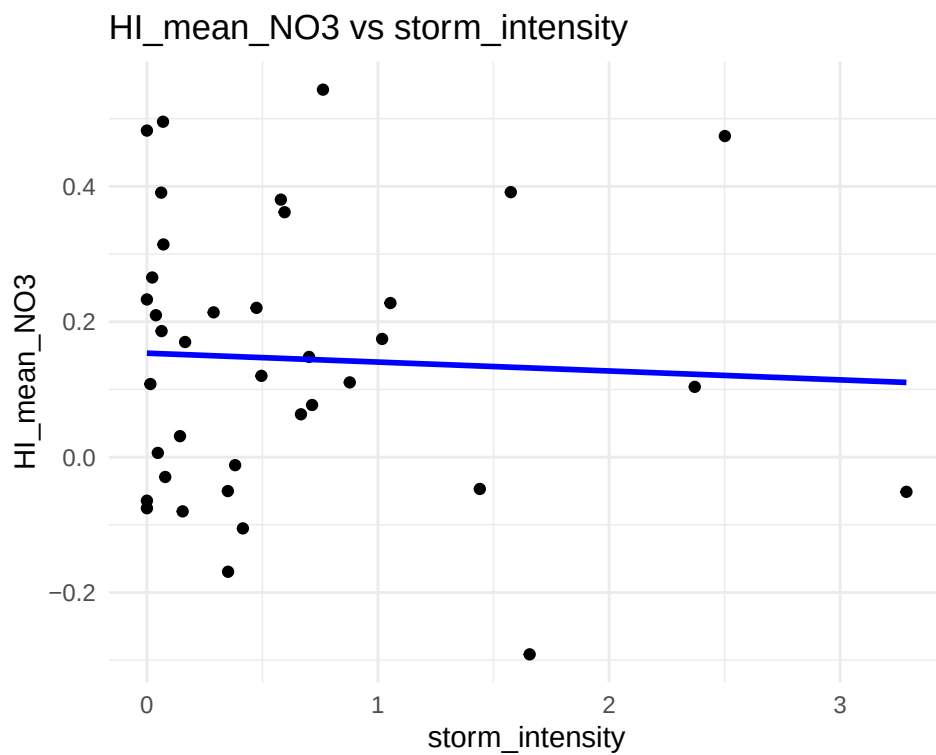


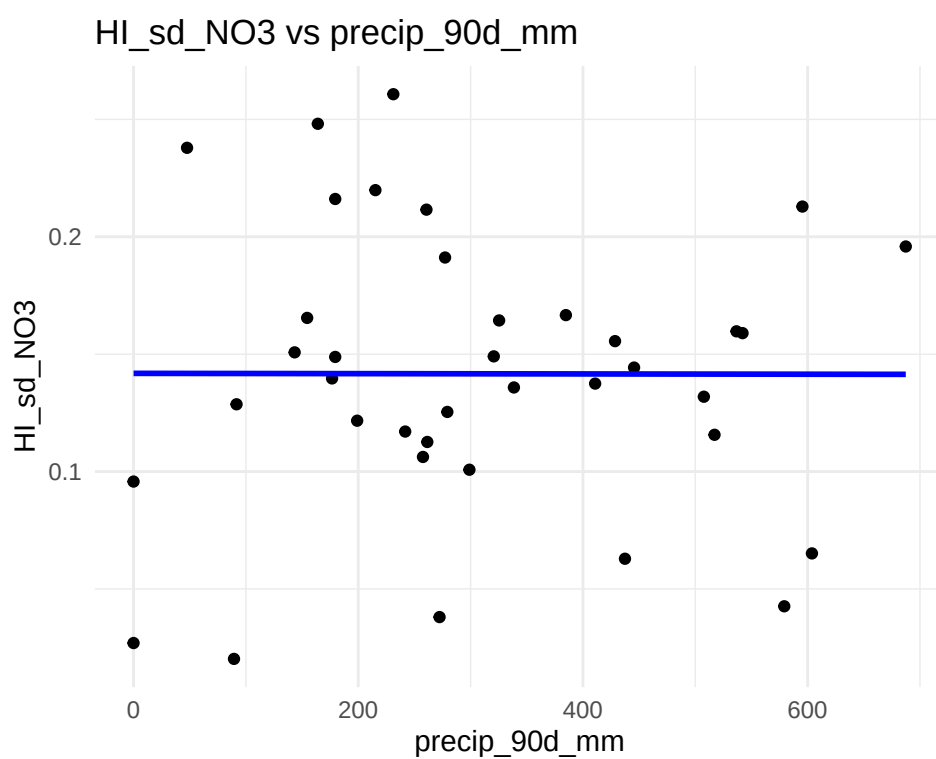
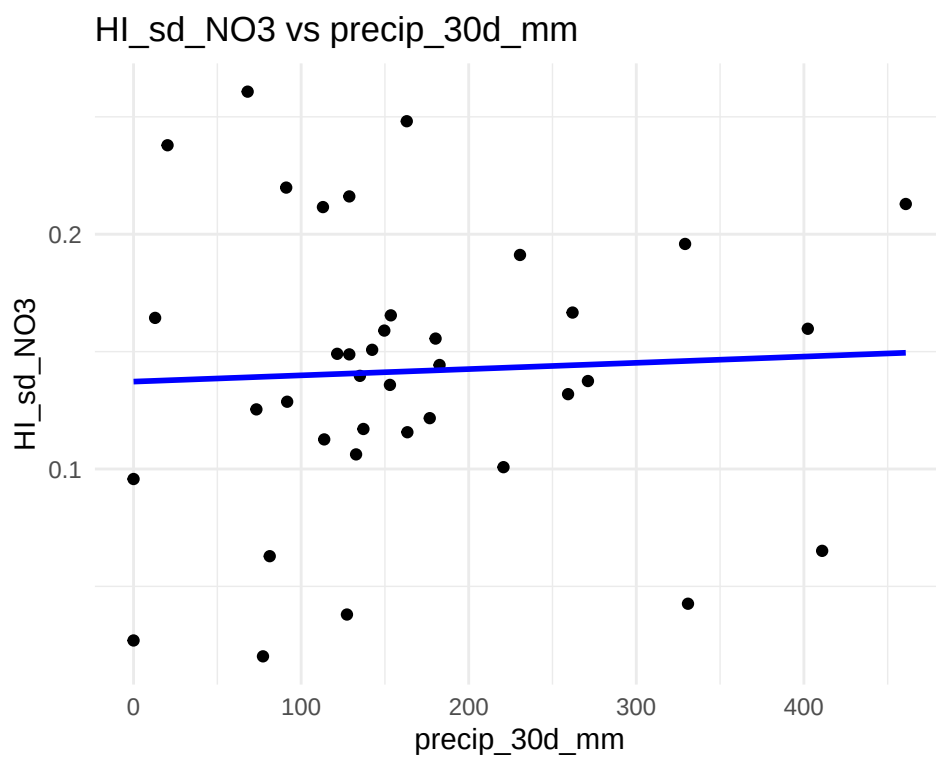
HI_mean_NO3 vs precip_90d_mm

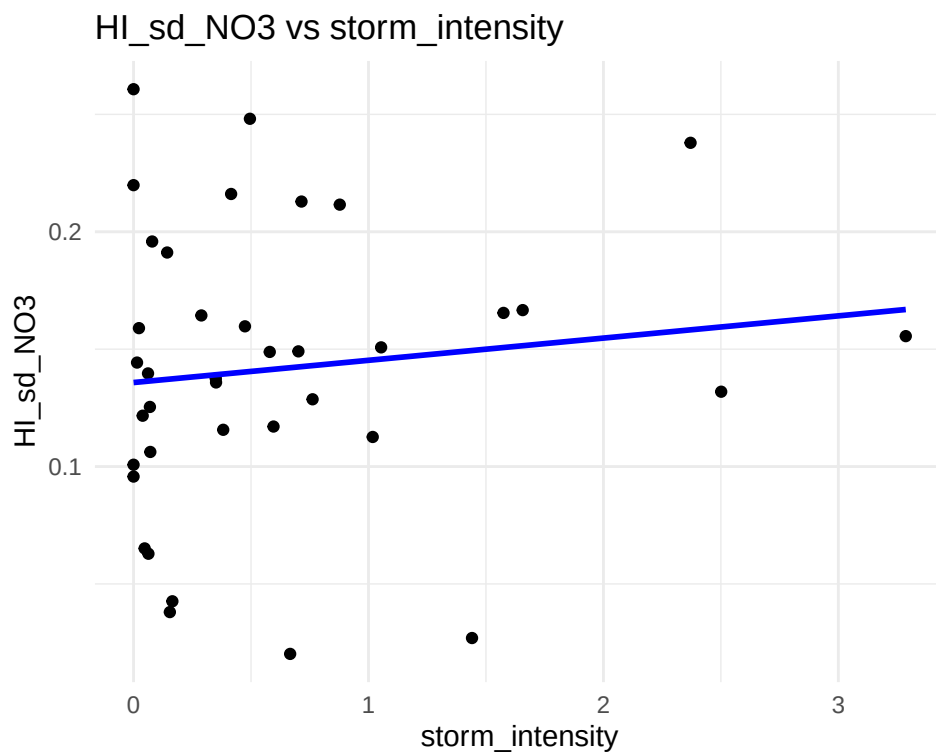
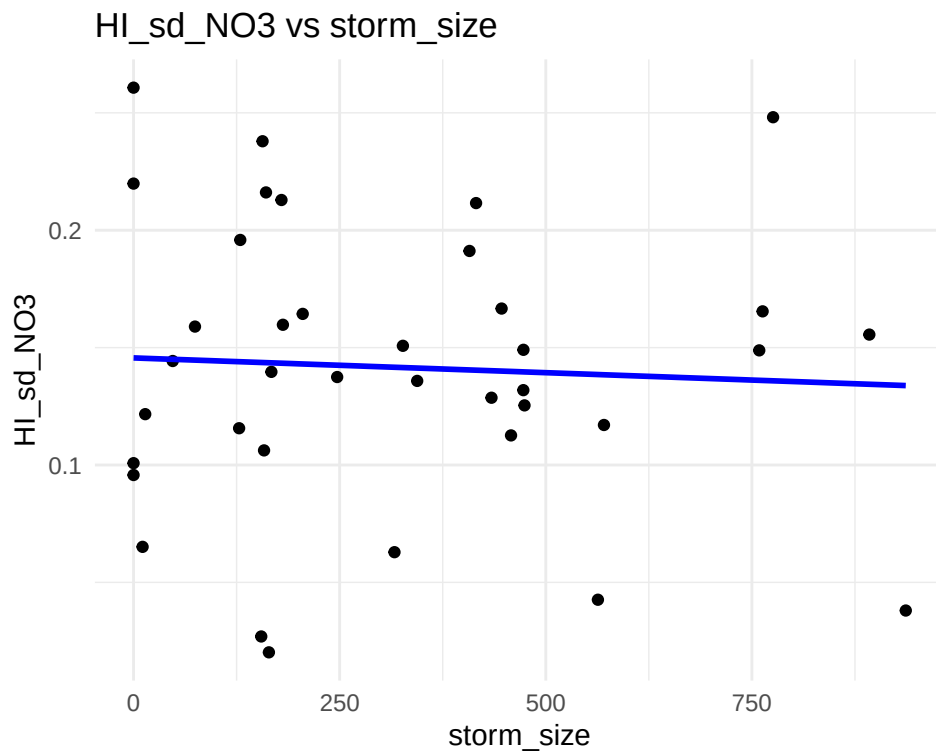


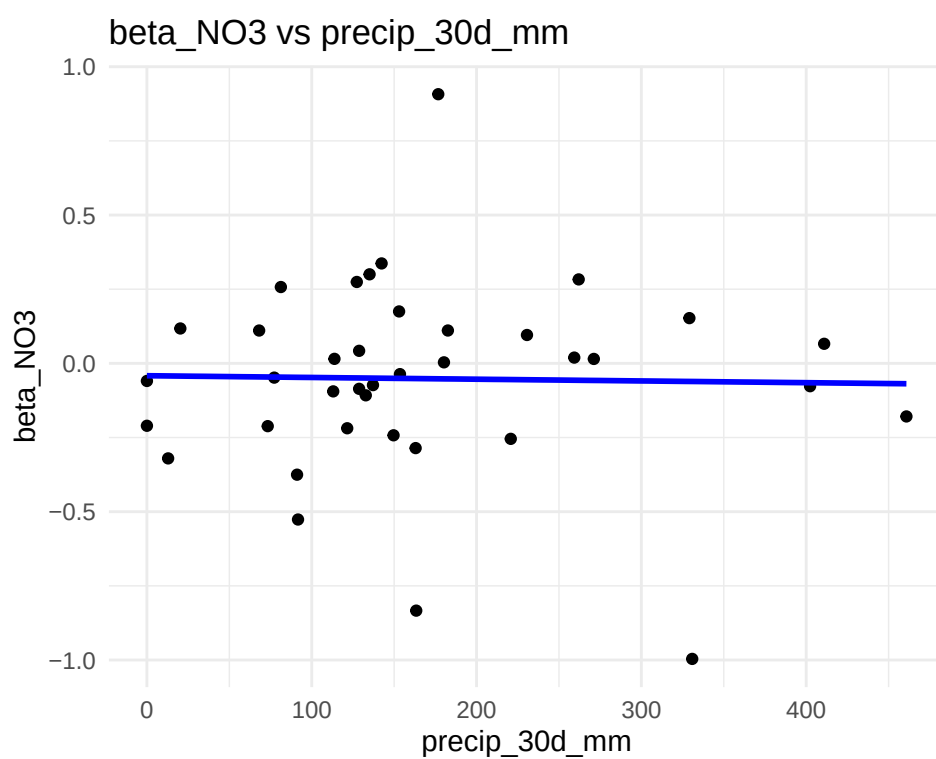
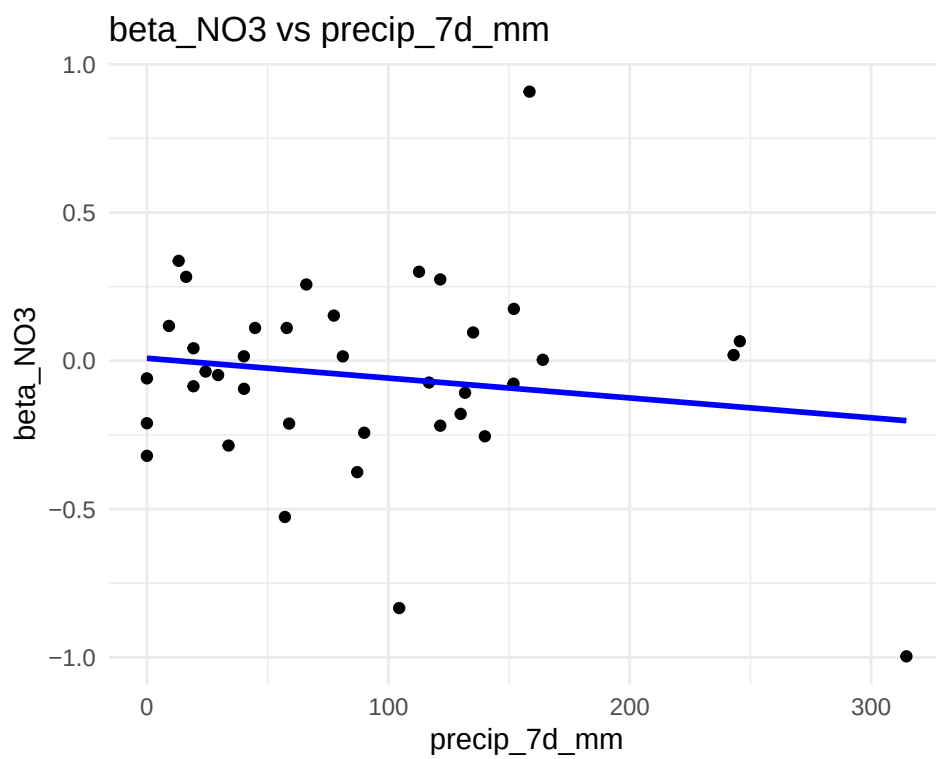
HI_mean_NO3 vs storm_size

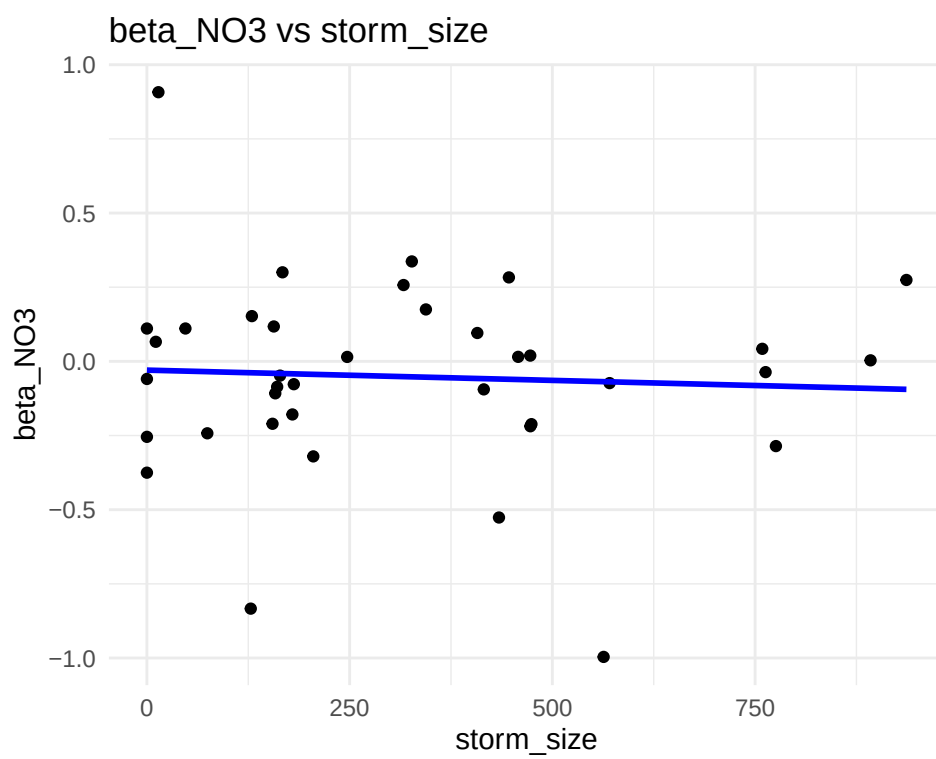
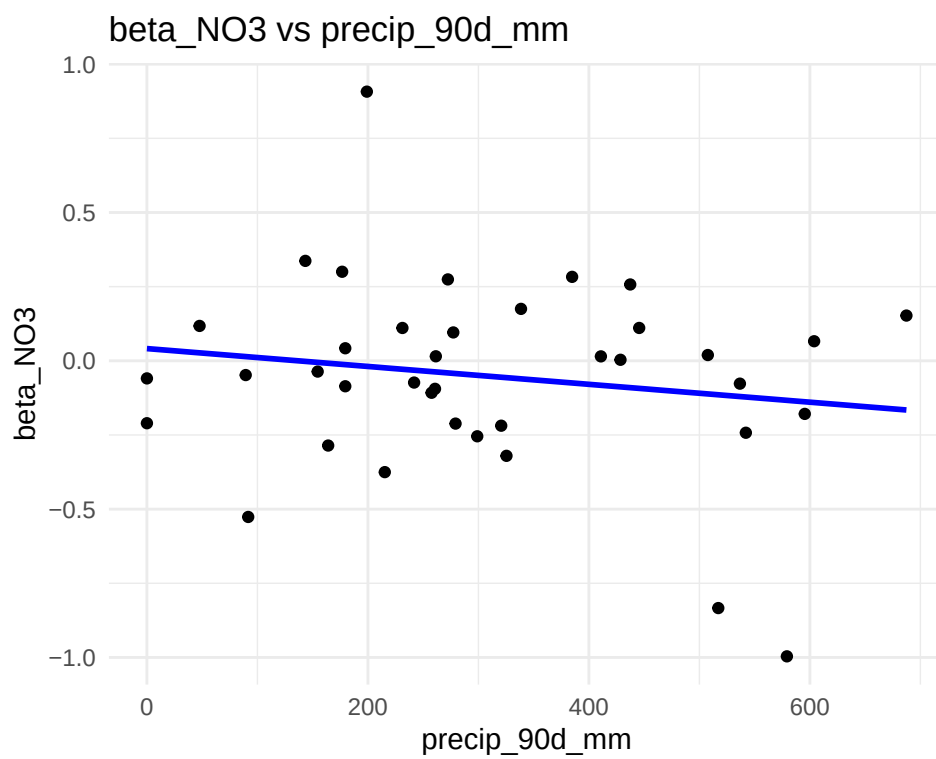


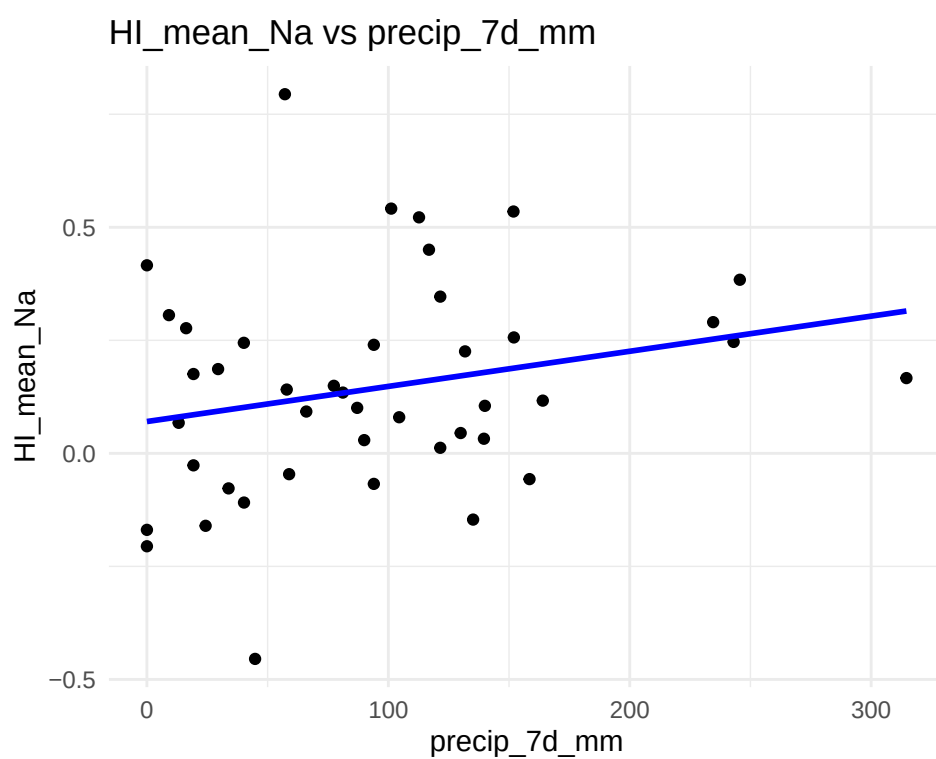
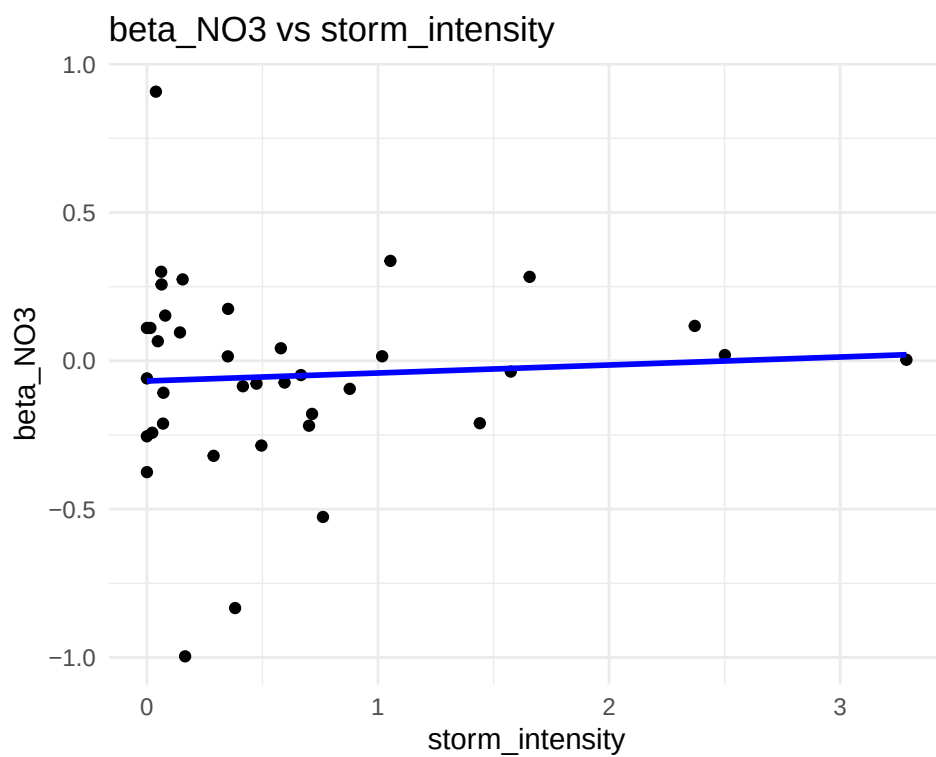


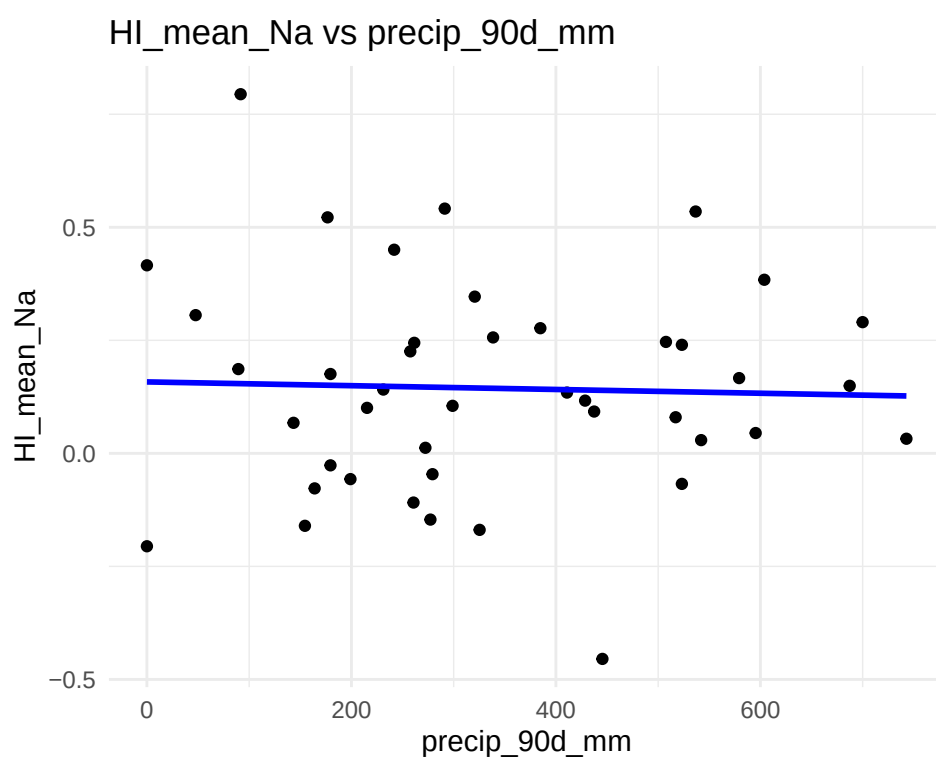
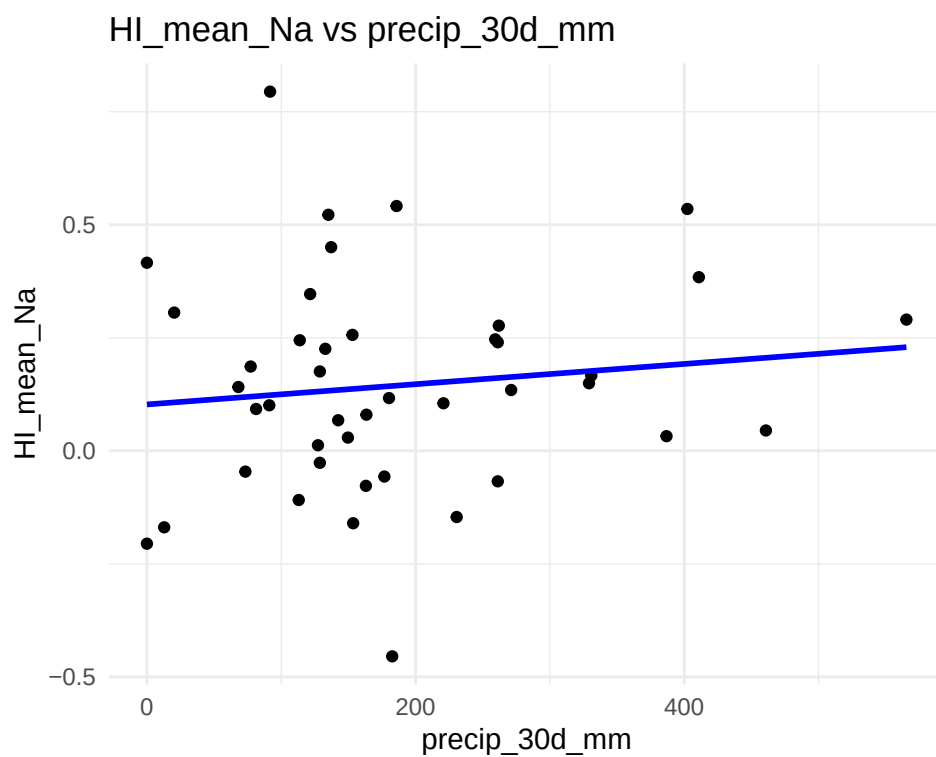




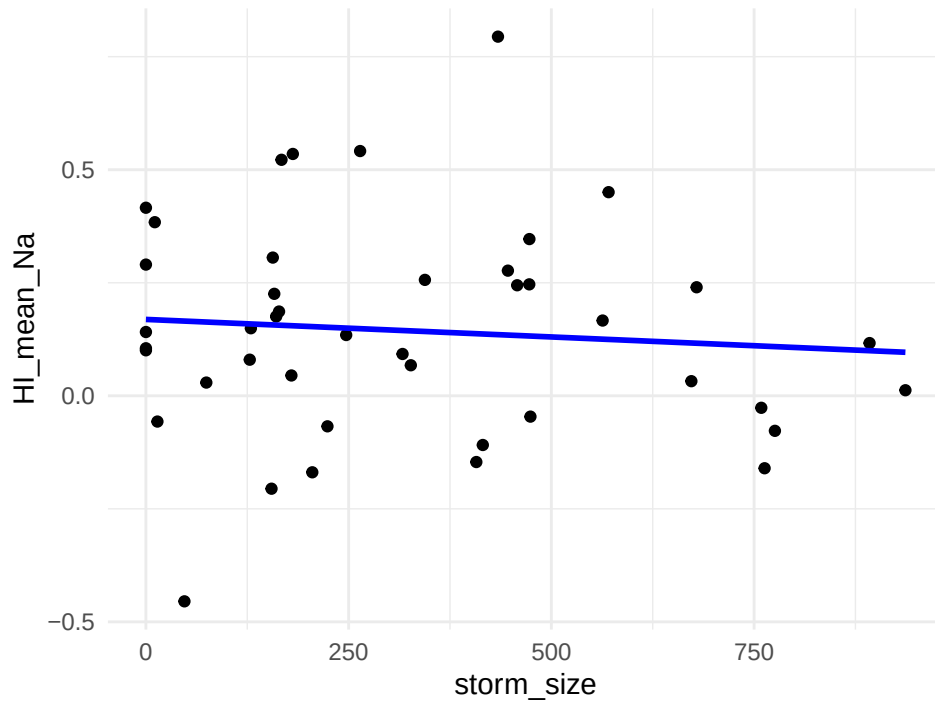




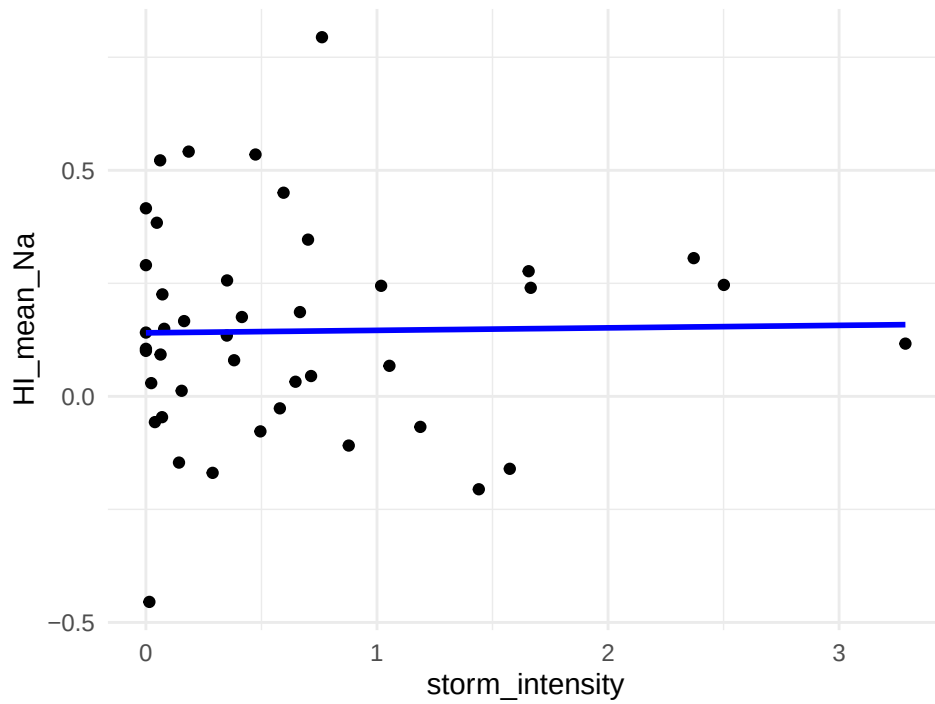


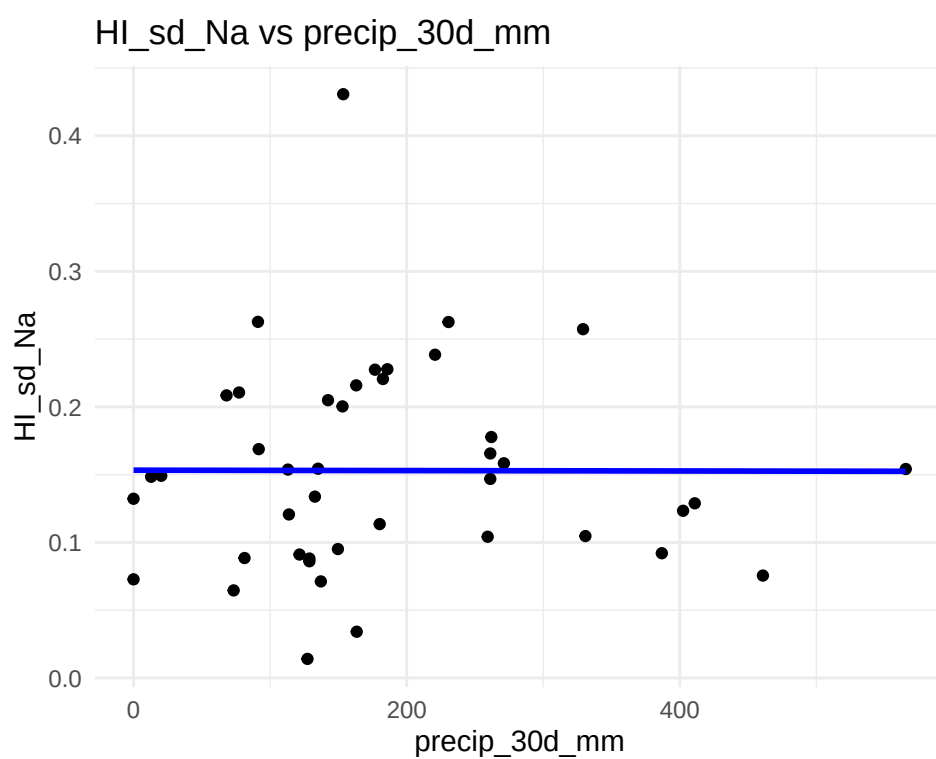
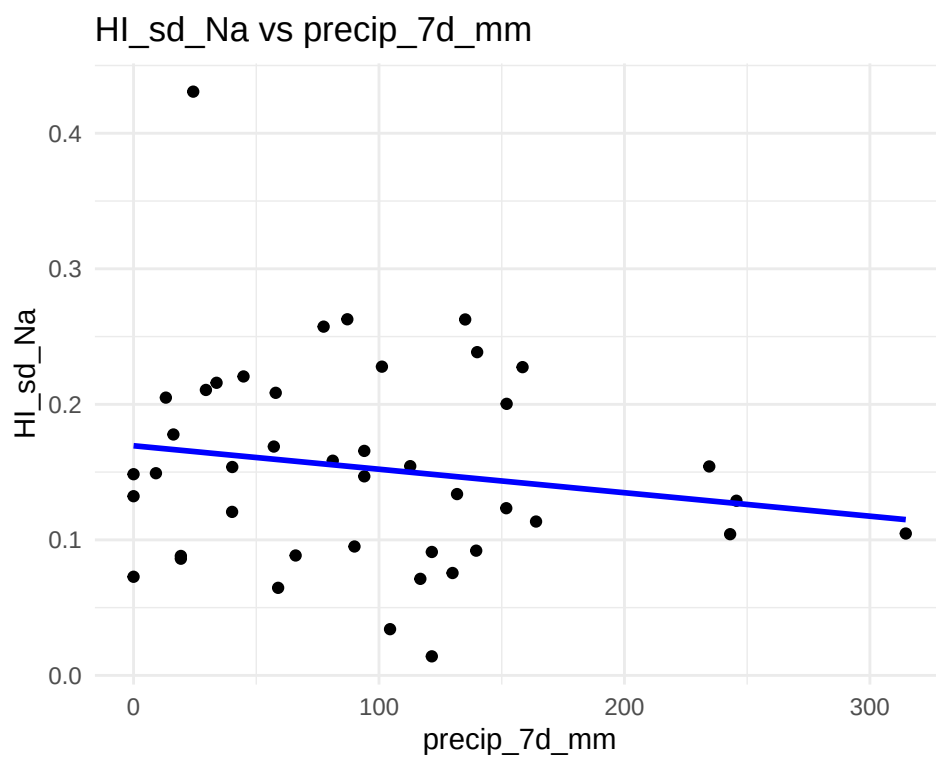


HI_mean_Na vs storm_size

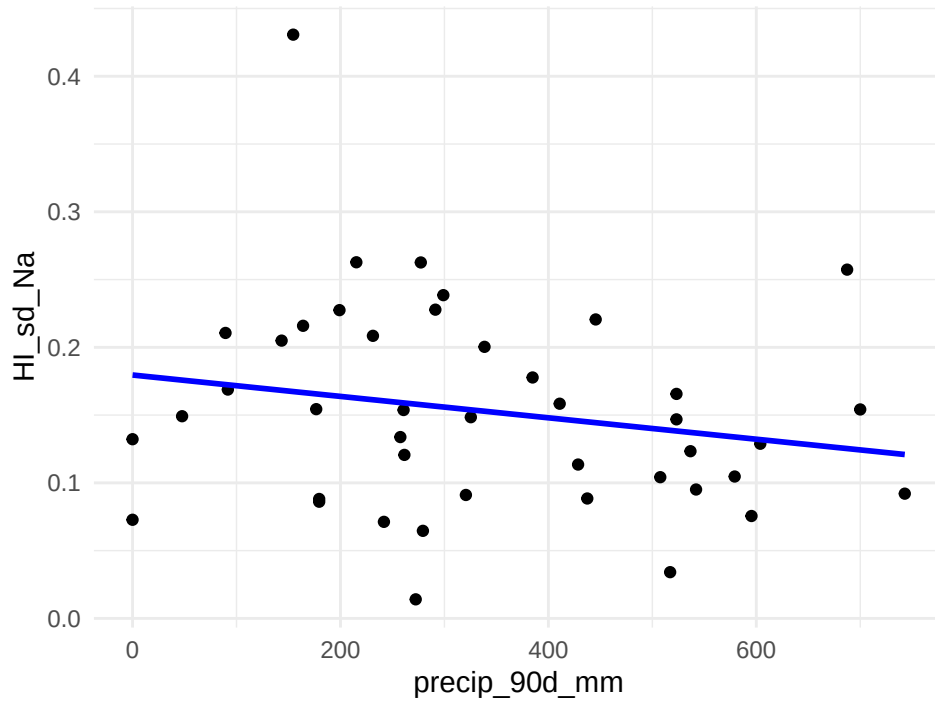


HI_mean_Na vs storm_intensity

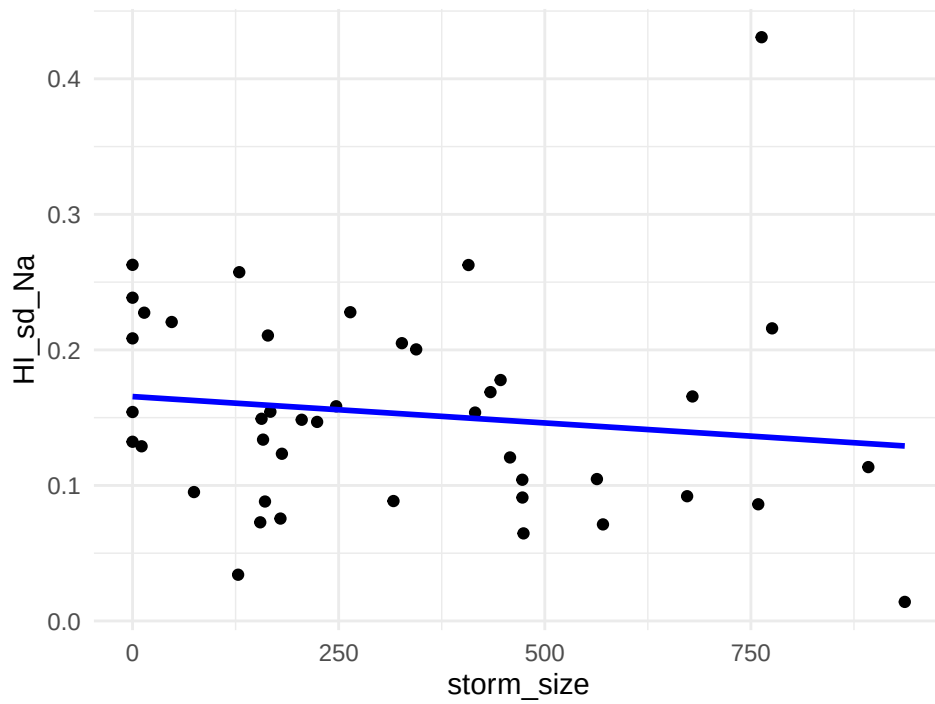


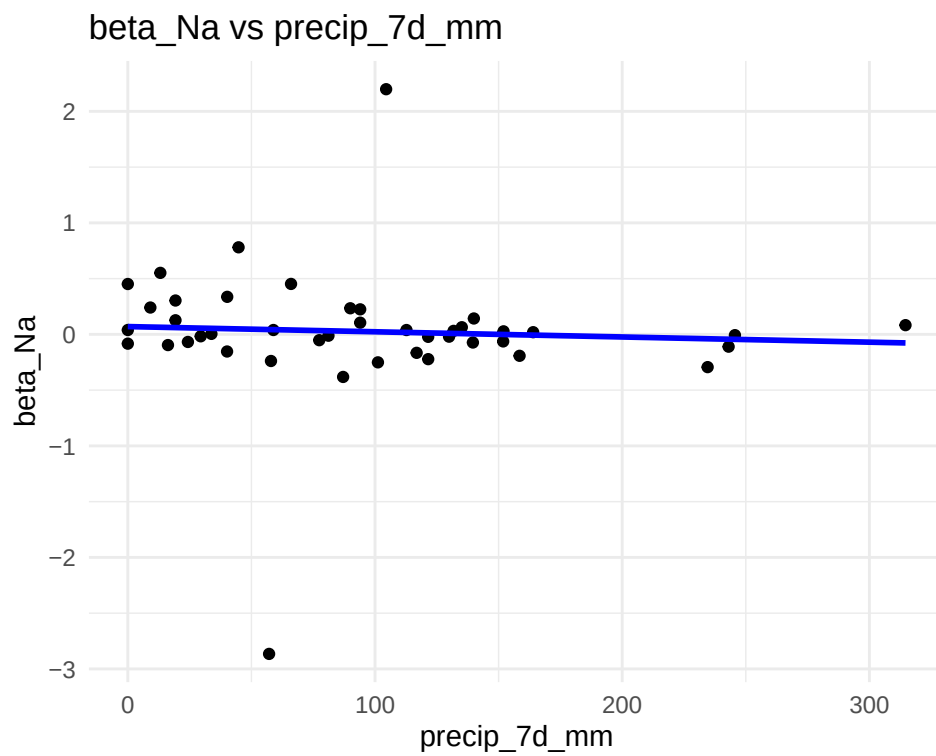
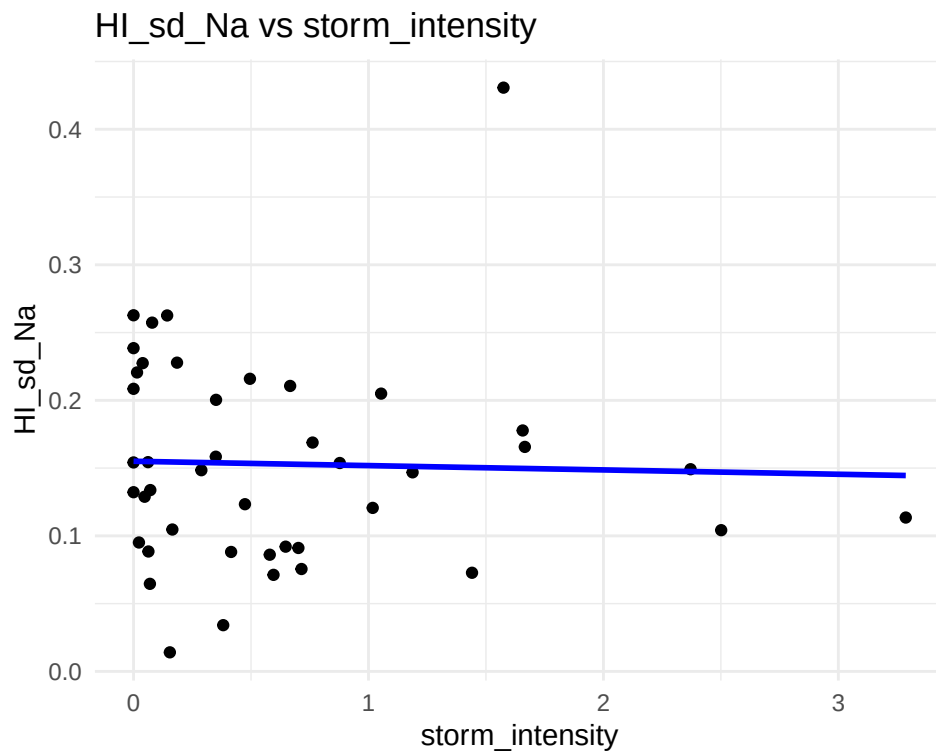


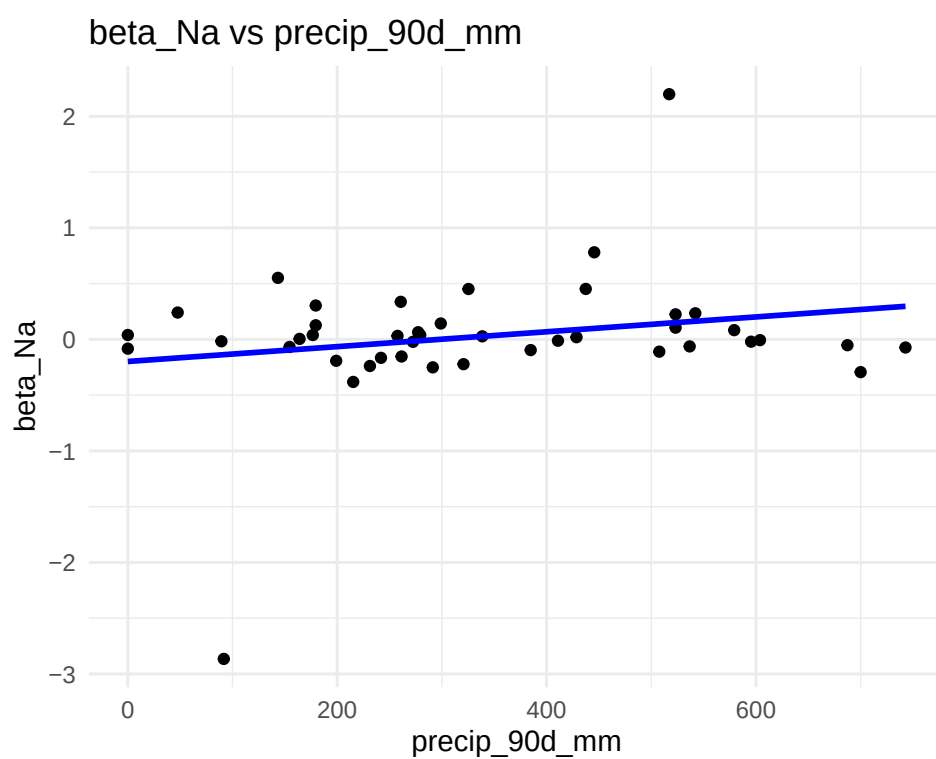
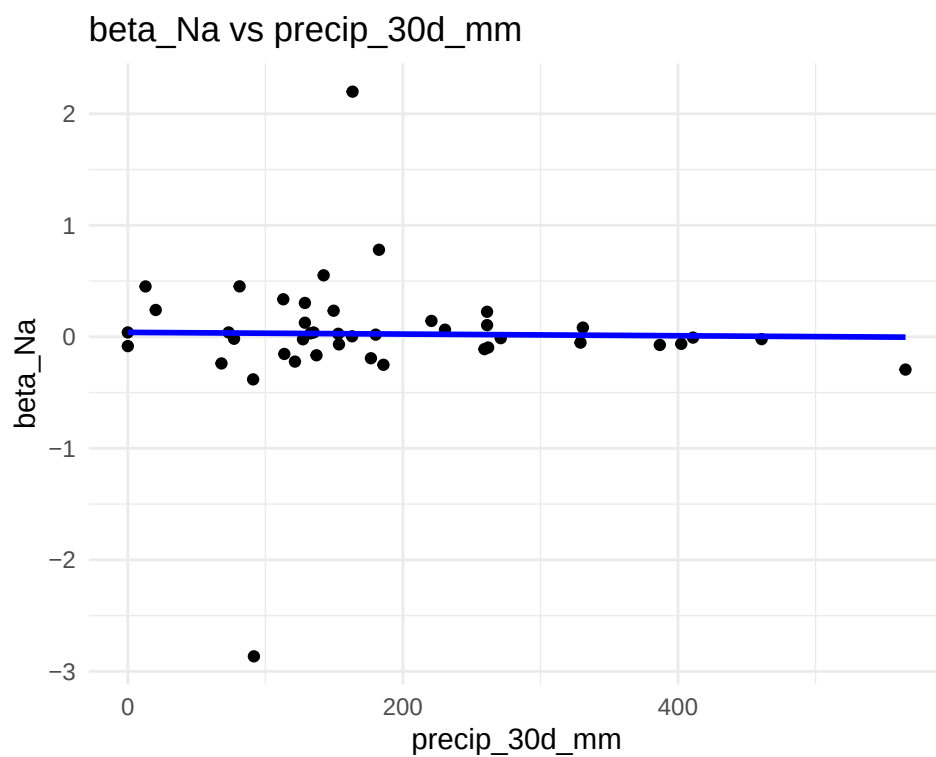
HI_sd_Na vs precip_90d_mm

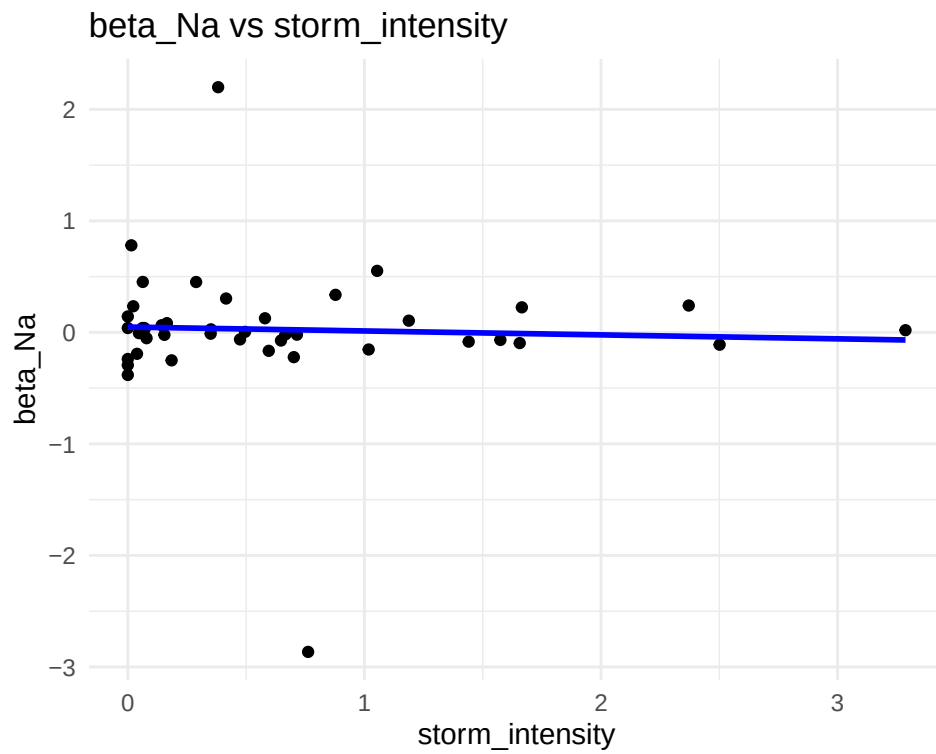
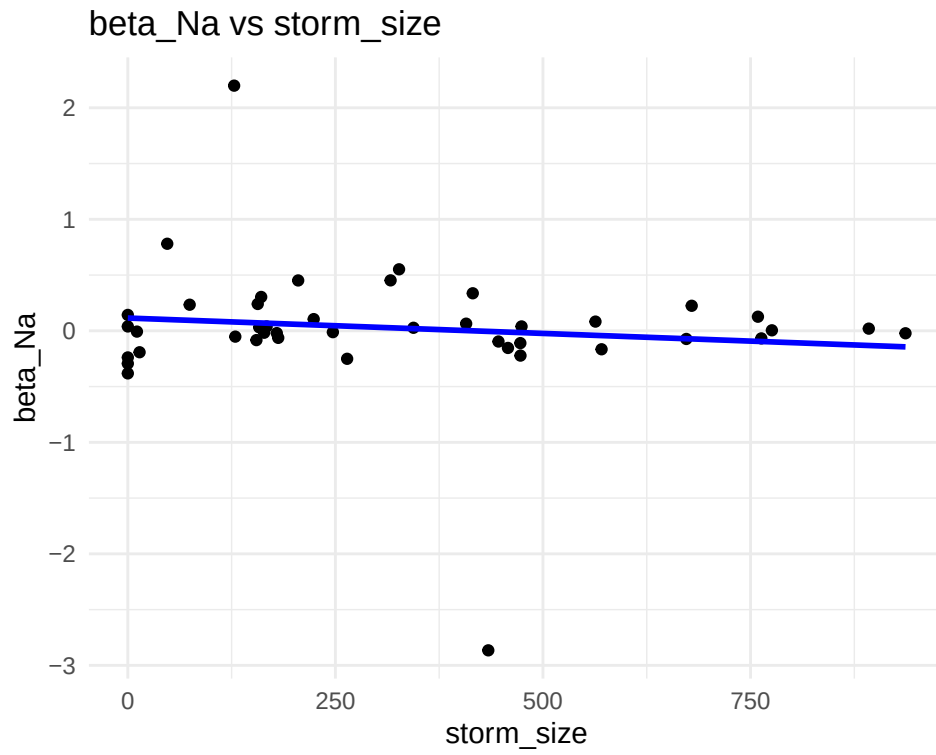


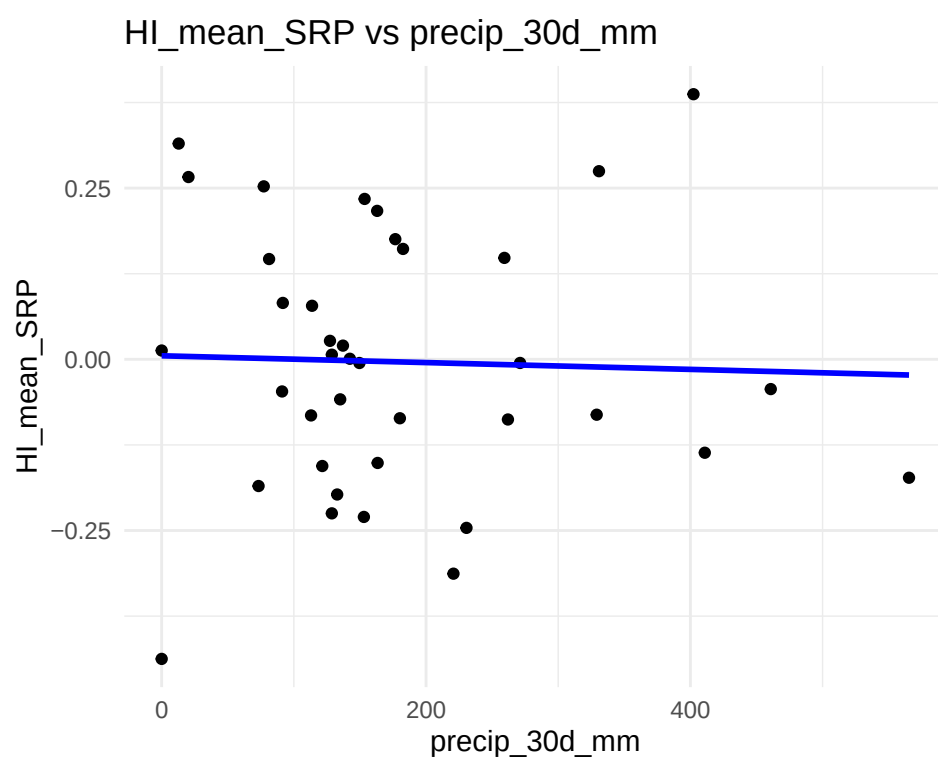
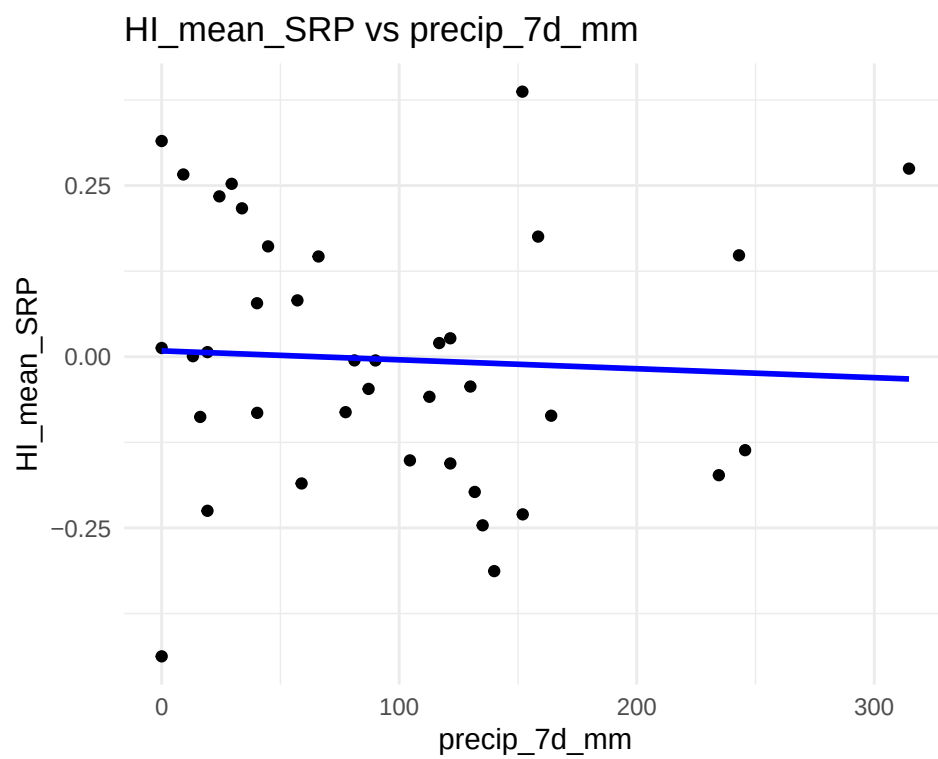
HI_sd_Na vs storm_size



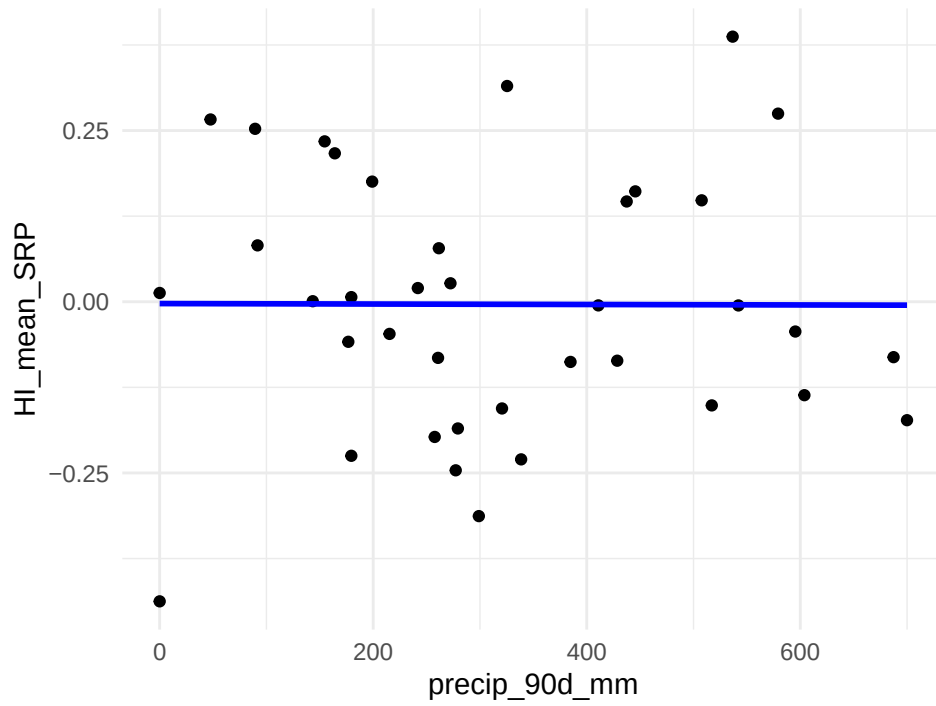




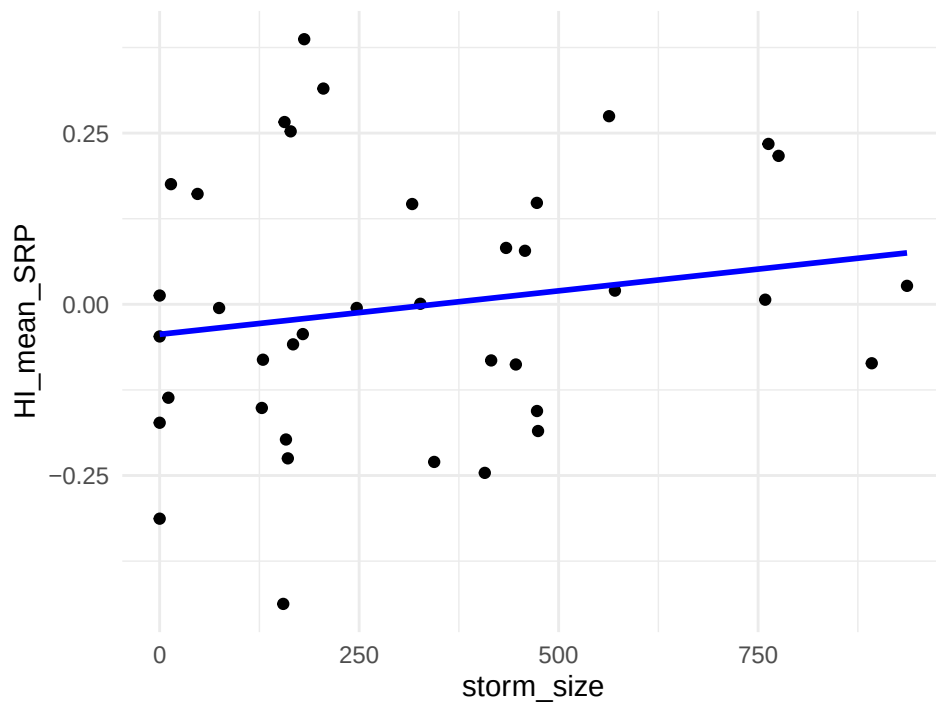


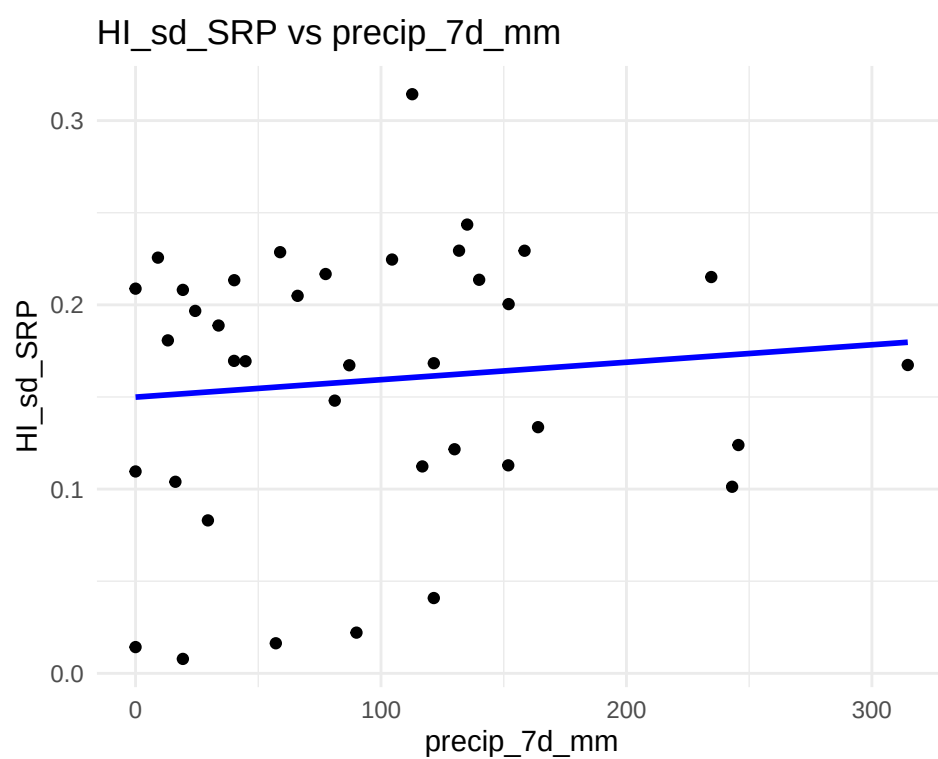
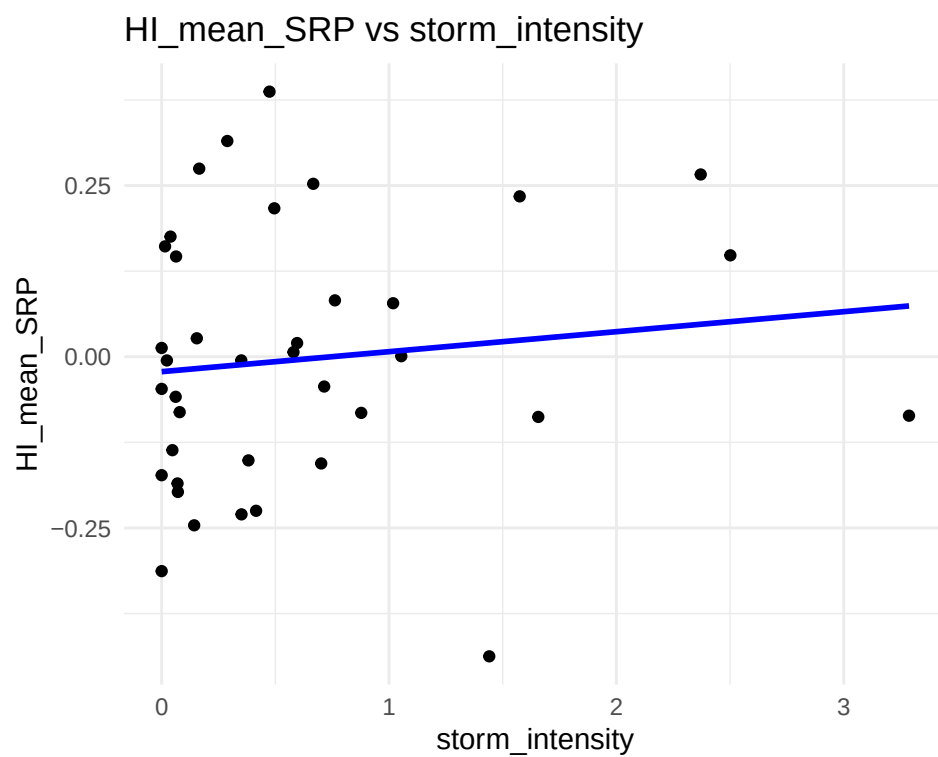


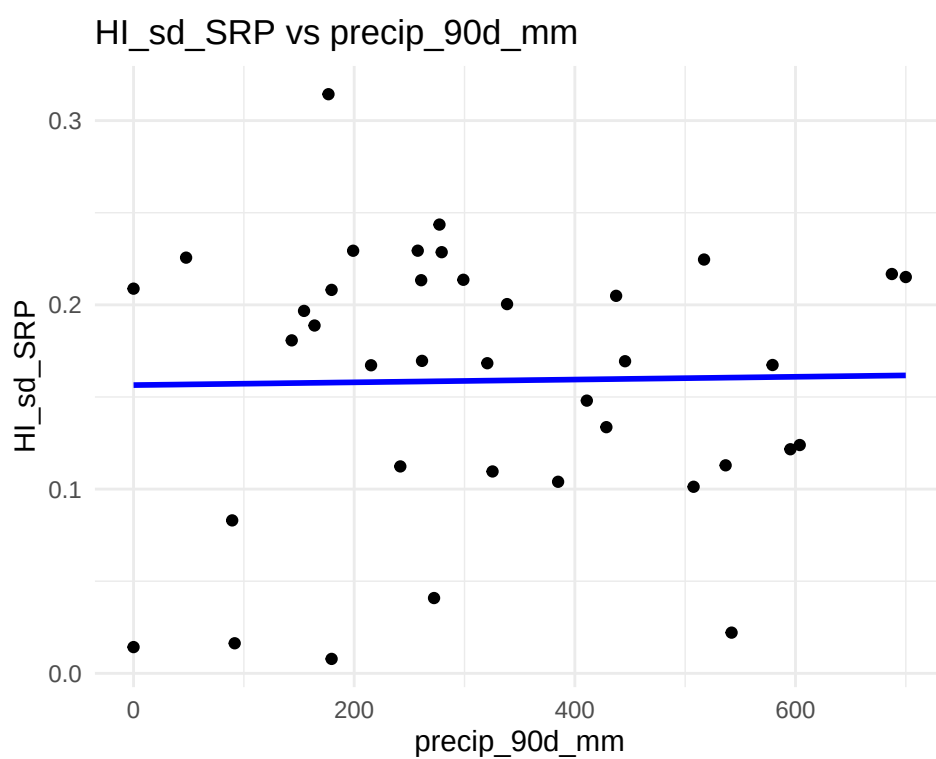
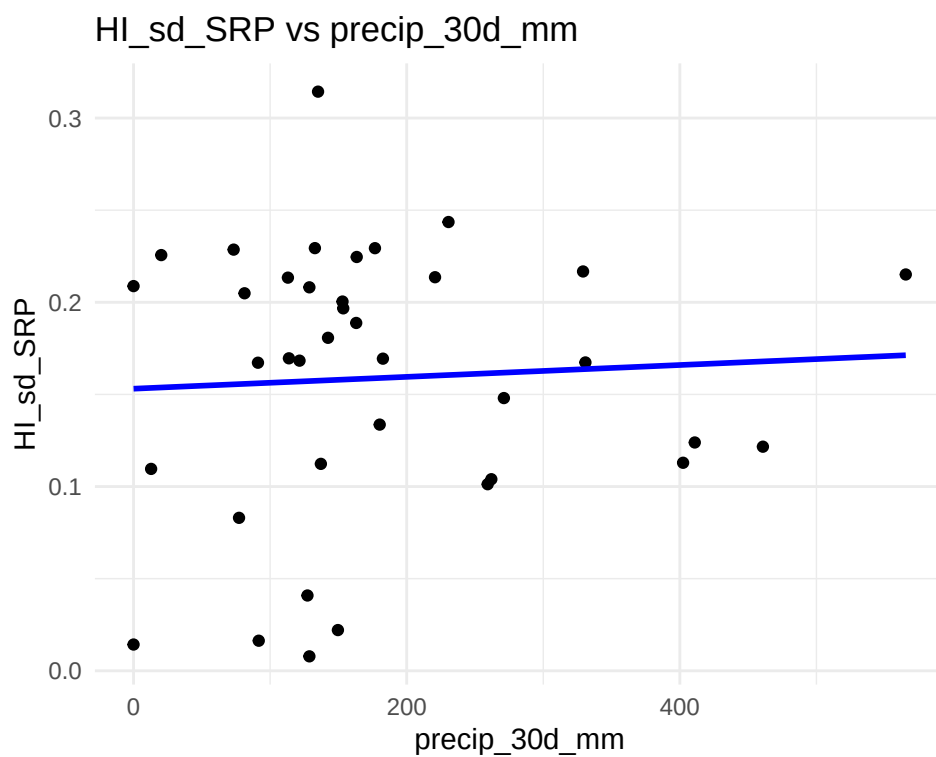
HI_mean_SRP vs precip_90d_mm

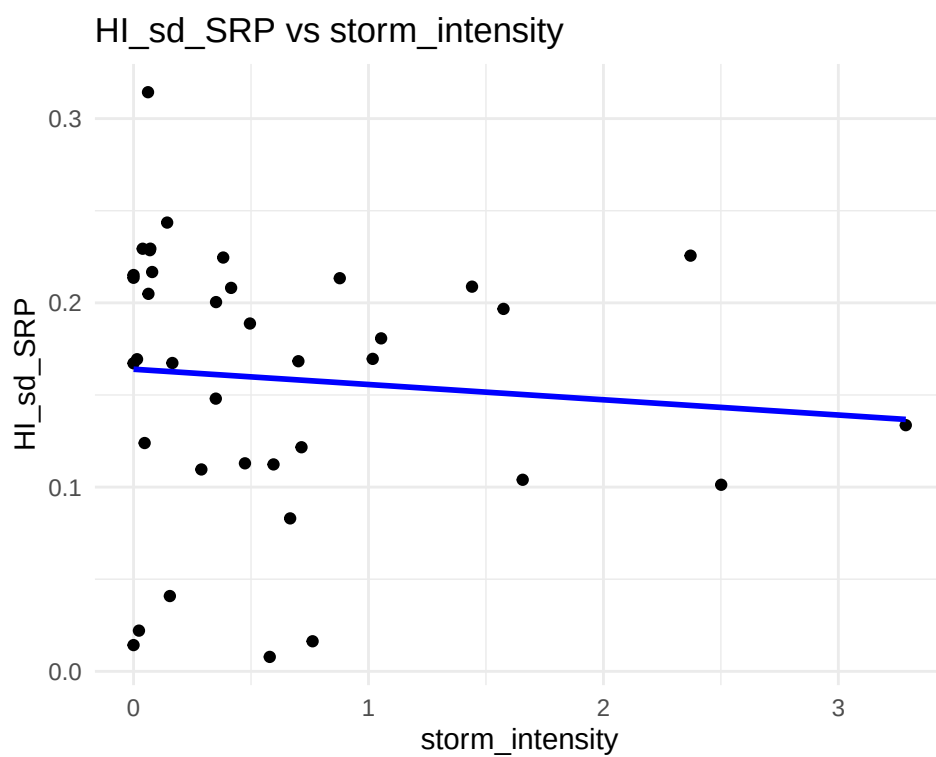
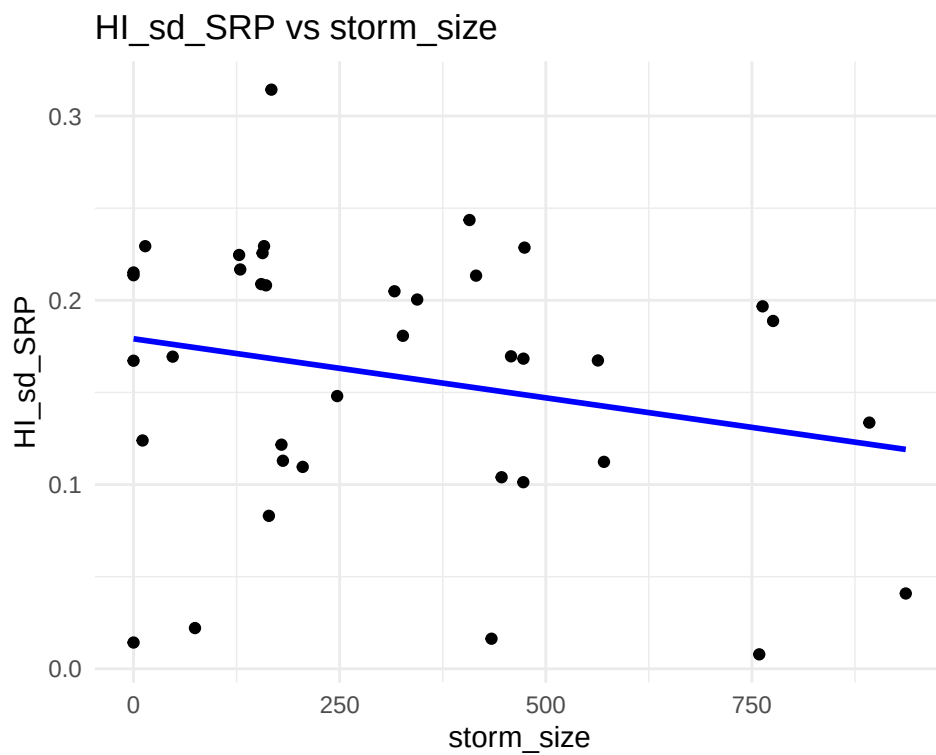


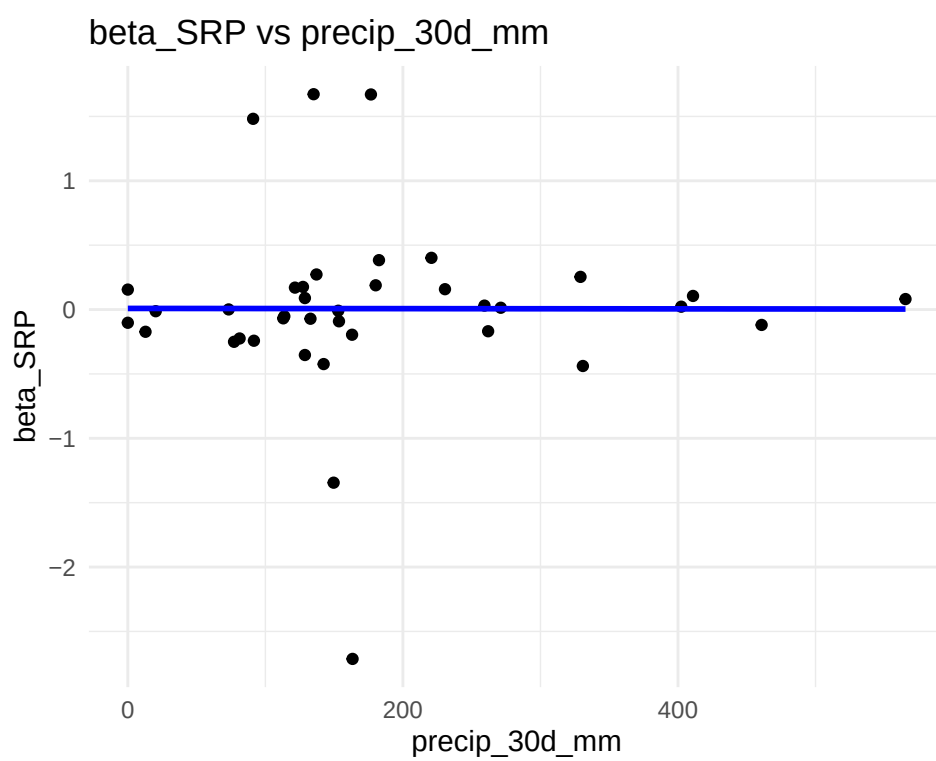
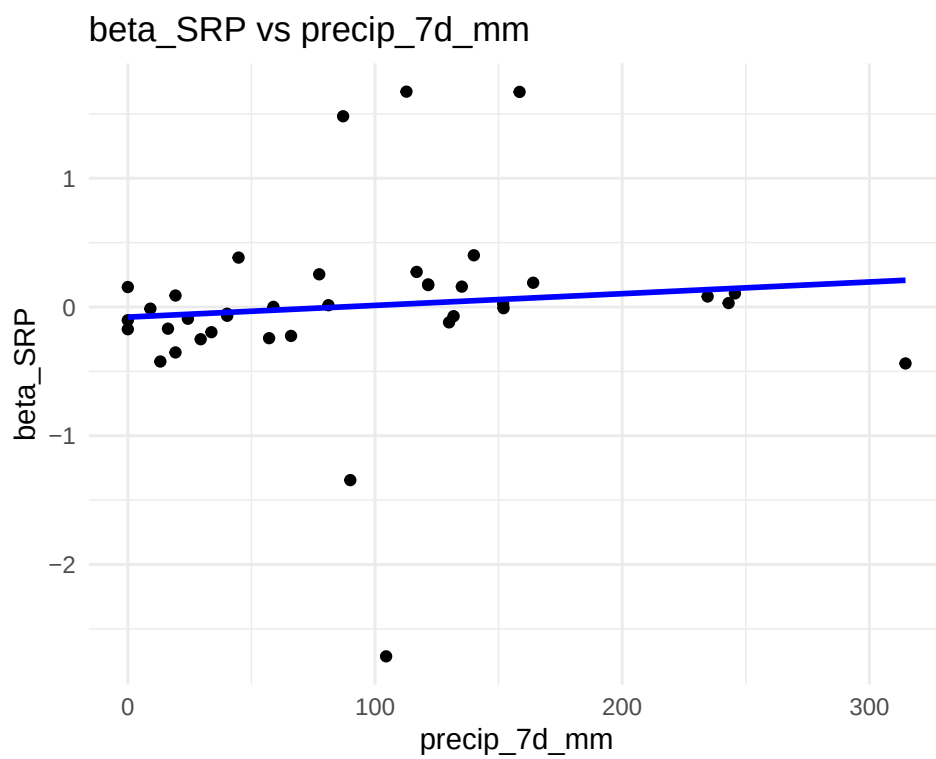
HI_mean_SRP vs storm_size



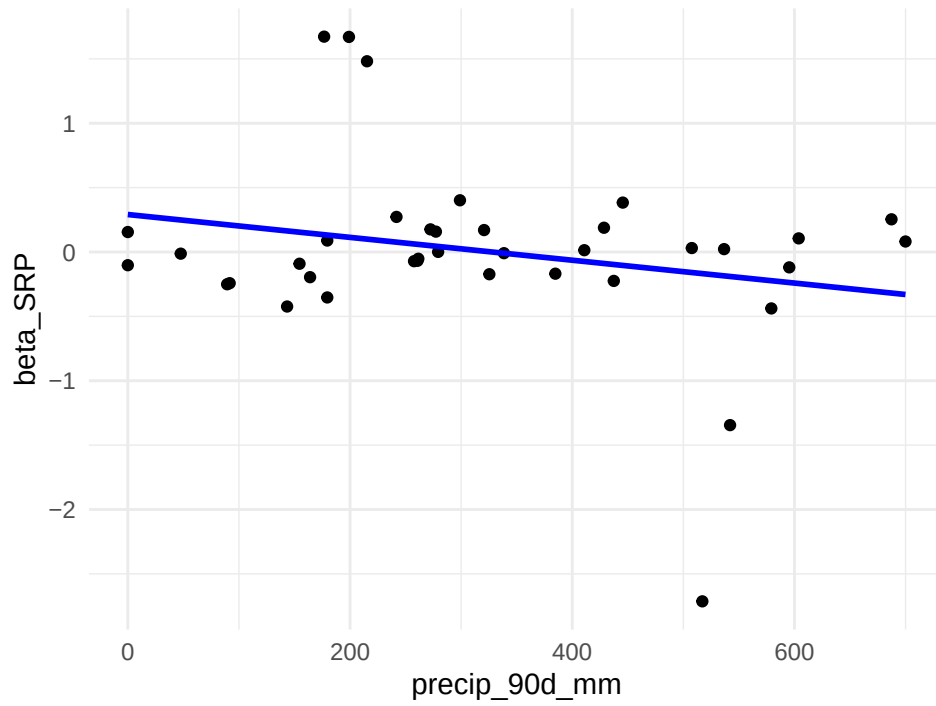




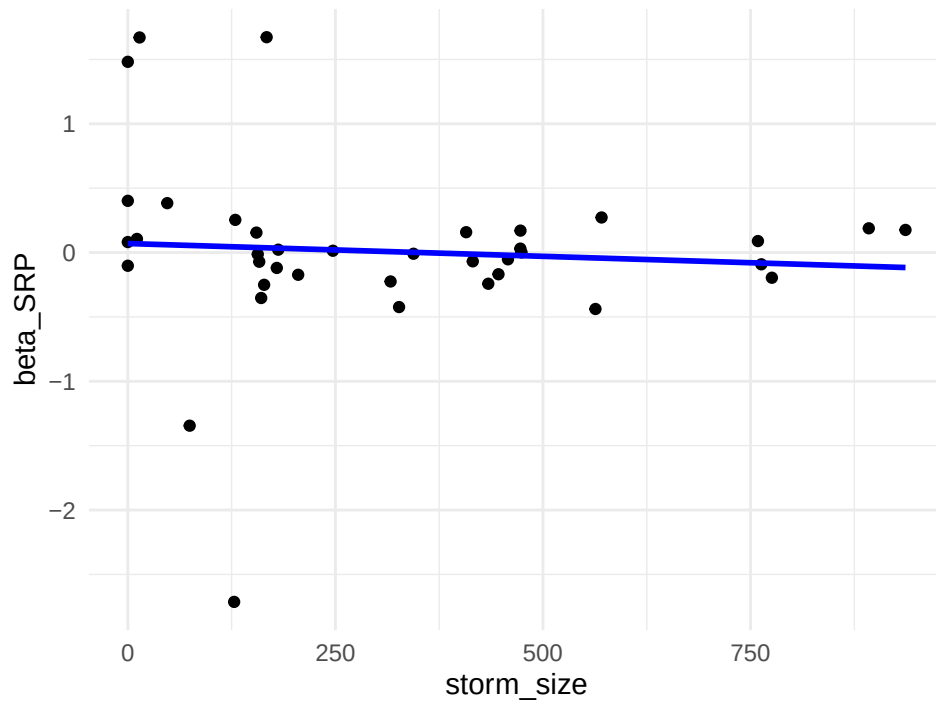


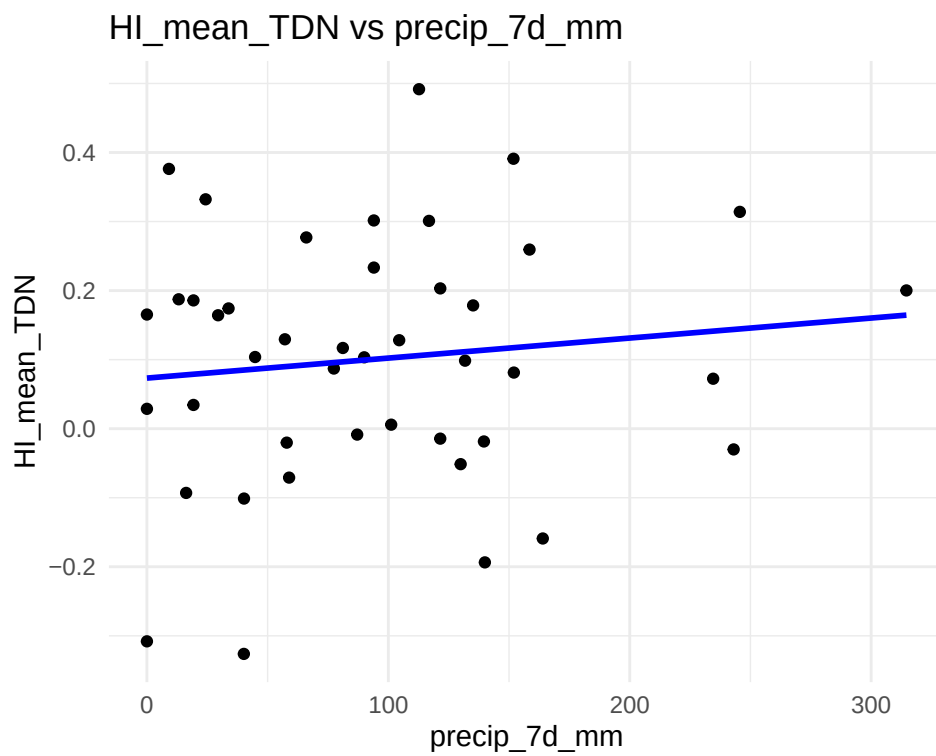
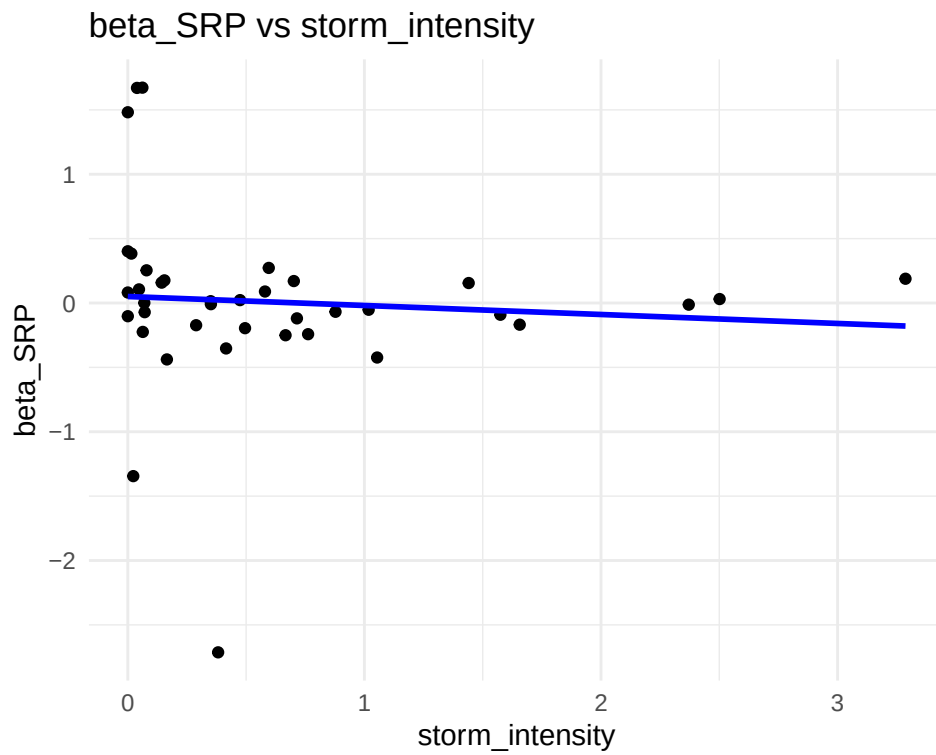


beta_SRP vs precip_90d_mm

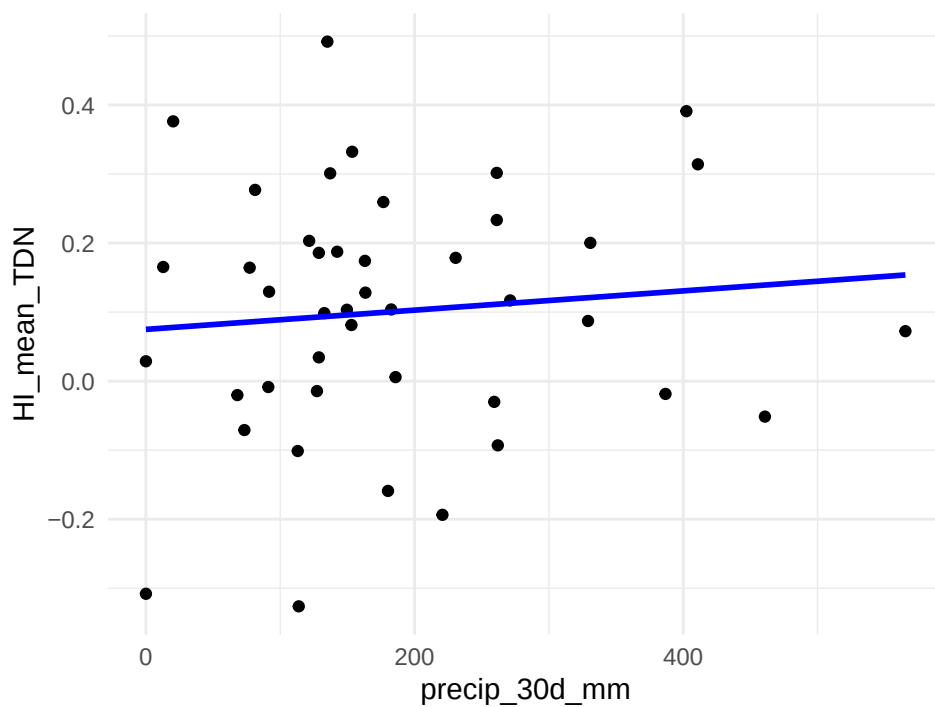


beta_SRP vs storm_size

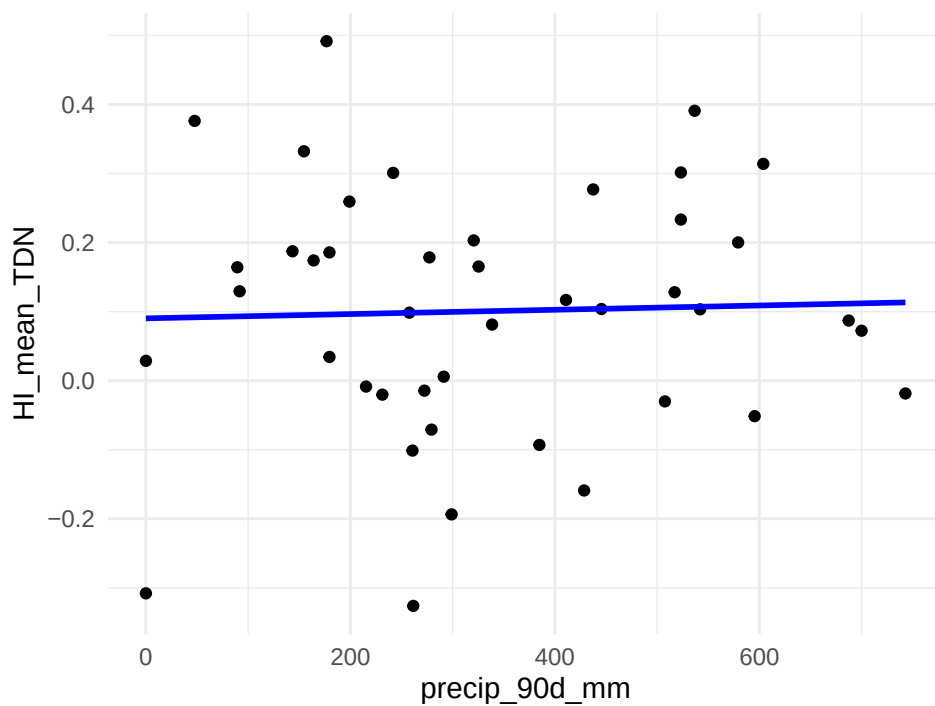




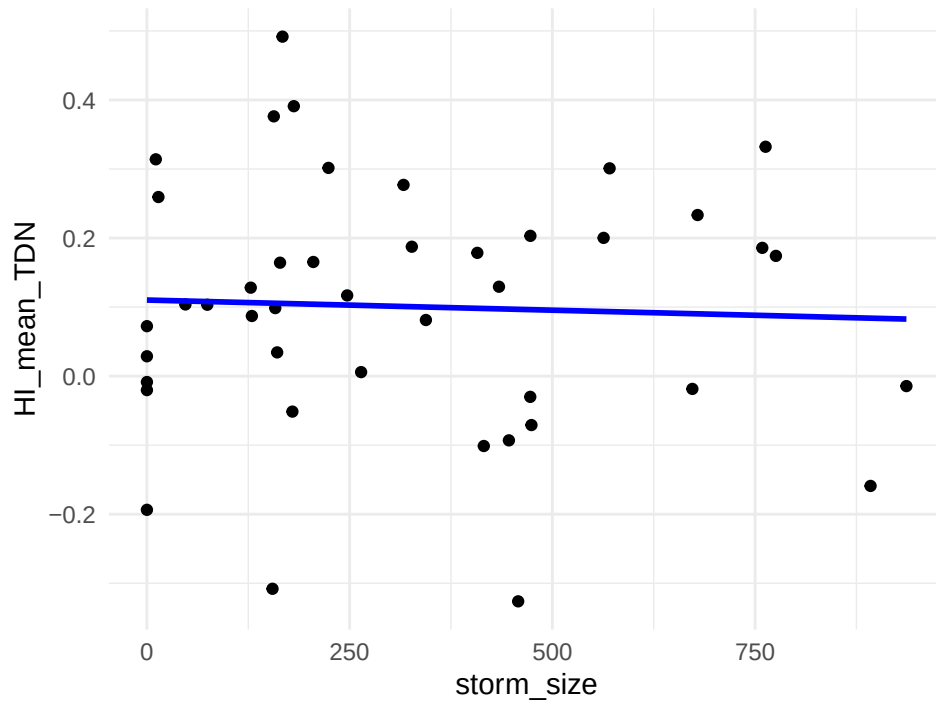
HI_mean_TDN vs precip_30d_mm



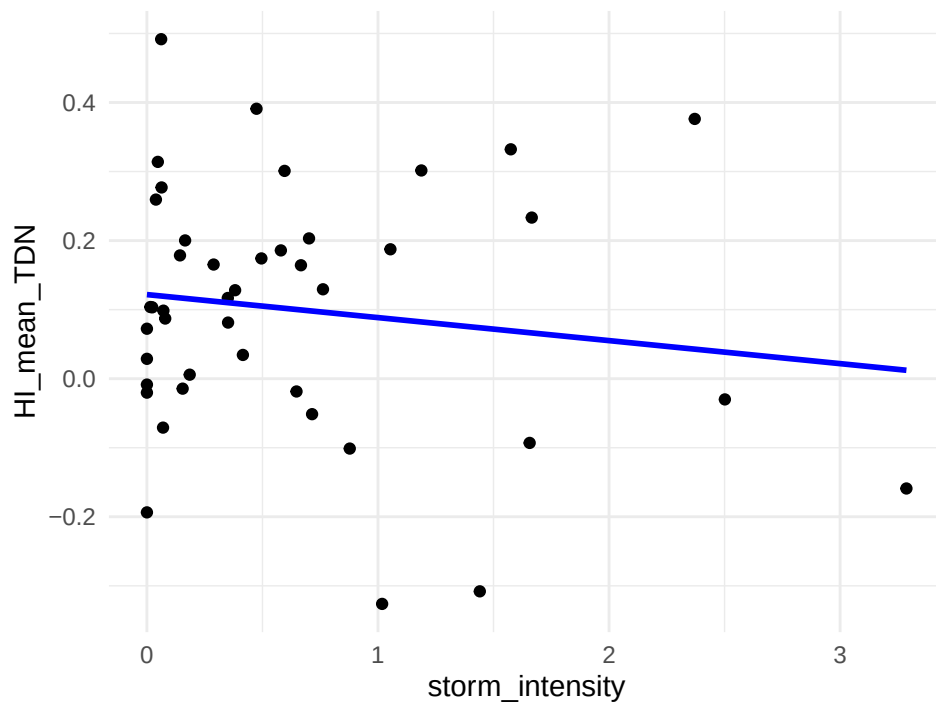
HI_mean_TDN vs precip_90d_mm

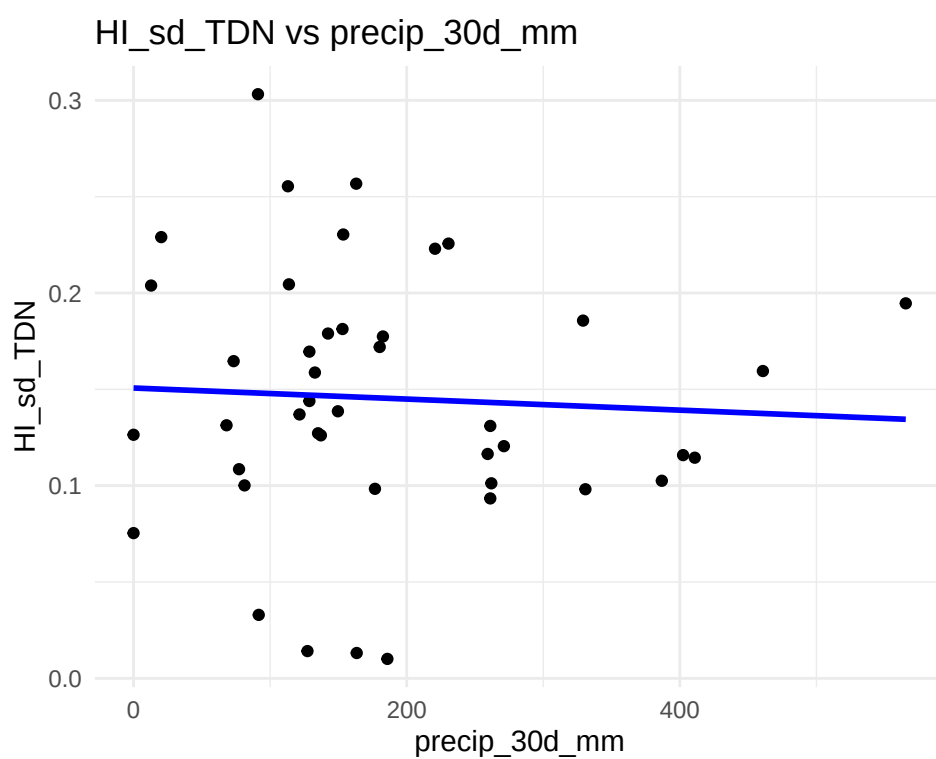
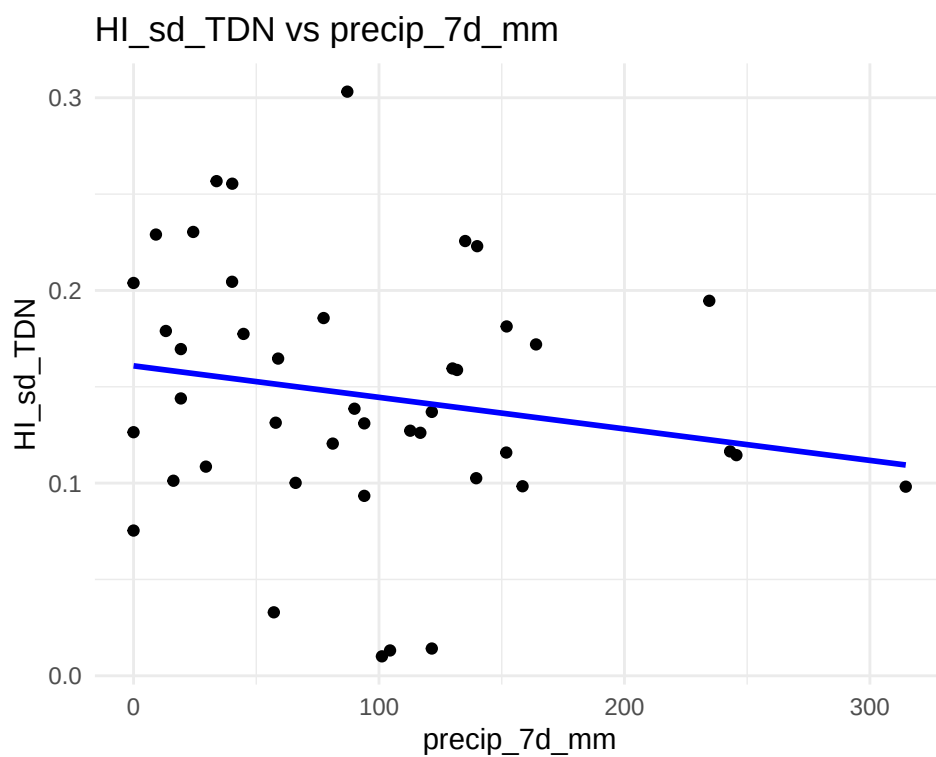


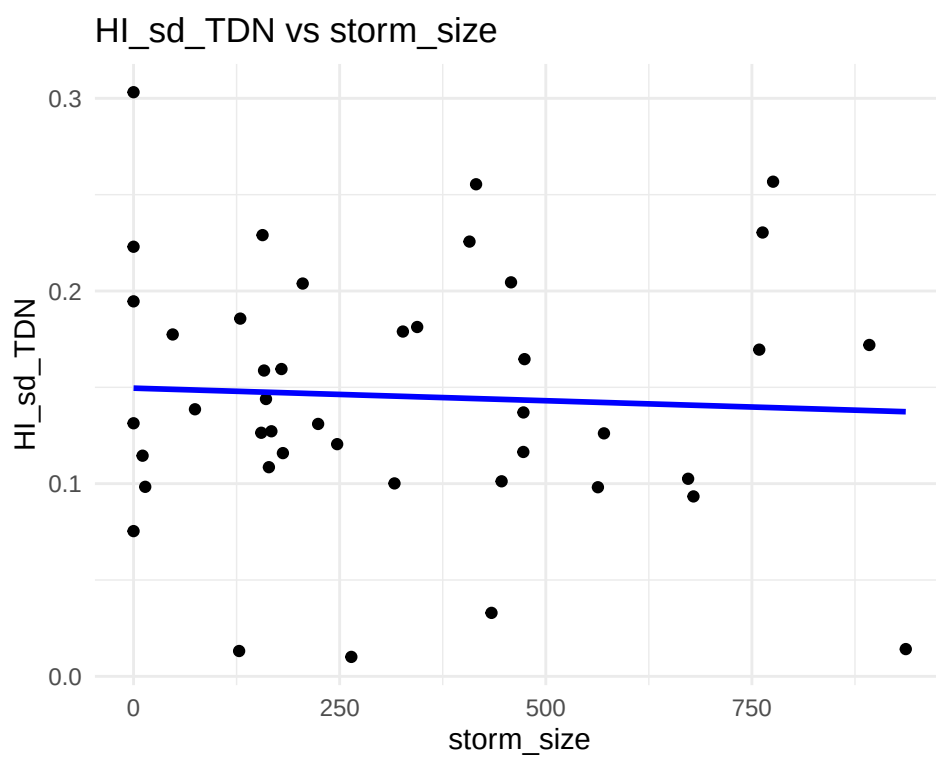
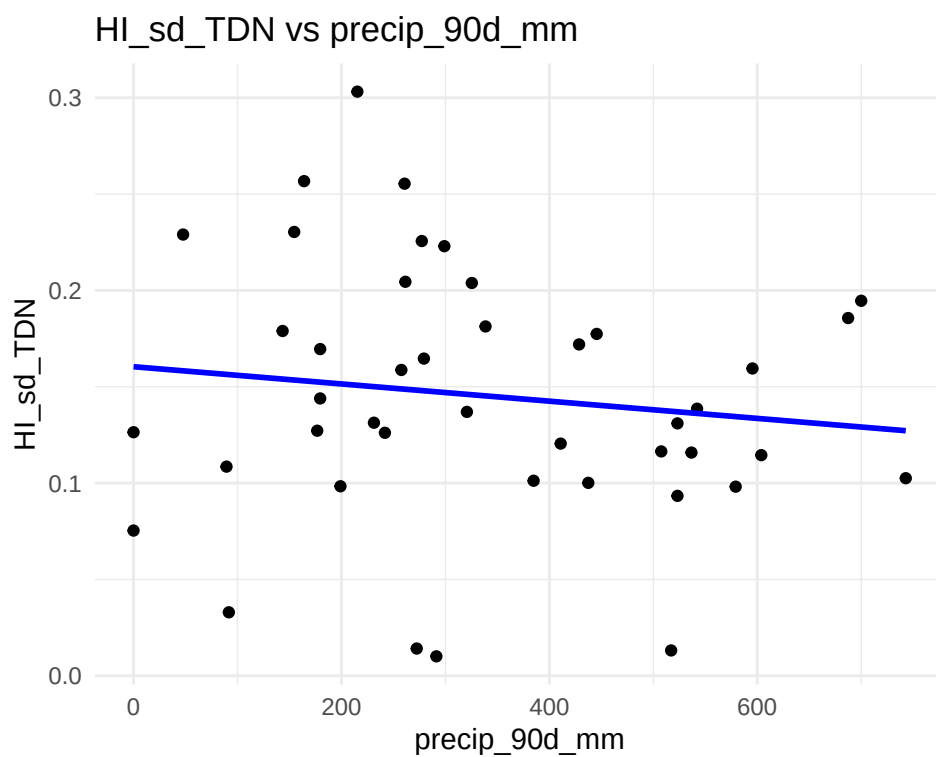
HI_mean_TDN vs storm_size

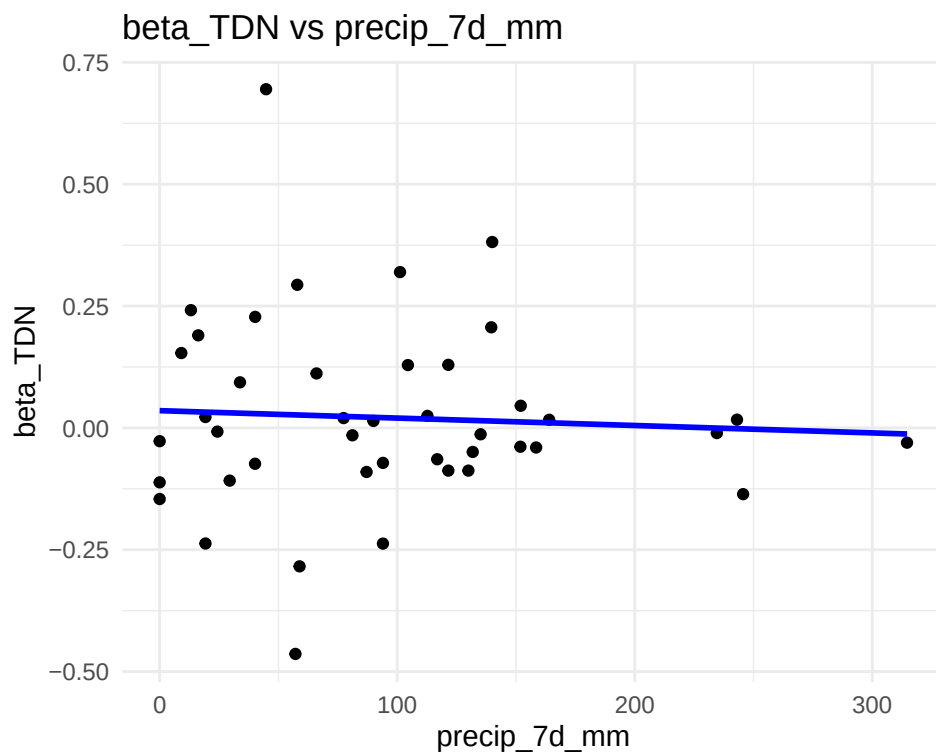
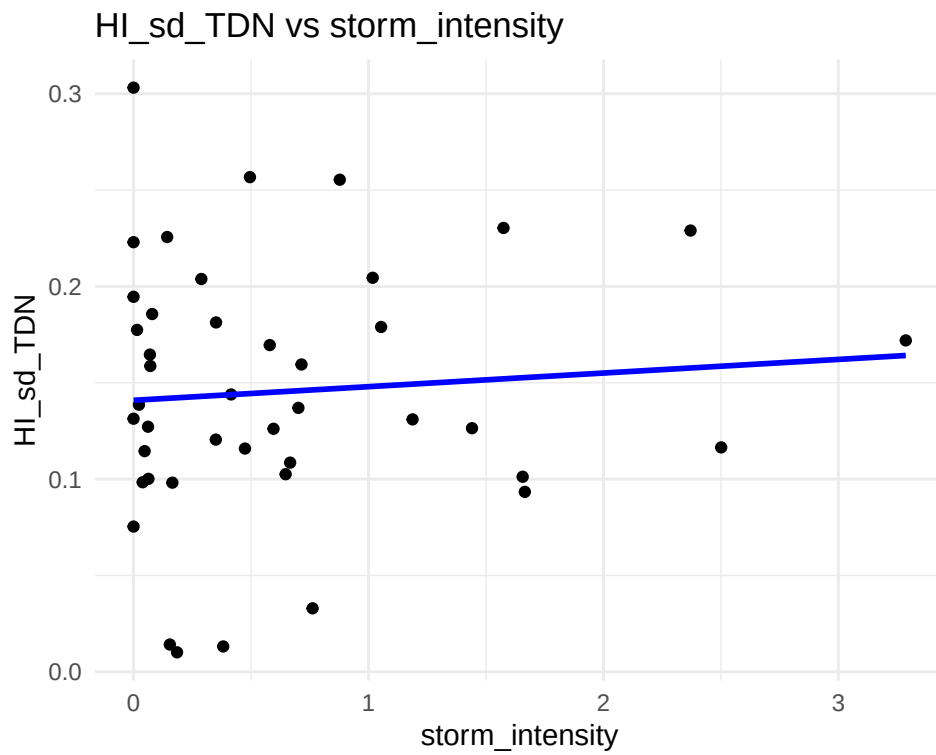


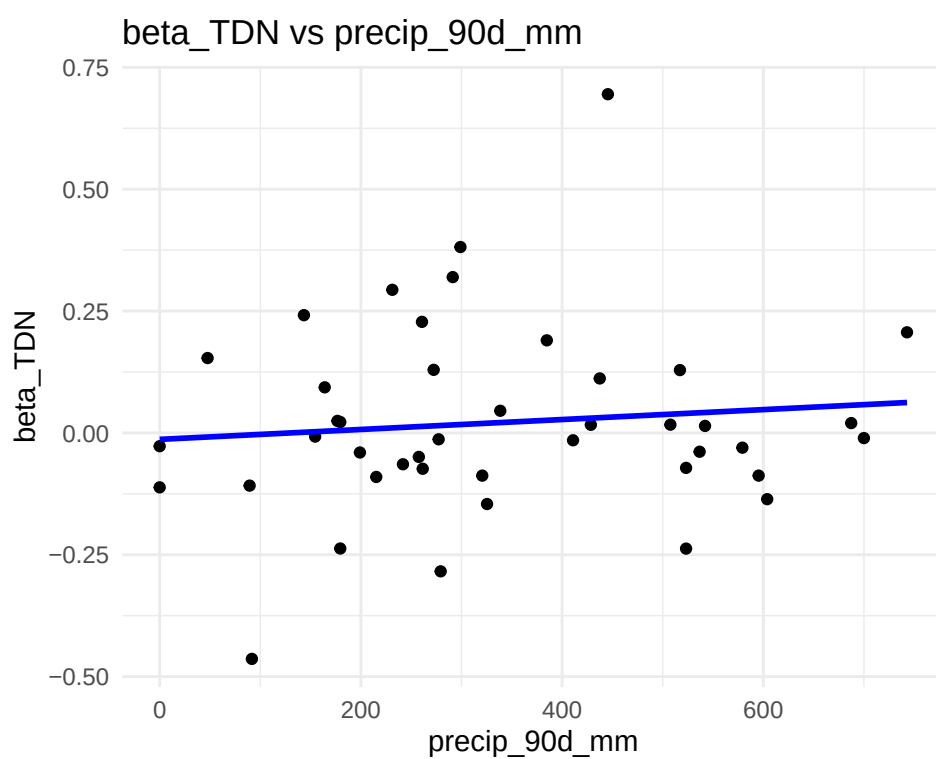
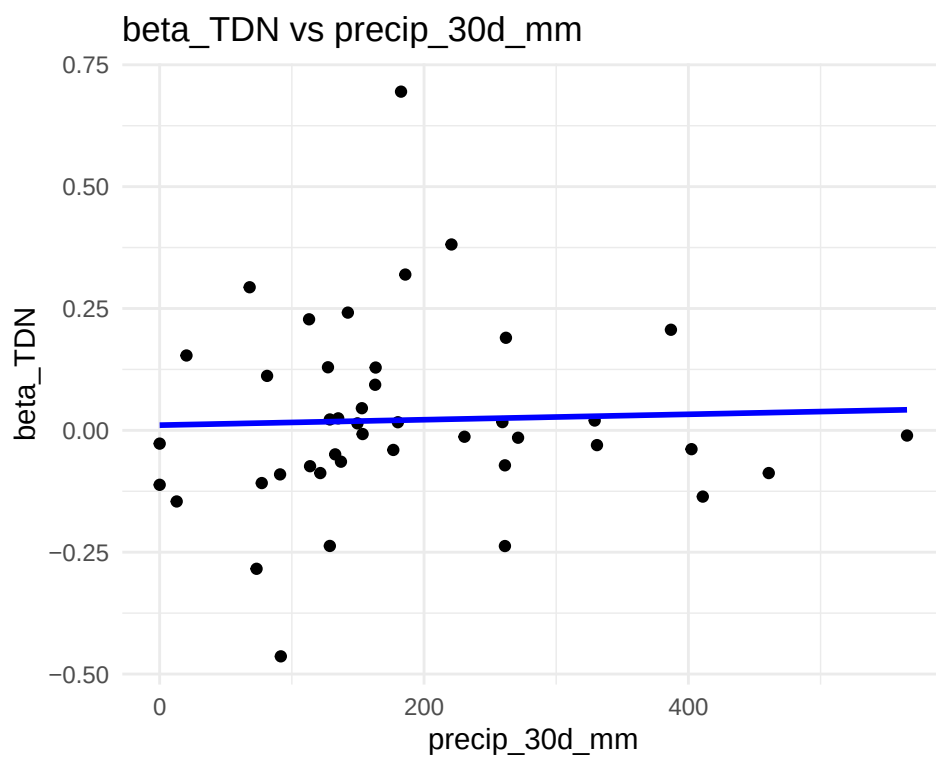
HI_mean_TDN vs storm_intensity

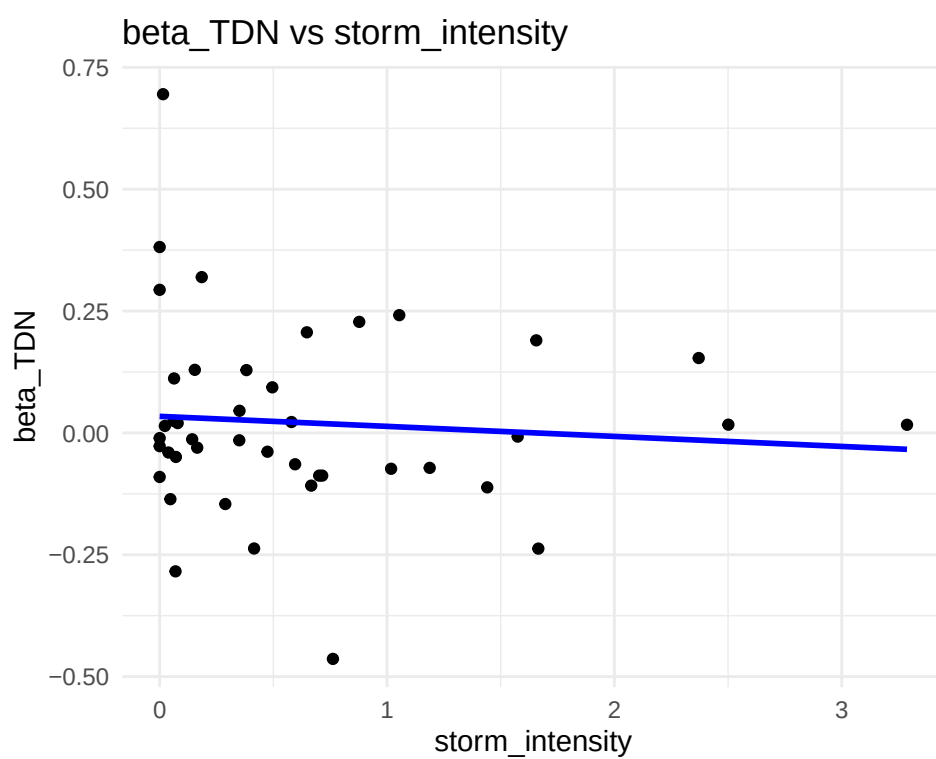
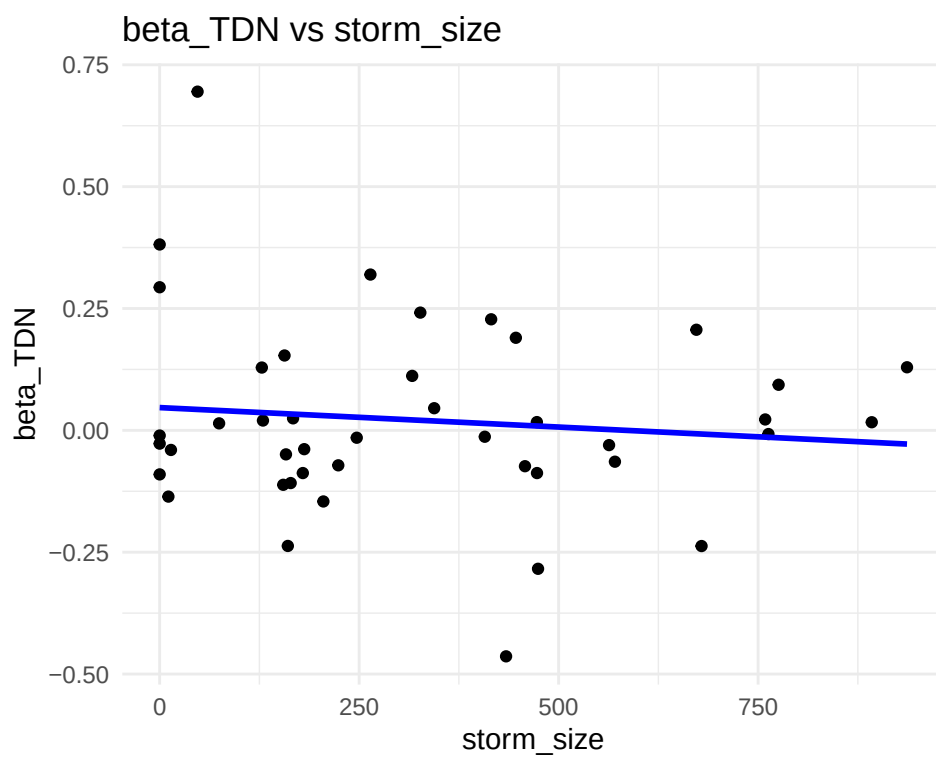


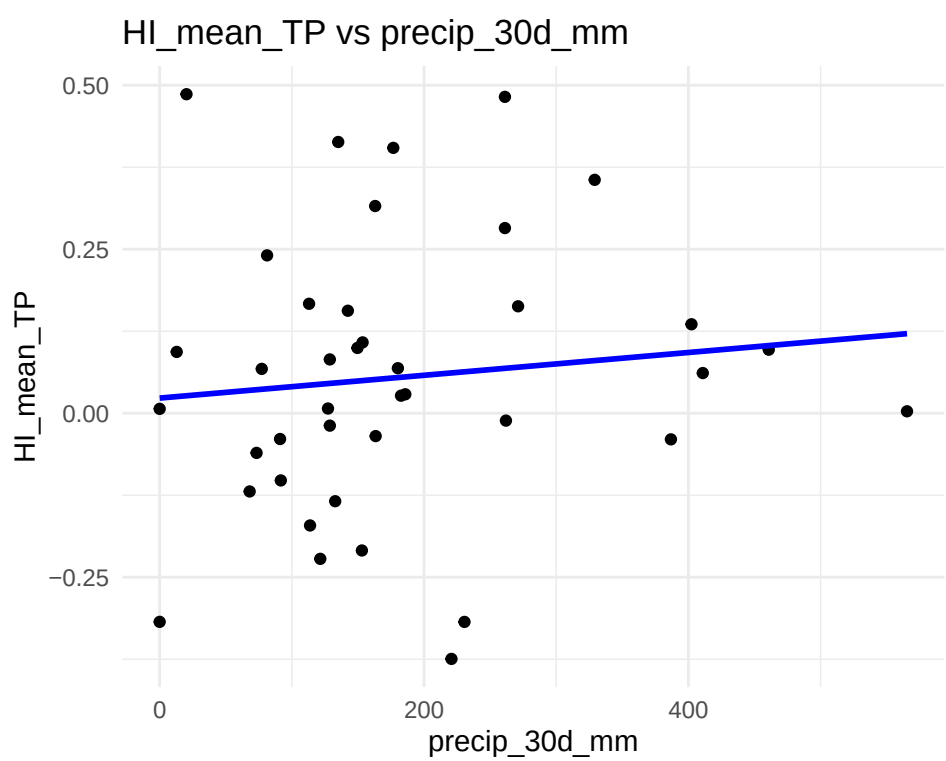
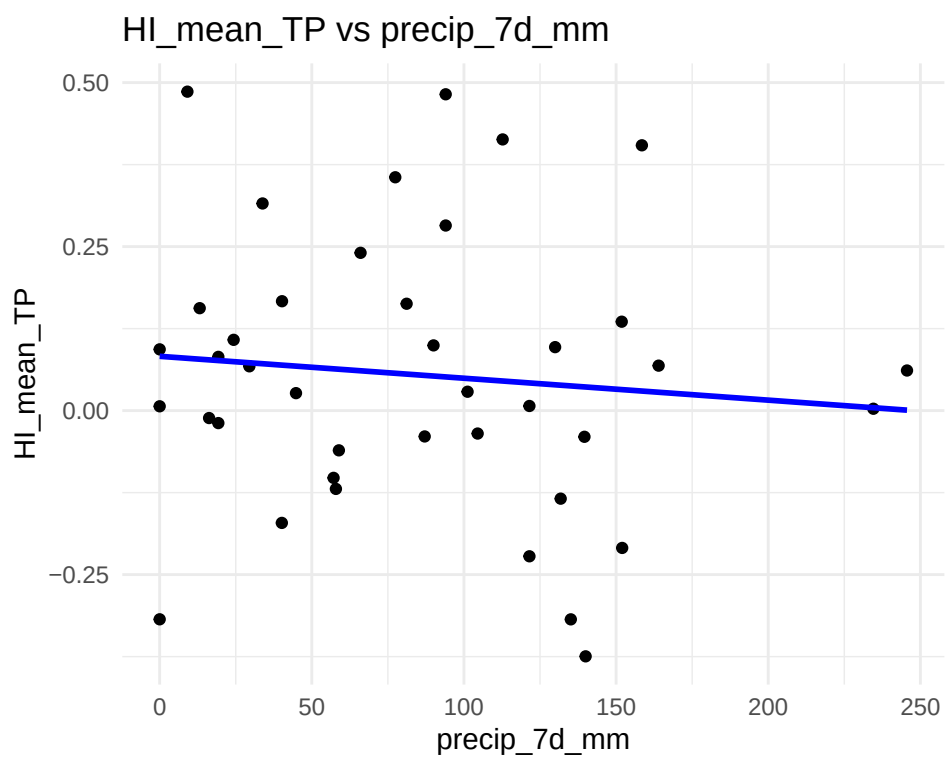


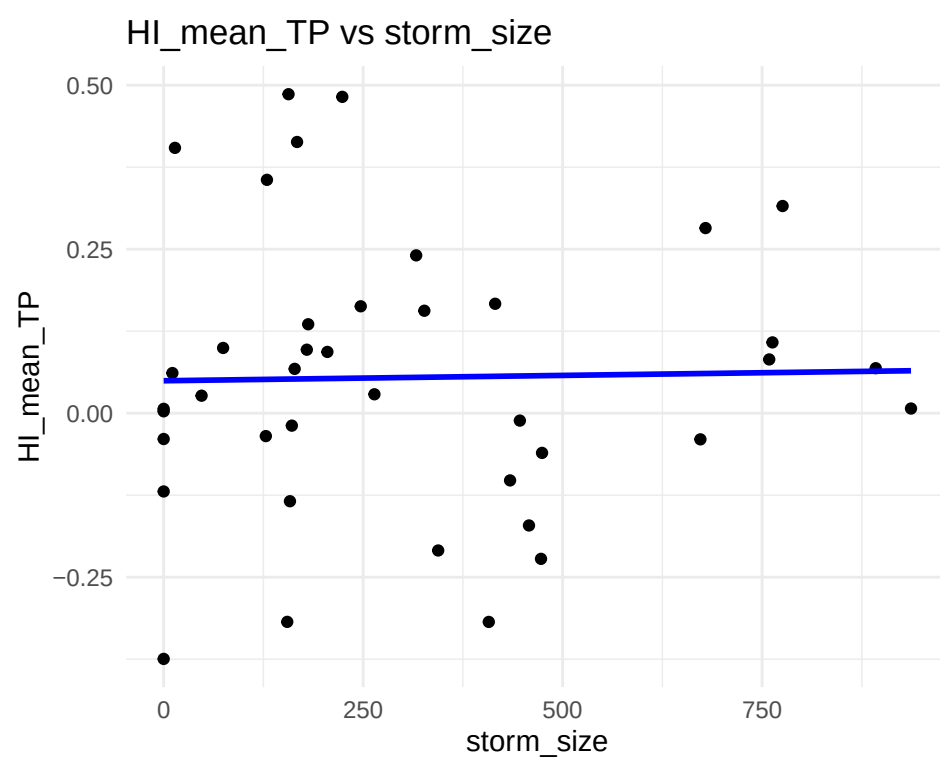
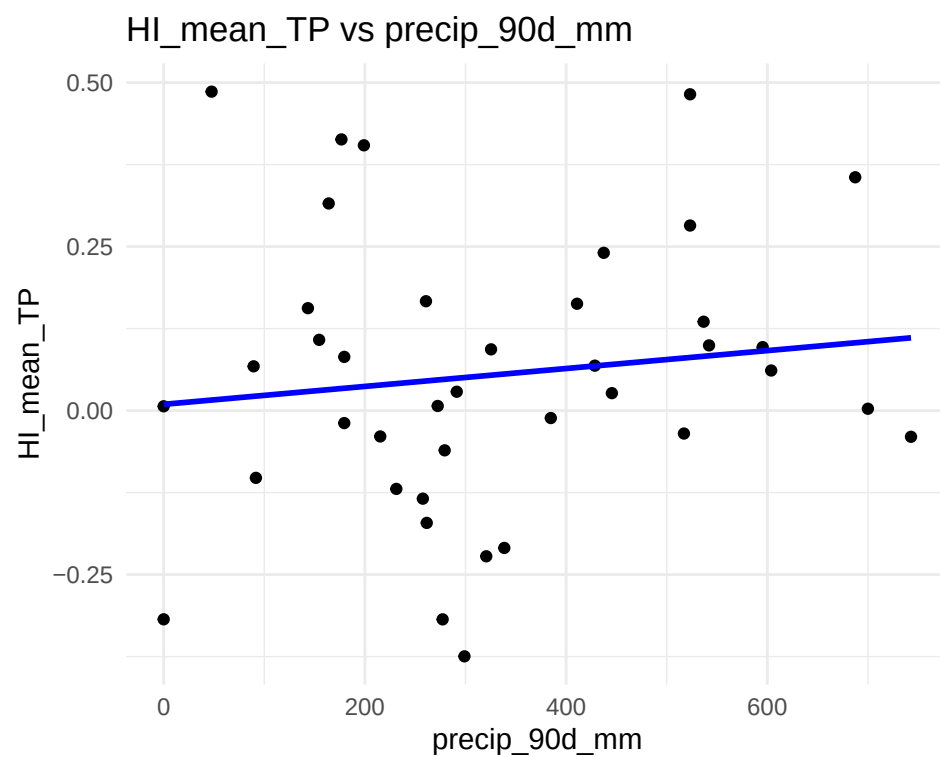


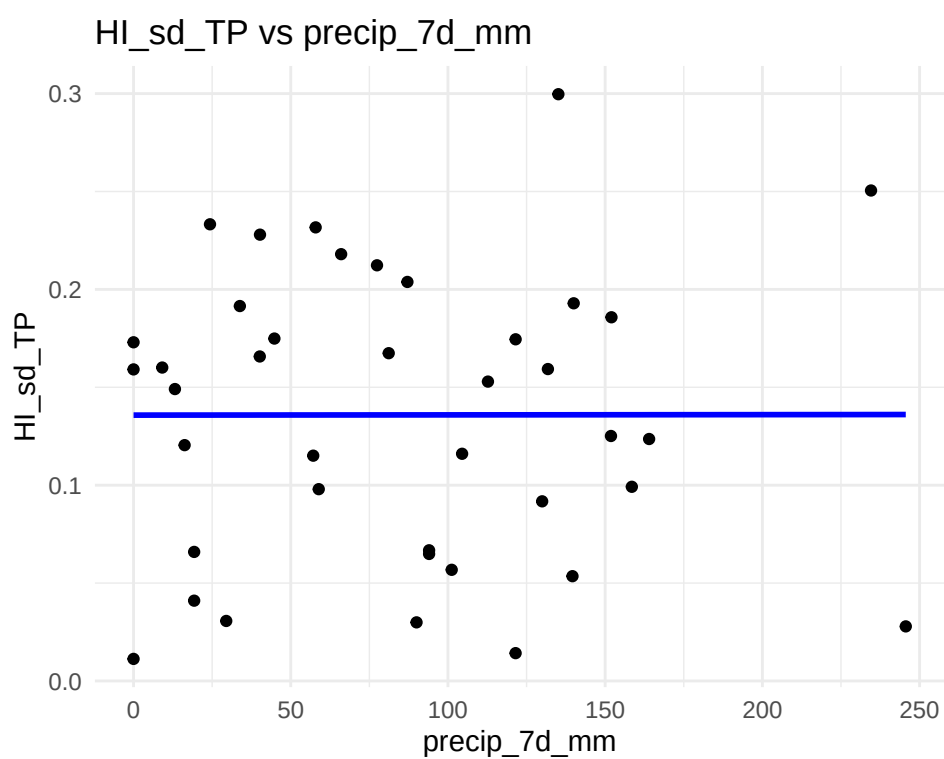
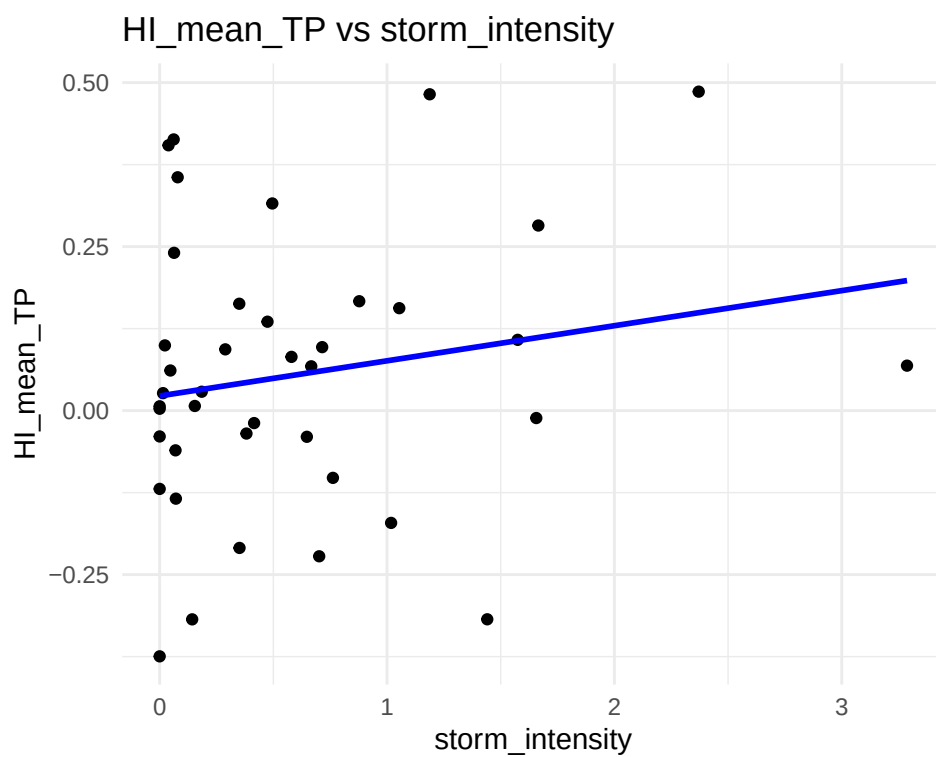


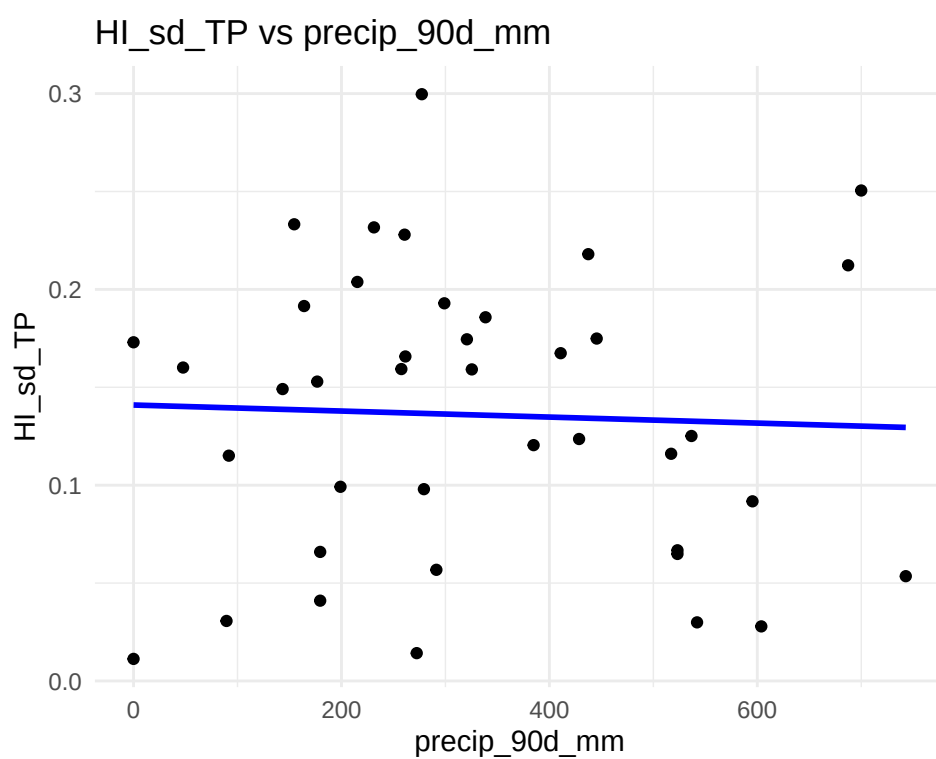
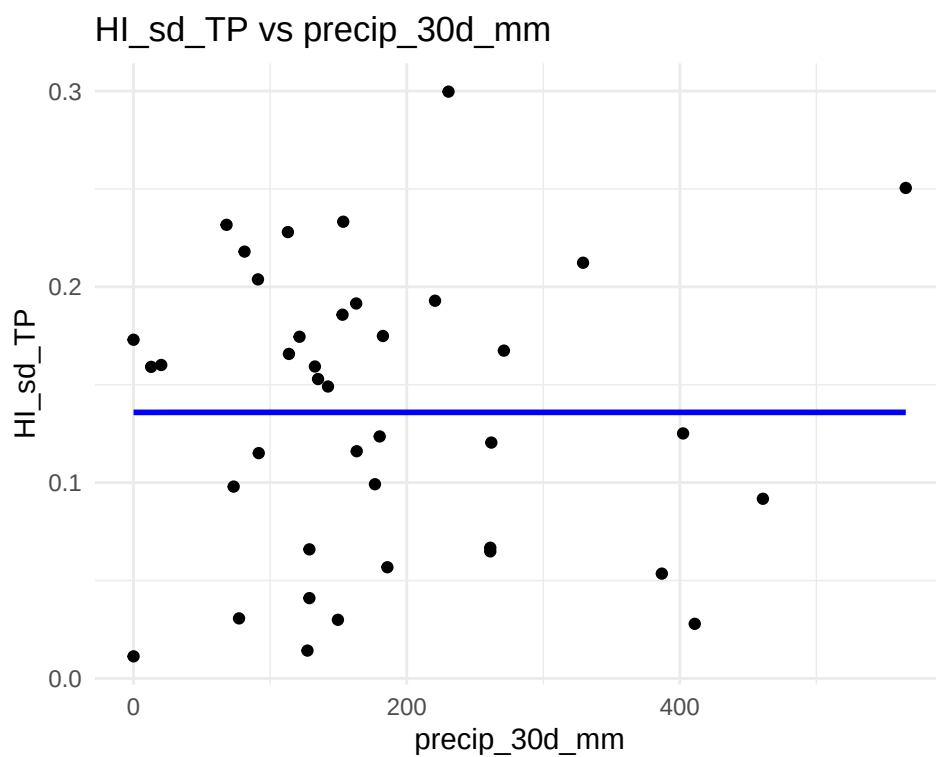


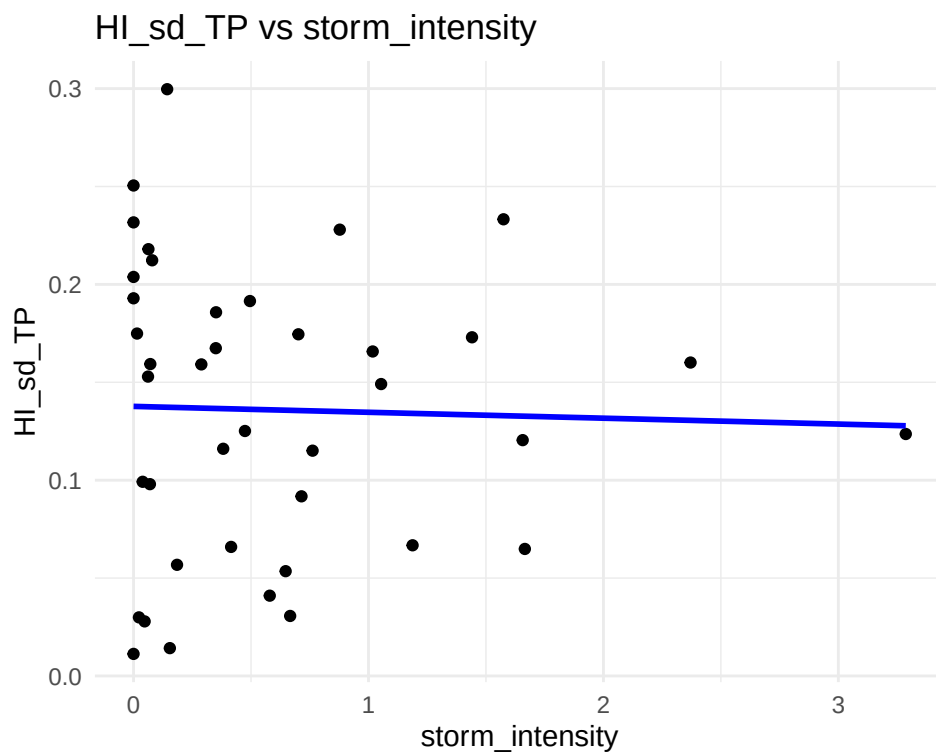
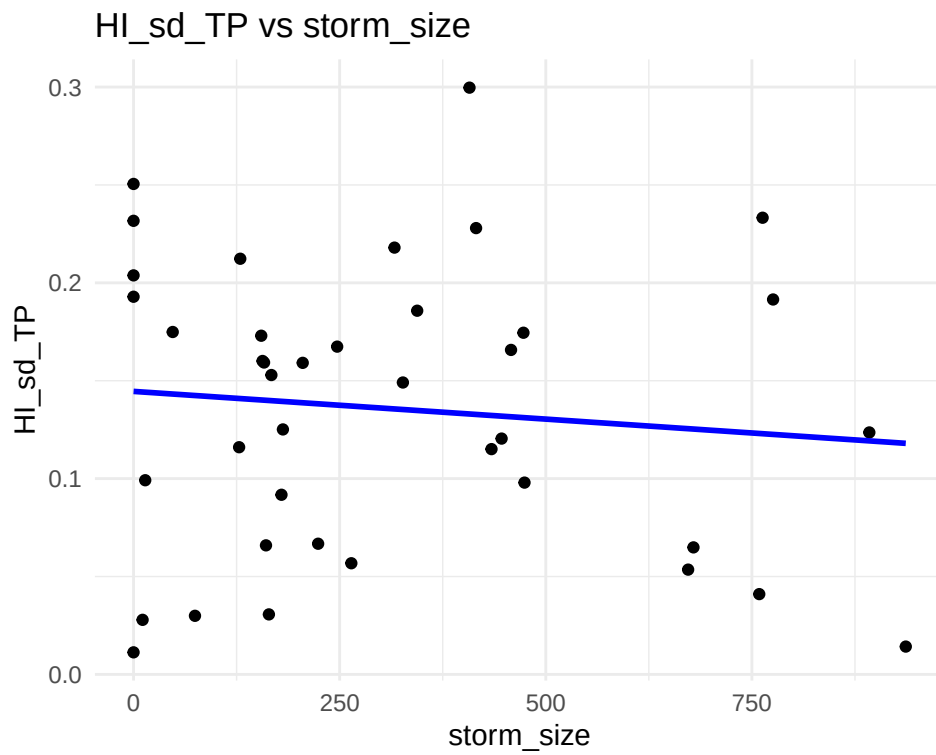




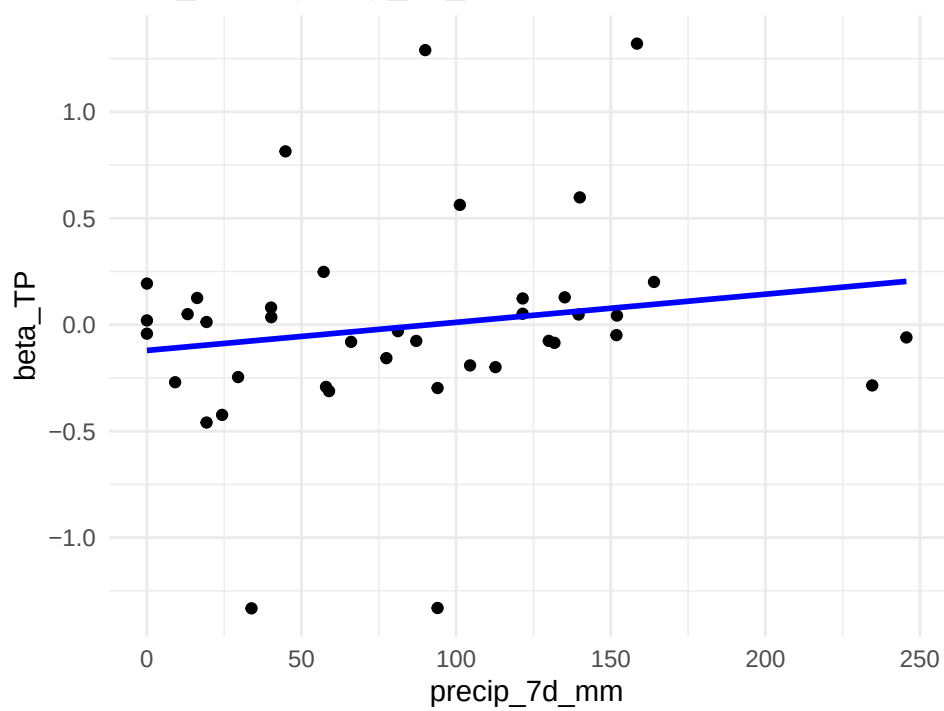




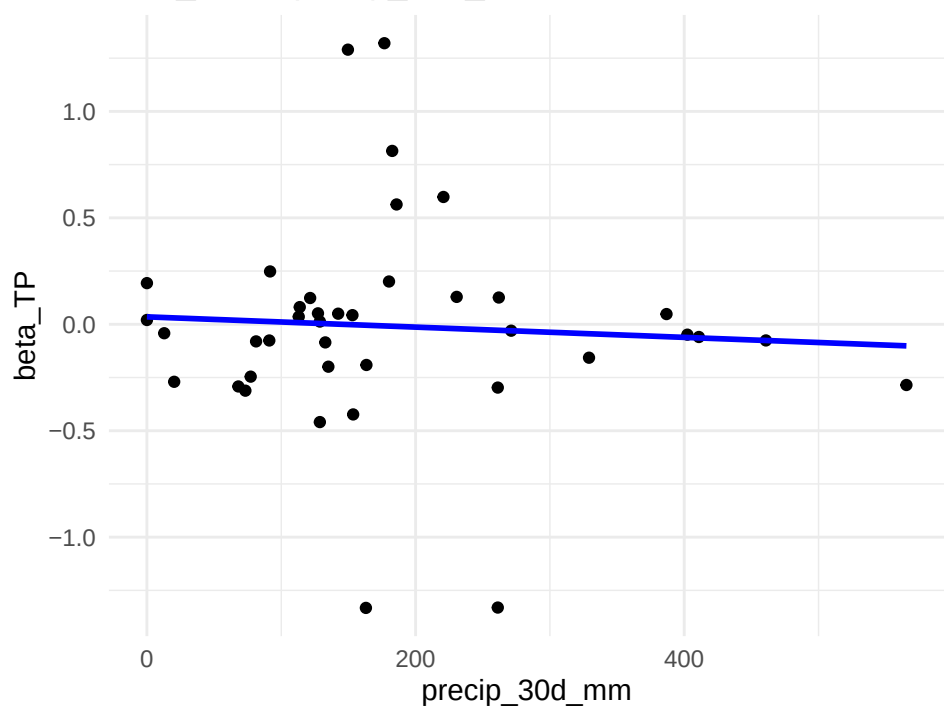


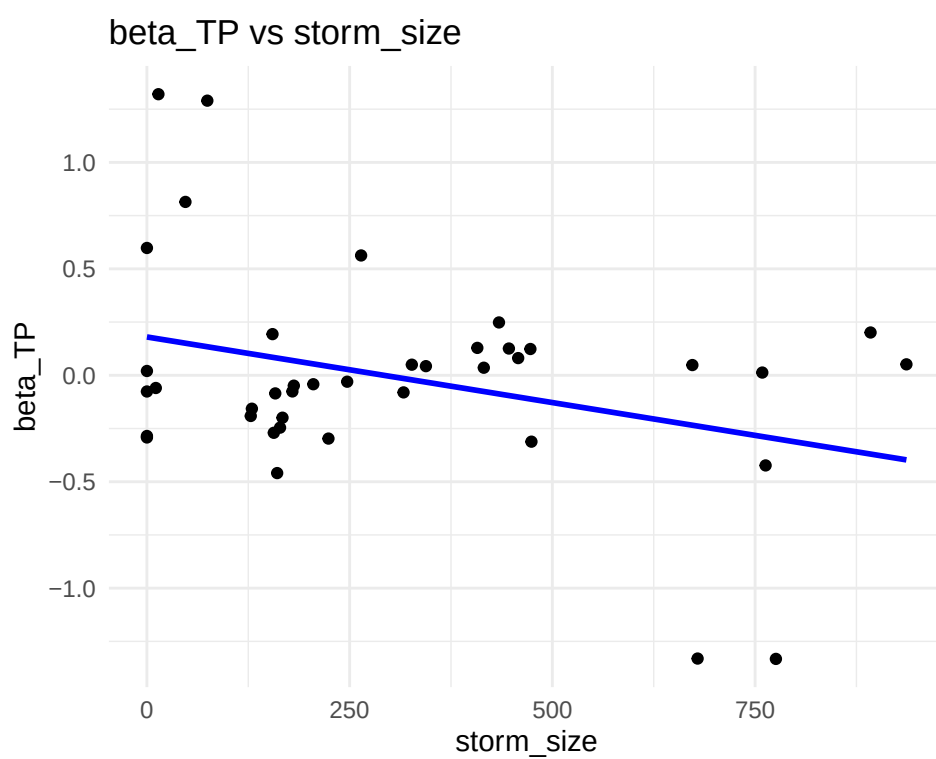
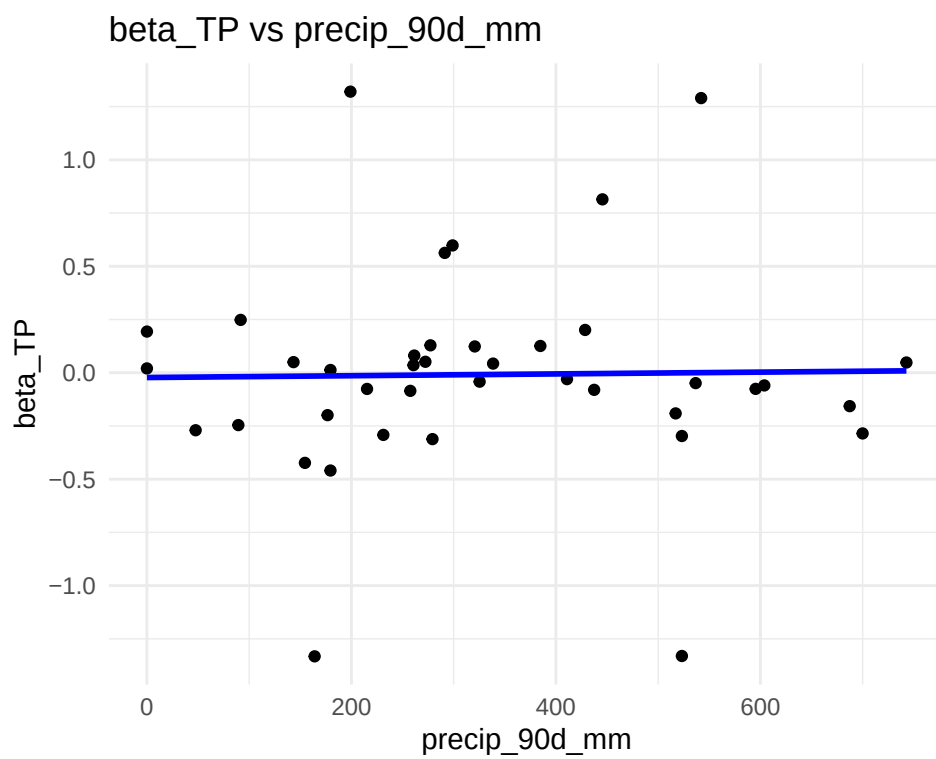


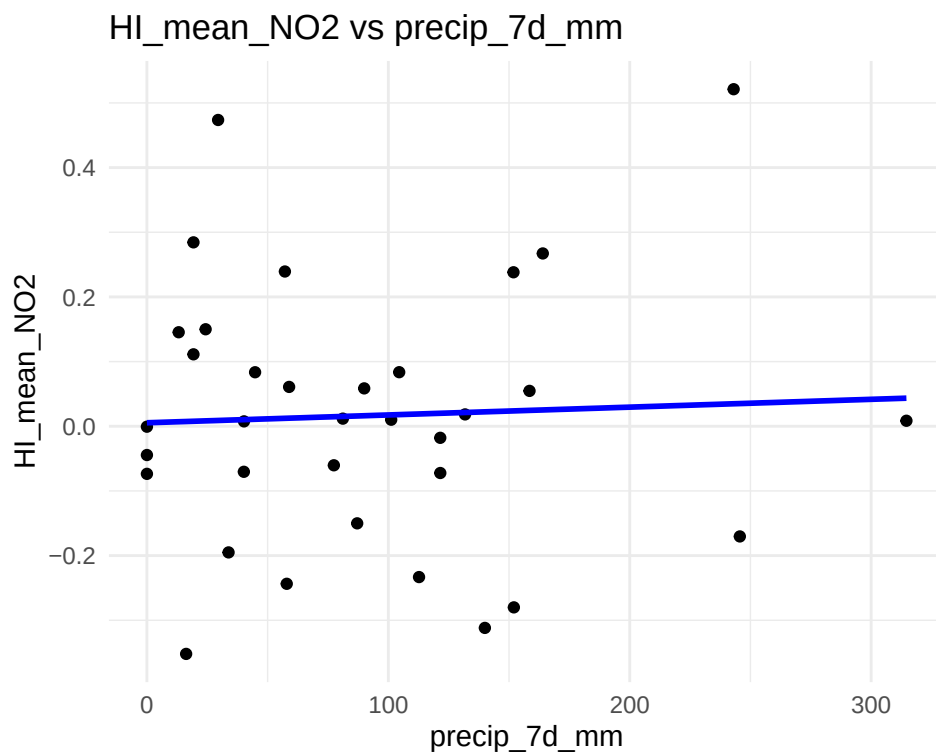
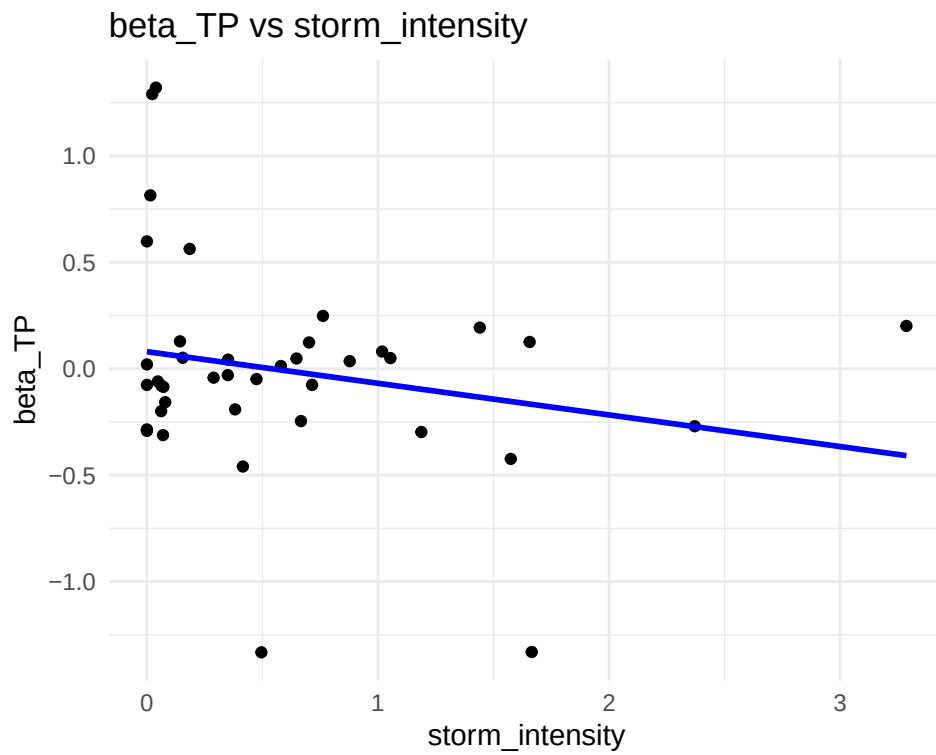
beta_TP vs precip_7d_mm

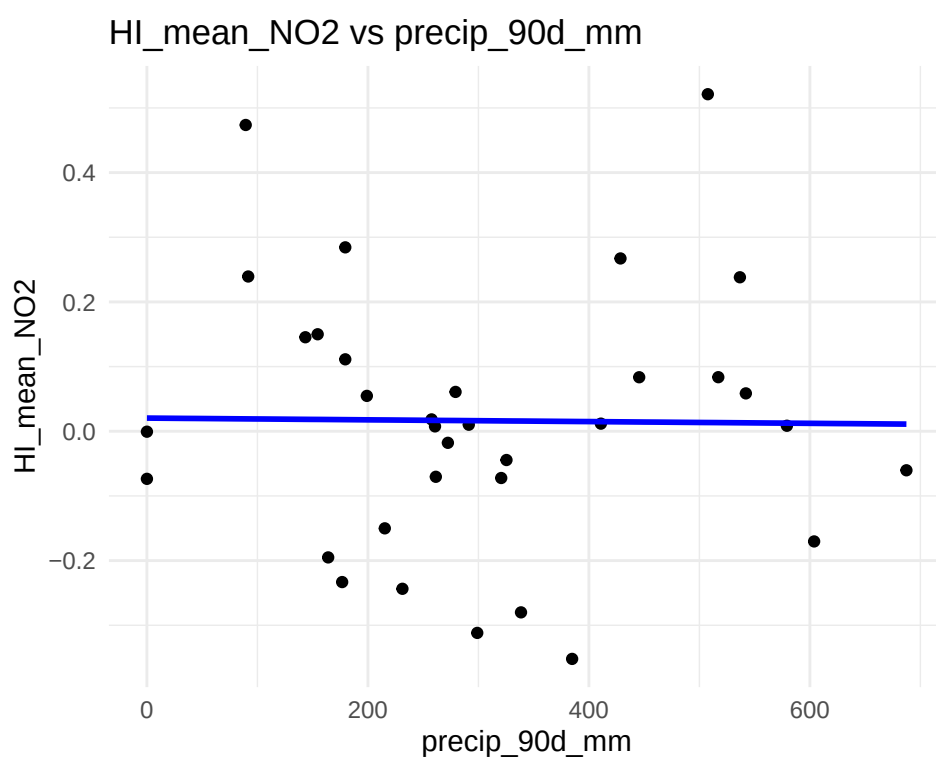
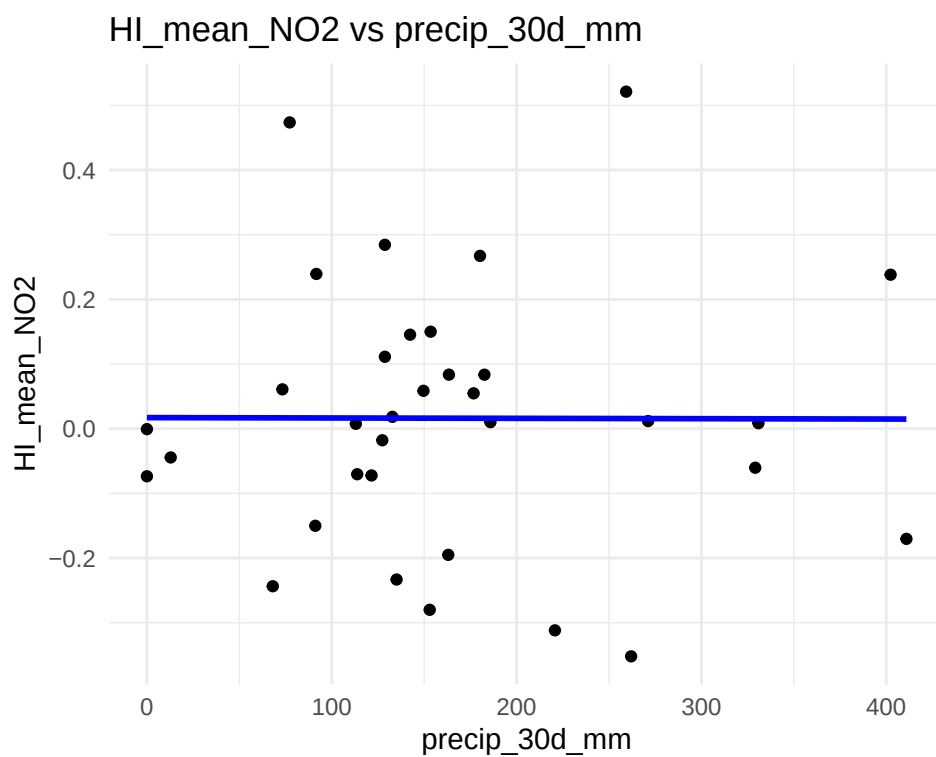


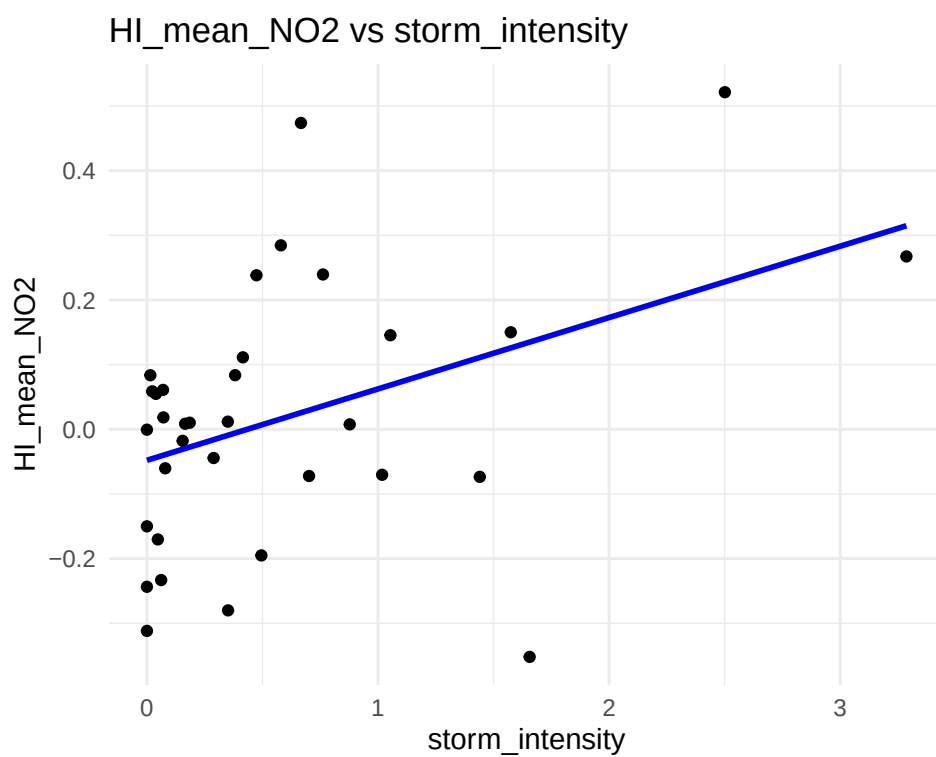
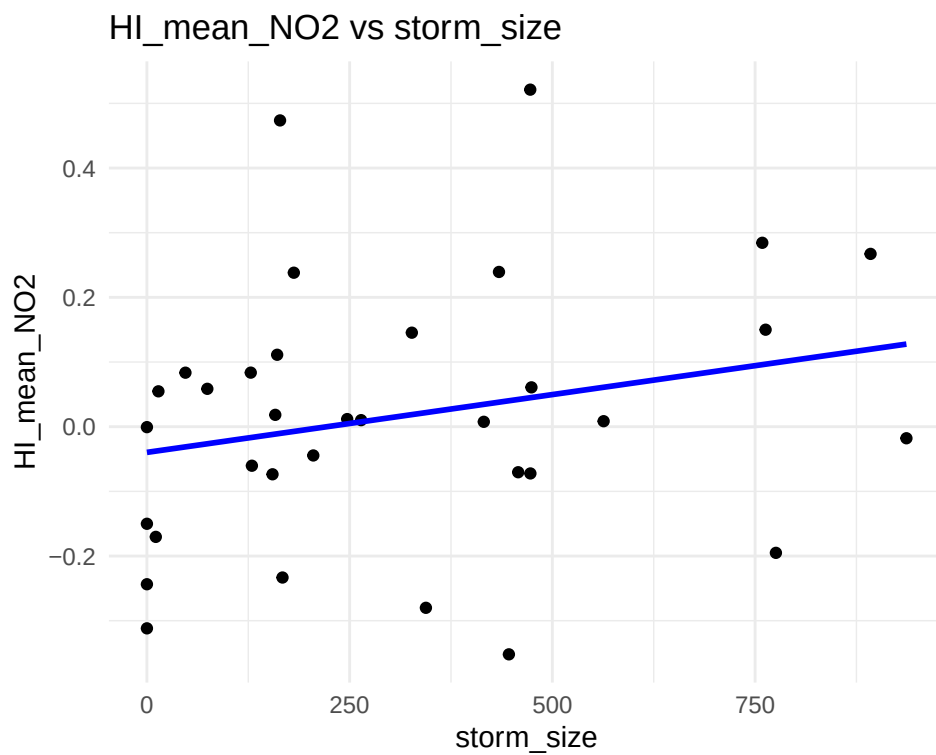
beta_TP vs precip_30d_mm

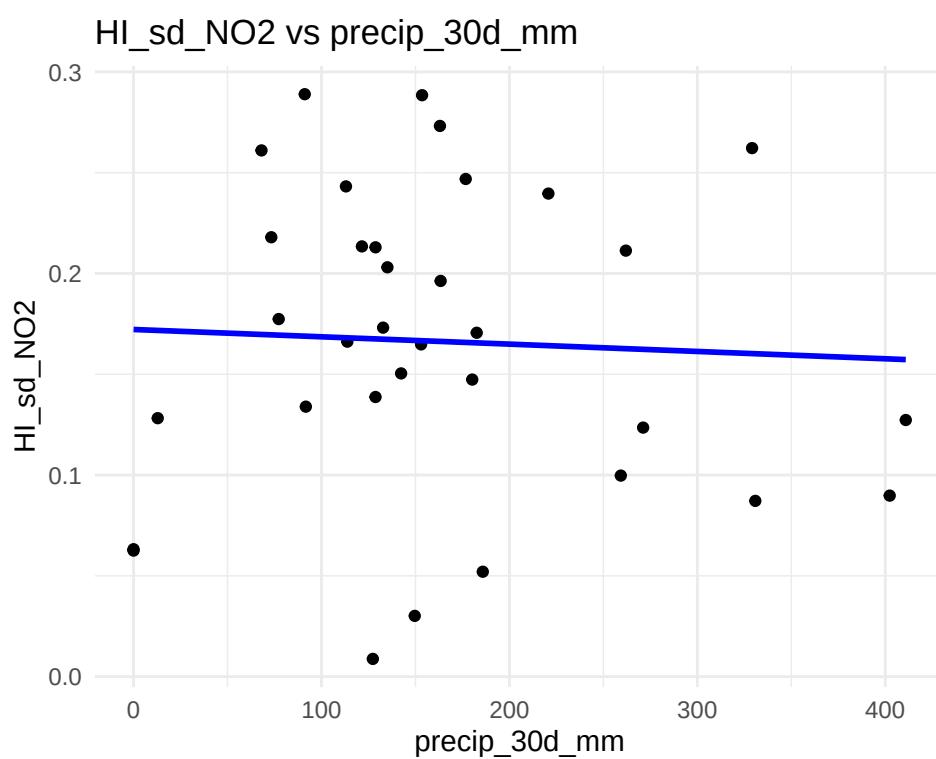
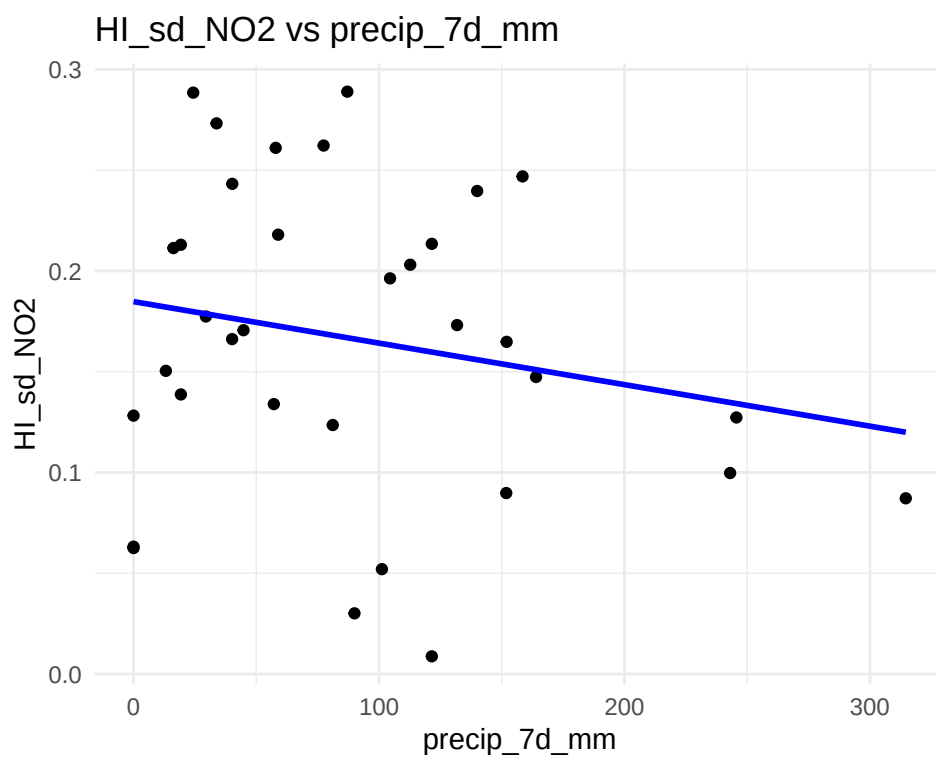


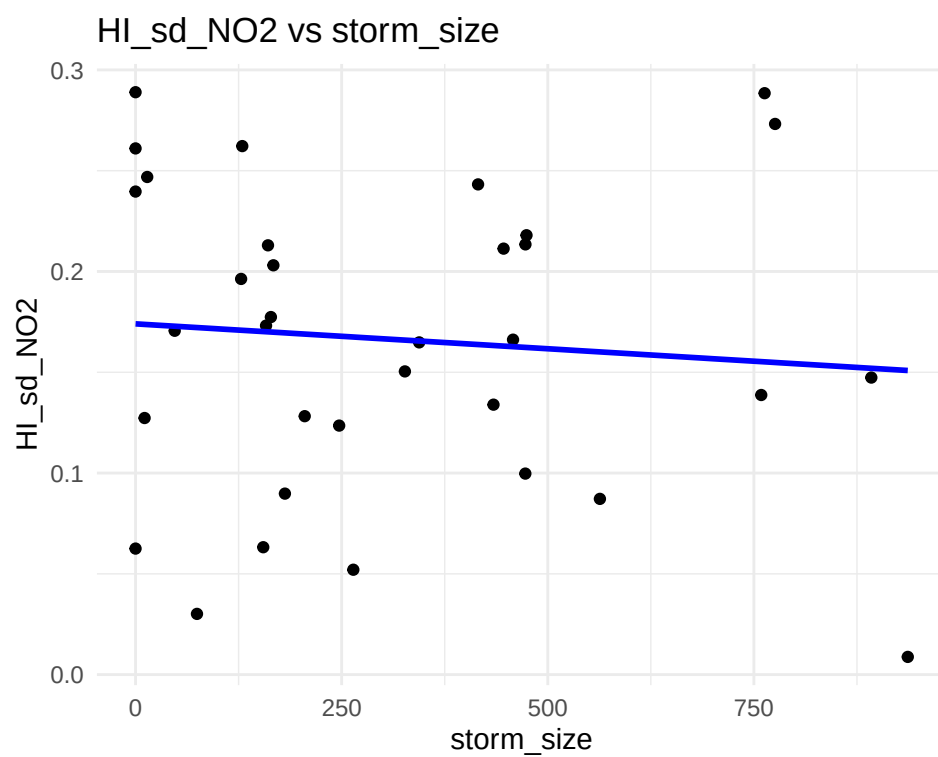
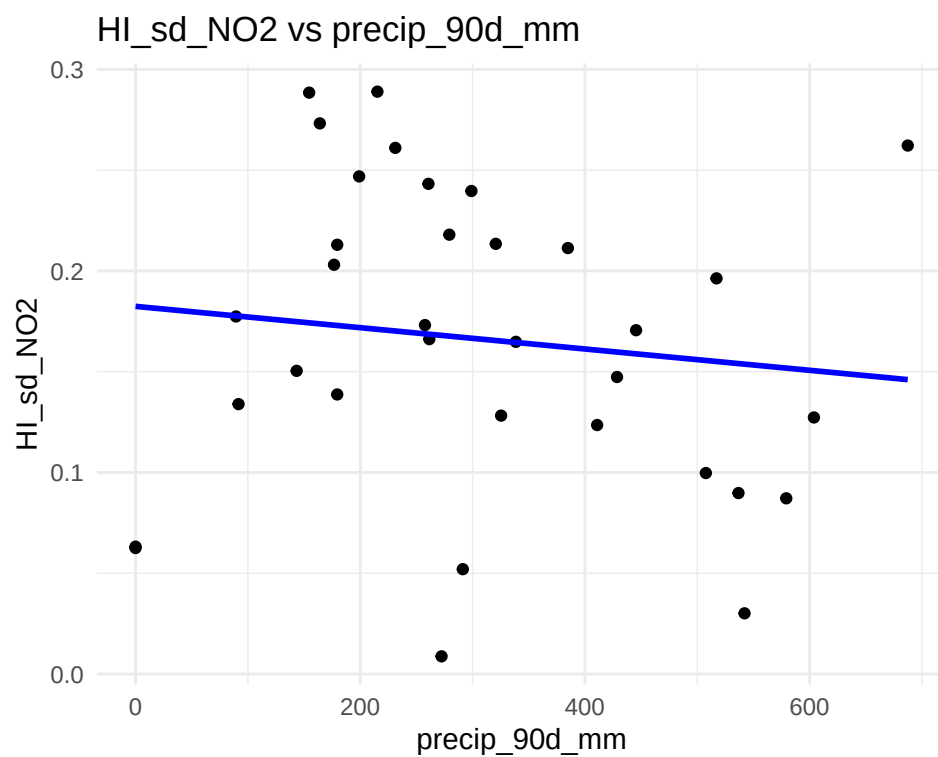


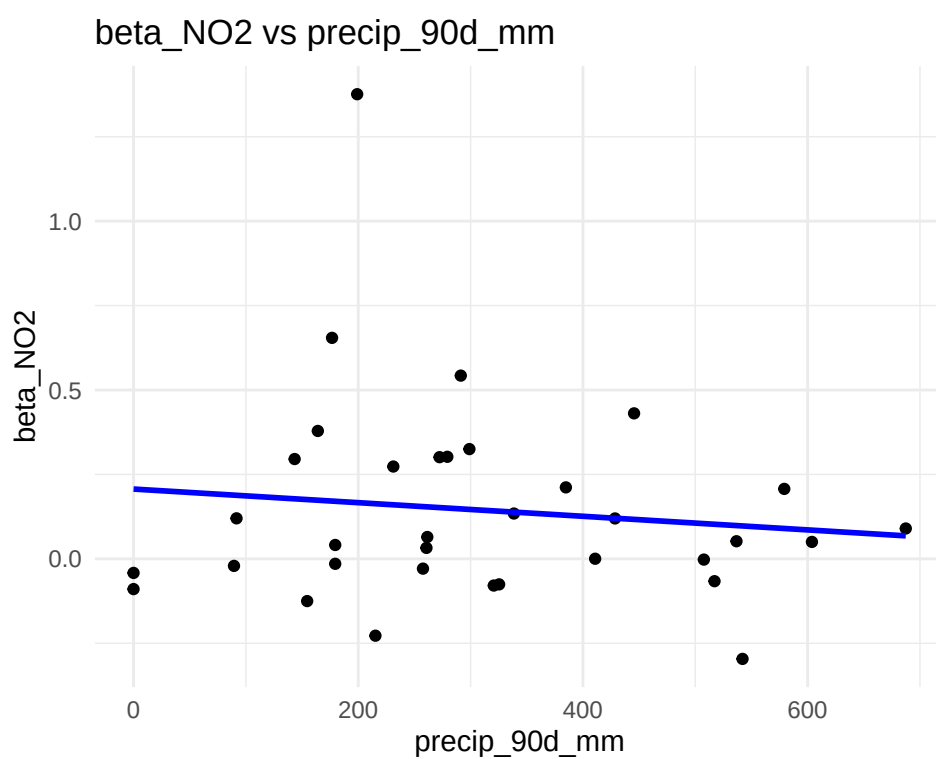
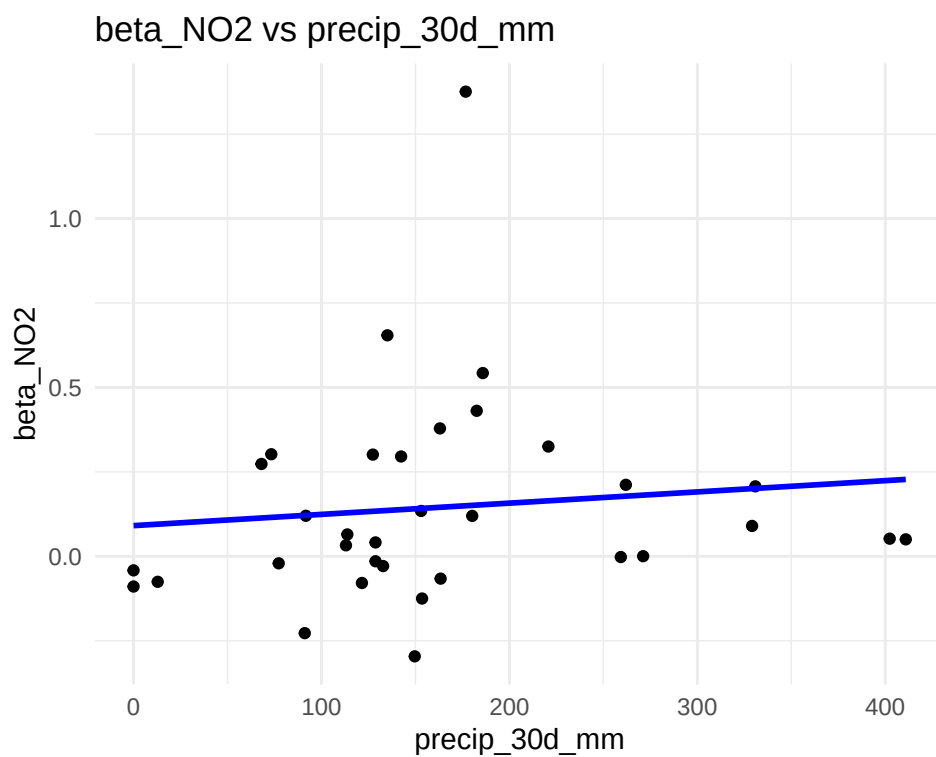


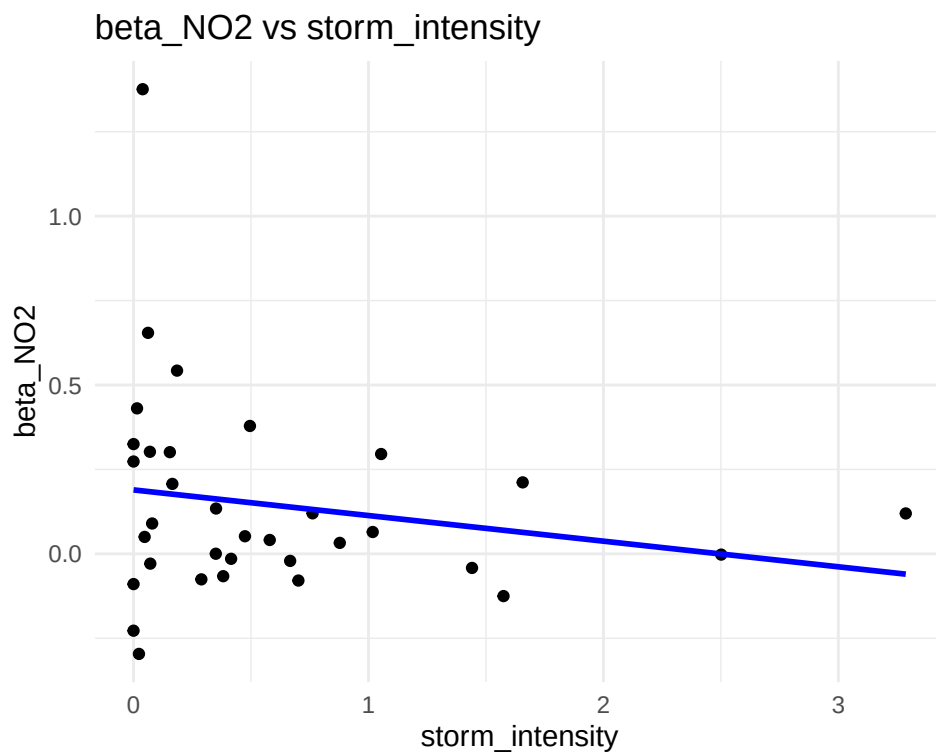
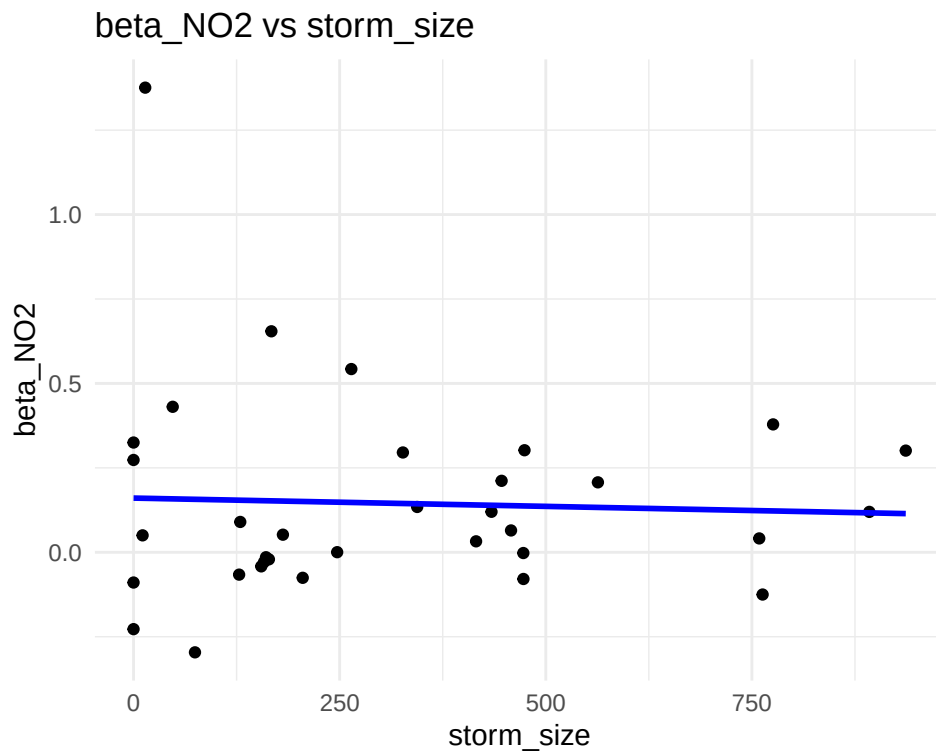




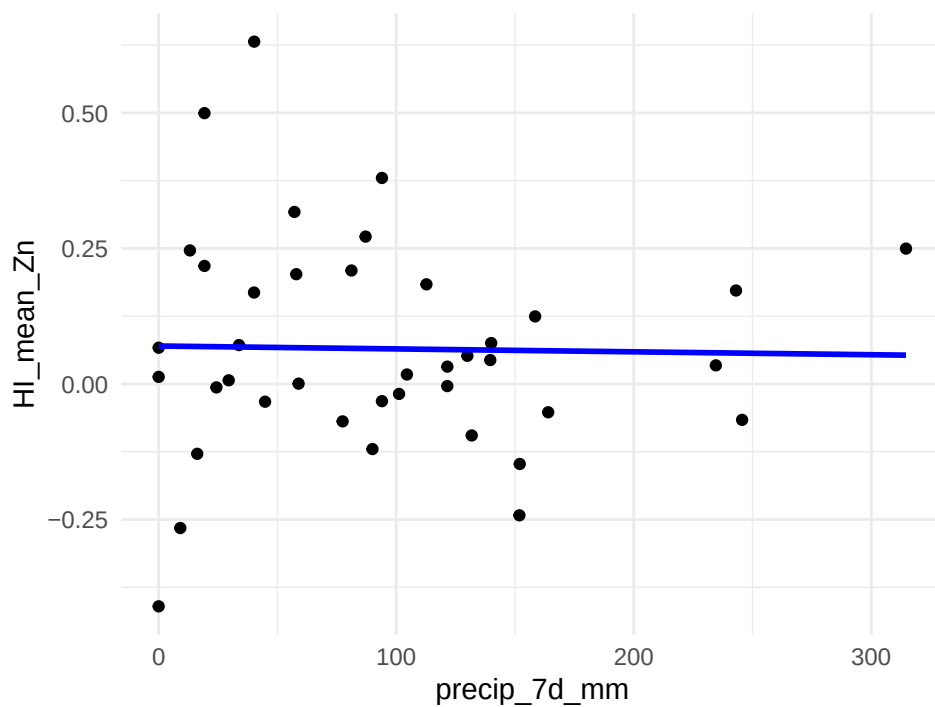




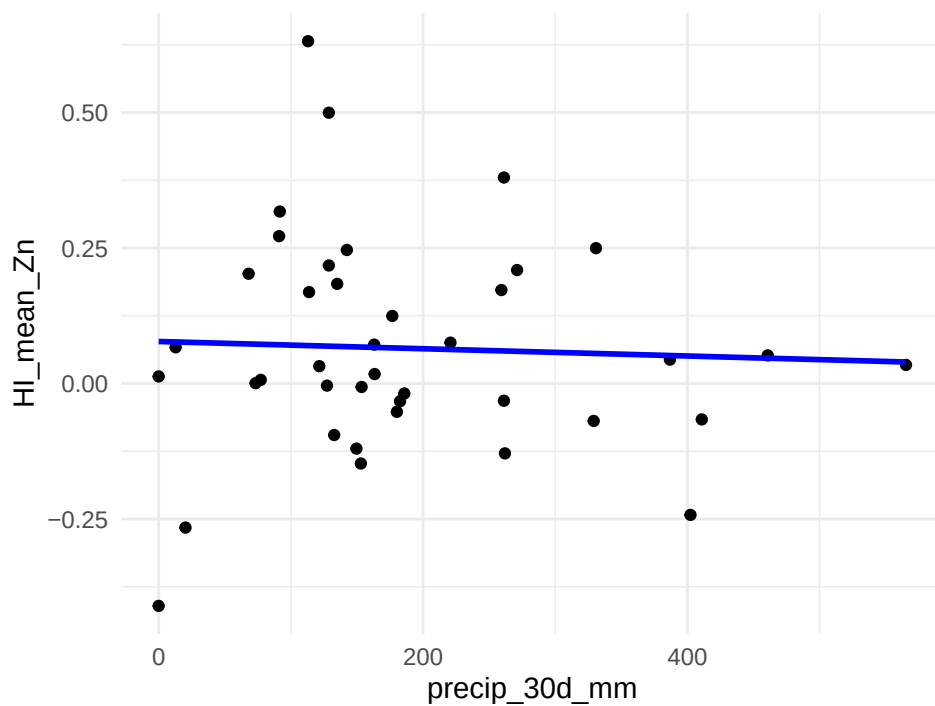




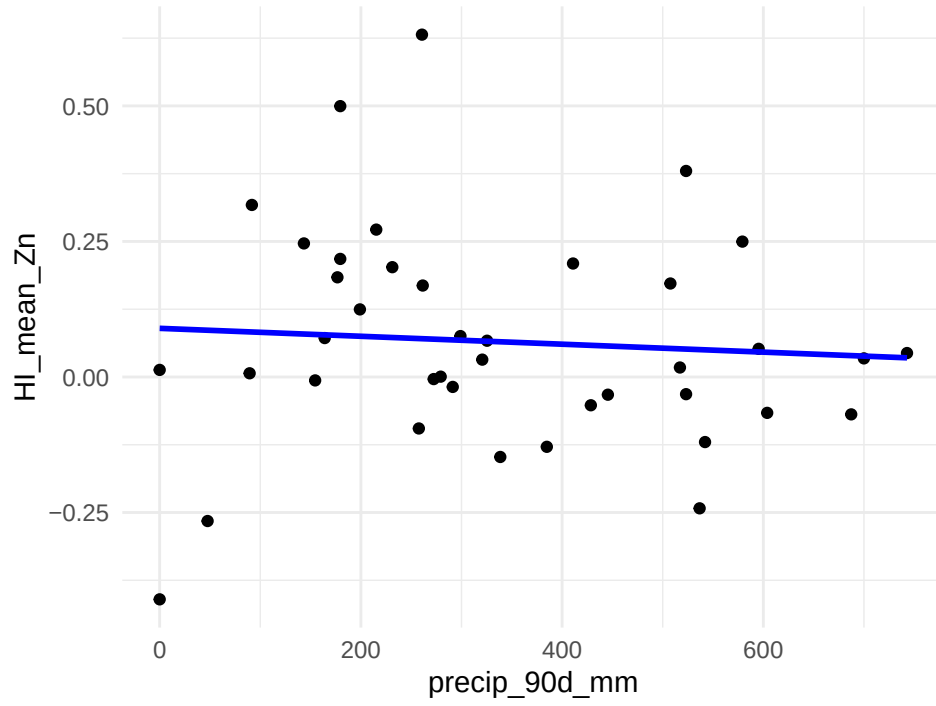
HI_mean_Zn vs precip_7d_mm



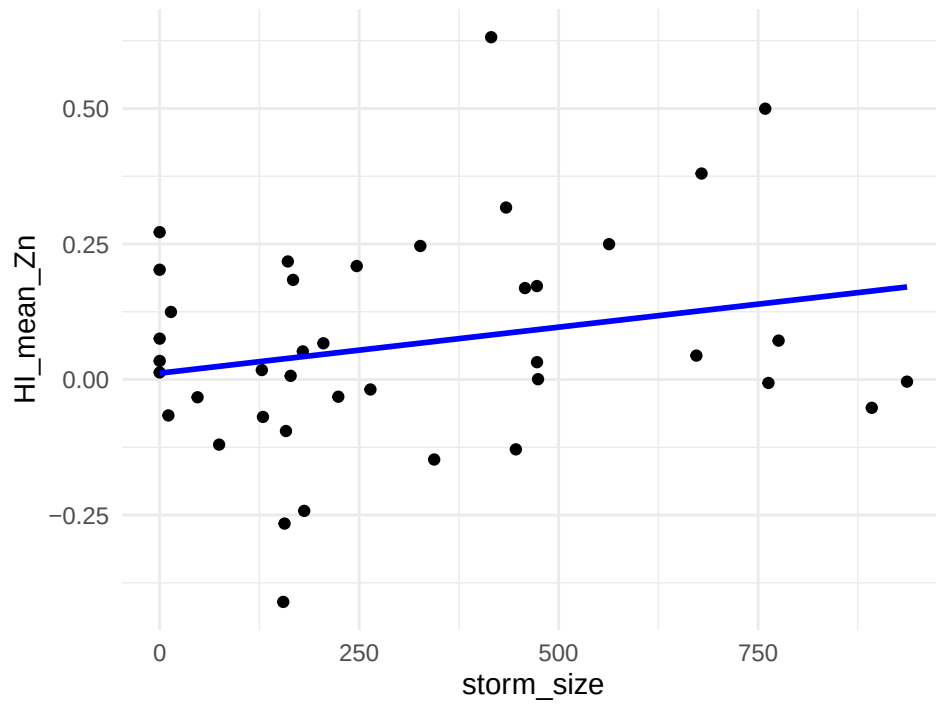
HI_mean_Zn vs precip_30d_mm

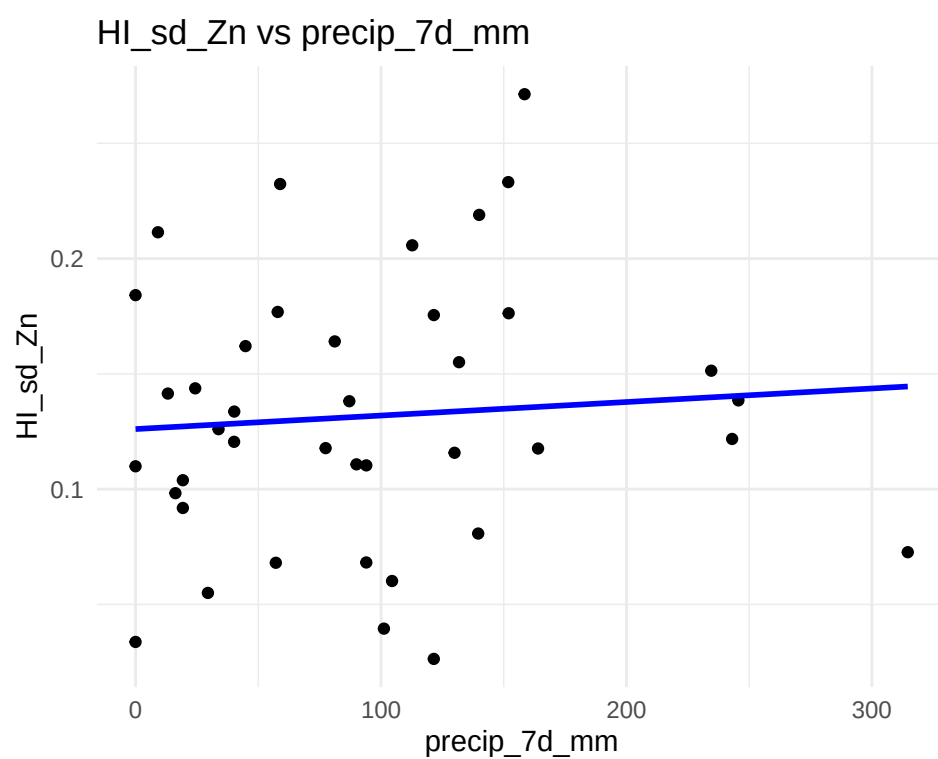
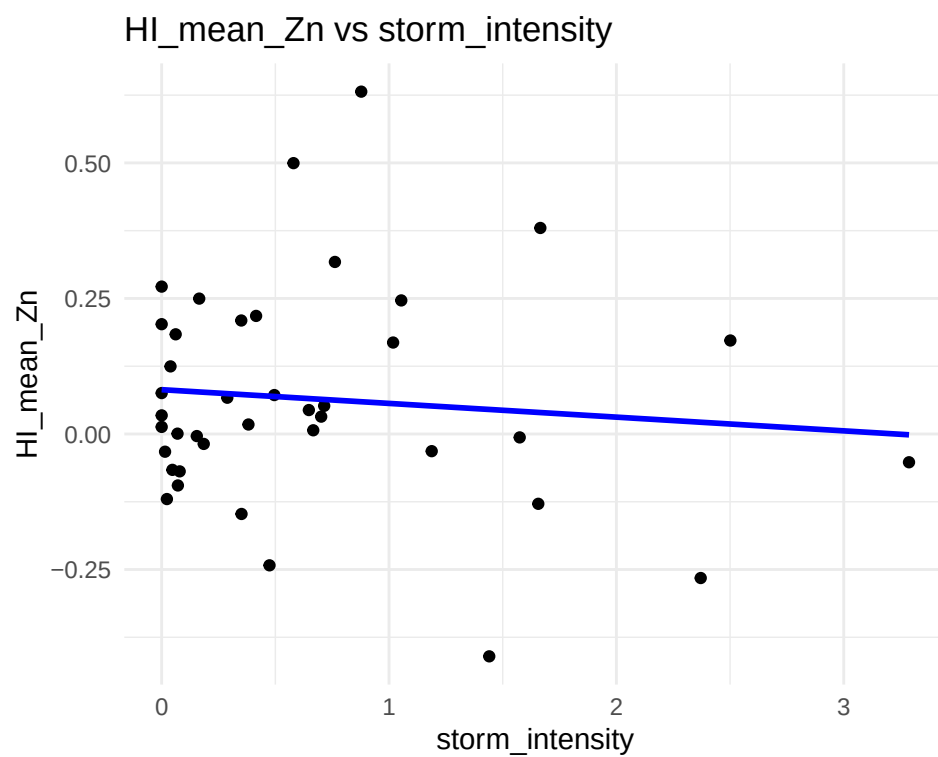


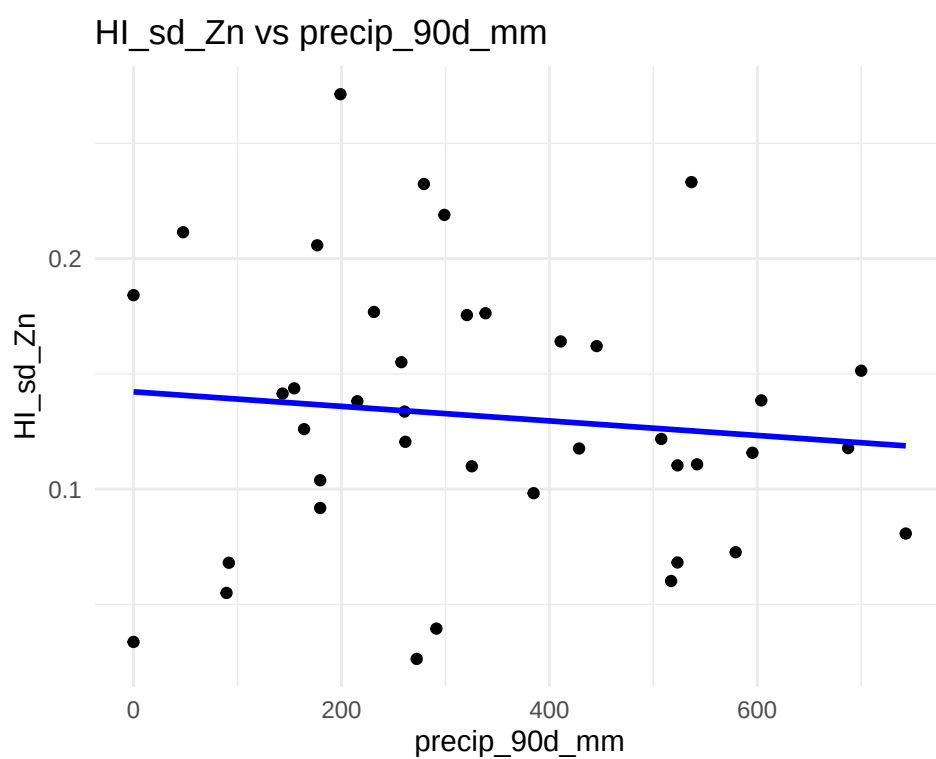
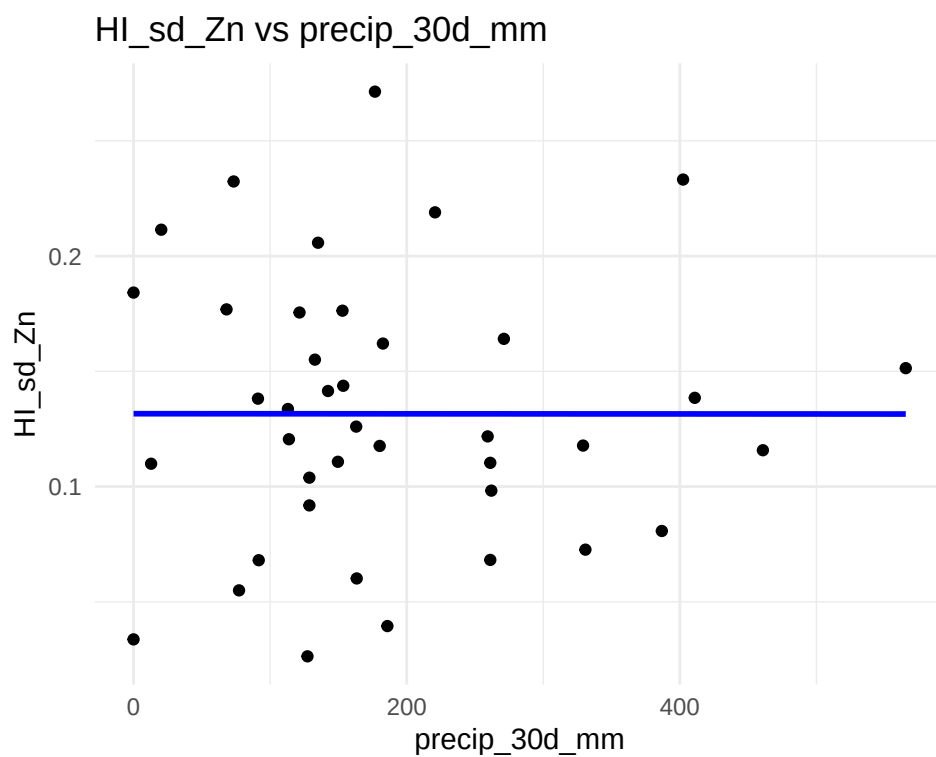
HI_mean_Zn vs precip_90d_mm



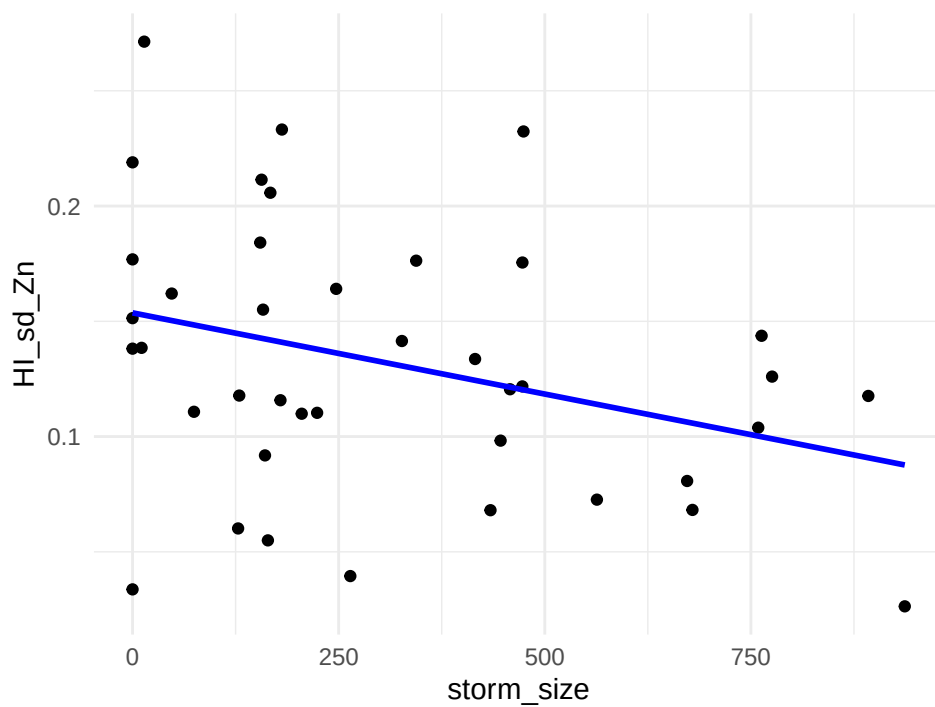
HI_mean_Zn vs storm_size



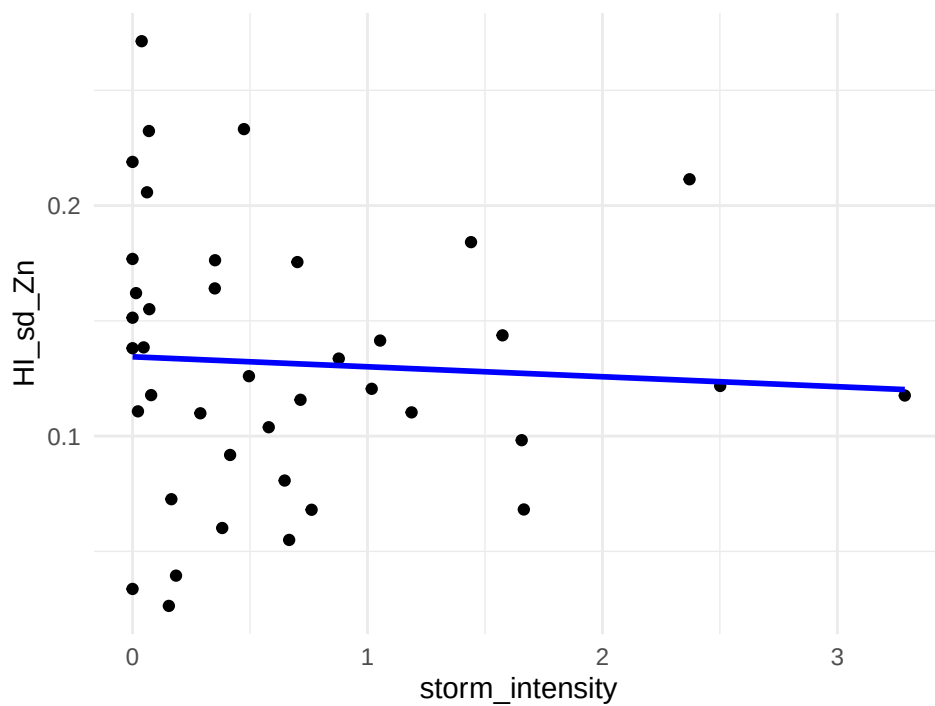


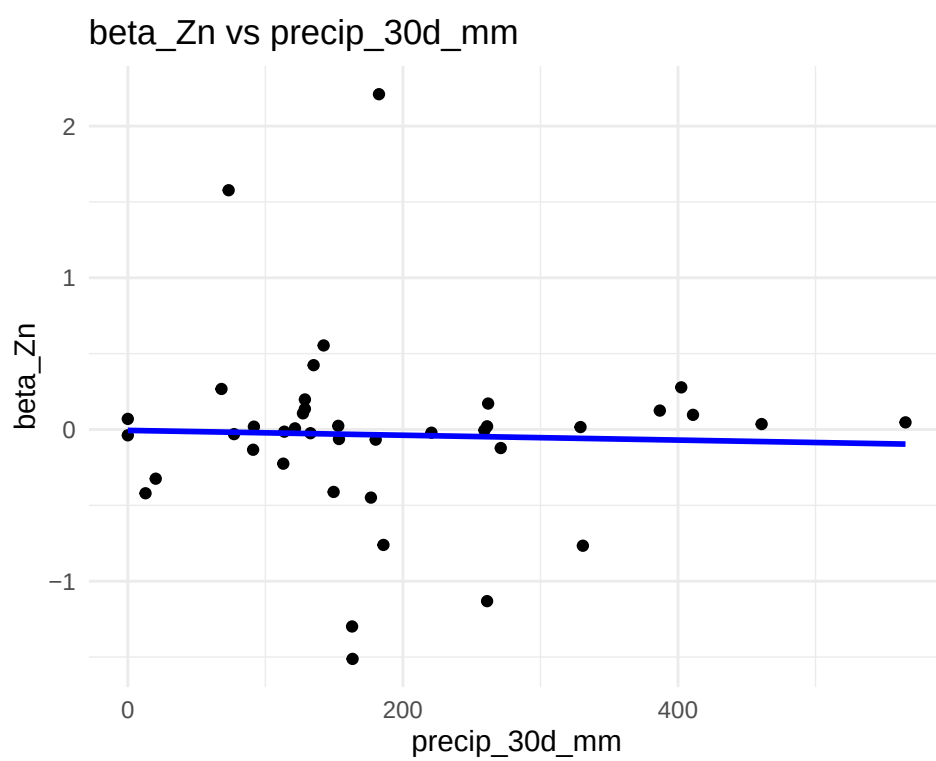
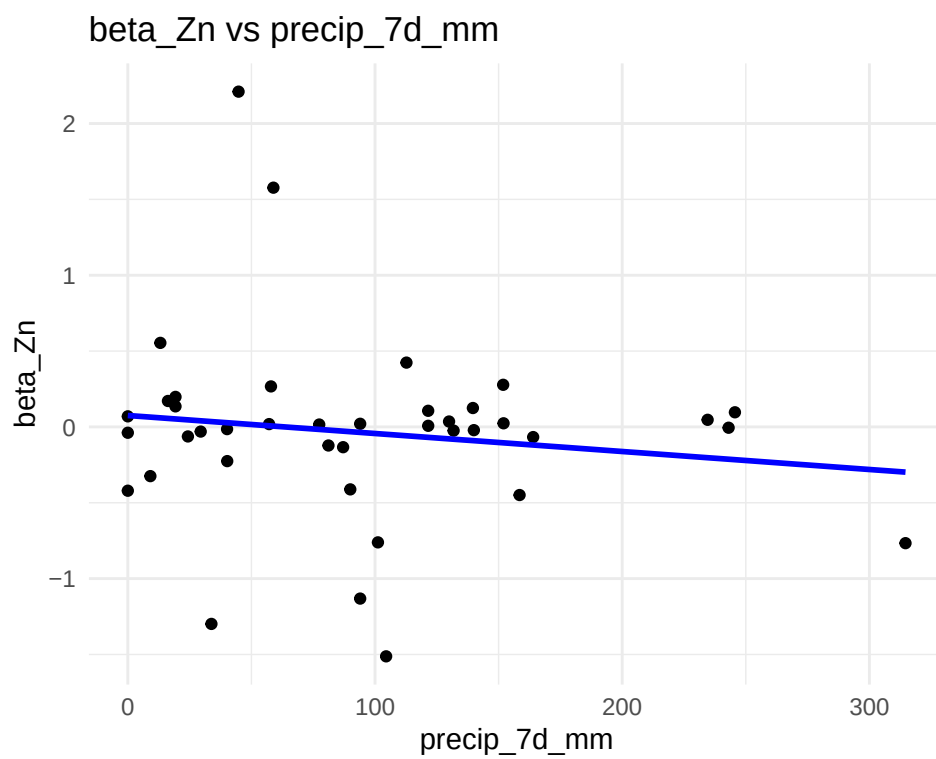


HI_sd_Zn vs storm_size

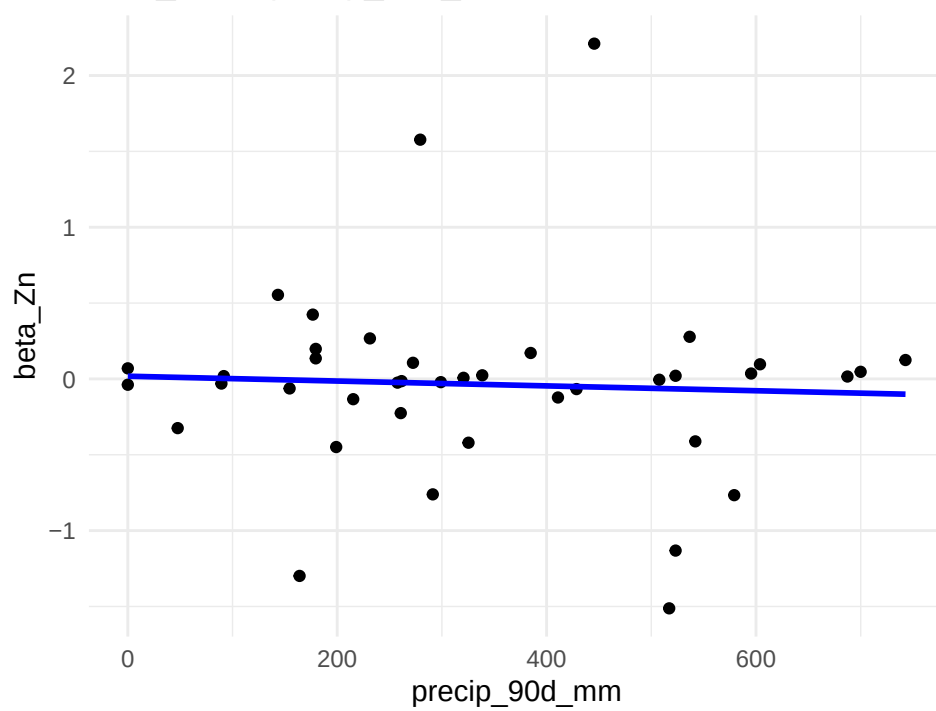


HI_sd_Zn vs storm_intensity

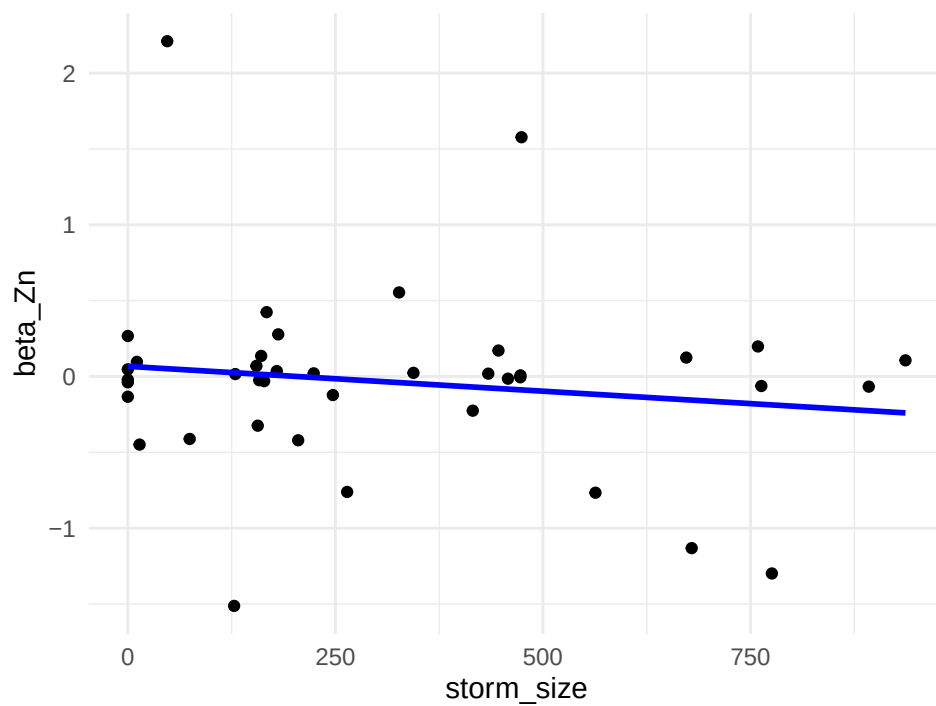


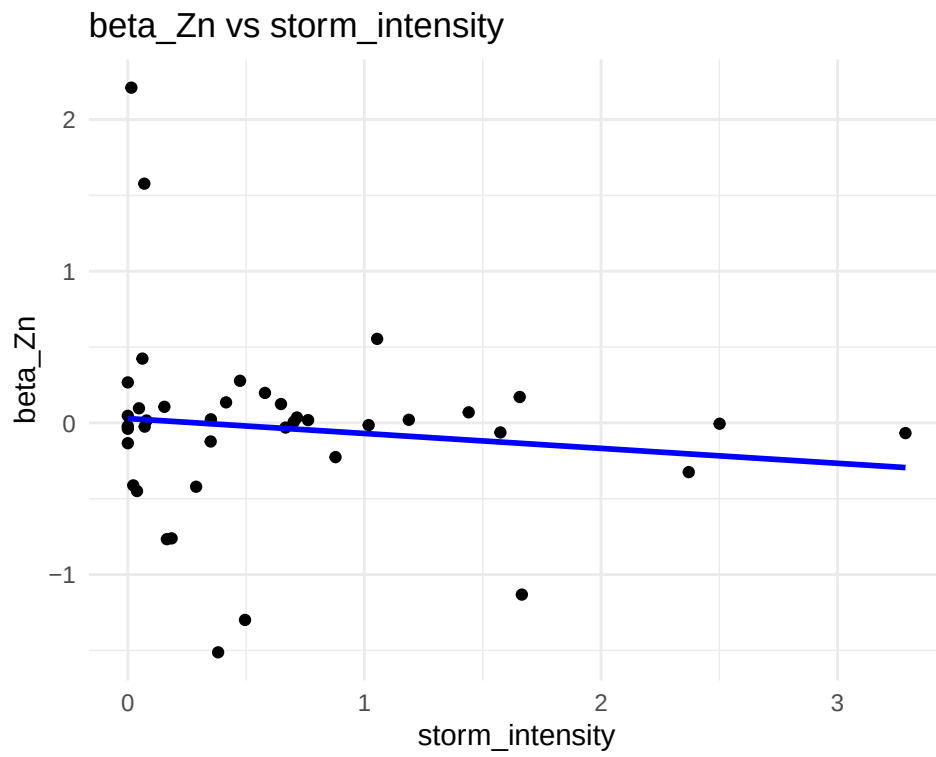


beta_Zn vs precip_90d_mm



beta_Zn vs storm_size



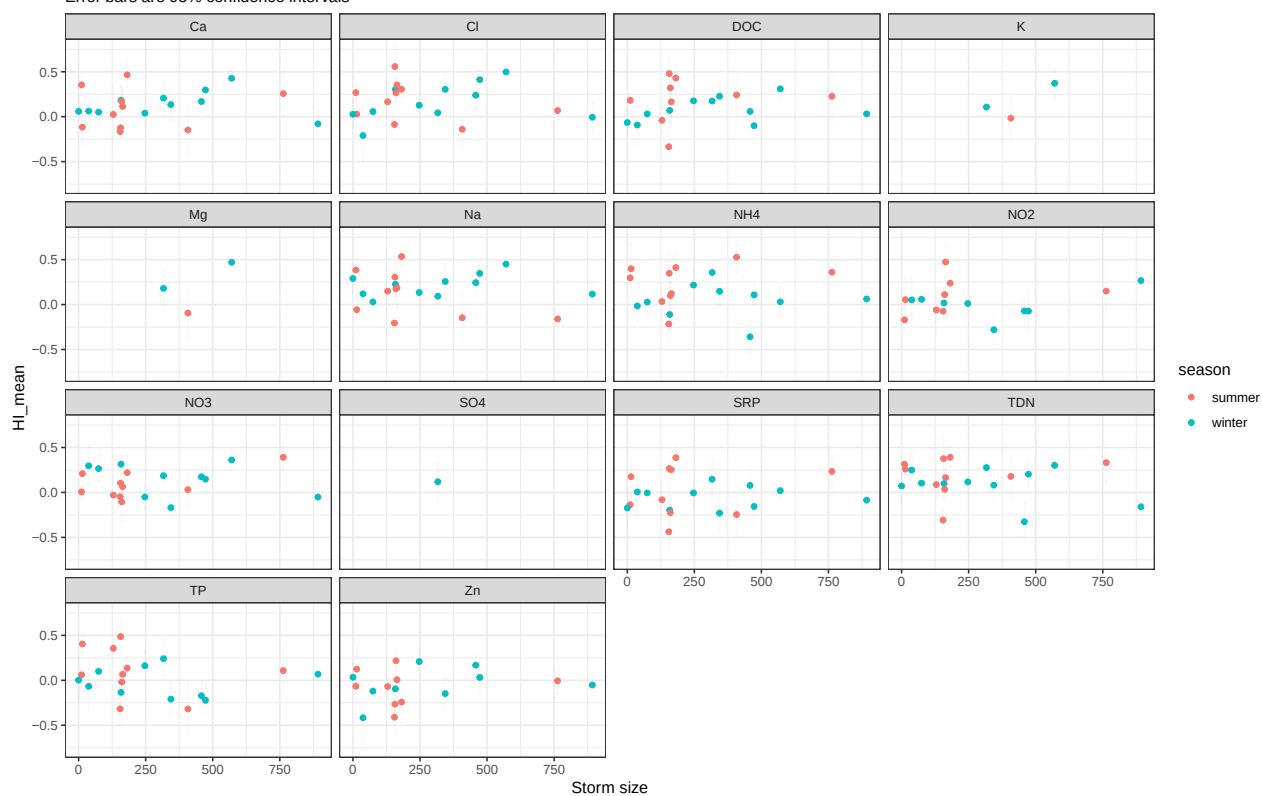


Precip and storm metrics

Storm size = cumulative precipitation (mm) per storm

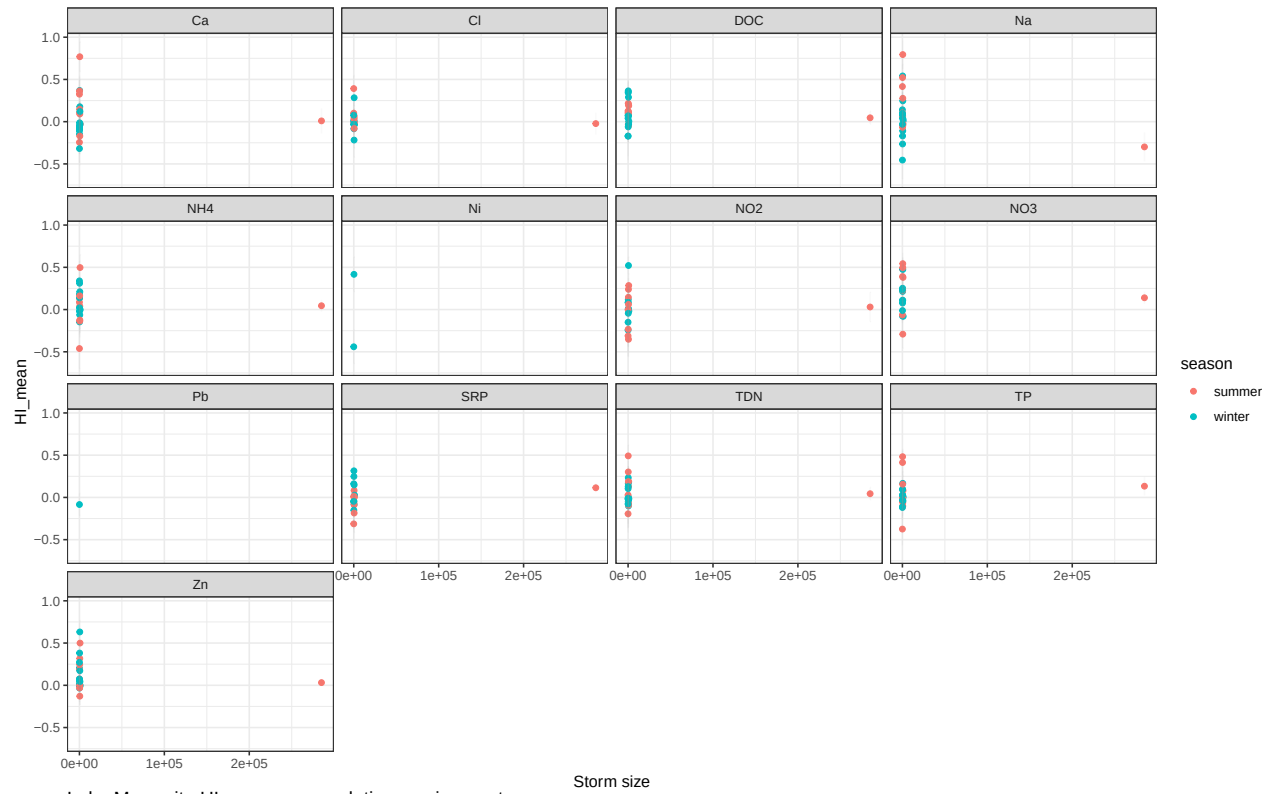
Curry HI mean vs cumulative precip per storm

Error bars are 95% confidence intervals



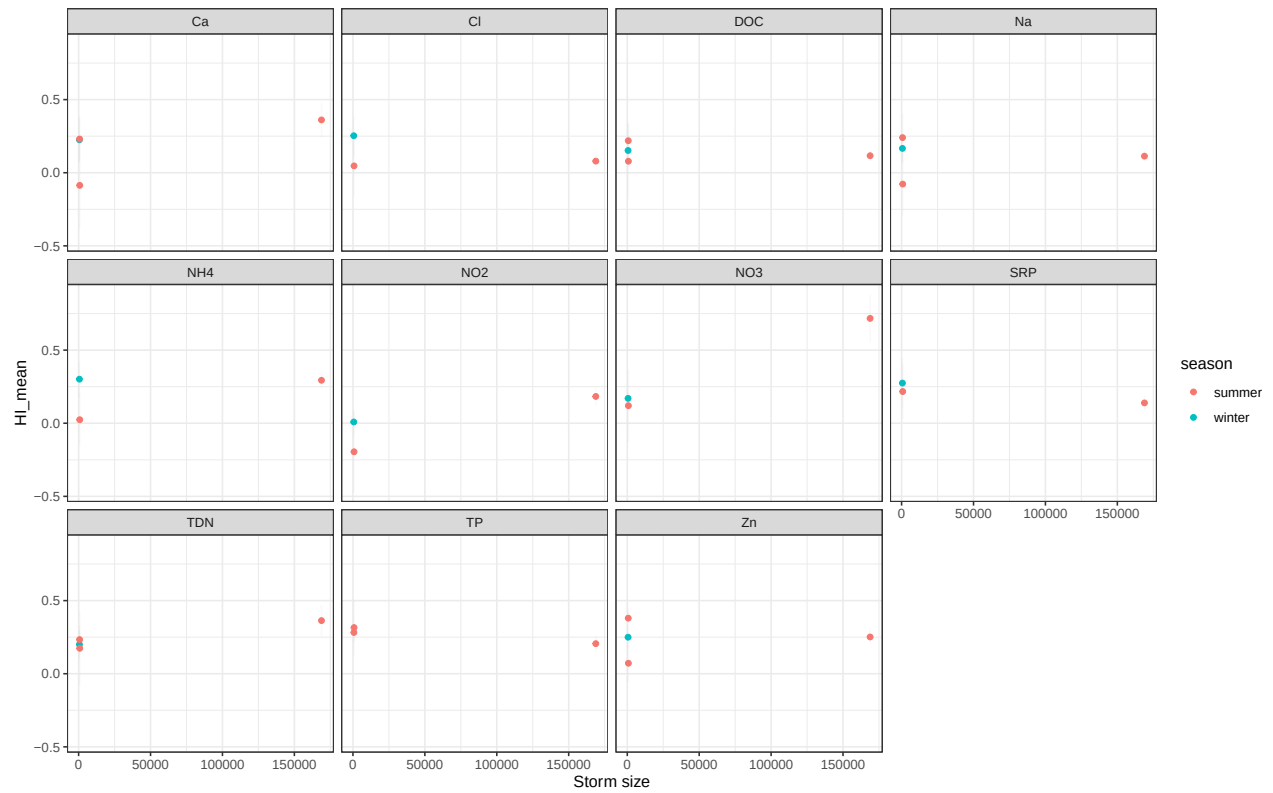
Silverado HI mean vs cumulative precip per storm

Error bars are 95% confidence intervals



Lake Margarita HI mean vs cumulative precip per storm

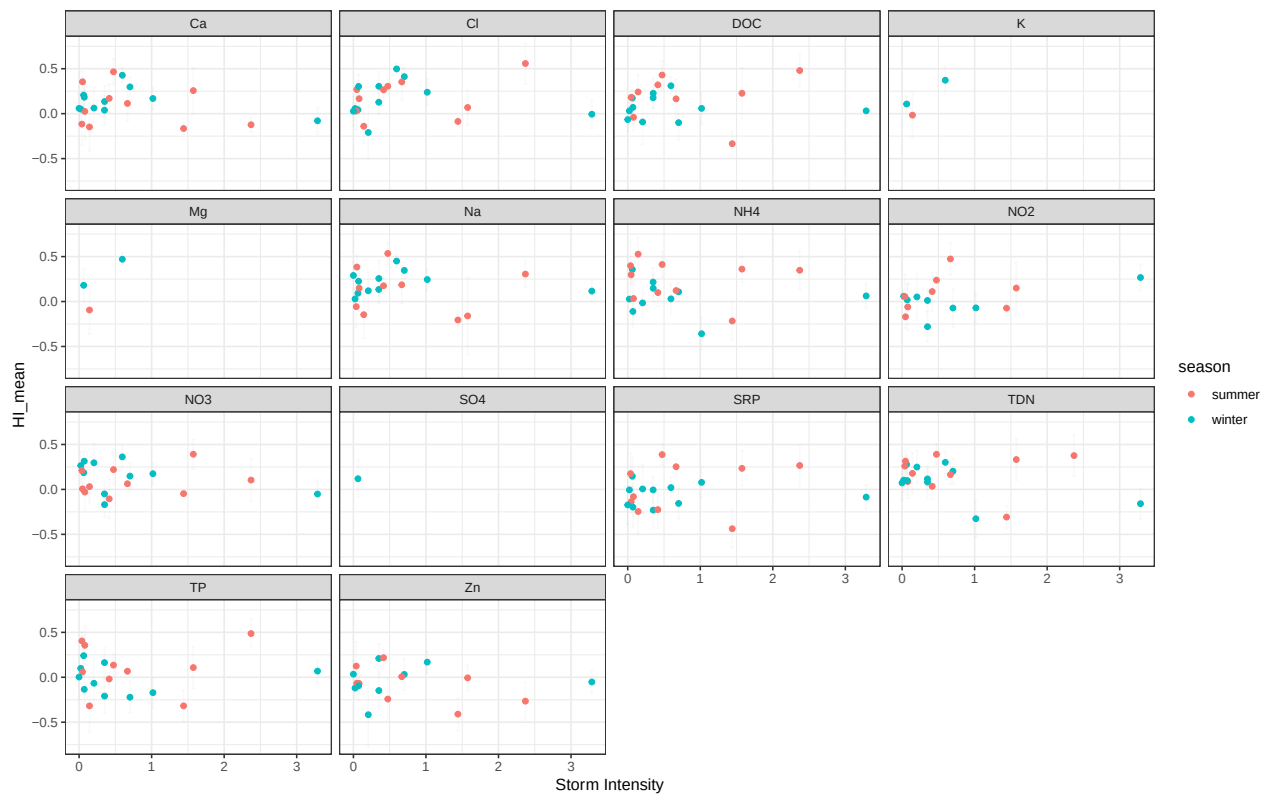
Error bars are 95% confidence intervals



Storm intensity = cumulative precipitation/storm duration

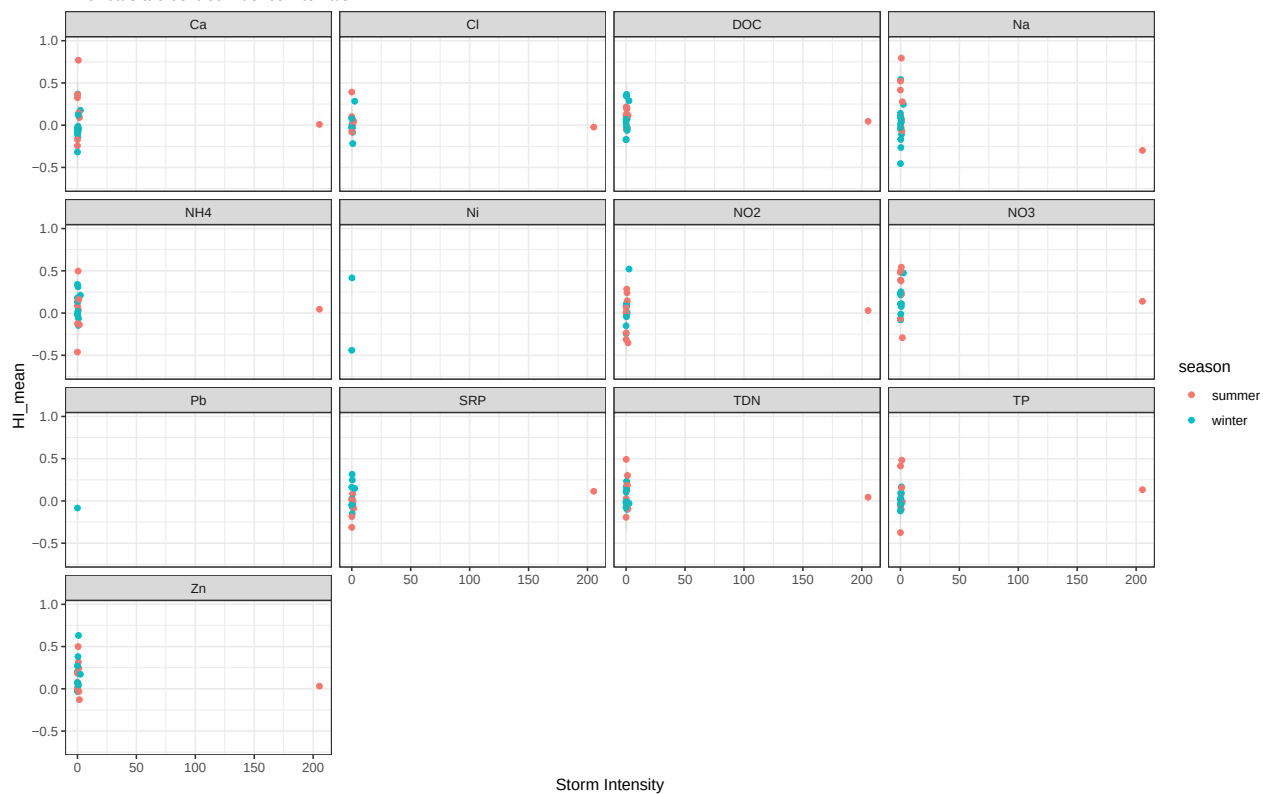
Curry HI mean vs storm intensity

Error bars are 95% confidence intervals



Silverado HI mean vs storm intensity

Error bars are 95% confidence intervals



Lake Margarita HI mean vs storm intensity

Error bars are 95% confidence intervals

